

———— // ZPWR-SYNTH // ————

zpwr-synth

v0.2.12

MODULE REFERENCE

wire your own voice from modular oscillators and filters

2026-06-20

MenkeTechnologies

———— MENKETECHNOLOGIES ————

Signal Sources

External sources you can cable into any block input or modulation route.

1 In L 2 In R 3 Noise Note Gate Velocity Mod Wheel Pitch Bend Aftertouch MPE Press MPE Slide MPE Bend Trigger

Oscillator

134 modules

Osc 3 inputs

Virtual-analog oscillator: saw/square/tri with PWM, sub and unison detune.

Param	Range	Default	Unit
Wave	Sine · Saw · Square · Tri · Ramp · Trapezoid · Rect Sine · Staircase · Soft Sq		—
Octave	-3 – 3	0	—
Fine	-100 – 100	0	ct
PW	0.05 – 0.95	0.5	—
Voices	1 – 11	1	—
Detune	0 – 50	15	ct
Phase	0 – 1	0	—
Retrig	0 – 1	0	—

Operator 3 inputs

Single FM operator: a sine tuned to Ratio times the note, phase-modulated by the in1 input (scaled by FM) plus its own Feedback - the patchable DX-style building block you chain to build any FM algorithm.

Param	Range	Default	Unit
Ratio	0.25 – 16	1	—
Fine	-100 – 100	0	ct
Level	0 – 1	1	—
Feedback	0 – 1	0	—
FM	0 – 8	1	—

FormantOsc 3 inputs

Vowel oscillator: a sawtooth driven through two resonant formant band-passes whose centre frequencies morph across the A-E-I-O-U vowels (Vowel), giving a singing/talking tone. Unlike FOF granular synthesis this is a buzz-through-resonators model; Res sets formant sharpness and Level the output.

Param	Range	Default	Unit
Vowel	A · E · I · O · U		—
Octave	-3 – 3	0	—
Fine	-100 – 100	0	ct
Res	1 – 20	8	—
Level	0 – 1	0.8	—

Vosim 3 inputs

VOSIM (VOcal SIMulation) oscillator: each fundamental period emits a burst of Pulses sin-squared pulses at FRatio times the pitch, each one Decay-scaled, then silence until the next period - the 1970s Kaegi-Tempelaars vocal/formant technique. FRatio places the formant region (vowel colour), Pulses sets the burst length and Decay the spectral tilt. Distinct from FOF granular and resonator formant synthesis; output is a unipolar pulse train.

Param	Range	Default	Unit
Octave	-3 – 3	0	—
Fine	-100 – 100	0	ct
FRatio	1 – 16	4	—
Pulses	1 – 16	4	—
Decay	0 – 1	0.6	—
Level	0 – 1	0.8	—

Shepard 0 inputs

Shepard-Risset tone: seven octave-spaced sine partials sweep continuously through the spectrum while a bell-shaped amplitude window fades each one in at the bottom and out at the top, producing the endless-glissando illusion of a pitch that rises (or falls) forever without ever changing register. Speed sets the glide rate, Direction up or down, Range the octave span the partials cover, and Level the output. Needs no note input - it is a self-running tone generator.

Param	Range	Default	Unit
Speed	0 – 4	0.5	oct/s
Direction	Up · Down	—	—
Range	2 – 9	7	oct
Level	0 – 1	0.6	—

Buzz 3 inputs

Buzzer / band-limited impulse train: sums Harmonics equal-amplitude harmonics of the note in closed form (the Dirichlet kernel $\sin(N\pi\text{phase})/(N\sin(\pi\text{phase}))$), giving a bright brassy pulse with a flat harmonic series and no aliasing - the classic Csound buzz generator. Harmonics sets the partial count (brightness), Octave and Fine tune it, and Level scales the output. A rich raw source to feed subtractive filters or formant shaping. The harmonic count is auto-capped so the top partial stays below Nyquist.

Param	Range	Default	Unit
Octave	-3 – 3	0	—
Fine	-100 – 100	0	ct
Harmonics	1 – 64	16	—
Level	0 – 1	0.6	—

PAF 3 inputs

Phase-Aligned Formant oscillator (IRCAM/Csound paf-style): a single formant built from a harmonic carrier centred on the Formant frequency, multiplied by a phase-aligned bell window whose width Bandwidth sets, all locked to the note's fundamental so the spectrum stays clean and clickless. Distinct from FOF (grain bursts), Vosim (sin-squared pulses) and FormantOsc (resonator-filtered saw): here one formant is a single phase-aligned carrier times window. Formant places the peak, Bandwidth its spectral width, Level the output.

Param	Range	Default	Unit
Octave	-3 – 3	0	—
Fine	-100 – 100	0	ct
Formant	100 – 4000	800	Hz
Bandwidth	0.05 – 1	0.4	—
Level	0 – 1	0.7	—

VCO 3 inputs

Analog-modeled VCO (Moog/classic voltage-controlled oscillator character): a band-limited (PolyBLEP, anti-aliased) triangle->saw->pulse morph with slow analog Drift detuning the pitch and a gentle waveshape Warmth, for the fat, slightly-unstable vintage tone

the clean base Osc does not model. Octave/Fine tune it, Shape sweeps triangle->saw->pulse, PWM sets the pulse width, Drift the analog pitch wander, Warmth a soft saturation, and Level the output. in1 = FM, in2 = pitch mod.

Param	Range	Default	Unit
Octave	-3 – 3	0	—
Fine	-100 – 100	0	ct
Shape	0 – 1	0.5	—
PWM	0.05 – 0.95	0.5	—
Drift	0 – 1	0.3	—
Warmth	0 – 1	0.2	—
Level	0 – 1	0.7	—

DCO 3 inputs

Analog-modeled DCO (Roland Juno digitally-controlled oscillator character): rock-stable pitch (no drift) with a band-limited (PolyBLEP) saw and variable-width pulse blended by Pulse, plus a square Sub oscillator one octave down - the clean, punchy Juno foundation that sits perfectly under a chorus. Octave/Fine tune it, PWM sets the pulse width, Pulse blends saw->pulse, Sub the sub-oscillator level, and Level the output. in1 = FM, in2 = pitch mod.

Param	Range	Default	Unit
Octave	-3 – 3	0	—
Fine	-100 – 100	0	ct
PWM	0.05 – 0.95	0.5	—
Pulse	0 – 1	0.5	—
Sub	0 – 1	0.3	—
Level	0 – 1	0.7	—

DualVCO 3 inputs

Analog-modeled dual VCO (the fat two-oscillator lead/bass, Moog/Prophet character): two band-limited (PolyBLEP) oscillators - osc2 offset by Interval semitones and Detune cents - summed for the classic thick two-VCO tone, with slow analog drift. Distinct from the 7-voice Susaw: this is the tuned 2-oscillator voice. Octave tunes both, Interval sets osc2's pitch offset, Detune the fine beating, Shape sweeps triangle->saw->pulse, Mix the osc1/osc2 balance, and Level the output. in1 = FM, in2 = pitch mod.

Param	Range	Default	Unit
Octave	-3 – 3	0	—
Interval	-24 – 24	0	st
Detune	-50 – 50	7	ct
Shape	0 – 1	0.5	—
Mix	0 – 1	0.5	—
Level	0 – 1	0.6	—

SyncVCO 3 inputs

Analog-modeled hard-sync oscillator (the classic sync-sweep lead): a master oscillator at the note pitch resets a slave whose frequency is Sync times higher, so sweeping Sync (from an envelope or LFO into in1) produces the screaming formant sweep of oscillator hard sync. Octave tunes the master, Sync sets the slave/master ratio, Shape the slave waveform (saw->pulse), and Level the output. in1 = sync-sweep modulation (added to Sync), in2 = pitch mod. Naive reset (bright/aliased by design - that is the sync sound).

Param	Range	Default	Unit
Octave	-3 – 3	0	—
Sync	1 – 12	2	—
Shape	0 – 1	0.3	—

Param	Range	Default	Unit
Level	0 – 1	0.6	—

Risset 2 inputs

Risset bell: a gate-struck additive bell from seven partials at the classic Jean-Claude Risset frequency ratios, each with its own amplitude and exponential decay (higher partials ring shorter), giving the shimmering slightly-detuned metallic bell tone that single-FM or harmonic-additive voices cannot. in0 is the note, in1 the strike gate. Decay sets the ring time, Inharm morphs from a harmonic stack (0) to the full inharmonic Risset ratios (1), and Level scales the output.

Param	Range	Default	Unit
Octave	-3 – 3	0	—
Fine	-100 – 100	0	ct
Decay	0.2 – 8	3	s
Inharm	0 – 1	1	—
Level	0 – 1	0.6	—

Bytebeat 3 inputs

Bytebeat oscillator: an algorithmic 8-bit generator that runs a bitwise integer formula over a pitch-driven sample counter, the low byte becoming the waveform - the chiptune/demoscene 'bytebeat' technique that turns shift-and-mask arithmetic into gnarly, ever-evolving digital timbres no analog or wavetable model can make. Formula selects the expression, Octave and Fine drive the counter rate (pitch), and Level scales the output. A world of aliased, glitchy, harmonically rich tones from one line of integer maths.

Param	Range	Default	Unit
Octave	-3 – 3	0	—
Fine	-100 – 100	0	ct
Formula	Classic · Sierpinski · Glitch · Crunch		—
Level	0 – 1	0.5	—

HardKick 2 inputs

Hardstyle/rawstyle kick generator: a gate-struck kick with a fast downward pitch sweep from the click transient to the body fundamental, an exponential amp decay and built-in tanh distortion for the signature hard punch. in0 = note (optional, transposes the body), in1 = gate (strike). Tune sets the body pitch, Punch the pitch-sweep depth, PSweep its time, Decay the body length, Click the attack transient, Drive the distortion. Patch a reverse-bass tail after it for the full hardstyle kick.

Param	Range	Default	Unit
Tune	20 – 200	50	Hz
Punch	0 – 60	36	st
PSweep	0.005 – 0.3	0.04	s
Decay	0.05 – 2	0.45	s
Click	0 – 1	0.4	—
Drive	0 – 1	0.6	—
Level	0 – 1	0.8	—

Screech 3 inputs

Hardstyle screech lead: a stack of detuned saws driven through heavy tanh distortion and a resonant formant band-pass, giving the screaming metallic gabber/hardstyle lead that a clean saw or single distortion cannot. in0 = note, in1 = formant sweep (added to Formant), in2 = pitch mod. Octave/Fine tune it, Detune spreads the saws, Drive the distortion, Formant the screech vowel peak, Reso its sharpness, Level the output.

Param	Range	Default	Unit
Octave	-2 – 3	1	—
Fine	-100 – 100	0	ct
Detune	0 – 50	18	ct
Drive	0 – 1	0.7	—
Formant	200 – 4000	1400	Hz
Reso	0.3 – 8	3	Q
Level	0 – 1	0.5	—

Hoover 3 inputs

Hoover / Mentasm lead (Alpha Juno 'What The...' rave stab): faithful to the original single-DCO patch - a sawtooth combed by a pulse (Roland 'sawtooth PWM', the signature hoover vowel), a square sub two octaves down, a light detuned ensemble standing in for the Juno chorus, and the note-on pitch sweep - the staple of rave/hardcore/hardstyle. Movement comes from the PWM comb, not a detuned saw stack. in0 = note, in1 = gate (arms the sweep), in2 = pitch mod. Octave/Fine tune it, PWM the comb width, Chorus the ensemble width, Sweep the note-on pitch-drop depth, SweepT its time, Level the output.

Param	Range	Default	Unit
Octave	-2 – 2	0	—
Fine	-100 – 100	0	ct
Chorus	0 – 30	8	ct
PWM	0.1 – 0.9	0.5	—
Sweep	0 – 24	7	st
SweepT	0.02 – 1	0.18	s
Level	0 – 1	0.5	—

Reese 2 inputs

DnB / neurofunk Reese bass: a stack of heavily detuned saws beating against each other (the classic Kevin Saunderson 'Reese' tone) through a low-pass and tanh drive, for the growling, metallic, ever-moving low end of drum & bass. in0 = note, in1 = pitch mod. Octave/Fine tune it, Detune sets the beating spread (the Reese character), Voices the saw count, Tone the low-pass, Drive the distortion, Level the output. Modulate Detune or Tone for the neuro morph.

Param	Range	Default	Unit
Octave	-3 – 1	-1	—
Fine	-100 – 100	0	ct
Detune	0 – 60	22	ct
Voices	2 – 7	3	—
Tone	100 – 12000	3000	Hz
Drive	0 – 1	0.4	—
Level	0 – 1	0.6	—

GeoMorph 3 inputs

Geometric oscillator (u-he Zebra GeoMorph-style): synthesises the waveform of a point tracing a regular polygon, so Order - the number of sides, continuously morphable - sets the harmonic richness, from a near-sine at low orders to bright buzzy formant-like spectra at high orders, while Teeth skews the geometry for asymmetric reedy timbres. A vertex/geometry-based oscillator distinct from wavetable and additive synthesis. Octave and Fine tune it, Level scales the output.

Param	Range	Default	Unit
Octave	-3 – 3	0	—
Fine	-100 – 100	0	ct
Order	2 – 16	3	—
Teeth	0 – 1	0	—

Param	Range	Default	Unit
Level	0 – 1	0.6	—

GeoBlend 3 inputs

Drawable geometric oscillator (u-he Zebra GeoBlend-style): trace two hand-drawn single-cycle geometries and Morph continuously between them. Draw Geo A and Geo B in the editor (drag points, double-click to add/remove); Morph crossfades A->B for evolving timbres no fixed wave can make. Octave/Fine tune it, Level scales the output.

Param	Range	Default	Unit
Octave	-3 – 3	0	—
Fine	-100 – 100	0	ct
Morph	0 – 1	0	—
Level	0 – 1	0.6	—

Wt 3 inputs

Wavetable oscillator: morphs a table bank with unison voices and Serum-style WARP (Bend/PWM/Sync/Mirror).

Param	Range	Default	Unit
Table	0 – 511	0	—
Position	0 – 1	0	—
Octave	-3 – 3	0	—
Fine	-100 – 100	0	ct
Voices	1 – 11	1	—
Detune	0 – 50	12	ct
Warp	Off · Bend+ · Bend- · PWM · Sync · Mirror · Quantize · FM · Flip · Asym		—
WarpAmt	0 – 1	0	—

Sample 0 inputs

Ableton-Simpler-style sample playback: pitch-tracked from the loaded slot with selectable interpolation (Linear/Cubic/None), playback Mode (Forward/Reverse/Ping-Pong), an independent loop region (Loop Start..End) with click-free loop Crossfade. Start/End set the played window; Loop wraps back to Loop Start.

Param	Range	Default	Unit
Slot	0 – 7	0	—
Start	0 – 1	0	—
Loop	0 – 1	0	—
Gain	0 – 2	1	—
End	0 – 1	1	—
Mode	Forward · Reverse · Ping-Pong		—
Interp	Linear · Cubic · None		—
LoopStart	0 – 1	0	—
Xfade	0 – 1	0	—
Warp	Repitch · Beats · Tones · Texture		—
Stretch	0.25 – 4	1	x
Grain	5 – 200	50	ms

SynthGranular 0 inputs

Granular oscillator: sprays overlapping grains from the sample slot. Pitch transposes each grain, Scan auto-advances the read position through the sample, and Dir plays grains in reverse.

Param	Range	Default	Unit
Slot	0 – 7	0	—
Position	0 – 1	0	—
Size	5 – 400	80	ms
Rate	1 – 200	30	Hz
Spray	0 – 1	0.05	—
Gain	0 – 2	1	—
Pitch	-24 – 24	0	st
Dir	0 – 1	0	—
Scan	-1 – 1	0	—

SynthNoise 0 inputs

Colored noise source (white/pink/brown).

Param	Range	Default	Unit
Color	0 – 1	0	—
Level	0 – 1	1	—

SynthSub 1 input

Sub oscillator: sine or square, octaves down, detuned.

Param	Range	Default	Unit
Octave	1 – 2	1	—
Wave	0 – 1	0	—
Level	0 – 1	1	—
Voices	1 – 11	1	—
Detune	0 – 50	8	ct

SynthFM 2 inputs

Two-operator FM with a selectable modulator waveform (Wave: sine / triangle / saw / square / half-sine / abs-sine / squared / pulse) for richer-than-sine spectra, and an audio input on in 2 that acts as a third FM operator. Ratio tunes the modulator, Index sets FM depth, Octave the pitch, and Voices/Detune stack a detuned unison.

Param	Range	Default	Unit
Ratio	0.5 – 8	1	—
Index	0 – 12	2	—
Octave	-3 – 3	0	—
Voices	1 – 11	1	—
Detune	0 – 50	10	ct
Wave	0 – 7	0	—

Karplus 1 input

Karplus-Strong plucked string: noise burst in a tuned damped delay.

Param	Range	Default	Unit
Octave	-3 – 3	0	—
Damping	0 – 1	0.5	—
Feedback	0.8 – 0.999	0.99	—

Supersaw 1 input

JP-8000 supersaw: seven detuned saws blended from centre to sides.

Param	Range	Default	Unit
Octave	-3 – 3	0	—
Detune	0 – 1	0.3	—
Mix	0 – 1	0.7	—
Voices	2 – 11	7	—

SynthAdditive 1 input

Additive oscillator: harmonic partials with rolloff and odd/even balance.

Param	Range	Default	Unit
Partial	1 – 16	8	—
Rolloff	0 – 1	0.6	—
Octave	-3 – 3	0	—
Odd/Even	0 – 1	0.5	—
Voices	1 – 11	1	—
Detune	0 – 50	12	ct

SynthSync 1 input

Hard-sync oscillator: a master phasor resets a detuned slave saw.

Param	Range	Default	Unit
Tune	-24 – 24	0	st
Sync	1 – 8	2	—
Octave	-3 – 3	0	—
Voices	1 – 11	1	—
Detune	0 – 50	8	ct

ChordOsc 1 input

Chord oscillator: one note drives a detuned chord of saws.

Param	Range	Default	Unit
Type	0 – 5	0	—
Octave	-3 – 3	0	—
Detune	0 – 1	0.1	—

SynthVector 1 input

Vector oscillator: XY morph across sine/saw/square/triangle.

Param	Range	Default	Unit
Octave	-3 – 3	0	—
X	0 – 1	0.5	—
Y	0 – 1	0.5	—
Detune	-50 – 50	0	ct

Wavetable 0 inputs

Wavetable oscillator: it advances a phase at Freq and reads single-cycle frames loaded from the node config, interpolating both within and between frames. Position (0 to 1) morphs across the frame stack, Warp (-1 to +1) phase-distorts the read pointer for timbral bending, Freq sets pitch (20 to 8000 Hz), and Level scales output. A pure generator (no inputs) that sweeps from one waveform to another as Position moves.

Param	Range	Default	Unit
Freq	20 – 8000	110	Hz
Position	0 – 1	0	—
Warp	-1 – 1	0	—
Level	0 – 1	0.5	—

Granular 1 input

Granular oscillator/texture: continuously captures the input into a ~2 s ring buffer and sprays overlapping Hann-windowed grains read back from it. Size sets grain length in ms, Density the spawn rate in grains/sec, Pitch transposes each grain in semitones, and Jitter randomizes the read origin for a diffuse, smeared texture. Mix blends the grain cloud against the dry input -- low density gives a sparse stutter, high density a thick wash.

Param	Range	Default	Unit
Size	5 – 500	120	ms
Density	1 – 200	40	—
Pitch	-24 – 24	0	st
Jitter	0 – 1	0.3	—
Mix	0 – 1	1	—

Spectral 1 input

FFT spectral-resynthesis oscillator: a streaming 1024-pt STFT phase vocoder (75% overlap) that FFTs each frame, reshapes the spectrum, then iFFTs. Stretch scales the spectral envelope by resampling bins, Shift moves bins up/down for a frequency shift, and Blur smears each bin's magnitude over time. Mix crossfades the resynthesis against the dry signal -- pushes toward metallic, harmonically reshaped tones.

Param	Range	Default	Unit
Blur	0 – 1	0.3	—
Shift	-50 – 50	0	bin
Stretch	0.5 – 2	1	—
Mix	0 – 1	1	—

Susaw 1 input

Massive X super-saw: sums 7 saws spread across a detune range, with a slow random drift added to the base frequency, then scaled by Level. Detune sets the +/-3% spread between voices, Blend lowers the 6 side voices relative to the loud centre voice, and Drift sets how much the wandering frequency offset is applied. Higher Blend gives a wider unison wash; lower Blend keeps the centre pitch dominant for a tighter lead.

Param	Range	Default	Unit
Freq	20 – 8000	110	Hz
Detune	0 – 1	0.4	—
Blend	0 – 1	0.7	—
Drift	0 – 1	0.1	—
Level	0 – 1	0.5	—

Sync 1 input

Hard-sync oscillator: a slave saw runs at Sync times the master frequency and is reset to zero each time the master phase wraps, producing the classic sync-sweep timbre, then scaled by Level. Freq sets the master pitch and Sync sets the slave ratio (1 to 6). Higher Sync ratios add brighter, more formant-like harmonics from the forced phase resets.

Param	Range	Default	Unit
Freq	20 – 8000	110	Hz
Sync	1 – 6	2	—
Level	0 – 1	0.5	—

Warp 1 input

Phase-warping oscillator: a master phase feeds a base waveform but is reshaped by one of several Mode warps (Neutral, Wrap, Grain, Hardsync, Hard Up-Down, Bend, Mirror, Random, Jitter) before the wave is read. Wave picks the base shape (sine, saw, square, triangle), Mode selects the warp, Amount sets warp depth, and Ratio scales the slave frequency for the sync modes. Pushing Amount drives anything from subtle phase bend to per-sample jitter noise depending on Mode.

Param	Range	Default	Unit
Freq	20 – 8000	110	Hz
Wave	0 – 3	1	—
Mode	0 – 8	0	—
Amount	0 – 1	0.5	—
Ratio	1 – 8	2	—

Sub 1 input

Dedicated sub oscillator: runs a single phase at Freq dropped by Octave, outputting either a sine or a square, then scaled by Level. Octave shifts down 0, 1, or 2 octaves, Wave selects sine or square, and Level sets the output gain. Use it to add low-end weight under a main oscillator, with sine for clean sub and square for a harder, more present low end.

Param	Range	Default	Unit
Freq	20 – 4000	110	Hz
Octave	-2 – 0	-1	—
Wave	0 – 1	0	—
Level	0 – 1	0.6	—

Spire 1 input

Spire-style multimode oscillator: sums up to 9 unison voices, each rendered by the selected Mode (Classic saw, Noise, FM, AMSync, SawPWM, HardFM, Vowel) with the sum normalized by the voice count. Mode picks the algorithm, Shape is the per-mode control (FM index, sync ratio, pulse width, or vowel position), Voices sets unison count (up to 9, matching Spire), Detune the fine +/-3% spread, and Interval fans the voices out over a musical interval (0-12 semitones, e.g. octave/fifth stacks). The Vowel mode adds three morphing formant band-passes over the summed saw for talking-synth timbres.

Param	Range	Default	Unit
Freq	20 – 8000	110	Hz
Mode	0 – 6	0	—
Shape	0 – 1	0.5	—
Voices	1 – 9	1	—
Detune	0 – 1	0.2	—
Interval	0 – 12	0	st

PMOsc 1 input

Phase-modulation oscillator: a sine modulator running at Ratio times Freq offsets the carrier phase by depth Index, and the warped phase is read through the selected carrier waveform, then scaled by Level. Wave selects the carrier shape (Sine, Tri, three bent/saturated triangle variants, narrowed sine), Ratio sets the modulator-to-carrier frequency, and Index sets modulation depth. Raising Index and Ratio builds increasingly bright, inharmonic FM-style spectra.

Param	Range	Default	Unit
Freq	20 – 8000	110	Hz
Wave	0 – 5	0	—
Ratio	0.5 – 8	1	—
Index	0 – 8	2	—
Level	0 – 1	0.5	—

PhaseDist 1 input

Casio CZ phase-distortion oscillator: a phase ramp drives a cosine through a warped read so DCW morphs a sine toward the chosen shape, then scales by Level. Shape selects Saw or Pulse (true phase distortion) or three resonant modes (ResoSaw, ResoTri, ResoTrap) that are windowed hard-sync, and DCW sets the distortion amount or, in resonant modes, the resonant multiple (1 to 16). Low DCW stays near a clean sine while high DCW opens into bright sawtooth, pulse, or formant-like resonant tones.

Param	Range	Default	Unit
Freq	20 – 8000	110	Hz
Shape	0 – 4	0	—
DCW	0 – 1	0.5	—
Level	0 – 1	0.5	—

Geiger 0 inputs

Sparse-impulse noise: each sample has a probability set by Rate of firing a random-amplitude impulse, which then decays exponentially, all scaled by Level. Rate sets the average clicks per second and Decay sets how long each impulse rings before fading. Low Rate with short Decay gives sparse ticks like a Geiger counter, while higher Rate and longer Decay thicken it toward a crackling texture.

Param	Range	Default	Unit
Rate	0.1 – 200	20	Hz
Decay	0 – 1	0.5	—
Level	0 – 1	0.5	—

Multisampler 3 inputs

Multisampler: on each note-on (from the note, gate, and velocity inputs) it strikes every region whose key and velocity zones match, resampling each by note-vs-root plus fine tune so it plays at the right pitch. Slot selects which loaded .sfz region set to use (0 to 7), Loop forces looping of the regions' loop or play windows, and Level scales output; velocity crossfades and round-robin sequencing come from the region data. Use it for realistic multisampled instruments where timbre tracks key and velocity.

Param	Range	Default	Unit
Slot	0 – 7	0	—
Loop	0 – 1	0	—
Level	0 – 1	0.6	—

Pluck 2 inputs

Karplus-Strong plucked-string physical model: a gate edge fills a delay line with a white-noise burst, then each pass averages neighbouring samples and scales by Decay so the noise settles into a pitched, decaying tone. Tune shifts the note in semitones (sets the delay length from the in0 note), Damp blends in more of the lowpass average to dull the highs, Decay sets how slowly the string rings out, and Level scales the output. Plucks on the in1 gate; short Decay and high Damp give a soft muted pluck, long Decay and low Damp a bright sustained string.

Param	Range	Default	Unit
Tune	-24 – 24	0	st
Damp	0 – 1	0.5	—
Decay	0.8 – 0.999	0.99	—
Level	0 – 1	0.6	—

Modal 1 input

Modal synthesis: a bank of 6 tuned decaying resonators struck by a gate (bells, mallets, struck metal). Structure morphs harmonic to inharmonic-bar partials.

Param	Range	Default	Unit
Freq	20 – 4000	220	Hz
Decay	0.05 – 8	1.5	s
Structure	0 – 1	0.3	—
Bright	0 – 1	0.6	—
Level	0 – 1	0.6	—

Blown 1 input

Waveguide wind instrument (STK-clarinet): a bore delay line with a nonlinear reed reflection, driven by breath from the gate. The blown counterpart to Karplus-Strong.

Param	Range	Default	Unit
Freq	20 – 2000	220	Hz
Breath	0 – 1	0.5	—
Reed	0 – 1	0.5	—
Tone	0 – 0.9	0.4	—
Level	0 – 1	0.6	—

Bowed 1 input

Bowed-string waveguide (STK-bowed): two delay lines for the string halves with a stick-slip bow-table at the bow point, driven by the gate. Violin/cello family.

Param	Range	Default	Unit
Freq	20 – 2000	220	Hz
Force	0 – 1	0.5	—
Pos	0.02 – 0.5	0.12	—
Tone	0 – 0.95	0.5	—
Level	0 – 1	0.6	—

FOF 0 inputs

Formant-wave-function synthesis (CHANT): each fundamental period retriggers a bank of decaying formant grains, generating vocal/formant tones. Vowel morphs A-E-I-O-U.

Param	Range	Default	Unit
Freq	20 – 800	110	Hz
Vowel	0 – 4	0	—
Bandwidth	0.2 – 4	1	—
Level	0 – 1	0.5	—

WaveTerrain 0 inputs

Wave-terrain synthesis: a Lissajous orbit traverses a 2D polynomial surface $z=f(x,y)$; the height is the output. Warp tilts the orbit, Ratio sets its y/x frequency.

Param	Range	Default	Unit
Freq	20 – 8000	110	Hz
Ratio	0.5 – 8	1	—
Warp	0 – 1	0	—
Level	0 – 1	0.5	—

Pulsar 0 inputs

Pulsar synthesis (Roads): a pulsaret (windowed sine) is emitted once per fundamental period, then silence. Freq sets pitch, Formant the pulsaret cycles, Width the duty.

Param	Range	Default	Unit
Freq	20 – 4000	110	Hz
Formant	1 – 16	3	—
Width	0.05 – 1	0.5	—
Level	0 – 1	0.5	—

Scanned 1 input

Scanned synthesis (Mathews/Verplank): a 16-mass spring ring evolves slowly after a gate hammer-strike; a scan reads the mass positions at audio rate as the waveform. Tension/Damping shape the timbre's evolution.

Param	Range	Default	Unit
Freq	20 – 4000	110	Hz
Tension	0.05 – 1	0.4	—
Damping	0 – 1	0.2	—
Level	0 – 1	0.6	—

Kick 2 inputs

Analog kick drum: a trigger on in0 fires a sine whose pitch sweeps from a high click down to the body Tune, with a percussive amp Decay. Tune sets the fundamental, Punch the pitch-drop time (snap), Decay the body length, Drive adds saturation for harder kicks, and Level the output - the synthesized 808/909-style kick. Patch a Clock, Euclid or sequencer gate into in0.

Param	Range	Default	Unit
Tune	20 – 200	50	Hz
Punch	0.005 – 0.2	0.04	s
Decay	0.05 – 2	0.4	s
Drive	0 – 1	0.2	—
Level	0 – 1	0.85	—

Snare 2 inputs

Analog snare drum: a trigger on in0 fires two detuned body tones plus a burst of band-passed noise (the snare wires). Tune sets the body pitch, Snap the noise level, Tone the body/noise balance, Decay the length, and Level the output.

Param	Range	Default	Unit
Tune	100 – 400	180	Hz
Snap	0 – 1	0.6	—
Tone	0 – 1	0.5	—
Decay	0.05 – 1	0.2	s
Level	0 – 1	0.7	—

Hat 1 input

Analog hi-hat: a trigger on in0 fires bright high-passed metallic noise with a short percussive decay. Tone sets the brightness, Decay the length (short = closed, long = open hat), Metal blends in inharmonic ringing for a more cymbal-like tone, and Level the output.

Param	Range	Default	Unit
Tone	0 – 1	0.7	—
Decay	0.02 – 1.5	0.08	s
Metal	0 – 1	0.3	—
Level	0 – 1	0.5	—

Clap 1 input

Analog hand-clap: a trigger on in0 fires band-passed noise shaped by a multi-burst envelope - three quick retriggers then a diffuse tail - the characteristic 909 clap. Tone sets the band-pass colour, Spread the gap between the bursts, Decay the tail length, and Level the output.

Param	Range	Default	Unit
Tone	0 – 1	0.5	—
Spread	0.005 – 0.05	0.012	s
Decay	0.05 – 1	0.25	s
Level	0 – 1	0.6	—

Tom 2 inputs

Analog tom drum: a trigger on in0 fires a sine whose pitch bends down from Tune with a gentler sweep than the Kick (no sub click), plus a short noise transient for stick attack. Tune sets the fundamental, Bend the pitch-drop amount, Decay the body length, Tone the attack-noise blend, and Level the output - the synthesized 808/909-style tom (low/mid/high by Tune). Patch a Clock, Euclid or sequencer gate into in0.

Param	Range	Default	Unit
Tune	60 – 300	120	Hz
Bend	0 – 1	0.4	—
Decay	0.05 – 1.5	0.5	s
Tone	0 – 1	0.2	—
Level	0 – 1	0.8	—

Cowbell 2 inputs

Analog cowbell (808-style): a trigger on in0 fires two detuned pulse oscillators a fixed ratio apart, band-passed and shaped by a short percussive decay for the classic metallic clank. Tune sets the lower oscillator, Detune the ratio of the upper one (the cowbell interval), Decay the ring length, and Level the output. Patch a Clock, Euclid or sequencer gate into in0.

Param	Range	Default	Unit
Tune	300 – 1200	540	Hz
Detune	1.2 – 1.8	1.48	—
Decay	0.05 – 1	0.3	s
Level	0 – 1	0.5	—

Rimshot 2 inputs

Analog rimshot (808-style): a trigger on in0 fires a very short burst of two high resonant tones (a bright tap plus a lower ring) with an extremely fast decay for the sharp click of a rim hit. Tune sets the bright tone (the lower ring tracks it), Decay the click length (very short), and Level the output. Patch a Clock, Euclid or sequencer gate into in0.

Param	Range	Default	Unit
Tune	800 – 2500	1700	Hz
Decay	0.01 – 0.2	0.04	s
Level	0 – 1	0.6	—

Cymbal 2 inputs

Analog cymbal (808-style metal): a trigger on in0 fires six square oscillators tuned to a fixed inharmonic ratio set, summed and high-passed, with a fast onset plus a long metallic tail. Distinct from the Hat (which filters noise) - this is the additive square-oscillator method for ride/crash timbres. Tune sets the base frequency of the cluster, Decay the tail length, Tone the high-pass brightness, and Level the output. Patch a Clock, Euclid or sequencer gate into in0; patch a Note source into in1 to keytrack the cluster.

Param	Range	Default	Unit
Tune	200 – 1200	540	Hz
Decay	0.1 – 3	1	s
Tone	0 – 1	0.6	—
Level	0 – 1	0.45	—

Marimba 2 inputs

Modal marimba: a trigger on in0 strikes a wooden bar modelled as inharmonic decaying sine partials (the characteristic 1 : 3.9 : 9.2 bar modes), warm with a soft mallet onset. Tune sets the pitch, Decay the ring length, Mallet the brightness (how long the upper modes sustain), and Level the output. A melodic mallet voice to pair with the drum kit; gate it from a Clock, Euclid or sequencer.

Param	Range	Default	Unit
Tune	50 – 2000	220	Hz
Decay	0.05 – 2	0.5	s
Mallet	0 – 1	0.5	—
Level	0 – 1	0.7	—

Glockenspiel 2 inputs

Modal glockenspiel: a trigger on in0 strikes a metal bar modelled as bright, long-ringing inharmonic partials (1 : 2.7 : 5.4 : 8.9), the clear bell-like tone of a glock/celesta. Tune sets the pitch, Decay the (long) ring, Mallet the brightness, and Level the output.

Param	Range	Default	Unit
Tune	200 – 4000	880	Hz
Decay	0.1 – 3	1.2	s
Mallet	0 – 1	0.6	—
Level	0 – 1	0.6	—

Kalimba 2 inputs

Modal kalimba (thumb-piano tine): a trigger on in0 plucks a metal tine modelled as a strong fundamental plus a couple of mellow inharmonic partials (1 : 2.0 : 3.9) with a soft buzzy onset. Tune sets the pitch, Decay the ring length, Buzz the attack-noise transient, and Level the output.

Param	Range	Default	Unit
Tune	80 – 1500	330	Hz
Decay	0.05 – 1.5	0.4	s
Buzz	0 – 1	0.3	—
Level	0 – 1	0.7	—

Vibraphone 2 inputs

Modal vibraphone: a trigger on in0 strikes an aluminium bar (inharmonic partials 1 : 4 : 10, a metal cousin of the marimba) with a long shimmering sustain, plus the characteristic motor tremolo - an amplitude vibrato. Tune sets the pitch, Decay the (long) ring, Vibrato the tremolo depth, and Level the output.

Param	Range	Default	Unit
Tune	80 – 2000	440	Hz
Decay	0.2 – 4	2	s
Vibrato	0 – 1	0.4	—
Level	0 – 1	0.6	—

MusicBox 2 inputs

Modal music box: a trigger on in0 plucks a steel comb tine - a bright bell-like tone from a strong fundamental plus high inharmonic partials (1 : 5.4 : 13) with a fast tinkling attack and medium decay, the unmistakable music-box timbre. Tune sets the pitch, Decay the ring, Tone the brightness of the upper partials, and Level the output.

Param	Range	Default	Unit
Tune	200 – 4000	880	Hz
Decay	0.1 – 2	0.6	s
Tone	0 – 1	0.6	—
Level	0 – 1	0.55	—

Steelpan 2 inputs

Modal steelpan (steel drum): a trigger on in0 strikes a tuned note on an oil-drum face - a bright shimmering tone with near-harmonic partials (1 : 2 : 3 : 4) and a metallic edge, medium sustain. Tune sets the pitch, Decay the ring, Shimmer the upper-partial brightness, and Level the output.

Param	Range	Default	Unit
Tune	100 – 1500	330	Hz
Decay	0.1 – 2	0.8	s
Shimmer	0 – 1	0.5	—
Level	0 – 1	0.55	—

Electric 2 inputs

Electric piano (Rhodes/Wurlitzer-style physical model): a gate on in1 strikes a tuned tine - an exponentially-decaying sine fundamental plus a bright inharmonic 'bark' partial that decays faster - run through a soft pickup nonlinearity, the classic electric-piano bell-and-body tone. in0 is the note (keyboard-tracked), Transpose tunes it, Decay the note length, Bark the metallic attack, Tone the pickup brightness, and Level the output.

Param	Range	Default	Unit
Transpose	-24 – 24	0	st
Decay	0.1 – 4	1.2	s
Bark	0 – 1	0.5	—
Tone	0 – 1	0.5	—
Level	0 – 1	0.7	—

ComplexOsc 1 input

Complex oscillator (Make Noise DPO / Buchla 259): a modulation oscillator thru-zero-FMs a primary oscillator, whose output is then run through a sine wavefolder for rich, evolving harmonics. Freq sets the primary pitch, Ratio the modulator's frequency relative to it, FM the thru-zero modulation depth (inharmonic clang at high settings), and Fold the wavefolder drive (adds bright folded harmonics). One voice spanning clean tones to metallic, animated timbres. Distinct from the separate ThruZero and Wavefolder blocks.

Param	Range	Default	Unit
Freq	20 – 5000	110	Hz
Ratio	0.25 – 8	2	—
FM	0 – 1	0.2	—
Fold	0 – 1	0.2	—

Basimilus 2 inputs

Harmonic percussion synth (Noise Engineering Basimilus Iteritas): a trigger on in0 fires a stack of six sine partials whose spacing morphs from harmonic toward inharmonic, run through a wavefolder and shaped by a fast percussive decay - the digital, metallic counterpart to the analog drum voices, good for kicks, snares, zaps and bells. Tune sets the fundamental, Harmonics the partial spread, Fold the wavefolder crunch, Decay the length, and Level the output. Gate it from a Clock, Euclid or sequencer.

Param	Range	Default	Unit
Tune	20 – 2000	80	Hz
Harmonics	0 – 1	0.3	—
Fold	0 – 1	0.3	—
Decay	0.02 – 2	0.3	s
Level	0 – 1	0.7	—

Plaits 2 inputs

Macro-oscillator (Mutable Instruments Plaits): one voice, six synthesis Models selected by the Model param - Analog (saw/square), FM (2-op), Grain (formant), Chord (stacked intervals), Wavetable (sine-saw-square morph) and Modal (struck bar) - each shaped by Timbre and Morph. Tune sets pitch; an internal decay envelope (Decay) fires when a gate hits in0, so it plays as a drone unpatched or a pinged percussive voice when triggered. The Swiss-army oscillator covering most synthesis techniques in a single block.

Param	Range	Default	Unit
Tune	20 – 4000	110	Hz
Model	Analog · FM · Grain · Chord · Wavetable · Modal		—
Timbre	0 – 1	0.5	—
Morph	0 – 1	0.5	—
Decay	0.01 – 4	2	s

Elements 2 inputs

Modal synthesis voice (Mutable Instruments Elements): an exciter feeds a tuned modal resonator bank. A gate on in0 fires the exciter, which Exciter morphs from blown (filtered noise) to struck (impulse); the six-mode resonator imposes the ringing of a physical object, with Geometry morphing its partials from a harmonic series toward inharmonic bell/plate ratios and Damping

setting the ring time. Tune sets pitch, Level the output. Bowed strings, struck bars, blown tubes and alien resonances from one voice. Distinct from the self-struck Modal and the resonator-only Rings.

Param	Range	Default	Unit
Tune	30 – 2000	110	Hz
Exciter	0 – 1	0.5	—
Geometry	0 – 1	0.3	—
Damping	0 – 1	0.7	—
Level	0 – 1	0.6	—

Speech 1 input

Vowel/speech synthesizer (Mutable Instruments / SAM-style): a buzzing saw source is shaped by three formant band-passes tuned to vowel sounds, so it sings. Vowel morphs through A-E-I-O-U; Speak, when above zero, auto-advances the vowel at that rate for talking/chanting textures; Tune sets the pitch. A formant voice for choirs, robots and vocal pads. Distinct from the Formant filter (which colours external audio) and Vosim (a single grain pair) - this sweeps a vowel sequence over its own oscillator.

Param	Range	Default	Unit
Tune	40 – 600	110	Hz
Vowel	0 – 1	0.5	—
Speak	0 – 8	0	Hz
Level	0 – 1	0.6	—

Noise 0 inputs

Colored-noise generator: Color morphs from white (flat spectrum) toward a low-passed pink-ish tone, scaled by Level. Use it as a percussion/wind source, a sample-and-hold input, or to add air and dither. It takes no signal input - it is a pure generator.

Param	Range	Default	Unit
Color	0 – 1	0	—
Level	0 – 1	1	—

Oscillator 1 input

Audio-rate tone generator with sine, saw, square and triangle waves (Wave) at Freq. Use it as a ring-mod or FM carrier, a drone, a sub layer, or a test tone. It takes no signal input - patch its output into RingMod, a filter, or straight to Out.

Param	Range	Default	Unit
Freq	20 – 8000	220	Hz
Wave	0 – 3	0	—
Level	0 – 1	0.5	—

VAnalog 1 input

Virtual-analog oscillator: saw or pulse (Wave) with adjustable pulse width (PWM), a square sub-oscillator an octave down (Sub), and slow random Drift for analog pitch instability. Level scales the output. A warm, slightly unstable VA voice - pair with the Mini/Jup-8 analog filters for a classic synth chain.

Param	Range	Default	Unit
Freq	20 – 8000	110	Hz
Wave	0 – 1	0	—
PWM	0.05 – 0.95	0.5	—

Param	Range	Default	Unit
Sub	0 – 1	0	—
Drift	0 – 1	0.1	—
Level	0 – 1	0.5	—

FM 1 input

Two-operator FM voice (the DX7 building block): a sine carrier is phase-modulated by a sine modulator at Ratio times the frequency, with Index setting depth and modulator self-Feedback adding harmonic edge toward noise. Integer ratios give harmonic, organ/bass timbres; non-integer ratios give bells and metallic tones.

Param	Range	Default	Unit
Freq	20 – 8000	110	Hz
Ratio	0.5 – 8	2	—
Index	0 – 12	2	—
Feedback	0 – 0.9	0	—
Level	0 – 1	0.5	—

Additive 1 input

Additive oscillator that sums up to 16 sine partials (Partials) with a spectral Tilt: negative Tilt rolls off the upper harmonics for a soft, mellow tone, positive brightens toward a buzzy saw. Partials above Nyquist are skipped, so it stays alias-free. A clean, controllable harmonic source from pure sine to bright buzz.

Param	Range	Default	Unit
Freq	20 – 4000	110	Hz
Partials	1 – 16	8	—
Tilt	-2 – 2	-0.7	—
Level	0 – 1	0.5	—

Weierstrass 0 inputs

Fractal-waveform oscillator (a first as a synth block): synthesises the Weierstrass function - the textbook curve that is continuous everywhere but differentiable nowhere - as a self-similar audio tone, summing eight cosine octaves whose frequencies climb by Lacunarity and amplitudes fall by Detail. The result is infinitely jagged: zoom in at any scale and the same roughness repeats. Detail sets the high-frequency weight (roughness/brightness) and Lacunarity the gap between layers. A genuinely fractal timbre, not a filtered or wavefolded one.

Param	Range	Default	Unit
Freq	20 – 2000	110	Hz
Detail	0.3 – 0.9	0.5	—
Lacunarity	2 – 7	3	—

PrimeOrgan 0 inputs

Prime-harmonic oscillator (a first as a synth block): an additive tone built from the fundamental plus only the PRIME harmonics - the 2nd, 3rd, 5th, 7th, 11th... partials, with the composite ones (4th, 6th, 8th, 9th...) left out entirely. Because every overtone is at a prime multiple, the usual octave and fifth reinforcements are missing, giving a hollow, slightly bell-like organ timbre that no harmonic or subtractive source produces. Partials sets how many prime overtones to stack, alias-free up to Nyquist. A number-theoretic spectrum, distinct from the tilt-shaped Additive oscillator.

Param	Range	Default	Unit
Freq	20 – 4000	110	Hz
Partials	2 – 16	8	—

FibonacciOsc ▯ inputs

Fibonacci-harmonic oscillator (a first as a synth block): an additive tone whose overtones land only on the Fibonacci numbers - the 1st, 2nd, 3rd, 5th, 8th, 13th, 21st... harmonics. The gaps between partials widen by the golden ratio as you climb, so the low end is dense and the top sparse, giving a warm body with a thin, glassy shimmer on top - a spectrum tuned to phi rather than the even comb of a saw. Partials sets how many Fibonacci overtones to stack, alias-free up to Nyquist. Distinct from the prime-harmonic and tilt-based oscillators.

Param	Range	Default	Unit
Freq	20 – 4000	110	Hz
Partials	2 – 14	8	—

ThueOsc ▯ inputs

Thue-Morse oscillator (a first as a synth block): an additive tone that keeps harmonic k only when k is a Thue-Morse 'odious' number - one whose binary digits sum to an ODD count (1, 2, 4, 7, 8, 11...). Because the keep/drop pattern is the self-similar Thue-Morse sequence, the spectrum is fractal: the same fair-share thinning repeats at every octave, producing a hollow, evolving timbre with a built-in self-similar comb. Partials sets the highest harmonic considered, alias-free up to Nyquist. A self-similar spectrum distinct from the prime and Fibonacci oscillators.

Param	Range	Default	Unit
Freq	20 – 4000	110	Hz
Partials	2 – 24	16	—

CantorOsc ▯ inputs

Cantor-spectrum oscillator (a first as a synth block): an additive tone that keeps harmonic k only when k written in base 3 contains NO digit 1 - the membership rule of the Cantor set, which leaves overtones at 2, 6, 8, 18, 20, 24... and removes the fundamental entirely. The result is a lacunary (gap-riddled) fractal spectrum, the audible cousin of the Cantor middle-thirds set, giving a pitched-but-rootless, metallic-hollow timbre with self-similar holes. Partials sets the highest harmonic considered. A fractal-set spectrum distinct from the other number-theoretic oscillators.

Param	Range	Default	Unit
Freq	20 – 4000	110	Hz
Partials	4 – 30	24	—

MobiusOsc ▯ inputs

Mobius oscillator (a first as a synth block): an additive tone where each harmonic k is weighted by the Mobius function $\mu(k)$ - so harmonics on a repeated-prime multiple (4, 8, 9, 12...) are silenced entirely, while the squarefree ones are added with a PLUS or MINUS sign depending on whether they have an even or odd number of prime factors. The mix of dropped partials and phase-flipped (inverted) partials carves a spectrum found nowhere in ordinary synthesis - a signed, squarefree-only comb. Partials sets the highest harmonic considered. Distinct from the other number-theoretic oscillators.

Param	Range	Default	Unit
Freq	20 – 4000	110	Hz
Partials	2 – 24	16	—

SubharmonicOsc ▯ inputs

Undertone-series oscillator (a first as a synth block): the mirror image of an ordinary tone - instead of stacking overtones ABOVE the fundamental it stacks the UNDERTONE series below it (f , $f/2$, $f/3$, $f/4$...), the descending ratios that Harry Partch and the old harmoniums explored. The fundamental sits at the TOP and the partials descend, giving a deep, hollow, slightly unsettling body quite unlike any overtone-based saw or square. Partial sets how many undertones to stack. A subharmonic spectrum distinct from every overtone oscillator. Needs no input.

Param	Range	Default	Unit
Freq	40 – 4000	220	Hz
Partial	2 – 16	8	—

GoldenOsc ▯ inputs

Golden-ratio oscillator (a first as a synth block): an INHARMONIC tone whose partials are spaced not by whole numbers but by powers of the golden ratio (1, 1.62, 2.62, 4.24, 6.85...). Because no partial is a simple multiple of another, nothing fuses into a clear pitch the way a harmonic tone does - the result is a shimmering, metallic, bell-like timbre that sits between a pitch and a noise, the most irrationally-spaced spectrum there is. Partial sets how many golden overtones to stack, alias-free up to Nyquist. A phi-spaced inharmonic spectrum distinct from the integer-harmonic oscillators. Needs no input.

Param	Range	Default	Unit
Freq	40 – 2000	220	Hz
Partial	2 – 12	8	—

StretchOsc ▯ inputs

Stretched-harmonic oscillator (a first as a synth block): models the inharmonicity of a real struck or plucked string, where stiffness makes each overtone sharper than a perfect multiple - partial k sits at $k \cdot \sqrt{1 + \text{Stretch} \cdot k^2}$ instead of exactly k . That tiny upward stretch, growing with each partial, is what gives a piano its warm, slightly detuned brilliance and a struck metal bar its character. Stretch sets the amount (0 = perfectly harmonic, up = ever more bell-like). Partial sets how many to stack. A physically-modelled inharmonic spectrum distinct from the golden and integer oscillators. Needs no input.

Param	Range	Default	Unit
Freq	40 – 4000	220	Hz
Partial	2 – 16	10	—
Stretch	0 – 0.003	0.001	—

TritaveOsc ▯ inputs

Bohlen-Pierce (tritave) oscillator (a first as a synth block): a tone built from ODD harmonics only (1, 3, 5, 7, 9, 11...), with no even partials at all - the spectrum that fits the Bohlen-Pierce scale, which divides the TRITAVE (a 3:1 ratio, an octave plus a fifth) into 13 steps instead of dividing the octave. With only odd harmonics it has a hollow, clarinet-like purity, and it consonates with the strange non-octave intervals of BP music. Partial sets how many odd harmonics to stack, alias-free up to Nyquist. A non-octave timbre distinct from the usual harmonic oscillators. Needs no input.

Param	Range	Default	Unit
Freq	40 – 4000	220	Hz
Partial	2 – 12	8	—

WarpOsc ▯ inputs

Spectral-warp oscillator (a first as a synth block): a single knob that bends the whole harmonic series. Each partial sits at k raised to the power Warp, so at Warp = 1 it is an ordinary harmonic saw; below 1 the partials crowd together (a compressed, dense, organ-like spectrum) and above 1 they spread apart into an ever-more-inharmonic, bell-to-clangorous timbre. Sweeping Warp morphs

continuously from harmonic to inharmonic - a one-control spectral stretch unlike any fixed waveform. Partial sets how many to stack, alias-free up to Nyquist. Needs no input.

Param	Range	Default	Unit
Freq	40 – 2000	220	Hz
Partial	2 – 16	10	—
Warp	0.5 – 2	1	—

Walsh 0 inputs

Walsh-function oscillator (a first as a synth block): builds its waveform from the Walsh functions - the SQUARE-wave orthogonal basis that does for digital signals what sines do for analogue, used in the Hadamard transform. Each one is a product of doubling square waves, so the output is a hard +/-1 pattern with exactly Sequency sign-changes per cycle (sequency being the square-wave equivalent of frequency). The result is a buzzy, bit-exact, distinctly digital timbre - a complete non-sinusoidal basis rather than a filtered square. Sequency picks which Walsh function to sound. Needs no input.

Param	Range	Default	Unit
Freq	20 – 2000	110	Hz
Sequency	0 – 31	5	—

Blancmange 0 inputs

Blancmange (Takagi) oscillator (a first as a synth block): synthesises the blancmange curve - a fractal made by stacking ever-smaller TRIANGLE waves, each half the height and twice the rate of the last. Like the Weierstrass function it is continuous everywhere but jagged at every scale, yet its triangle (rather than cosine) building blocks give a softer, more granular roughness - a self-similar timbre that thickens as Detail adds layers. Detail sets how many triangle octaves to sum. A triangle-based fractal waveform distinct from the cosine-based Weierstrass. Needs no input.

Param	Range	Default	Unit
Freq	20 – 2000	110	Hz
Detail	2 – 12	8	—

Devil 0 inputs

Devil's-staircase oscillator (a first as a synth block): synthesises the Cantor function - the 'devil's staircase' that climbs from 0 to 1 entirely on the Cantor set, staying perfectly FLAT across every gap yet still rising overall, with no jumps. As a repeating waveform it is a fractal ramp: a sawtooth assembled from infinitely many flat plateaus and self-similar steps, a singular curve whose spectrum is unlike any band-limited ramp. A monotone fractal staircase distinct from the jagged Weierstrass and Blancmange. Needs no input.

Param	Range	Default	Unit
Freq	20 – 2000	110	Hz

ByteBeat 0 inputs

Bytebeat oscillator (a first as a synth block): the famous 'one-line music' - feed a running sample counter through a short expression of bit-shifts, ORs, ANDs and multiplies, keep the low 8 bits, and a whole chiptune-like texture falls out of pure integer arithmetic. The bitwise interference of the counter against its own shifted copies produces self-similar, glitchy, rhythmic patterns for free. Formula selects among classic bytebeat expressions and Speed sets the counter rate (its pitch and tempo). A bit-arithmetic waveform unlike any oscillator or sample. Needs no input.

Param	Range	Default	Unit
Speed	0.1 – 8	1	—

Param	Range	Default	Unit
Formula	Sierpinski · Classic · Squares · Mask		

Gendy ▯ inputs

Dynamic-stochastic (GENDYN) oscillator (a first as a synth block): Xenakis's radical idea of synthesising sound directly from probability - the waveform is a handful of breakpoints joined by straight lines, and after every cycle each breakpoint takes a small RANDOM WALK in height. So the wave keeps its rough shape but never repeats exactly, slowly mutating from one cycle to the next - the engine behind his GENDY pieces, sound built bottom-up from chance rather than from notes. Points sets the number of breakpoints and Step how far they wander each cycle (0 = frozen, high = noisy). A stochastic waveform distinct from every fixed-shape oscillator. Needs no input.

Param	Range	Default	Unit
Freq	20 – 2000	110	Hz
Points	3 – 8	6	—
Step	0 – 0.3	0.08	—

Theta ▯ inputs

Jacobi-theta oscillator (a first as a synth block): synthesises the theta function - a periodic version of the Gaussian bell, built as a cosine series whose harmonic k is weighted by q to the power k-squared. As Width (the q value) rises toward 1 those weights stay strong for many harmonics, so the bell sharpens into a narrow spike-train; as it falls the upper harmonics vanish and it rounds back toward a pure sine. A continuous sine-to-pulse morph rooted in one of the deepest functions in mathematics, distinct from a PWM or a filtered pulse. Needs no input.

Param	Range	Default	Unit
Freq	20 – 4000	110	Hz
Width	0.1 – 0.9	0.6	—

Lorentzian ▯ inputs

Lorentzian-pulse oscillator (a first as a synth block): emits the Lorentzian (Cauchy) bell - $1 / (1 + (t/\text{Width})^2)$ - once per cycle, a peak with a tall sharp tip and long, slowly-fading shoulders, the lineshape of a resonance in physics. Its fatter tails (compared with a Gaussian or cosine pulse) give a brighter, buzzier spectrum that rolls off exponentially, while Width sets how narrow and piercing the spike is. A resonance-shaped pulse train distinct from the Theta and window pulses. Needs no input.

Param	Range	Default	Unit
Freq	20 – 4000	110	Hz
Width	0.01 – 0.3	0.06	—

Minkowski ▯ inputs

Minkowski question-mark oscillator (a first as a synth block): synthesises Minkowski's $?(x)$ function - the 'question-mark' curve that warps the number line by walking the Stern-Brocot tree, sending the quadratic irrationals to the rationals and flattening the golden ratio's run of 1s into a plain binary fraction. As a waveform it is a singular staircase like the devil's staircase but woven from continued fractions instead of base 3, giving a different self-similar ramp with its own fractal spectrum. A continued-fraction singular function distinct from the Cantor (Devil) and the smooth ramps. Needs no input.

Param	Range	Default	Unit
Freq	20 – 2000	110	Hz

Expo ▯ inputs

Exponential-ramp oscillator (a first as a synth block): a sawtooth whose ramp is not straight but EXPONENTIALLY curved - it rises (or, with the curve reversed, decays) like a charging capacitor rather than a linear slope, the natural shape of an analogue envelope or an RC circuit. Curve sets how bent the ramp is: near zero it is an ordinary saw, high and it hugs the floor then snaps up at the end. The curvature reshapes the harmonic rolloff, giving a rounder, more organic edge than the buzzy linear saw. A capacitor-curve waveform distinct from the geometric saw. Needs no input.

Param	Range	Default	Unit
Freq	20 – 4000	110	Hz
Curve	0.5 – 6	3	—

Sigmoid ▯ inputs

Sigmoid sine-to-square oscillator (a first as a synth block): takes a pure sine and pushes it through a tanh sigmoid, so as Shape rises the sine's curves are progressively squashed flat - rounded at low Shape, then a soft-clipped near-square, then a hard square at the top. One knob sweeps the entire continuum from sine to square WITHOUT the abrupt harmonic jumps of switching waveforms or the aliasing of a raw square, because the edges stay smoothly rounded. A continuous waveform morph distinct from the fixed geometric shapes. Needs no input.

Param	Range	Default	Unit
Freq	20 – 4000	110	Hz
Shape	0.1 – 8	2	—

TriangularOsc ▯ inputs

Triangular-number oscillator (a first as a synth block): an additive tone whose overtones land on the triangular numbers - the 1st, 3rd, 6th, 10th, 15th, 21st... harmonics ($k(k+1)/2$). The gaps between partials grow by one each step, so the spectrum starts dense and thins out smoothly as it climbs, giving a clear-but-airy body that sits between a full harmonic saw and a sparse sine stack. Partial sets how many triangular overtones to sound, alias-free up to Nyquist. A figurate-number spectrum distinct from the prime and Fibonacci oscillators. Needs no input.

Param	Range	Default	Unit
Freq	20 – 4000	110	Hz
Partial	2 – 16	8	—

SquareNumOsc ▯ inputs

Square-number oscillator (a first as a synth block): an additive tone (NOT a square wave) whose overtones sit on the perfect squares - the 1st, 4th, 9th, 16th, 25th... harmonics. Because the partials thin out quadratically, the few that remain are widely and unevenly spaced, giving a clear, glassy, almost-pure tone with a faint metallic ring - the spectral fingerprint of the squares. Partial sets how many square overtones to sound, alias-free up to Nyquist. A figurate spectrum distinct from the triangular and prime oscillators. Needs no input.

Param	Range	Default	Unit
Freq	20 – 4000	110	Hz
Partial	2 – 12	7	—

OctaveOsc ▯ inputs

Octave-stack oscillator (a first as a synth block): an additive tone whose overtones are only the powers of two - the 1st, 2nd, 4th, 8th, 16th... harmonics, i.e. the fundamental plus a stack of pure octaves above it. Because every partial is an octave of the one below, they fuse into a single, enormously reinforced pitch with the bright, hollow body of a full organ stop or a Shepard-tone layer,

no in-between harmonics to muddy it. Partial sets how many octaves to stack, alias-free up to Nyquist. A pure-octave spectrum distinct from the dense harmonic saw. Needs no input.

Param	Range	Default	Unit
Freq	20 – 4000	110	Hz
Partial s	2 – 10	6	—

LucasOsc 0 inputs

Lucas-number oscillator (a first as a synth block): an additive tone whose overtones land on the Lucas numbers - 1, 3, 4, 7, 11, 18, 29... - the Fibonacci sequence's companion, built on the same golden-ratio recurrence but from a different start. Its partials share Fibonacci's phi-spaced widening yet sit on different harmonics, so it has a related warm-but-glassy character with its own distinct colour - a sibling timbre to the FibonacciOsc. Partial sets how many Lucas overtones to sound, alias-free up to Nyquist. Needs no input.

Param	Range	Default	Unit
Freq	20 – 4000	110	Hz
Partial s	2 – 14	8	—

CatalanOsc 0 inputs

Catalan-number oscillator (a first as a synth block): an additive tone whose overtones land on the Catalan numbers - the 1st, 2nd, 5th, 14th, 42nd... harmonics. Because the Catalan numbers explode almost as fast as 4^n , only a few partials fit below Nyquist and they are spaced ever further apart, so the tone is nearly a pure sine with two or three widely-separated, ringing upper partials - a strikingly sparse, bell-clear spectrum. Partial sets how many Catalan overtones to attempt, alias-free up to Nyquist. A combinatorial spectrum distinct from the other number-theoretic oscillators. Needs no input.

Param	Range	Default	Unit
Freq	20 – 2000	110	Hz
Partial s	2 – 8	5	—

Bongo 2 inputs

Modal bongo: a trigger on in0 strikes a small tuned hand drum - a tight high-pitched membrane whose inharmonic modes (about 1 : 1.6 : 2.3) ring briefly with a quick downward pitch drop on the hit and a touch of finger-slap noise. Tune sets the head pitch (high or low bongo), Decay the short ring, Slap the attack-noise amount, Level the output. Trigger from a Clock, Euclid or sequencer.

Param	Range	Default	Unit
Tune	150 – 600	320	Hz
Decay	0.05 – 0.6	0.18	s
Slap	0 – 1	0.3	—
Level	0 – 1	0.6	—

Conga 2 inputs

Modal conga: a trigger on in0 strikes a tall hand drum - a deeper, woodier membrane than the Bongo, with modes near 1 : 1.6 : 2.2, a longer open ring and a pronounced pitch drop. Tune sets the head pitch, Decay the ring length, Tone moves from an open tone toward a tighter heel-slap, Level the output. Trigger from a Clock, Euclid or sequencer.

Param	Range	Default	Unit
Tune	80 – 350	160	Hz
Decay	0.1 – 1.2	0.45	s

Param	Range	Default	Unit
Tone	0 – 1	0.4	—
Level	0 – 1	0.6	—

Clave 2 inputs

Modal clave / woodblock: a trigger on in0 strikes a solid piece of wood - two tight high resonant modes (about 1 : 1.5) that ring for only a few milliseconds, the bright dry 'tok' that cuts through any mix. Tune sets the pitch (clave up high, woodblock lower), Decay the very short ring, Level the output. No pitch bend, no body - just wood. Trigger from a Clock, Euclid or sequencer.

Param	Range	Default	Unit
Tune	600 – 2600	1200	Hz
Decay	0.01 – 0.15	0.05	s
Level	0 – 1	0.5	—

Tabla 2 inputs

Modal tabla: a trigger on in0 strikes a tuned Indian hand drum whose loaded membrane gives near-harmonic modes (1 : 2 : 3 : 4) plus its signature pitch-bend - the fundamental swoops as it rings, the singing 'tun' of the dayan. Tune sets the pitch, Decay the ring, Bend how far the pitch glides during the hit, Level the output. Trigger from a Clock, Euclid or sequencer.

Param	Range	Default	Unit
Tune	100 – 500	200	Hz
Decay	0.1 – 1.5	0.5	s
Bend	0 – 1	0.5	—
Level	0 – 1	0.55	—

Djembe 2 inputs

Modal djembe: a trigger on in0 strikes a West-African goblet drum - a deep resonant bass membrane (modes near 1 : 1.5 : 2.1) with a sharp bright finger-slap on top, so it spans from a booming bass tone to a cracking slap. Tune sets the head pitch, Decay the ring, Slap the bright attack-noise amount, Level the output. Trigger from a Clock, Euclid or sequencer.

Param	Range	Default	Unit
Tune	60 – 250	110	Hz
Decay	0.1 – 1	0.4	s
Slap	0 – 1	0.45	—
Level	0 – 1	0.6	—

Cajon 2 inputs

Modal cajon: a trigger on in0 plays the box drum - a low wooden front-plate resonance (the bass 'thump' when struck in the centre) layered with a bright buzzing rattle of internal snare wires that rings on after the tone. Tune sets the box pitch, Decay the body, Snare the wire-buzz amount and brightness, Level the output. Trigger from a Clock, Euclid or sequencer.

Param	Range	Default	Unit
Tune	60 – 250	95	Hz
Decay	0.05 – 0.6	0.18	s
Snare	0 – 1	0.5	—
Level	0 – 1	0.6	—

Timbale 2 inputs

Modal timbale: a trigger on in0 strikes a shallow metal-shell drum - a tuned head plus the bright metallic inharmonic ring of the steel shell (modes near 1 : 2.7 : 4.2), the cutting crack of salsa and Latin percussion. Tune sets the pitch, Decay the ring, Ring the metallic shell brightness, Level the output. Trigger from a Clock, Euclid or sequencer.

Param	Range	Default	Unit
Tune	150 – 600	300	Hz
Decay	0.05 – 0.8	0.25	s
Ring	0 – 1	0.5	—
Level	0 – 1	0.55	—

Surdo 2 inputs

Modal surdo: a trigger on in0 strikes the deep Brazilian samba bass drum - a very low membrane (modes near 1 : 1.8) with a long booming decay and a slow pitch drop, the heartbeat under a samba bateria. Tune sets the low pitch, Decay the long boom, Mute tightens it toward a damped stroke, Level the output. Trigger from a Clock, Euclid or sequencer.

Param	Range	Default	Unit
Tune	40 – 160	65	Hz
Decay	0.2 – 2	0.8	s
Mute	0 – 1	0.2	—
Level	0 – 1	0.7	—

Shaker 1 input

Shaker / maraca: a trigger on in0 fires a soft burst of band-passed noise with a quick 'tsh' envelope - the egg-shaker and maraca sound, all air and no pitch. Tone sets the noise band (low and dark to high and crisp), Decay the burst length, Level the output. Needs only a trigger; chain two at offset times for a swung shaker groove.

Param	Range	Default	Unit
Tone	0 – 1	0.5	—
Decay	0.01 – 0.3	0.06	s
Level	0 – 1	0.5	—

Cabasa 1 input

Cabasa: a trigger on in0 fires the steel-bead cabasa - a brighter, raspier, grittier rattle than the Shaker, with a sharp metallic-bead attack and a fast scrape-like decay. Tone shifts the bead brightness, Decay the rasp length, Level the output. Needs only a trigger.

Param	Range	Default	Unit
Tone	0 – 1	0.5	—
Decay	0.008 – 0.2	0.05	s
Level	0 – 1	0.5	—

Tambourine 2 inputs

Modal tambourine: a trigger on in0 shakes a ring of jingles - a cluster of high, bright, inharmonic metallic partials that shimmer and ring, struck with a noise transient. Tune sets the jingle pitch, Decay the shimmer length, Jingle the metallic-partial richness, Level the output. Trigger from a Clock, Euclid or sequencer for tambourine hits.

Param	Range	Default	Unit
Tune	1500 – 6000	3200	Hz
Decay	0.05 – 0.6	0.2	s
Jingle	0 – 1	0.6	—
Level	0 – 1	0.45	—

Guiro 1 input

Guiro scraper: a trigger on in0 plays one stroke across the ridges of a guiro - a 'rrrip' of band-passed noise rhythmically chopped by the teeth at Rate, the rasping scrape of Latin and cumbia percussion. Rate sets how fast the ridges pass, Length the stroke duration, Tone the brightness, Level the output. Re-trigger for the up-down stroke pattern.

Param	Range	Default	Unit
Rate	0 – 1	0.5	—
Length	0.02 – 0.5	0.15	s
Tone	0 – 1	0.5	—
Level	0 – 1	0.5	—

Triangle 2 inputs

Modal triangle: a trigger on in0 strikes a metal triangle - a cluster of high inharmonic metallic partials with a long, bright, shimmering ring that sits on top of a mix. Tune sets the pitch, Decay the long sustain, Shimmer the upper-partial brightness, Level the output. A clean, sustained metallic ping, unlike the short noisy shakers. Trigger from a Clock, Euclid or sequencer.

Param	Range	Default	Unit
Tune	800 – 4000	2000	Hz
Decay	0.3 – 4	1.5	s
Shimmer	0 – 1	0.6	—
Level	0 – 1	0.4	—

Cuica 2 inputs

Modal cuica: a trigger on in0 plays the Brazilian friction drum - a resonant tone whose pitch swoops up or down as a wet thumb drags the internal stick, the squeaking, talking, almost human voice of a samba bateria. Tune sets the base pitch, Decay the ring, Glide the direction and depth of the pitch swoop (bipolar), Level the output. Trigger from a Clock, Euclid or sequencer; modulate Glide for talking lines.

Param	Range	Default	Unit
Tune	120 – 800	350	Hz
Decay	0.1 – 1	0.4	s
Glide	-1 – 1	0.6	—
Level	0 – 1	0.55	—

Vibraslap 1 input

Modal vibraslap: a trigger on in0 cracks the ball against the wooden box, then the loose pins rattle and buzz as they ring out - the sharp 'crack-brrrrr' heard at the top of so many percussion breaks. Tone sets the rattle brightness, Decay how long the pins buzz, Level the output. Needs only a trigger.

Param	Range	Default	Unit
Tone	0 – 1	0.5	—
Decay	0.1 – 1.2	0.5	s

Param	Range	Default	Unit
Level	0 – 1	0.5	—

Berimbau 2 inputs

Modal berimbau: a trigger on in0 strikes the steel string of the Brazilian musical bow - a bright metallic plucked tone (near-harmonic partials) with a gourd-resonated body and the buzz of the caxixi rattle, the hypnotic pulse of capoeira. Tune sets the string pitch (open/pressed via Pitch CV), Decay the ring, Buzz the rattle amount, Level the output. Trigger from a Clock, Euclid or sequencer.

Param	Range	Default	Unit
Tune	80 – 500	160	Hz
Decay	0.1 – 1.5	0.6	s
Buzz	0 – 1	0.3	—
Level	0 – 1	0.5	—

Gong 2 inputs

Modal gong: a trigger on in0 strikes a large gong - a dense cluster of low inharmonic metal partials that wash together into a shimmering, slowly-decaying roar with a long sustain. Tune sets the pitch, Decay the long tail, Shimmer the upper-partial brightness and clangour, Level the output. Distinct from the noise-based Cymbal - this is tuned, blooming metal. Trigger from a Clock, Euclid or sequencer.

Param	Range	Default	Unit
Tune	50 – 400	120	Hz
Decay	0.5 – 6	2.5	s
Shimmer	0 – 1	0.6	—
Level	0 – 1	0.45	—

Anvil 2 inputs

Modal anvil: a trigger on in0 strikes iron - a hard, bright, dense burst of high inharmonic metal partials with a fast attack and a ringing metallic clang, the blacksmith's hammer or industrial-percussion hit. Tune sets the pitch, Decay the ring, Ring the metallic richness, Level the output. Trigger from a Clock, Euclid or sequencer.

Param	Range	Default	Unit
Tune	300 – 2000	800	Hz
Decay	0.05 – 1	0.3	s
Ring	0 – 1	0.6	—
Level	0 – 1	0.4	—

Whip 1 input

Whip crack / slapstick: a trigger on in0 fires a single sharp crack - a fast burst of bright band-passed noise with a quick downward zip, the slapstick thwack and whip-crack of cartoon and orchestral percussion. Tone sets the brightness of the crack, Decay its (very short) length, Level the output. Needs only a trigger.

Param	Range	Default	Unit
Tone	0 – 1	0.6	—
Decay	0.005 – 0.08	0.02	s
Level	0 – 1	0.5	—

Flute 2 inputs

Breathy flute voice: a gate on in0 sounds a soft nearly-pure tone (fundamental with a whisper of upper harmonics) wrapped in breath noise and a gentle vibrato, with a slow soft attack. Freq sets the note, Breath the airy-noise amount, Vibrato the pitch waver, Level the output. Hold the gate to sustain; in1 is the Pitch CV.

Param	Range	Default	Unit
Freq	80 – 2000	440	Hz
Breath	0 – 1	0.25	—
Vibrato	0 – 1	0.3	—
Level	0 – 1	0.5	—

Sax 2 inputs

Saxophone voice: a gate on in0 sounds a reedy harmonically-rich tone (a soft sawtooth) with vibrato - honking low, singing high. Freq sets the note, Tone the brightness and reediness (number of harmonics), Vibrato the waver, Level the output. Hold the gate to sustain; in1 is the Pitch CV.

Param	Range	Default	Unit
Freq	80 – 1500	330	Hz
Tone	0 – 1	0.5	—
Vibrato	0 – 1	0.3	—
Level	0 – 1	0.35	—

Trumpet 2 inputs

Trumpet voice: a gate on in0 sounds a bright brassy tone - harmonically rich with a brass-like formant peak that opens up as the note swells in. Freq sets the note, Bright the brassiness (upper-harmonic weight), Vibrato the waver, Level the output. Hold the gate to sustain; in1 is the Pitch CV.

Param	Range	Default	Unit
Freq	100 – 1200	440	Hz
Bright	0 – 1	0.6	—
Vibrato	0 – 1	0.25	—
Level	0 – 1	0.35	—

Didgeridoo 2 inputs

Didgeridoo voice: a gate on in0 drones a deep buzzing tone whose overtones swell and fade under a slow wah-like formant sweep, the circular-breathing rhythm of the instrument. Freq sets the drone pitch (very low), Wobble the speed and depth of the formant sweep, Tone the overall brightness, Level the output. Hold the gate to drone; in1 is the Pitch CV.

Param	Range	Default	Unit
Freq	40 – 160	70	Hz
Wobble	0 – 1	0.5	—
Tone	0 – 1	0.6	—
Level	0 – 1	0.4	—

Harmonica 2 inputs

Harmonica voice: a gate on in0 sounds two slightly-detuned free reeds beating together into the bright wheezing tone of a blues harp, with vibrato. Freq sets the note, Detune the reed beating (thin to thick), Vibrato the waver, Level the output. Hold the gate to sustain; in1 is the Pitch CV.

Param	Range	Default	Unit
Freq	100 – 2000	440	Hz
Detune	0 – 1	0.4	—
Vibrato	0 – 1	0.4	—
Level	0 – 1	0.35	—

Accordion 2 inputs

Accordion voice: a gate on in0 sounds a stack of two or three slightly-detuned reed voices into the warm wheezy musette of an accordion or bandoneon. Freq sets the note, Musette the detuning between the reed banks (dry to wide wet musette), Tone the brightness, Level the output. Hold the gate to sustain; in1 is the Pitch CV.

Param	Range	Default	Unit
Freq	80 – 1500	330	Hz
Musette	0 – 1	0.5	—
Tone	0 – 1	0.5	—
Level	0 – 1	0.3	—

Organ 1 input

Additive drawbar organ oscillator that sums four sine partials at the first four harmonics of Freq. Bar1, Bar2 and Bar3 set the drawbar levels of the fundamental, second and third harmonics (the fourth harmonic tracks Bar2 at half level), and Level sets the overall output. A self-running tone generator with no audio input, voiced from sine-only partials for a clean drawbar timbre.

Param	Range	Default	Unit
Freq	20 – 4000	220	Hz
Bar1	0 – 1	1	—
Bar2	0 – 1	0.6	—
Bar3	0 – 1	0.4	—
Level	0 – 1	0.5	—

ColorNoise 0 inputs

Noise generator producing white, pink or brown noise from an xorshift source filtered to taste. Color selects the spectrum: flat white, a one-pole pink-tilted blend, or a heavily integrated brown noise with a steeper low-frequency emphasis. Level sets the output amplitude, making it a self-running source for textures, test signals or modulation.

Param	Range	Default	Unit
Color	0 – 2	1	—
Level	0 – 1	0.5	—

Pulsaret 1 input

Pulsar synthesis oscillator: at each fundamental period a Hann-windowed sinusoidal grain (the pulsaret) is retriggered, with silence filling the rest of the cycle. Freq sets the fundamental (grain repetition rate), Formant sets the sine frequency inside the grain (formant pitch), Width sets the fraction of the period the grain occupies, and Level scales the output. Narrow widths and high formants give buzzy, formant-rich tones.

Param	Range	Default	Unit
Freq	20 – 2000	110	Hz
Formant	1 – 16	4	—
Width	0.1 – 1	0.5	—

Param	Range	Default	Unit
Level	0 – 1	0.5	—

WaveScan 1 input

Additive harmonic scanner: sums a series of sine partials whose amplitudes follow a power-law roll-off, then normalizes the result. Freq sets the fundamental, Harmonics sets how many partials are stacked, Scan tilts the spectrum from bright (all harmonics equal at 0) toward dark (fundamental only as it approaches 1), and Level scales the output. Sweep Scan for a sawtooth-to-sine timbral morph.

Param	Range	Default	Unit
Freq	20 – 2000	110	Hz
Scan	0 – 1	0.3	—
Harmonics	1 – 16	8	—
Level	0 – 1	0.5	—

SuperPWM 1 input

Supersaw built from detuned pulse waves: stacks several pulse oscillators spread symmetrically in tuning and averages them for a thick unison. Freq sets the center pitch, Detune spreads the voices apart in tuning, Width sets the pulse duty cycle (timbre), Voices sets how many oscillators stack, and Level scales the output. High detune and voice counts give wide, rich pad and lead tones.

Param	Range	Default	Unit
Freq	20 – 2000	110	Hz
Detune	0 – 1	0.3	—
Width	0.05 – 0.95	0.5	—
Voices	1 – 7	5	—
Level	0 – 1	0.5	—

BellFM 1 input

Two-operator FM bell: a sine carrier is phase-modulated by a sine modulator running at a non-integer multiple of the carrier frequency, producing inharmonic metallic partials. Freq sets the carrier pitch, Ratio sets the modulator-to-carrier frequency ratio (typically fractional for bell tones), Index sets the modulation depth (brightness and partial spread), and Level scales the output. Higher Index and non-integer Ratio yield clangorous, bell-like timbres.

Param	Range	Default	Unit
Freq	20 – 2000	220	Hz
Ratio	0.5 – 8	3.5	—
Index	0 – 8	3	—
Level	0 – 1	0.5	—

Wind 0 inputs

Wind texture from filtered noise: white noise is fed through a resonant TPT band-pass whose center frequency is modulated by a slow 0.3 Hz gust LFO. Cutoff sets the band-pass center frequency, Reso sets the resonance (how whistling/pitched the noise becomes), Gust scales the LFO cutoff sweep depth, and Level scales the output. Produces ambient wind, breath and atmospheric beds.

Param	Range	Default	Unit
Cutoff	100 – 8000	800	Hz
Reso	1 – 20	8	—

Param	Range	Default	Unit
Gust	0 – 1	0.5	—
Level	0 – 1	0.5	—

Filter

94 modules

SynthFilter 3 inputs

State-variable filter (low/high/band) with resonance and pitch mod.

Param	Range	Default	Unit
Cutoff	20 – 20000	1000	Hz
Reso	0.1 – 10	0.7	Q
Mode	0 – 2	0	—
Mod	0 – 4	0	oct

DiodeLadder 3 inputs

Diode-ladder low-pass filter (TB-303 lineage) with resonance.

Param	Range	Default	Unit
Cutoff	20 – 18000	1000	Hz
Reso	0 – 1	0.4	—
Mod	0 – 4	0	oct
Drive	1 – 8	1	—

Vocoder 2 inputs

Analysis/synthesis vocoder (Pigments-style): splits the carrier at in0 and modulator at in1 into log-spaced bandpass bands, tracks each modulator band envelope, and multiplies the matching carrier band by it before summing, all blended by Mix against the dry carrier. Bands sets the number of analysis bands (4 to 40 - more bands give more intelligible speech imprinting), Q sets band sharpness, Attack sets the envelope follower attack time in ms (shorter tracks transients tighter), Mode picks the band character (Vintage soft to 6 kHz / Modern bright to 10 kHz / Dirty saturated), and Shift formant-shifts the carrier synthesis bands against the modulator analysis bands for chipmunk or deep-throat colours.

Param	Range	Default	Unit
Bands	4 – 40	16	—
Q	1 – 16	6	—
Attack	1 – 50	5	ms
Mix	0 – 1	1	—
Mode	Vintage · Modern · Dirty		—
Shift	0.25 – 4	1	—

PZFilter 1 input

Drawable magnitude-response filter: samples a frequency-to-gain curve drawn in node config at 16 log-spaced bands and bakes one peaking-EQ biquad per band, passing the input at in0 through the series to reproduce the drawn shape. Amount morphs between the dry input and the fully filtered output. Draw boosts and cuts across the spectrum to build custom EQ curves or resonant filter shapes.

Param	Range	Default	Unit
Amount	0 – 1	1	—

PeakFilter 1 input

A biquad peaking EQ band that boosts a single frequency region, followed by a tanh soft limiter to tame the boosted peaks. Freq sets the center frequency (40..16000 Hz), Gain sets the boost amount (0..18 dB), Edge sets the band Q/sharpness, and Mix blends the filtered signal against dry. Useful for emphasizing a resonant frequency without hard clipping when pushed.

Param	Range	Default	Unit
Freq	40 – 16000	1000	Hz
Gain	0 – 18	9	dB
Edge	0.5 – 12	3	Q
Mix	0 – 1	1	—

FilterLFO 1 input

A 4-pole resonant ladder lowpass swept by an internal LFO that morphs between triangle and sawtooth shapes. Rate sets the LFO speed (Hz), Range sets the sweep depth around the cutoff (~60 Hz..15.6 kHz), Reso sets the resonant feedback, Shape morphs the LFO from triangle to saw, and Mix blends wet against dry. High Reso adds squelch while Shape changes the sweep from symmetric to ramped.

Param	Range	Default	Unit
Rate	0.02 – 12	0.5	Hz
Range	0 – 1	0.7	—
Reso	0 – 1	0.5	—
Shape	0 – 1	0	—
Mix	0 – 1	1	—

LowPassGate 2 inputs

Buchla west-coast low-pass gate: a vactrol couples a VCA and VCF so a ping opens both with the characteristic nonlinear ringing decay. Mode picks VCA / VCF / both.

Param	Range	Default	Unit
Ping	0.01 – 2	0.4	s
Color	0 – 1	0.5	—
Mode	0 – 2	2	—
Level	0 – 1	0.8	—

SplitEQ 1 input

A peaking EQ that knows transients from tone: a fast/slow envelope split weighs the band gain toward the attack or the sustain, so you can brighten transients without harshening the body.

Param	Range	Default	Unit
Freq	40 – 16000	2000	Hz
Transient	-18 – 18	4	dB
Tonal	-18 – 18	0	dB
Q	0.3 – 8	1.5	—
Mix	0 – 1	1	—

Mini 1 input

Minimoog-style 4-pole transistor ladder: per-stage saturation thins the bass as resonance climbs to self-oscillation.

Param	Range	Default	Unit
Cutoff	20 – 18000	1200	Hz
Reso	0 – 1	0.3	—
Mod	0 – 4	0	oct
Drive	1 – 8	1	—

Jup-8 1 input

Jupiter-8-style 4-pole OTA ladder: per-stage OTA transconductance saturation with smooth resonance that keeps the low end (no Moog bass loss).

Param	Range	Default	Unit
Cutoff	20 – 18000	1200	Hz
Reso	0 – 1	0.3	—
Mod	0 – 4	0	oct
Drive	1 – 8	1	—

MS-20 1 input

MS-20-style 2-pole Sallen-Key low-pass: input op-amp saturation and a saturated resonance integrator plus diode clipping in the resonance path - screams and distorts at high resonance.

Param	Range	Default	Unit
Cutoff	20 – 18000	1000	Hz
Reso	0 – 1	0.3	—
Mod	0 – 4	0	oct
Drive	1 – 8	1	—

SEM 1 input

Oberheim SEM-style 2-pole state-variable filter: VCA-transistor saturation in the resonance integrator, continuously morphing low-pass through notch to high-pass.

Param	Range	Default	Unit
Cutoff	20 – 18000	1000	Hz
Reso	0 – 1	0.3	—
Mod	0 – 4	0	oct
Morph	0 – 1	0	—

EMS 1 input

EMS VCS3-style 3-pole (18 dB) diode ladder: asymmetric diode shaping gives a sharp, acidic resonance.

Param	Range	Default	Unit
Cutoff	20 – 18000	1000	Hz
Reso	0 – 1	0.4	—
Mod	0 – 4	0	oct
Drive	1 – 8	1	—

Wasp 1 input

EDP Wasp-style 12 dB multimode filter: the CMOS-inverter gain-stage outputs hard-clip at the rails for a fizzy, gritty bite. Mode 0 LP / 1 BP / 2 HP.

Param	Range	Default	Unit
Cutoff	20 – 18000	1000	Hz
Reso	0 – 1	0.4	—
Mod	0 – 4	0	oct
Mode	LP · BP · HP		—

Acido 1 input

TB-303-style 4-pole diode ladder: per-stage diode soft-clip and a diode-shaped feedback stage for high-resonance acid squelch.

Param	Range	Default	Unit
Cutoff	20 – 18000	800	Hz
Reso	0 – 1	0.6	—
Mod	0 – 4	0	oct
Drive	1 – 8	1.5	—

Pultec 1 input

Pultec EQP-1A-style passive EQ: low boost and cut use offset corners so they do not cancel (the boost-with-dip trick), plus gentle tube warmth.

Param	Range	Default	Unit
Low Boost	0 – 14	0	dB
Low Cut	0 – 17	0	dB
Low Freq	20 – 150	60	Hz
High Boost	0 – 18	0	dB
High Freq	3000 – 16000	5000	Hz
High Cut	0 – 16	0	dB

API 550 1 input

API 550-style 3-band EQ with proportional-Q (bands widen at low boost, tighten at high boost) and op-amp soft-clip bite.

Param	Range	Default	Unit
Low	-12 – 12	0	dB
Low Freq	30 – 400	100	Hz
Mid	-12 – 12	0	dB
Mid Freq	200 – 5000	800	Hz
High	-12 – 12	0	dB
High Freq	2500 – 20000	8000	Hz

N1073 1 input

Neve 1073-style EQ: fixed 12 kHz HF shelf, sweepable mid bell and LF shelf, into an iron-core transformer whose flux saturates the low frequencies first for genuine LF harmonic warmth.

Param	Range	Default	Unit
Low	-16 – 16	0	dB
Low Freq	35 – 220	60	Hz
Mid	-18 – 18	0	dB
Mid Freq	360 – 7200	1600	Hz
High	-16 – 16	0	dB

SSL E 1 input

SSL E-series-style 4-band parametric EQ: constant-Q (bandwidth stays put as you boost or cut) for a clean, surgical response.

Param	Range	Default	Unit
Low	-15 – 15	0	dB
Low Freq	30 – 450	80	Hz
LMF	-15 – 15	0	dB
LMF Freq	200 – 2500	500	Hz
HMF	-15 – 15	0	dB
HMF Freq	600 – 7000	3000	Hz
High	-15 – 15	0	dB

SSL G 1 input

SSL G-series-style 4-band parametric EQ: proportional-Q (the more you boost or cut, the narrower the band) for a musical, self-limiting curve.

Param	Range	Default	Unit
Low	-15 – 15	0	dB
Low Freq	30 – 450	80	Hz
LMF	-15 – 15	0	dB
LMF Freq	200 – 2500	500	Hz
HMF	-15 – 15	0	dB
HMF Freq	600 – 7000	3000	Hz
High	-15 – 15	0	dB

Manley 1 input

Manley Massive Passive-style EQ: wide bandwidth limits the available gain (opposite of proportional-Q), passive LC curves into an output transformer with LF-flux iron harmonics.

Param	Range	Default	Unit
Low	-11 – 11	0	dB
Low Freq	22 – 1000	100	Hz
Bandwidth	0 – 1	0.5	—
High	-11 – 11	0	dB
High Freq	560 – 20000	8000	Hz

BBE 1 input

BBE Sonic Maximizer-style processor: a phase-alignment network advances the highs ahead of the (allpass-delayed) lows and adds exciter-style HF harmonics for clarity and punch.

Param	Range	Default	Unit
Process	0 – 1	0.4	—
LoContour	0 – 1	0.3	—
Mix	0 – 1	1	—

Helios 1 input

Helios Type 69-style console EQ: a low bell, a switchable peak-or-trough mid at stepped frequencies, and a 10 kHz high shelf - punchy and musical.

Param	Range	Default	Unit
Low	-15 – 15	0	dB
Low Freq	60 – 400	100	Hz
Mid	-10 – 15	0	dB
Mid Freq	700 – 6000	1400	Hz
High	-16 – 12	0	dB

GML 1 input

GML 8200-style parametric EQ: ultra-clean, transparent constant-Q bands with no saturation - the surgical reference where boost and cut cancel exactly.

Param	Range	Default	Unit
Low	-15 – 15	0	dB
Low Freq	30 – 800	120	Hz
Mid	-15 – 15	0	dB
Mid Freq	120 – 8000	1000	Hz
High	-15 – 15	0	dB
High Freq	1000 – 20000	8000	Hz

Trident 1 input

Trident A-Range-style console EQ: four fixed musical bands (LF shelf, two mid bells, HF shelf) with an aggressive, forward voicing.

Param	Range	Default	Unit
Low	-15 – 15	0	dB
Lo Mid	-15 – 15	0	dB
Hi Mid	-15 – 15	0	dB
High	-15 – 15	0	dB

Chandler 1 input

Chandler Curve Bender-style EQ (EMI/Abbey Road): musical stepped bands with transformer/valve harmonic colour.

Param	Range	Default	Unit
Low	-15 – 15	0	dB
Low Freq	30 – 300	60	Hz
Mid	-15 – 15	0	dB
Mid Freq	200 – 6000	1000	Hz
High	-15 – 15	0	dB
High Freq	3000 – 20000	10000	Hz

API 560 1 input

API 560-style graphic EQ: fixed octave bands with proportional-Q so adjacent bands sum smoothly (4-band).

Param	Range	Default	Unit
31 Hz	-12 – 12	0	dB
125 Hz	-12 – 12	0	dB
1 kHz	-12 – 12	0	dB
8 kHz	-12 – 12	0	dB

MEQ5 1 input

Pultec MEQ-5-style midrange passive EQ: a low-mid boost, a mid dip and a high-mid boost for vocal/guitar presence.

Param	Range	Default	Unit
Low Mid	0 – 12	0	dB
Dip	0 – 12	0	dB
Hi Mid	0 – 12	0	dB

Maag 1 input

Maag EQ4-style EQ: musical low and mid bands plus the famous Air Band - a very high shelf for open, silky top.

Param	Range	Default	Unit
Low	-10 – 10	0	dB
Mid	-10 – 10	0	dB
Mid Freq	200 – 5000	1000	Hz
Air	0 – 10	0	dB

Avalon 1 input

Avalon VT-737-style tube channel EQ: clean, musical bell and shelf bands with gentle tube warmth.

Param	Range	Default	Unit
Low	-14 – 14	0	dB
Low Freq	30 – 300	80	Hz
Mid	-14 – 14	0	dB
Mid Freq	200 – 6000	1000	Hz
High	-14 – 14	0	dB
High Freq	3000 – 20000	10000	Hz

Matrix12 1 input

Oberheim Xpander / Matrix-12 multimode VCF (CEM3372): a 4-pole OTA ladder whose Mode taps a binomial mix of the four pole outputs to produce low-pass, high-pass and band-pass responses at 1-, 2- and 4-pole slopes from a single filter -- the signature Oberheim pole-mixing the SEM's 2-pole SVF cannot do. Per-stage tanh models the OTA soft clip and Reso feeds the 4-pole output back for resonance up to self-oscillation. Cutoff sweeps it, Drive pushes the stages.

Param	Range	Default	Unit
Cutoff	20 – 18000	1200	Hz
Reso	0 – 1	0.3	—
Mode	4P LP · 2P LP · 1P LP · 4P HP · 2P HP · 1P HP · 2P BP · 4P BP	—	—

Param	Range	Default	Unit
Drive	1 – 8	1	—

Polivoks 1 input

Soviet Polivoks 2-pole OTA filter: its resonance path clips like a comparator, so at high Reso it does not ring sweetly -- it slams into a hard square-wave self-oscillation and screams. A 2x-oversampled two-integrator SVF whose band-pass feedback is hard-clipped (not soft tanh) gives that brutal, buzzy Eastern-bloc character no Moog/SEM filter has. Cutoff, Reso, Drive into the core, and Mode taps LP / BP / HP.

Param	Range	Default	Unit
Cutoff	20 – 16000	800	Hz
Reso	0 – 1	0.4	—
Drive	1 – 16	2	—
Mode	LP · BP · HP		—

CS80 1 input

Yamaha CS-80 dual resonant filter: a 2-pole resonant high-pass in series with a 2-pole resonant low-pass, each swept independently for the lush, vocal, brass-and-strings voice (Vangelis / Blade Runner) a single multimode filter cannot make. HP Cutoff lifts the bottom, LP Cutoff closes the top, Reso peaks both corners (soft-saturated, musical -- not a screaming self-oscillator), Drive pushes the input.

Param	Range	Default	Unit
LP Cutoff	60 – 16000	4000	Hz
HP Cutoff	20 – 6000	80	Hz
Reso	0 – 1	0.3	—
Drive	1 – 8	1	—

Odyssey 1 input

ARP Odyssey 2-pole filter (4023): brighter and reedier than the 4-pole ladders, with a loud, aggressive resonance that bites and sings -- the thin, vocal ARP lead voice. A soft-saturated 2-pole state-variable filter tapping LP / BP / HP via Mode. Cutoff sweeps it, Reso peaks the corner toward self-oscillation, Drive pushes the input.

Param	Range	Default	Unit
Cutoff	30 – 16000	1200	Hz
Reso	0 – 1	0.4	—
Mode	LP · BP · HP		—
Drive	1 – 12	2	—

ARP2600 1 input

ARP 2600-style 4072 four-pole ladder: aggressive, clean self-oscillating resonance with a brighter, wetter bite than a Moog ladder.

Param	Range	Default	Unit
Cutoff	20 – 18000	1200	Hz
Reso	0 – 1	0.4	—
Mod	0 – 4	0	oct
Drive	1 – 8	1	—

Polysix 1 input

Korg Polysix-style SSM2044 four-pole low-pass: gentle OTA saturation gives a smooth, creamy resonance that stays musical right up to self-oscillation.

Param	Range	Default	Unit
Cutoff	20 – 18000	1100	Hz
Reso	0 – 1	0.4	—
Mod	0 – 4	0	oct
Drive	1 – 8	1	—

Korg35 1 input

Korg MS-10-style Korg35 Sallen-Key low-pass: hard diode-clipped resonance that screams and squares up at high settings - more aggressive than the later MS-20 OTA revision.

Param	Range	Default	Unit
Cutoff	20 – 18000	1000	Hz
Reso	0 – 1	0.4	—
Mod	0 – 4	0	oct
Drive	1 – 8	1	—

Synthacon 1 input

Steiner-Parker Synthacon-style diode multimode filter: an asymmetric diode-clipped resonance path gives a snarly, vocal bite, and Mode sweeps the output continuously LP -> BP -> HP - rougher than a clean SVF.

Param	Range	Default	Unit
Cutoff	20 – 18000	1000	Hz
Reso	0 – 1	0.4	—
Mod	0 – 4	0	oct
Mode	0 – 2	0	—
Drive	1 – 8	1	—

VactrolLP6 1 input

Buchla 292-style low-pass gate: a vactrol (LED + photocell) opens a 2-pole low-pass and a VCA together, and the photocell's slow nonlinear light-decay gives the plucky 'bongo' ring as transients fade. Mode picks gate (VCF+VCA), filter-only, or amp-only.

Param	Range	Default	Unit
Cutoff	100 – 16000	6000	Hz
Decay	20 – 2000	300	ms
Amount	0 – 1	1	—
Mode	Both · VCF · VCA		—

MuTron3 1 input

Mu-Tron III-style envelope auto-wah: an OTA state-variable filter whose cutoff is swept by an envelope follower of the playing dynamics. Sens sets the sweep depth, Direction flips up/down, Peak sets resonance, Mode picks LP/BP/HP.

Param	Range	Default	Unit
Sens	0 – 1	0.5	—

Param	Range	Default	Unit
Peak	0 – 1	0.5	—
Range	200 – 2000	500	Hz
Direction	Up · Down		—
Mode	LP · BP · HP		—

CEM3320 3 inputs

Curtis CEM3320-style 4-pole OTA filter (Prophet-5 rev3 / SH-101 / Pro-One): four OTA integrator poles with per-stage soft clip and a bass-compensated resonance path, so it stays fat where a Moog ladder thins out - cleaner, brighter, more aggressive into self-oscillation. Cutoff, Reso to self-osc, Mod-CV depth (in 3), Drive pushes the cells.

Param	Range	Default	Unit
Cutoff	20 – 18000	1200	Hz
Reso	0 – 1	0.3	—
Mod	0 – 4	0	oct
Drive	1 – 6	1	—

CryBaby 1 input

Dunlop Cry Baby / Vox inductor wah: a resonant inductor-cap band-pass swept by Pedal over the throaty 350 Hz - 2.2 kHz range, with the Q peaking through mid-sweep the way a real wah inductor does and falling at the heel/toe. Pedal sets the position (modulate it for auto-wah), Q the bite, Mix the blend.

Param	Range	Default	Unit
Pedal	0 – 1	0.5	—
Q	0 – 1	0.6	—
Mix	0 – 1	1	—

Talkbox 1 input

Vowel formant filter (talkbox-style): passes the input through three resonant formant band-passes whose centre frequencies morph across the A-E-I-O-U vowels, imposing a singing/talking vocal colour on any sound so a guitar, synth or drum loop 'speaks' a vowel. Vowel selects and morphs the formant, Res sets the formant sharpness, and Mix blends against the dry. Single-input (unlike the Vocoder, which needs a separate modulator) and it resonates the formants rather than shifting existing ones.

Param	Range	Default	Unit
Vowel	A · E · I · O · U		—
Res	1 – 20	10	—
Mix	0 – 1	1	—

Cluster 1 input

Cluster filter (Pigments-style): places up to five resonant band-pass filters around a single Cutoff, fanned outward by Spread, and sums them into one animated, vowel-like comb of peaks. Sweeping Cutoff glides the whole cluster; widening Spread pulls the peaks apart from a single resonance into a chord of formants; Reso sharpens each peak toward self-oscillation; Mix blends against dry. Distinct from the harmonic SMR bank (whose bands track a fundamental's overtones) - here the peaks are symmetric octave-spread satellites of one cutoff.

Param	Range	Default	Unit
Cutoff	40 – 8000	600	Hz
Spread	0 – 1	0.5	—

Param	Range	Default	Unit
Bands	1 – 5	5	—
Reso	1 – 30	8	—
Mix	0 – 1	1	—

LoFi 1 input

Lo-fi filter (Pigments-style): degrades the signal by undersampling (Downsample sample-and-holds the input at a reduced rate for aliasing grit), bit-reducing (Bits quantises the amplitude for digital crunch), injecting Noise (a hiss/dirt floor), then smoothing with a Cutoff low-pass - from subtle vintage-sampler warmth to extreme broken-converter mangling. Mix blends against dry. A degradation filter rather than a resonant one: use it to age pads, crush drums or add converter character.

Param	Range	Default	Unit
Cutoff	200 – 16000	6000	Hz
Downsample	1 – 50	4	—
Bits	2 – 16	12	—
Noise	0 – 1	0.1	—
Mix	0 – 1	1	—

Sherman 2 inputs

Sherman Filterbank (circuit model): a faithful model of the Belgian dual analog filter / distortion box. The input hits a VCA-overdrive stage (asymmetric soft clip -> even-harmonic grind, hotter Drive = more distortion and a stronger envelope), then TWO two-integrator (Chamberlin) state-variable filters whose resonance feedback path is saturated, so high Reso screams and self-oscillates instead of blowing up - the Sherman's signature. A rectifier + RC envelope follower tracks the (hot) input and sweeps the cutoff by Env, Freq sets filter 1's base cutoff, Harmonics offsets filter 2 in semitones, and Mode routes the pair Serial (1->2 LP/BP/HP), Parallel (LP+HP or BP+BP) or Ring (the two band-passes ring-modulated). in1 is the FM input (patch an LFO/envelope/audio to frequency-modulate the cutoff). Mix is the Bypass/Effect blend.

Param	Range	Default	Unit
Drive	1 – 64	8	—
Freq	40 – 8000	600	Hz
Reso	0 – 1	0.7	—
Env	-1 – 1	0.5	—
Harmonics	-24 – 24	12	st
Mode	Serial LP · Serial BP · Serial HP · Par LP+HP · Par BP · Ring		
Mix	0 – 1	1	—

XMf 1 input

Cross-modulation filter (u-he Zebra XMf-style): two resonant state-variable filters run in parallel, but each one's cutoff is modulated by the OTHER filter's output, so they chase each other into FM-like sidebands, growl and screaming resonant textures a single filter cannot make. Cutoff sets the base frequency, Spread offsets the two filters, XMod the cross-modulation depth, Reso the resonance, and Mix the wet blend. At high XMod and Reso it self-oscillates into metallic, vocal, clangorous tones.

Param	Range	Default	Unit
Cutoff	40 – 8000	800	Hz
Spread	0 – 24	7	st
XMod	0 – 1	0.3	—
Reso	0 – 0.95	0.5	—
Mix	0 – 1	1	—

Pipe 1 input

Waveguide pipe resonator (Absynth Pipe-style): a tuned feedback comb that rings at the Tune frequency, switchable between an Open pipe (full harmonic series, positive feedback) and a Closed pipe (only odd harmonics, inverted feedback) for a hollow, clarinet-like colour. Feedback sets the resonance and ring time, Damp rolls off the highs in the loop for a softer pipe, and Mix blends the resonated signal against the dry input. Excite it with a transient or noise burst for plucked/blown-tube tones.

Param	Range	Default	Unit
Tune	40 – 4000	220	Hz
Feedback	0 – 0.99	0.9	—
Mode	Open · Closed		—
Damp	0 – 1	0.3	—
Mix	0 – 1	0.5	—

Filter 3 inputs

Multimode state-variable (TPT) filter switchable between low-pass, high-pass and band-pass via Mode. Cutoff tracks an exponential mod input (in 3) scaled by Mod in octaves, while Reso sets the Q up to self-oscillation and an optional pre-Drive (tanh) dirties the resonance. Mix blends the filtered result back against the dry signal.

Param	Range	Default	Unit
Cutoff	20 – 20000	1000	Hz
Reso	0.1 – 10	0.7	Q
Mode	0 – 2	0	—
Mod	0 – 4	0	oct
Drive	0 – 24	0	dB
Mix	0 – 1	1	—

SMR 1 input

Spectral multiband resonator (4ms SMR): six resonant band-pass filters tuned in a harmonic series excite the in0 input into a ringing, vocal chord - an oscillator bank, resonator and vocoder-like effect in one. Tune sets the lowest band, Spread the spacing of the six bands (harmonic to stretched), Reso the resonance/ring time of each band, and Mix the wet blend. Feed it drums or noise for struck metallic chords, or audio for formant/vocoder colour. Distinct from the single-band Resonator and the modal Rings.

Param	Range	Default	Unit
Tune	40 – 2000	110	Hz
Spread	0.5 – 3	1	—
Reso	0 – 1	0.7	—
Mix	0 – 1	0.8	—

ThreeSisters 1 input

Triple resonant filter (Mannequins Three Sisters): three filters - Low, Centre, High - move together by Freq while Span spreads their cutoffs apart. Quality sweeps from a gentle resonant emphasis up to self-oscillating ringing sine tones (a three-voice resonance choir); Mode switches between the spread filter and a Formant voicing. Feed it audio for sweepable formants and resonant drones. Distinct from the single-band filters and the SMR resonator bank.

Param	Range	Default	Unit
Freq	30 – 8000	500	Hz
Span	0 – 1	0.4	—
Quality	0 – 1	0.3	—
Mode	Filter · Formant		—

AutoWah 1 input

Envelope-following resonant band-pass (auto-wah): the input level sweeps a band-pass from Base upward by Range, with Sens setting how strongly playing dynamics open it and Attack the follow speed. Reso sets the peak sharpness and Mix the blend. Funky on guitar, clav and bass; raise Reso for a vocal quack.

Param	Range	Default	Unit
Sens	0 – 1	0.6	—
Base	100 – 2000	400	Hz
Range	200 – 8000	3000	Hz
Reso	0.3 – 8	3	Q
Attack	1 – 200	15	ms
Mix	0 – 1	1	—

Formant 1 input

Morphing vowel filter: three parallel band-pass resonators tuned to formant frequencies are swept continuously across the A-E-I-O-U vowels by Vowel, with Q setting the formant sharpness and Mix the blend. Produces talking, vocal timbres. Sweep Vowel with an LFO or envelope for a wah-like vowel morph.

Param	Range	Default	Unit
Vowel	0 – 4	0	—
Q	2 – 20	10	—
Mix	0 – 1	1	—

EQ 1 input

Single parametric peaking EQ band (RBJ biquad): Gain boosts or cuts a bell centred at Freq, with Q setting its width - narrow for surgical notches, wide for broad tone shaping. Chain several for a full parametric EQ. See GraphicEQ for fixed bands and the analog pack (Pultec/Neve/SSL) for coloured curves.

Param	Range	Default	Unit
Freq	30 – 18000	1000	Hz
Gain	-18 – 18	0	dB
Q	0.3 – 8	1	—

ParaEQ 2 inputs

Fully-parametric EQ band with stereo / mid-side / L / R processing - stack several for an Ableton EQ-Eight-style mid-side parametric EQ. Type selects the RBJ filter shape (low cut, low shelf, bell, high shelf, high cut, notch), Freq the centre/corner, Gain the boost or cut (shelf/bell), Q the width, and Mode whether the band acts on the full Stereo signal, only the Mid, only the Side, or one channel (L/R). Out picks L or R. Reads in0 = L, in1 = R; place two instances (Out=L and Out=R) for a stereo band.

Param	Range	Default	Unit
Type	Low Cut · Low Shelf · Bell · High Shelf · High Cut · Notch		—
Freq	20 – 20000	1000	Hz
Gain	-24 – 24	0	dB
Q	0.1 – 10	1	—
Mode	Stereo · Mid · Side · L · R		—
Out	0 – 1	0	—

Ladder 3 inputs

Moog-style 4-pole transistor-ladder low-pass: four cascaded one-poles with a tanh input Drive and global resonance feedback that self-oscillates near maximum Reso. Cutoff tracks an exponential mod input (in 3) in octaves. Warmer and more characterful than the clean Filter - push Drive into the ladder for growl; see the Mini analog model for a Minimoog-voiced variant.

Param	Range	Default	Unit
Cutoff	20 – 18000	1200	Hz
Reso	0 – 1	0.3	—
Mod	0 – 4	0	oct
Drive	1 – 8	1	—

Shelf 1 input

Low- or high-shelf EQ (RBJ shelving biquad; Type selects low or high): boosts or cuts everything below/above Freq by Gain at a fixed slope. Use for broad tonal tilt - warming the lows or adding air to the highs. Pair two for a tilt curve, or see TiltEQ for a single-knob version.

Param	Range	Default	Unit
Freq	30 – 18000	2000	Hz
Gain	-18 – 18	0	dB
Type	0 – 1	0	—

DJFilter 3 inputs

One-knob DJ filter: with Cutoff at centre the signal is open, turning down sweeps in a low-pass and turning up sweeps in a high-pass, while Reso adds bite at the corner. The Mod input (in 3) offsets the knob for automation. The classic build-and-drop sweep on a single control.

Param	Range	Default	Unit
Cutoff	0 – 1	0.5	—
Reso	0.1 – 8	0.7	Q
Mod	-1 – 1	0	—

Notch 1 input

Band-reject (notch) filter that removes a narrow band around Freq, with Q setting its width and Mix the blend. Use it to kill hum, a resonance or a feedback frequency, or sweep it for phaser-like effects. Narrow Q for surgical removal, wider for broad scooping.

Param	Range	Default	Unit
Freq	30 – 18000	1000	Hz
Q	0.3 – 16	4	—
Mix	0 – 1	1	—

Wah 3 inputs

Manually-swept resonant band-pass: Pedal sweeps the peak across Range, with Reso setting the quack and Mix the blend. Drive Pedal from the mod input (in 3) - an LFO, envelope or mod wheel - for auto-wah, talk-box or pedal-wah effects. See AutoWah for a built-in envelope follower.

Param	Range	Default	Unit
Pedal	0 – 1	0.5	—

Param	Range	Default	Unit
Range	200 – 3000	1800	Hz
Reso	1 – 10	5	Q
Mod	-1 – 1	0	—
Mix	0 – 1	1	—

TiltEQ 1 input

One-knob spectral tilt that splits the signal at a pivot frequency with a one-pole low-pass and rocks the low and high halves in opposite directions. Tilt sets the slope from -1 (low band up, high band down) through 0 (flat) to +1 (high band up, low band down), Freq places the pivot the split happens around, and Range scales the maximum boost/cut in dB applied to each half. Useful as a fast tone balancer for brightening or warming a source without reaching for a full EQ.

Param	Range	Default	Unit
Tilt	-1 – 1	0	—
Freq	100 – 8000	1000	Hz
Range	0 – 12	6	dB

Telephone 1 input

Band-limits the input to roughly 300 Hz-3.4 kHz by cascading a one-pole low-pass at 3.4 kHz into a one-pole high-pass at 300 Hz, leaving only the narrow midrange band that survives a phone line. Mix crossfades between the dry signal and the band-passed result. Use it for lo-fi telephone, intercom and AM-radio voice effects.

Param	Range	Default	Unit
Mix	0 – 1	1	—

SallenKey 3 inputs

Resonant low-pass built on a TPT state-variable core taking the low-pass output, with the cutoff frequency exponentially modulated by in 3 scaled by Mod in octaves. Cutoff sets the corner, Reso raises the Q toward self-oscillation, and Mix blends the filtered low-pass back against the dry input. The Sallen-Key topology gives a smooth, gentle resonance suited to warm synth and bass filtering.

Param	Range	Default	Unit
Cutoff	20 – 20000	1000	Hz
Reso	0 – 1	0.3	—
Mod	0 – 4	0	oct
Mix	0 – 1	1	—

Steiner 1 input

Steiner-Parker style multimode filter whose TPT core derives low-pass, high-pass, band-pass and notch responses simultaneously, with Mode selecting which one is output. Cutoff sets the corner frequency, Reso controls the Q toward self-oscillation, and Mix crossfades the selected response against the dry signal. Switching modes covers everything from dark sweeps to hollow notch tones for synth and effect duty.

Param	Range	Default	Unit
Cutoff	20 – 20000	1000	Hz
Reso	0 – 1	0.3	—
Mode	0 – 3	0	—
Mix	0 – 1	1	—

Crossover 1 input

Two-band splitter that derives a low band from two cascaded one-pole low-passes (an LR2 response) and forms the high band by subtracting it from the input, outputting one band at a time. Freq sets the split frequency and Band chooses whether the low band or the high band is passed. Use it to route or isolate a frequency region, or to feed separate processing chains per band.

Param	Range	Default	Unit
Freq	50 – 5000	800	Hz
Band	0 – 1	0	—

Resonator 1 input

High-Q band-pass tuned to a musical pitch, converting the Pitch parameter in semitones to a centre frequency and ringing the input through it like a struck body. Pitch sets the resonant note, Reso sets the Q (sharpness and ring length), and Mix blends the doubled resonant output against the dry signal. Drive it with transients or noise for tuned percussion, pinged-body and comb-resonance textures.

Param	Range	Default	Unit
Pitch	24 – 108	60	st
Reso	1 – 40	12	Q
Mix	0 – 1	1	—

Fizz 1 input

Resonant TPT band-pass filter whose output is mixed with a rectified copy of itself, adding fizzy upper harmonics that grow with resonance. Cutoff sets the band centre (100-12000 Hz), Reso sets both the filter Q and the amount of rectified harmonic content, and Mix blends the result against the dry signal. Use it to add bright, buzzy edge and emphasis to a narrow frequency band.

Param	Range	Default	Unit
Cutoff	100 – 12000	2000	Hz
Reso	0 – 1	0.6	—
Mix	0 – 1	1	—

Rasp 1 input

Resonant TPT band-pass filter followed by an asymmetric tanh waveshaper that drives positive half-cycles harder than negative ones, producing a rasping, lopsided distortion. Cutoff sets the band centre (100-12000 Hz), Reso sets the Q and scales the drive so higher resonance increases the snarl, and Mix blends against the dry signal. Ranges from a soft growl at low settings to an aggressive scream at high resonance.

Param	Range	Default	Unit
Cutoff	100 – 12000	1500	Hz
Reso	0 – 1	0.6	—
Mix	0 – 1	1	—

Ripple 1 input

Tuned comb resonator built from a short delay line whose length tracks a musical pitch, with a low-passed feedback path that makes it ring at that frequency. Pitch sets the resonant note as a MIDI value (24-108 st), Reso sets the feedback amount up to 0.98 for longer, sharper ringing, Damp sets the high-frequency damping in the feedback path (a darker, woodier ring as it rises), and Mix blends the resonant tone against the dry signal. Pushes input toward a pitched, metallic ring keyed to the chosen note.

Param	Range	Default	Unit
Pitch	24 – 108	60	st
Reso	0 – 0.98	0.7	—
Mix	0 – 1	1	—
Damp	0 – 1	0.8	—

PrimeResonator 1 input

Prime-tuned resonator bank (a first as a synth block): runs the input through a bank of sharp resonators tuned to the PRIME harmonics of Base ($\text{Base} \times 2, *3, *5, *7, *11 \dots$), so whatever you feed it - noise, a drum loop, a voice - rings out as a pitched drone whose overtones are the primes, the hollow bell-organ colour of the prime spectrum imposed on any source. Sharpness sets the resonance Q (how long each prime rings), and Mix blends the resonated tone against the dry. A number-tuned resonator distinct from a plain comb or formant filter.

Param	Range	Default	Unit
Base	20 – 1000	110	Hz
Sharpness	0.9 – 0.999	0.99	—
Partial	2 – 12	8	—
Mix	0 – 1	0.7	—

FibResonator 1 input

Fibonacci-tuned resonator bank (a first as a synth block): rings the input through resonators tuned to the Fibonacci harmonics of Base ($\text{Base} \times 1, *2, *3, *5, *8, *13 \dots$), so any source is recoloured into a drone whose overtones widen by the golden ratio - the warm, glassy Fibonacci spectrum imposed on whatever passes through. Sharpness sets the resonance Q and Mix the wet/dry blend. A golden-spaced resonator distinct from the prime-tuned and harmonic resonators.

Param	Range	Default	Unit
Base	20 – 1000	110	Hz
Sharpness	0.9 – 0.999	0.99	—
Partial	2 – 12	8	—
Mix	0 – 1	0.7	—

OddResonator 1 input

Odd-harmonic resonator bank (a first as a synth block): rings the input through resonators on the ODD harmonics of Base only ($\text{Base} \times 1, *3, *5, *7, *9 \dots$), the spectrum of a square wave or a stopped pipe, so any source takes on a hollow, woody, clarinet-like resonance with no even overtones. Sharpness sets the resonance Q and Mix the wet/dry blend. An odd-harmonic resonator distinct from the prime and Fibonacci banks.

Param	Range	Default	Unit
Base	20 – 1000	110	Hz
Sharpness	0.9 – 0.999	0.99	—
Partial	2 – 12	8	—
Mix	0 – 1	0.7	—

OctaveResonator 1 input

Octave-stack resonator bank (a first as a synth block): rings the input through resonators tuned to pure octaves of Base ($\text{Base} \times 1, *2, *4, *8, *16 \dots$), so any source is pulled toward a single strongly-reinforced pitch with the bright, hollow body of an organ stop - every resonance an octave of the last, nothing in between. Sharpness sets the resonance Q and Mix the wet/dry blend. A pure-octave resonator distinct from the dense harmonic and prime banks.

Param	Range	Default	Unit
Base	20 – 1000	110	Hz
Sharpness	0.9 – 0.999	0.99	—
Partial	2 – 10	6	—
Mix	0 – 1	0.7	—

GoldenResonator 1 input

Golden-ratio resonator bank (a first as a synth block): rings the input through resonators spaced by powers of the golden ratio (Base, Base*1.618, Base*2.618, Base*4.236...) - an INHARMONIC bank, so no resonance is a simple multiple of another and the source is recoloured into a shimmering, metallic, bell-like resonance that hangs between a pitch and a clangour. Sharpness sets the resonance Q and Mix the wet/dry blend. The inharmonic counterpart of the prime and octave resonators, ideal for turning drums or noise into struck-metal textures.

Param	Range	Default	Unit
Base	20 – 1000	110	Hz
Sharpness	0.9 – 0.999	0.99	—
Partial	2 – 12	8	—
Mix	0 – 1	0.7	—

DrumheadReson 1 input

Drumhead resonator (a first as a synth block): rings the input through a bank tuned to the modes of a circular DRUM MEMBRANE - the inharmonic Bessel-function overtones (1, 1.59, 2.14, 2.30, 2.65...) that give a struck skin its pitch-ambiguous, tom-like 'boing' rather than a clear note. Drop a drum loop, voice or noise through it and the source takes on the body of a tuned membrane. Freq sets the membrane pitch, Sharpness the ring time, Mix the wet/dry blend.

Param	Range	Default	Unit
Freq	20 – 2000	110	Hz
Sharpness	0.9 – 0.999	0.99	—
Mix	0 – 1	0.7	—

BarReson 1 input

Bar resonator (a first as a synth block): rings the input through the modes of a free-free metal or wooden BAR - the widely-stretched overtones (1, 2.76, 5.40, 8.93...) of a marimba, glockenspiel or music-box tine. Because the partials leap apart so fast it imposes a clear, bright, struck-mallet pitch on whatever passes through, turning drums or noise into tuned-percussion tones. Freq sets the bar's pitch, Sharpness the ring time, Mix the wet/dry blend.

Param	Range	Default	Unit
Freq	20 – 2000	110	Hz
Sharpness	0.9 – 0.999	0.99	—
Mix	0 – 1	0.7	—

PlateReson 1 input

Plate resonator (a first as a synth block): rings the input through the dense, inharmonic modes of a vibrating METAL PLATE (about 1, 2.08, 3.41, 3.89, 5.00, 6.45) - the crowded overtone field behind plate reverbs and the shimmer of a cymbal or gong. Its many close, non-integer partials smear any source into a bright metallic wash. Freq sets the plate's tuning, Sharpness the ring time, Mix the wet/dry blend. A 2-D plate spectrum distinct from the 1-D bar and string resonators.

Param	Range	Default	Unit
Freq	20 – 2000	110	Hz
Sharpness	0.9 – 0.999	0.99	—
Mix	0 – 1	0.7	—

BellReson 1 input

Bell resonator (a first as a synth block): rings the input through the partials of a cast BELL - the hum, prime, tierce, quint and nominal tones (0.5, 1, 1.2, 1.5, 2, 2.5, 3) whose deliberately-tuned MINOR THIRD gives a bell its dark, mournful clang. Feeding a source through it casts that bronze, struck-bell colour over it. Freq sets the strike tone, Sharpness the ring time, Mix the wet/dry blend. A tuned-bell spectrum distinct from the bar and plate resonators.

Param	Range	Default	Unit
Freq	20 – 2000	110	Hz
Sharpness	0.9 – 0.999	0.99	—
Mix	0 – 1	0.7	—

GlassReson 1 input

Glass resonator (a first as a synth block): rings the input through the sparse, nearly-pure modes of a rubbed wine GLASS or glass-harmonica bowl (about 1, 2.32, 4.25, 6.63, 9.38) - a few widely-spaced, slow-decaying partials that give glass its clear, singing, almost-sine voice. Passing a source through it lends that fragile crystalline ring. Freq sets the glass pitch, Sharpness the (long) ring time, Mix the wet/dry blend. A glassy sparse spectrum distinct from the dense plate and bell resonators.

Param	Range	Default	Unit
Freq	20 – 2000	110	Hz
Sharpness	0.9 – 0.999	0.99	—
Mix	0 – 1	0.7	—

MajorReson 1 input

Major-scale resonator (a first as a synth block): rings the input through a bank of resonators tuned to the seven notes of a MAJOR scale (plus the octave), so any source - a drum loop, noise, a voice - is pulled into a bright, consonant, major-key wash, an instant ambient pad. Root sets the key, Sharpness the ring time (turn it up for a sustaining drone), and Mix the wet/dry blend. A scale-tuned resonator distinct from the harmonic and object resonators.

Param	Range	Default	Unit
Root	20 – 1000	110	Hz
Sharpness	0.9 – 0.999	0.99	—
Mix	0 – 1	0.7	—

MinorReson 1 input

Minor-scale resonator (a first as a synth block): rings the input through resonators tuned to the seven notes of a NATURAL MINOR scale (plus the octave), recolouring any source into a darker, melancholic, minor-key drone. Root sets the key, Sharpness the ring time, Mix the wet/dry blend. The minor counterpart of the major resonator, ideal for turning drums or noise into a brooding ambient pad.

Param	Range	Default	Unit
Root	20 – 1000	110	Hz
Sharpness	0.9 – 0.999	0.99	—
Mix	0 – 1	0.7	—

PentatonicReson 1 input

Pentatonic resonator (a first as a synth block): rings the input through resonators tuned to the five notes of a MAJOR PENTATONIC scale (across two octaves) - the gap-toothed, no-semitone scale of folk and gamelan that has no dissonant intervals, so ANYTHING fed through it comes out consonant and open. Root sets the key, Sharpness the ring time, Mix the wet/dry blend. The most foolproof scale resonator for instant harmonious drones from any source.

Param	Range	Default	Unit
Root	20 – 1000	110	Hz
Sharpness	0.9 – 0.999	0.99	—
Mix	0 – 1	0.7	—

WholeToneReson 1 input

Whole-tone resonator (a first as a synth block): rings the input through resonators tuned to the six equal WHOLE-TONE steps (plus the octave) - the symmetrical, rootless scale of Debussy and dream sequences, with no perfect fifths to anchor it. Any source takes on a floating, ambiguous, shimmering quality with no clear home note. Root sets the centre, Sharpness the ring time, Mix the wet/dry blend. A symmetrical scale resonator distinct from the major and minor banks.

Param	Range	Default	Unit
Root	20 – 1000	110	Hz
Sharpness	0.9 – 0.999	0.99	—
Mix	0 – 1	0.7	—

ChromaticReson 1 input

Chromatic resonator (a first as a synth block): rings the input through resonators tuned to all twelve equal-tempered SEMITONES of the octave - a comb of every chromatic pitch, so it imposes a fixed twelve-tone pitch grid on whatever passes through, quantising noise or percussion toward an equal-tempered cluster. Root sets the lowest note, Sharpness the ring time, Mix the wet/dry blend. The densest scale resonator, a chromatic pitch-grid distinct from the consonant major/minor/pentatonic banks.

Param	Range	Default	Unit
Root	20 – 1000	110	Hz
Sharpness	0.9 – 0.999	0.99	—
Mix	0 – 1	0.7	—

SympatheticStrings 1 input

Sympathetic-string bank (a first as a synth block): rings the input through a bank of tuned, plucked-string loops (Karplus-Strong) set to a major chord, so any source - a drum hit, a breath, a noise burst - excites the strings into a rich, sustaining sympathetic resonance, the way the undamped tarab strings of a sitar or sarangi shimmer behind every note. Unlike the thin 2-pole resonators each string rings its WHOLE harmonic series, giving a dense, lively, plucked-metal sustain. Root sets the chord, Decay the sustain, Damp the brightness, Mix the wet/dry blend.

Param	Range	Default	Unit
Root	20 – 1000	110	Hz
Decay	0.8 – 0.999	0.98	—
Damp	0 – 0.9	0.4	—
Mix	0 – 1	0.6	—

Tanpura 1 input

Tanpura-drone string bank (a first as a synth block): rings the input through four sympathetic strings tuned to the classic Indian tanpura drone (Sa, Sa, Pa, high Sa), so any source sets off the endless, buzzing, overtone-rich Sa-Pa-Sa drone that underpins all

Hindustani music. The long feedback loops and gentle damping give the characteristic shimmering, never-quite-decaying jivari sustain. Root sets the tonic, Decay the (long) drone length, Damp the brightness, Mix the wet/dry blend. A specific drone tuning distinct from the chordal sympathetic bank.

Param	Range	Default	Unit
Root	20 – 1000	110	Hz
Decay	0.8 – 0.999	0.98	—
Damp	0 – 0.9	0.4	—
Mix	0 – 1	0.6	—

HarpReson 1 input

Harp-string bank (a first as a synth block): rings the input through eight plucked-string loops tuned to a full DIATONIC scale (a whole octave of harp strings), so any source ripples across the strings and rings back as cascading, scale-tuned harp resonance - feed it transients and it answers in glissando-like washes. Root sets the key, Decay the string sustain, Damp the brightness, Mix the wet/dry blend. A diatonic string bank distinct from the chordal and drone banks.

Param	Range	Default	Unit
Root	20 – 1000	110	Hz
Decay	0.8 – 0.999	0.98	—
Damp	0 – 0.9	0.4	—
Mix	0 – 1	0.6	—

KotoReson 1 input

Koto-string bank (a first as a synth block): rings the input through nine plucked-string loops tuned to a PENTATONIC scale across two octaves, the open, gap-toothed tuning of a Japanese koto or a guzheng. Because the pentatonic scale has no dissonant intervals, any source - even noise - comes back as a consonant, bright, plucked-string cascade. Root sets the key, Decay the string sustain, Damp the brightness, Mix the wet/dry blend. A pentatonic string bank distinct from the diatonic harp and chordal banks.

Param	Range	Default	Unit
Root	20 – 1000	110	Hz
Decay	0.8 – 0.999	0.98	—
Damp	0 – 0.9	0.4	—
Mix	0 – 1	0.6	—

PluckReson 1 input

Single-string resonator (a first as a synth block): rings the input through ONE tuned plucked-string loop (Karplus-Strong), turning any source into a sustained, pitched, plucked-string resonance at the Root note - a string reverb. Transients excite the string into a clear plucked ring; sustained input drives it like a bowed or e-bowed string. Root sets the pitch, Decay the sustain (toward infinite at the top), Damp the brightness, Mix the wet/dry blend. The single-string core of the sympathetic banks, a tuned string-reverb distinct from the multi-string drones.

Param	Range	Default	Unit
Root	20 – 1000	110	Hz
Decay	0.8 – 0.999	0.98	—
Damp	0 – 0.9	0.4	—
Mix	0 – 1	0.6	—

LPG 2 inputs

Vactrol-style low-pass gate: the control input (in 2) simultaneously opens a low-pass filter and a VCA, so brightness and level rise together as the control climbs. Response curves the control law from snappy to sluggish, emulating the vactrol's lag, and Mix blends against dry. Driven by an envelope or trigger it gives the plucky, organic dynamics of Buchla-style patches.

Param	Range	Default	Unit
Response	0 – 1	0.5	—
Mix	0 – 1	1	—

GraphicEQ 1 input

Three-band graphic EQ: one-pole filters at 250 Hz and 4 kHz split the input into low, mid and high bands, each scaled by its own +/-12 dB gain and summed back. Low, Mid and High set the per-band gain in dB and Mix blends against dry. A simple tonal shaping tool for broad low/mid/high balance rather than surgical correction.

Param	Range	Default	Unit
Low	-12 – 12	0	dB
Mid	-12 – 12	0	dB
High	-12 – 12	0	dB
Mix	0 – 1	1	—

Rings 1 input

Modal resonator bank: three TPT band-pass resonators tuned around a base pitch are excited by the input, summed to produce a struck/plucked resonant body. Pitch sets the fundamental in semitones, Structure morphs the partial spacing from harmonic (1, 2, 3) to inharmonic (1, 2.76, 5.40), Decay sets the resonator Q (ring length), and Bright scales the summed output level. Mix blends the resonant tone with the dry signal for physical-modeling and bell/string textures.

Param	Range	Default	Unit
Pitch	24 – 96	48	st
Structure	0 – 1	0.3	—
Decay	0 – 0.99	0.9	—
Bright	0 – 1	0.5	—
Mix	0 – 1	0.5	—

Wobble 1 input

Tempo-coherent resonant low-pass wobble: an LFO phased from c.time sweeps a TPT low-pass cutoff between 150 Hz and roughly 6 kHz for dubstep-style movement. Rate sets the LFO frequency, Depth scales the cutoff sweep range, Reso sets the filter resonance, and Shape selects sine, ramp or square LFO contour. Mix blends the wobbled output with the dry signal.

Param	Range	Default	Unit
Rate	0.25 – 8	2	Hz
Depth	0 – 1	0.8	—
Reso	0.5 – 15	8	—
Shape	0 – 2	0	—
Mix	0 – 1	1	—

Sweep 1 input

Onset-triggered filter sweep: an amplitude envelope detects input onsets and restarts a ramp that drives a TPT low-pass cutoff from 100 Hz upward for riser/faller effects. Time sets the ramp duration, Range scales how far the cutoff travels, Reso sets the filter resonance, and Dir flips the ramp between rising and falling. Mix blends the swept output with the dry signal.

Param	Range	Default	Unit
Time	0.1 – 5	1	s
Range	0 – 1	1	—

Param	Range	Default	Unit
Reso	0.5 – 15	3	—
Dir	0 – 1	0	—
Mix	0 – 1	1	—

Distortion

78 modules

Array 1 input

Drawable waveshaper: maps the input at in0 (folded from bipolar into a 0 to 1 index) through a transfer curve drawn in node config via linear interpolation, then back to bipolar, blended by Mix against the dry input. The drawn breakpoints define the input-to-output mapping and Mix sets the wet/dry balance. A diagonal curve is a clean pass-through while steeper or folded curves act as custom distortion and waveshaping.

Param	Range	Default	Unit
Mix	0 – 1	1	—

Mulholland 1 input

Tanh overdrive feeding a resonant Chamberlin state-variable lowpass, with a slow internal LFO wobbling the cutoff. Drive sets the input gain into the saturator (1..30), Cutoff sets the filter frequency (~80 Hz..14 kHz), Reso sets the filter Q, Mod sets how much the LFO modulates cutoff, and Mix blends wet against dry. Goes from gentle filtered grit to a screaming self-resonant squeal as Drive and Reso climb.

Param	Range	Default	Unit
Drive	1 – 30	8	—
Cutoff	0 – 1	0.5	—
Reso	0.5 – 12	6	—
Mod	0 – 1	0.2	—
Mix	0 – 1	1	—

NevePre 1 input

Neve 1073-style preamp: a class-A triode gain stage (grid conduction) into an iron-core output transformer whose LF flux saturates first - the bass 'sings' with iron warmth.

Param	Range	Default	Unit
Drive	1 – 12	2	—
Bias	0 – 0.3	0.08	—
Tone	2000 – 18000	12000	Hz
Mix	0 – 1	1	—

ApiPre 1 input

API 312-style preamp: a discrete 2520 op-amp (level-dependent knee) into an API output transformer with LF-flux iron, plus a bright top; the harmonic tilt shifts upward as level rises.

Param	Range	Default	Unit
Drive	1 – 12	2	—
Bias	0 – 0.2	0.04	—
Tone	3000 – 16000	8000	Hz

Param	Range	Default	Unit
Mix	0 – 1	1	—

TubePre 1 input

Vintage tube console-style preamp: two cascaded triode stages with an interstage coupling cap (2nd-harmonic-dominant asymmetric clipping) and a warm low shelf.

Param	Range	Default	Unit
Drive	1 – 16	3	—
Bias	0 – 0.4	0.15	—
Tone	0 – 6	2	dB
Mix	0 – 1	1	—

BusSat 1 input

SSL bus-style VCA saturation into an output transformer: a thin 'glue' veneer for the mix bus - subtle thickening, not obvious distortion.

Param	Range	Default	Unit
Drive	0 – 1	0.3	—
Bias	0 – 0.2	0.06	—
Mix	0 – 1	0.25	—

Studer 1 input

Studer A800-style tape: magnetic hysteresis (B/H saturation + remanence), a firm low-frequency head bump and gentle head-gap HF loss - punchy track-level glue (pro deck: flutter negligible by design).

Param	Range	Default	Unit
Drive	1 – 8	2	—
Head Bump	0 – 8	4	dB
HF Loss	4000 – 18000	12000	Hz
Mix	0 – 1	1	—

Ampex 1 input

Ampex ATR-102-style mastering tape: gentle magnetic hysteresis (B/H + remanence), a soft wide head bump and smooth head-gap HF roll-off - subtle mix-bus polish (pro deck: flutter negligible by design).

Param	Range	Default	Unit
Drive	1 – 6	1.5	—
Head Bump	0 – 6	2	dB
HF Loss	6000 – 18000	14000	Hz
Mix	0 – 1	0.7	—

TubeScreamer 1 input

Ibanez Tube Screamer-style overdrive: a real op-amp feedback diode clipper - the mid-focused clean gain is clamped by an antiparallel Shockley diode pair, Newton-solved and 4x oversampled (not a tanh) - with the famous ~720 Hz mid hump. Asym mismatches the two diodes for the asymmetric-clip mod, adding the even harmonics a symmetric clipper can't.

Param	Range	Default	Unit
Drive	1 – 50	15	—
Tone	1000 – 5000	2500	Hz
Level	0 – 1	0.5	—
Asym	0 – 1	0	—

RAT 1 input

ProCo RAT-style distortion: an LM308-slewed gain stage into a Newton-solved silicon diode pair clamping to ground (the real shunt clipper, 4x oversampled - not a hard jlimit), tamed by a single low-pass Filter knob.

Param	Range	Default	Unit
Drive	1 – 80	20	—
Filter	500 – 6000	3000	Hz
Level	0 – 1	0.4	—

BigMuff 1 input

Big Muff Pi-style fuzz: two cascaded silicon diode clipping stages (Newton-solved Shockley pairs, 4x oversampled) that square up into a wall of sustain, with a scooped-mid tone.

Param	Range	Default	Unit
Sustain	1 – 40	15	—
Tone	0 – 1	0.5	—
Level	0 – 1	0.5	—

FuzzFace 1 input

Fuzz Face-style germanium fuzz: two cascaded germanium common-emitter transistor stages (Ebers-Moll, 4x oversampled) whose asymmetric cutoff/saturation gives the gated, spitty fuzz; Bias starves the second stage toward the splatty/gated voicing, and it cleans up as you lower Input - just like rolling back the guitar volume.

Param	Range	Default	Unit
Fuzz	1 – 60	20	—
Bias	0.4 – 0.9	0.65	—
Input	0 – 1	1	—
Level	0 – 1	0.5	—

Klon 1 input

Klon Centaur-style overdrive: a Newton-solved germanium diode pair (4x oversampled) summed back with the clean signal - 'transparent' via the dual-path blend, not soft clipping.

Param	Range	Default	Unit
Gain	1 – 30	8	—
Treble	0 – 6	2	dB
Blend	0 – 1	0.6	—
Level	0 – 1	0.6	—

DS1 1 input

Boss DS-1-style distortion: a Newton-solved symmetric silicon diode pair (4x oversampled - the real shunt clipper, not a hard jlimit) with an active tilt-tone control - brash and cutting.

Param	Range	Default	Unit
Dist	1 – 60	20	—
Tone	0 – 1	0.5	—
Level	0 – 1	0.4	—

MXRDist 1 input

MXR Distortion+-style overdrive: a Newton-solved ASYMMETRIC germanium diode pair (4x oversampled) - mismatched diodes give the even-harmonic, woolier-than-a-DS-1 crunch - with an output low-pass.

Param	Range	Default	Unit
Dist	1 – 50	15	—
Level	0 – 1	0.5	—

Octavia 1 input

Octavia-style octave-up fuzz: 4x-oversampled full-wave rectification doubles the pitch, then heavy fuzz - the ring-mod-like upper-octave scream.

Param	Range	Default	Unit
Drive	1 – 40	15	—
Tone	1000 – 8000	3500	Hz
Level	0 – 1	0.5	—

Rangemaster 1 input

Dallas Rangemaster-style treble booster: a high-pass into a real germanium common-emitter transistor stage (Ebers-Moll, Newton-free closed form, 4x oversampled) - its asymmetric cutoff/saturation adds the bright, singing even+odd harmonics on top of the dry signal.

Param	Range	Default	Unit
Boost	1 – 20	6	—
Range	200 – 2000	700	Hz
Level	0 – 1	0.6	—

Aphex 1 input

Aphex Aural Exciter-style enhancer: a tuned high-pass side chain into an asymmetric class-A harmonic generator (2nd+3rd) and a phase-shift network, blended back for air and presence.

Param	Range	Default	Unit
Tune	1000 – 8000	3000	Hz
Harmonics	0 – 1	0.4	—
Mix	0 – 1	0.5	—

SuperFuzz 1 input

Univox Super-Fuzz-style octave fuzz: 4x-oversampled full-wave rectification for the octave-up into hard germanium square clipping, with a switchable 1 kHz mid-scoop notch.

Param	Range	Default	Unit
Drive	1 – 40	18	—
Tone	Flat · Scoop		—
Level	0 – 1	0.5	—

OCD 1 input

Fulltone OCD-style overdrive: 4x-oversampled asymmetric MOSFET hard clipping (smooth-edged) with a bright/dark voicing switch.

Param	Range	Default	Unit
Drive	1 – 40	12	—
Tone	1500 – 9000	4000	Hz
Voicing	LP · HP		—
Level	0 – 1	0.5	—

MetalZone 1 input

Boss MT-2 Metal Zone-style high-gain distortion: two 4x-oversampled clip stages into an active dual-peak mid scoop with a sweepable parametric mid.

Param	Range	Default	Unit
Gain	1 – 60	30	—
Mid Freq	240 – 5000	800	Hz
Mid Level	-15 – 15	0	dB
Scoop	0 – 1	0.6	—
Level	0 – 1	0.4	—

HeliosPre 1 input

Helios console-style preamp: punchy class-A transistor saturation into a console transformer (LF-flux iron) with an asymmetric Bias and a forward ~1.5 kHz midrange push. Drive sets the grit, Tone rolls the top, Mix blends dry/wet - aggressive British-console colour.

Param	Range	Default	Unit
Drive	1 – 12	2	—
Bias	0 – 0.2	0.05	—
Tone	2000 – 16000	9000	Hz
Mix	0 – 1	1	—

V76 1 input

Telefunken V76-style tube preamp: two cascaded tube stages with interstage coupling into an output transformer - rich, warm 2nd-harmonic saturation with a low-mid body.

Param	Range	Default	Unit
Drive	1 – 16	3	—
Bias	0 – 0.4	0.18	—

Param	Range	Default	Unit
Body	0 – 6	2	dB
Mix	0 – 1	1	—

ChandlerPre 1 input

Chandler TG2-style preamp (EMI/Abbey Road): an input stage into an iron-core transformer (LF flux saturation) - punchy with a distinctive forward presence.

Param	Range	Default	Unit
Drive	1 – 12	2	—
Bias	0 – 0.2	0.06	—
Tone	2000 – 16000	8000	Hz
Mix	0 – 1	1	—

ToneBender 1 input

Tone Bender MkII germanium fuzz: three cascaded germanium transistor stages, each asymmetric (germanium clips harder on one rail) with interstage coupling-cap high-pass between them, for the thick, sustaining, slightly-broken vintage fuzz. Bias starves the supply so quiet/decaying notes GATE into the famous spitty sputter; Drive sets the gain, Tone the output roll-off, Level the volume. Distinct from FuzzFace (2 transistors) and Big Muff (4 silicon stages + diodes).

Param	Range	Default	Unit
Drive	1 – 60	20	—
Bias	0 – 1	0.3	—
Tone	700 – 9000	3000	Hz
Level	0 – 1	0.7	—

CultureVulture 1 input

Thermionic Culture Vulture valve distortion: Triode mode is an asymmetric, even-harmonic soft saturation (warm, musical thickening); Pentode mode is a harder odd-harmonic clip with a push-pull crossover dip near zero for the snarling, aggressive 'P' destruction. Bias shifts the valve operating point (asymmetry / gate), Drive pushes it from fattening to full disintegration, Tone tilts the output, Level trims it. A valve character/destruction unit, not a pedal or amp.

Param	Range	Default	Unit
Drive	1 – 40	6	—
Mode	Triode · Pentode	—	—
Bias	0 – 1	0.3	—
Tone	-12 – 12	0	dB
Level	0 – 1	0.7	—

UA610 1 input

Universal Audio 610 tube preamp (Bill Putnam): a 12AX7 valve gain stage between input and output transformers. Push Drive to overdrive the tube into warm asymmetric saturation while Level pads the output -- the '610 move' of cranking the valve and taming the volume -- with transformer low-end thickening and the 610's Low/High program EQ. Distinct from the generic tube preamp by the transformer saturation and the gain/level interaction.

Param	Range	Default	Unit
Drive	1 – 40	4	—
Low	-12 – 12	0	dB

Param	Range	Default	Unit
High	-12 – 12	0	dB
Level	0 – 1	0.7	—

Hiwatt 1 input

Hiwatt DR103 amp (full circuit): three could-be-cleaner cascaded triode stages with a stiff power supply and a bright, high-headroom voice -- the clean-but-LOUD British military-grade tone (Townshend / Gilmour) that stays articulate where a Marshall breaks up. Gain drives the preamp, Tone tilts the stack, Master pushes the power stage, Level trims. Cascaded triode transfer + voiced tone stack + supply sag, not a single tanh + fixed EQ.

Param	Range	Default	Unit
Gain	0 – 1	0.5	—
Tone	-1 – 1	0	—
Master	0 – 1	0.5	—
Level	0 – 1	0.7	—

Orange 1 input

Orange OR-series amp (full circuit): mid-forward, thick and woody with more preamp gain and a spongier supply than the Hiwatt -- the chunky British crunch. Gain, Tone, Master, Level. Shares the cascaded-triode + voiced-tone-stack + supply-sag circuit model, voiced for Orange's mid push.

Param	Range	Default	Unit
Gain	0 – 1	0.55	—
Tone	-1 – 1	0	—
Master	0 – 1	0.5	—
Level	0 – 1	0.7	—

Dumble 1 input

Dumble Overdrive Special amp (full circuit): smooth, singing high-gain with a spongy supply for bloom and sustain -- the holy-grail boutique lead voice. Gain, Tone, Master, Level. Same cascaded-triode + voiced-tone-stack + supply-sag model, voiced for Dumble's compressed sustain.

Param	Range	Default	Unit
Gain	0 – 1	0.6	—
Tone	-1 – 1	0	—
Master	0 – 1	0.5	—
Level	0 – 1	0.7	—

Mark IIC 1 input

Mesa/Boogie Mark IIC+ amp (full circuit): tight, scooped, high-gain lead with a stiff supply for fast attack and a singing sustain (the 80s metal lead voice). Gain, Tone, Master, Level. Same cascaded-triode + voiced-tone-stack + supply-sag model, voiced for the Mark's tight low end and scooped mids -- distinct from the looser, fatter Rectifier.

Param	Range	Default	Unit
Gain	0 – 1	0.7	—
Tone	-1 – 1	0	—
Master	0 – 1	0.5	—
Level	0 – 1	0.7	—

SD1 1 input

Boss SD-1 Super Overdrive: an op-amp gain stage into an ASYMMETRIC soft clipper (two diodes one way, one the other) for a more open, even-harmonic crunch than the symmetric Tube Screamer, with the familiar mid-hump tone that cuts through a band. Drive sets the gain, Tone the mid/treble balance, Level the output.

Param	Range	Default	Unit
Drive	1 – 50	12	—
Tone	0 – 1	0.5	—
Level	0 – 1	0.6	—

JC120 1 input

Roland JC-120 Jazz Chorus (solid-state clean): the transistor clean platform -- a stiff power supply (no sag), big bright headroom that stays glassy and articulate, and a harder, more brittle clip when finally pushed, unlike the soft, compressing tube amps. Drive sets the gain, Tone the brightness, Level the output. Pair with a Chorus block for the full JC stereo shimmer.

Param	Range	Default	Unit
Drive	0 – 1	0.4	—
Tone	0 – 1	0.6	—
Level	0 – 1	0.7	—

Peavey5150 1 input

Peavey 5150 / 6505 amp (full circuit): tight, aggressive, high-gain crunch with a stiff supply for percussive attack -- the metalcore / Van Halen rhythm and lead standard. Gain, Tone, Master, Level. Shares the cascaded-triode + voiced-tone-stack + supply-sag model, voiced for the 5150's tight low end and biting upper mids.

Param	Range	Default	Unit
Gain	0 – 1	0.75	—
Tone	-1 – 1	0	—
Master	0 – 1	0.5	—
Level	0 – 1	0.7	—

SunnModelT 1 input

Sunn Model T amp (full circuit): fat, dark, high-headroom and loose -- the wall-of-low-end doom / sludge / stoner voice, the opposite of a tight bright amp. Gain, Tone, Master, Level. Same cascaded-triode + voiced-tone-stack + supply-sag model, voiced for huge bass, dark top and a spongy, blooming low end.

Param	Range	Default	Unit
Gain	0 – 1	0.5	—
Tone	-1 – 1	0	—
Master	0 – 1	0.5	—
Level	0 – 1	0.7	—

HM2 1 input

Boss HM-2 Heavy Metal ('Swedish chainsaw'): a high-gain hard clipper into the HM-2's scooped dual-band Color EQ. Turn Color up and it guts the mids while pushing a resonant low end and a fizzy top into the unmistakable buzzsaw wall of Swedish death metal (Entombed). Dist sets the gain, Color sweeps flat -> full chainsaw scoop, Level the output. Distinct from the mid-scoop MetalZone by the dual-band buzzsaw voicing.

Param	Range	Default	Unit
Dist	1 – 60	30	—
Color	0 – 1	0.8	—
Level	0 – 1	0.5	—

FZ1 1 input

Maestro FZ-1 Fuzz-Tone: the first fuzz pedal (the 'Satisfaction' buzz). Three germanium stages give a thin, splatty, ripped-speaker square-wave fuzz that GATES hard as the note decays into the famous sputtering decay. Thinner, harder and more gated than the thick Tone Bender or the FuzzFace. Attack sets the gain, Bias starves the supply (more gate/sputter), Level the output.

Param	Range	Default	Unit
Attack	1 – 60	25	—
Bias	0 – 1	0.4	—
Level	0 – 1	0.5	—

EP3 1 input

Echoplex EP-3 preamp: the Echoplex's FET preamp section, famous as an always-on tone boost (Page, Van Halen). A touch of FET saturation plus a presence bump and slightly tightened lows make whatever runs through it fuller, clearer and more present -- a buffer/boost with character, not a distortion or amp. Gain sets the boost, Bright the presence bump, Level the output.

Param	Range	Default	Unit
Gain	1 – 10	2.5	—
Bright	0 – 1	0.5	—
Level	0 – 1	0.8	—

BlueBox 1 input

MXR Blue Box-style octave fuzz: cascaded flip-flops divide the pitch down two octaves into a gritty square sub-octave, blended with a fuzzed dry signal - the synthetic, glitchy -2-octave fuzz.

Param	Range	Default	Unit
Drive	1 – 40	12	—
Sub	0 – 1	0.6	—
Tone	800 – 6000	2500	Hz
Level	0 – 1	0.5	—

FuzzFactory 1 input

Z.Vex Fuzz Factory-style two-transistor fuzz: massive asymmetric gain into a hard clip, with Gate choking the decay into a sputtery rip and Stab starving the bias toward a self-oscillating squeal that ducks under the note - the pinched, gated, velcro fuzz. 4x-oversampled.

Param	Range	Default	Unit
Drive	1 – 60	20	—
Gate	0 – 1	0.3	—
Stab	0 – 1	0	—
Tone	800 – 6000	2800	Hz
Level	0 – 1	0.5	—

Percolator 1 input

Interfax Harmonic Percolator-style fuzz: mixed germanium/silicon asymmetric clipping generates strong even-order harmonics for a wooly, octave-tinged thickness unlike symmetric fuzzes. Balance skews the asymmetry from even-rich to odd-rich. 4x-oversampled.

Param	Range	Default	Unit
Drive	1 – 50	14	—
Balance	-1 – 1	0.4	—
Tone	700 – 6000	2600	Hz
Level	0 – 1	0.5	—

SansAmp 1 input

Tech 21 SansAmp-style amp/cab DI: a Character tilt into cascaded asymmetric soft-clipping (4x oversampled) and a speaker-cabinet low-pass, giving the recorded amp tone direct with no mic. Drive sets the gain into the stages, Character tilts presence vs warmth, Level the output.

Param	Range	Default	Unit
Drive	1 – 40	8	—
Character	0 – 1	0.5	—
Level	-24 – 12	0	dB

BluesDriver 1 input

Boss BD-2 Blues Driver-style overdrive: a JFET-buffered asymmetric soft-clipper that stays bright, dynamic and amp-like - cleans up with pick attack, breaks up when pushed. Gain sets the drive, Tone the top-end tilt, Level the output.

Param	Range	Default	Unit
Gain	1 – 60	10	—
Tone	0 – 1	0.5	—
Level	-24 – 6	-6	dB

Guvnor 1 input

Marshall Guv'nor-style distortion: hard symmetric diode clipping after a Marshall-voiced mid push with an active treble tilt - the 'amp in a box' crunch-to-roar. Gain sets the clipping, Tone the brightness, Body the low-mid push, Level the output.

Param	Range	Default	Unit
Gain	1 – 80	20	—
Tone	0 – 1	0.5	—
Body	0 – 1	0.5	—
Level	-30 – 6	-8	dB

Timmy 1 input

Paul C Timmy-style transparent overdrive: low-gain op-amp soft clipping with cut-only Bass and Treble shaped BEFORE the gain (so it tightens and de-fizzes without adding colour) - clean, dynamic, amp-like. Gain sets the drive, Bass/Treble cut, Level the output.

Param	Range	Default	Unit
Gain	1 – 30	6	—

Param	Range	Default	Unit
Bass	0 – 1	1	—
Treble	0 – 1	1	—
Level	-18 – 6	-3	dB

Fullldrive 1 input

Fulltone Fulldrive-style overdrive: a TS-derived circuit opened up with less mid-hump and more gain on tap, asymmetric soft clipping that stays articulate when pushed, plus a cleaner Boost stage. Gain sets the drive, Tone the brightness, Boost stacks gain, Level the output.

Param	Range	Default	Unit
Gain	1 – 60	14	—
Tone	0 – 1	0.5	—
Boost	0 – 1	0	—
Level	-24 – 6	-6	dB

Wavemult 1 input

West-coast wave multiplier (Serge/Buchla-style): cascades several triangle wavefolders so the signal is folded again and again as Drive rises, multiplying the harmonics far beyond a single Wavefolder for ringing, metallic, formant-rich timbres. Drive sets the fold depth, Stages the number of cascaded folders (more = denser), Bias the fold asymmetry, and Mix the dry/wet blend.

Param	Range	Default	Unit
Drive	0 – 10	2	—
Stages	1 – 4	2	—
Bias	-1 – 1	0	—
Mix	0 – 1	1	—

Roar 1 input

Feedback waveshaper (Ableton Roar-style): drives the signal through a selectable saturation curve, but wraps it in a resonant feedback loop - the distorted output is band-passed at Color and fed back into the input, so it rings, blooms and self-oscillates into the gnarly 'roar' that plain saturation can't make. Drive sets the push into the Shape (soft sine / tube / soft clip / diode / fold), Tone tilts the spectrum before the shaper, Feedback how much output returns, Color the feedback resonance frequency, and Mix the dry/wet. (Single-shaper core; the full device's serial/parallel/multiband/mid-side routings need a rack.)

Param	Range	Default	Unit
Drive	0 – 1	0.4	—
Shape	Soft Sine · Tube · Soft Clip · Diode · Fold		—
Tone	-1 – 1	0	—
Feedback	0 – 0.98	0.3	—
Color	60 – 8000	600	Hz
Mix	0 – 1	1	—

DrumBuss 1 input

Drum-bus processor (Ableton Drum Buss-style): a one-stop chain for drum groups - Transient sharpens or softens the attack, Drive adds tube grit, Crunch a harder harmonic edge, and Boom adds a tuned resonant low-end thump at BoomFreq. Mix blends it back. Glues and beefs up a drum bus in one block.

Param	Range	Default	Unit
Drive	0 – 1	0.3	—
Crunch	0 – 1	0.2	—
Transient	-1 – 1	0.2	—
Boom	0 – 1	0.3	—
BoomFreq	40 – 200	80	Hz
Mix	0 – 1	1	—

Drive 1 input

Waveshaping overdrive with three curves (Type: tanh soft-clip, sine fold, or hard clip) driven by up to 48 dB of gain plus a Bias offset for even-harmonic asymmetry. A post Tone low-pass tames fizz, Mix blends dry/wet, and Output trims the final level. Spans subtle warmth to aggressive distortion depending on Drive and curve.

Param	Range	Default	Unit
Drive	0 – 48	12	dB
Output	-24 – 6	-6	dB
Bias	-1 – 1	0	—
Tone	200 – 18000	18000	Hz
Mix	0 – 1	1	—
Type	0 – 2	0	—

Wavefolder 1 input

West-coast wavefolder that multiplies the signal by Fold and reflects it through triangle folding boundaries, adding bright harmonics that bloom as level rises. Bias shifts the fold symmetry for asymmetric timbres, Mix blends dry/wet and Output trims gain. Unlike clipping, folding keeps energy moving, so the tone stays lively even at extreme settings.

Param	Range	Default	Unit
Fold	1 – 12	2	—
Bias	-1 – 1	0	—
Mix	0 – 1	1	—
Output	-24 – 6	0	dB

Bitcrush 1 input

Digital degradation: Bits quantizes the amplitude to 2^{Bits} levels (quantization noise) while Downsample holds each sample for N frames (sample-rate reduction and aliasing). Mix blends against the dry signal. Low Bits gives gritty lo-fi crunch; high Downsample adds metallic aliasing for retro and 8-bit textures.

Param	Range	Default	Unit
Bits	1 – 16	8	—
Downsample	1 – 64	4	x
Mix	0 – 1	1	—

Waveshaper 1 input

Memoryless waveshaper with four transfer curves (Shape: tanh, atan, cubic soft-clip, sine fold) driven by Drive and blended via Mix. Each curve adds a different harmonic signature, from smooth saturation (tanh/atan) to a foldback character (sine). Lighter-weight than Drive when you just want one fixed shaping curve.

Param	Range	Default	Unit
Drive	1 – 20	2	—
Shape	0 – 3	0	—
Mix	0 – 1	1	—

Exciter 1 input

Harmonic exciter: splits off the band above Freq, generates new upper harmonics from it with a tanh Drive, and blends them back via Amount for added air and presence. Subtle settings add sheen and intelligibility without raising broadband level. Most effective on dull vocals, drums and full mixes.

Param	Range	Default	Unit
Freq	500 – 12000	3000	Hz
Drive	1 – 20	4	—
Amount	0 – 1	0.4	—

Lofi 1 input

Lo-fi degrader: sample-rate reduction (holds each sample for Rate frames) plus a Tone low-pass and added tape Hiss, blended via Mix. Unlike Bitcrush it quantizes in time rather than bit-depth, layering filtering and noise for a warmer vintage-sampler grime. Push Rate for aliasing; lower Tone and raise Hiss for cassette/SP-style character.

Param	Range	Default	Unit
Rate	1 – 64	8	x
Tone	500 – 12000	4000	Hz
Hiss	0 – 1	0.05	—
Mix	0 – 1	1	—

Tube 1 input

Tube-style saturation: an asymmetric transfer curve (offset by Bias) adds the even and odd harmonics of a valve stage, tamed by a Tone low-pass and blended via Mix. Warmer and rounder than the harder Drive/Waveshaper - subtle settings thicken and glue, higher Drive gives singing tube grit. See the TubePre and VariMu analog models for circuit-specific versions.

Param	Range	Default	Unit
Drive	0 – 36	12	dB
Bias	0 – 1	0.3	—
Tone	500 – 16000	8000	Hz
Mix	0 – 1	1	—

Clip 1 input

Clipper that morphs from soft (tanh) to hard clipping at the Threshold via Hardness, with Output trim and dry/wet Mix. Soft adds rounded saturation harmonics; hard adds bright odd harmonics and caps peaks dead. Use as a tone shaper or a transparent peak ceiling ahead of a limiter.

Param	Range	Default	Unit
Threshold	-24 – 0	-6	dB
Hardness	0 – 1	0.5	—
Output	-12 – 12	0	dB
Mix	0 – 1	1	—

Shaper 1 input

SynthMaster-style 12-mode waveshaper: Shape selects soft/hard clip, tube, diode, sine and triangle fold, full/half rectify, sine, crush, sync-wrap and Chebyshev modes, each driven by Drive with Bias for symmetry. Mix and Output blend and trim. Modulate Shape to morph between distortion characters - a one-stop tone-shaping distortion.

Param	Range	Default	Unit
Drive	0 – 36	12	dB
Shape	0 – 11	0	—
Bias	-1 – 1	0	—
Mix	0 – 1	1	—
Output	-24 – 6	0	dB

Rectify 1 input

Rectifier waveshaper: full-wave (Mode 0) folds the negative half upward to add strong octave-up and even harmonics, while half-wave (Mode 1) keeps only the positive half. Mix blends with the dry signal. The classic octave-fuzz colour - aggressive alone, useful in small amounts for harmonic lift.

Param	Range	Default	Unit
Mode	0 – 1	0	—
Mix	0 – 1	1	—

Fuzz 1 input

High-gain fuzz: heavy drive soft-saturates then hard-clips into a square wave for a thick, sustaining buzz, with a Tone low-pass to tame the fuzz and Level to set output. Mix blends dry/wet. Aggressive and compressed - see the BigMuff and FuzzFace analog models for circuit-specific fuzz voicings.

Param	Range	Default	Unit
Drive	0 – 48	24	dB
Tone	0 – 1	0.5	—
Level	-24 – 6	-8	dB
Mix	0 – 1	1	—

OctaveFuzz 1 input

Octave fuzz: blends a fuzz-distorted signal with a DC-corrected full-wave-rectified upper octave, the classic Hendrix/Octavia scream. Drive sets gain, Octave the upper-octave blend and Mix dry/wet. Tracks best on clean single notes near the 12th fret; chords inter-modulate, by design.

Param	Range	Default	Unit
Drive	0 – 48	24	dB
Octave	0 – 1	0.6	—
Mix	0 – 1	1	—

Chebyshev 1 input

Chebyshev-polynomial shaper that synthesizes a single chosen Harmonic (2nd-8th) from the input via the T_n recurrence, blended by Amount and Mix. Unlike broadband clipping it adds one targeted overtone - the 2nd for octave thickness, higher orders for metallic ring. Pairs well with a clean sine input for precise harmonic generation.

Param	Range	Default	Unit
Harmonic	2 – 8	2	—
Amount	0 – 1	0.5	—
Mix	0 – 1	1	—

SubEnhance 1 input

Sub-bass enhancer: a flip-flop toggled per rising zero crossing generates a half-frequency square tracked by the input envelope and low-passed, added under the dry signal at Sub. Reinforces weak low end on kicks, bass and 808s. Monophonic by nature - best on clean low-register sources.

Param	Range	Default	Unit
Sub	0 – 1	0.5	—
Mix	0 – 1	1	—

TapeSat 1 input

Tape-style saturation: a soft tanh stage with a touch of linear blend emulates magnetic-tape compression, with Warmth rolling off the highs for a rounder tone and Mix blending dry/wet. Gentle glue and harmonic thickening rather than overt distortion. See the Studer and Ampex analog models for machine-specific tape.

Param	Range	Default	Unit
Drive	0 – 36	12	dB
Warmth	0 – 1	0.4	—
Mix	0 – 1	1	—

XformerSat 1 input

Transformer saturation: asymmetric soft-clipping knees (different for positive and negative swings) add even-harmonic colour, plus a low-passed component for iron-core low-end thickening. Subtle weight and warmth on bass, drums and the mix bus. See the NevePre analog model for a transformer-preamp variant.

Param	Range	Default	Unit
Drive	0 – 36	12	dB
Mix	0 – 1	1	—

DynSat 1 input

Dynamic saturation: the drive amount tracks the input envelope, so louder passages distort more and quiet ones stay clean (Sens sets how strongly). Adds harmonics on transients and peaks without muddying sustained low-level material. Good for program-dependent grit on drums and bus.

Param	Range	Default	Unit
Drive	0 – 36	6	dB
Sens	0 – 1	0.5	—
Mix	0 – 1	1	—

Erosion 1 input

Noise-modulation distortion: sprays band-limited noise (Freq sets its colour) onto the signal by multiplying it with the input, eroding the tone into gritty granular texture. Amount sets intensity and Mix dry/wet. From subtle dirt to harsh destruction - a creative degrade rather than a clean overdrive.

Param	Range	Default	Unit
Amount	0 – 1	0.4	—
Freq	0 – 1	0.5	—
Mix	0 – 1	1	—

Character 1 input

Character drive with five voicings (Mode): Push (soft clip), Heat (S-clip), Soar (sine fold), Howl (triangle fold) and Shred (wrap fold), each driven by Drive and blended via Mix. One module spanning warm saturation to wild folding distortion. Modulate Mode or Drive for evolving textures.

Param	Range	Default	Unit
Drive	0 – 36	12	dB
Mode	0 – 4	0	—
Mix	0 – 1	1	—

Vinyl 1 input

Vinyl character that reads the signal through a slowly modulated delay for wow pitch-drift, injects random impulses for surface crackle, and pushes the result through a tanh saturator for groove distortion. Wow scales the pitch-drift depth, Crackle sets how often surface pops appear, Drive sets the saturation amount, and Mix sets dry/wet. Use it to age a clean source with turntable wobble, dust noise and warm grit.

Param	Range	Default	Unit
Crackle	0 – 1	0.4	—
Wow	0 – 1	0.3	—
Drive	0 – 1	0.3	—
Mix	0 – 1	1	—

AmpSim 1 input

Guitar amp model chaining an asymmetric tanh preamp, a tone-tilt filter and a speaker-cabinet low-pass with output compensation. Drive sets preamp gain (0-48 dB) into the asymmetric clipper, Tone tilts between darker and brighter response, and Cab sets the cabinet cutoff (brighter at higher values). Mix blends the modeled tone against the dry; ranges from edge-of-breakup grit to saturated lead tones.

Param	Range	Default	Unit
Drive	0 – 48	18	dB
Tone	0 – 1	0.5	—
Cab	0 – 1	0.7	—
Mix	0 – 1	1	—

Bitwise 1 input

Bitwise mangler / databender (a first as a synth block): converts each sample to a 16-bit integer and applies a raw bit operation - XOR, AND, OR or bit-rotate - against Mask, then back to audio. Where Bitcrush only quantises the amplitude, this rips into the actual bit pattern: flipping a high bit folds the waveform, toggling low bits adds fizzing digital grit. Op picks the operation, Mask the bits to act on, Mix the blend. Pure circuit-bent digital destruction with no analogue equivalent.

Param	Range	Default	Unit
Op	XOR · AND · OR · Rotate	—	—
Mask	0 – 65535	0	—

Param	Range	Default	Unit
Mix	0 – 1	1	—

SlewClip 1 input

Slew-rate distortion (a first as a synth block): limits how fast the output can move between samples, so when Drive pushes the input past that maximum slope the block cannot keep up - fast transients are sheared into ramps and the waveform is dragged toward a buzzing triangle. Where a clipper flattens the peaks, SlewClip mangles the slopes: a harsh, gritty, distinctly digital lo-fi grind that also drops level as it limits. Rate sets the maximum step per sample, Drive the input gain, Mix the blend.

Param	Range	Default	Unit
Rate	0.002 – 1	0.05	—
Drive	1 – 16	2	—
Mix	0 – 1	1	—

Schmitt 1 input

Schmitt-trigger fuzz (a first as a synth block): a comparator with memory - the output snaps fully high once the driven input rises past +Hyst and stays high until it falls past -Hyst, squaring the signal into a hard two-level wave that resists chattering near zero. The hysteresis band makes it 'stick', cleaning the buzz of a raw square-up while adding a gated, brutal edge. Drive sets the gain into the trigger, Hyst the dead-band width, Level the output, Mix the blend. A hysteretic square fuzz no clipper or fold reproduces.

Param	Range	Default	Unit
Drive	1 – 32	4	—
Hyst	0 – 0.5	0.05	—
Level	0 – 1	0.7	—
Mix	0 – 1	1	—

MultiDist 1 input

Three-band distortion: one-pole filters at 300 Hz and 3 kHz split the input into low, mid and high bands, each saturated through its own tanh stage before being summed. Low, Mid and High set the drive into each band's saturator and Mix blends the result against dry. Per-band drive lets you grind the highs without muddying the lows, or add weight to bass alone.

Param	Range	Default	Unit
Low	0 – 1	0.3	—
Mid	0 – 1	0.3	—
High	0 – 1	0.3	—
Mix	0 – 1	1	—

CabSim 1 input

Generic speaker-cabinet voicing: a 90 Hz high-pass trims sub rumble, a high-frequency rolloff tames fizz, and a peak around 1.5 kHz adds cabinet body, all blended by Mix. Body sets how much of the resonant midrange peak is added and Bright raises the rolloff corner for a more open top end. A lightweight tone-shaping voicing for amp signals, not an impulse-response capture of a real cabinet.

Param	Range	Default	Unit
Body	0 – 1	0.5	—
Bright	0 – 1	0.5	—
Mix	0 – 1	1	—

Diode 1 input

Asymmetric diode clipper that drives the input into an exponential soft-knee saturation, shaping the positive and negative halves differently. Drive sets the input gain in dB into the nonlinearity, while Bias increases the curvature of the negative half for stronger asymmetry and added even harmonics. Mix blends the clipped signal against the dry input for germanium/silicon-style warmth and grit.

Param	Range	Default	Unit
Drive	0 – 36	12	dB
Bias	0 – 1	0.3	—
Mix	0 – 1	1	—

VHS 1 input

Cassette/VHS tape emulation: the input runs through a short modulated delay whose time wobbles via an internal 0.7 Hz LFO, then a one-pole low-pass rolls off highs and random pseudo-noise hiss is summed in. Wow sets the LFO delay-modulation depth (pitch wobble), Hiss scales the added noise level, and Dropout raises the probability of brief signal dropouts driven by a per-sample random envelope. Mix blends the degraded signal against the dry input for subtle aging or full lo-fi destruction.

Param	Range	Default	Unit
Wow	0 – 1	0.4	—
Hiss	0 – 1	0.3	—
Dropout	0 – 1	0.2	—
Mix	0 – 1	1	—

Subharmonic 1 input

Subharmonic generator: zero-crossing detection on the input drives a divide-by-four flip-flop, multiplying the input's rectified amplitude to synthesize a tone two octaves below, then smooths it with a one-pole low-pass. Sub sets the level of the generated sub, Tone opens the low-pass cutoff (darker to brighter sub), and Mix sums the sub onto the dry signal. Adds weight and low-end body to bass, kicks and leads.

Param	Range	Default	Unit
Sub	0 – 1	0.6	—
Tone	0 – 1	0.5	—
Mix	0 – 1	1	—

Amp

13 modules

Tweed 1 input

Fender Tweed 5E3-style amp: two asymmetric triode stages into a physical B+ reservoir sag (droops and blooms under pick attack) and an output-transformer stage - warm, loose, early breakup.

Param	Range	Default	Unit
Gain	1 – 20	4	—
Tone	0 – 1	0.5	—
Sag	0 – 1	0.6	—
Level	0 – 1	0.5	—

Blackface 1 input

Fender Blackface-style amp: a triode stage, scooped-mid clean headroom (~500 Hz notch), a stiff (low-sag) power section and an output-transformer stage with a bright, glassy top.

Param	Range	Default	Unit
Gain	1 – 12	2	—
Treble	0 – 1	0.6	—
Bass	0 – 1	0.5	—
Bright	0 – 1	0.3	—
Level	0 – 1	0.6	—

Recto 1 input

Mesa Rectifier-style high-gain amp: three cascaded asymmetric triode stages with tight inter-stage high-passing, a deep mid scoop, selectable reservoir sag and an output-transformer stage.

Param	Range	Default	Unit
Gain	1 – 40	15	—
Scoop	0 – 1	0.6	—
Presence	0 – 1	0.5	—
Sag	0 – 1	0.3	—
Master	0 – 1	0.5	—

Ampeg 1 input

Ampeg SVT-style bass amp: two triode stages, thick low-mid focus, power-supply sag and an output-transformer stage with a dark cab - built for bass.

Param	Range	Default	Unit
Gain	1 – 20	3	—
Bass	0 – 1	0.6	—
Mid	0 – 1	0.5	—

Param	Range	Default	Unit
Master	0 – 1	0.6	—

Friedman 1 input

Friedman BE-100-style amp: hot-rodded Marshall - two cascaded triode stages, tight low end, singing mids, reservoir sag and an output-transformer stage.

Param	Range	Default	Unit
Gain	1 – 40	12	—
Mid	0 – 1	0.6	—
Presence	0 – 1	0.5	—
Master	0 – 1	0.5	—

Soldano 1 input

Soldano SLO-100-style amp: three cascaded triode stages for a smooth, saturated lead with singing sustain, reservoir sag and an output-transformer stage.

Param	Range	Default	Unit
Gain	1 – 40	15	—
Tone	0 – 1	0.5	—
Presence	0 – 1	0.5	—
Master	0 – 1	0.5	—

Diezel 1 input

Diezel VH4-style amp: three cascaded triode stages, ultra-tight aggressive high-gain with a scooped low end, reservoir sag and an output-transformer stage.

Param	Range	Default	Unit
Gain	1 – 40	18	—
Scoop	0 – 1	0.6	—
Presence	0 – 1	0.5	—
Master	0 – 1	0.5	—

ENGL 1 input

ENGL-style amp: three cascaded triode stages, tight scooped modern-metal high-gain, reservoir sag and an output-transformer stage with a dark cab.

Param	Range	Default	Unit
Gain	1 – 40	16	—
Scoop	0 – 1	0.7	—
Presence	0 – 1	0.5	—
Master	0 – 1	0.5	—

Supro 1 input

Supro 1624T-style octal amp: 6973 power tubes on a tube-rectifier supply with no negative feedback, so loud transients sag the rails - the amp ducks then blooms, breaking up asymmetrically and early. Sag sets how hard the supply droops.

Param	Range	Default	Unit
Drive	1 – 30	6	—
Sag	0 – 1	0.5	—
Tone	1500 – 9000	4000	Hz
Level	0 – 1	0.5	—

Bassman 1 input

Fender 5F6-A tweed Bassman / JTM45-lineage tube amp: two cascaded 12AX7 gain stages modeled with the Koren triode plate-current equation (real self-bias, grid-conduction asymmetry - not a tanh) into a Fender-style tone stack (wide ~650 Hz mid), then a class-AB EL34 push-pull power stage whose plate currents subtract through the output transformer (even harmonics cancel, odd-harmonic clip + supply sag emerge from the physics), 4x oversampled.

Param	Range	Default	Unit
Gain	0 – 1	0.5	—
Bass	0 – 1	0.5	—
Mid	0 – 1	0.5	—
Treble	0 – 1	0.6	—
Master	0 – 1	0.6	—
Level	-24 – 6	-6	dB

Plexi 1 input

Marshall Plexi 1959 Super Lead-style amp: two cascaded Koren 12AX7 stages into a scooped Marshall tone stack (~500 Hz mid) and a class-AB EL34 push-pull power stage - the bright, crunchy British roar, with real push-pull even-harmonic cancellation and supply sag. Same physics engine as the Bassman, Marshall voicing.

Param	Range	Default	Unit
Gain	0 – 1	0.5	—
Bass	0 – 1	0.5	—
Mid	0 – 1	0.5	—
Treble	0 – 1	0.6	—
Master	0 – 1	0.6	—
Level	-24 – 6	-6	dB

JCM800 1 input

Marshall JCM800 2203-style high-gain amp: three cascaded Koren 12AX7 stages (the extra gain stage = hot-rodded crunch) into a scooped Marshall tone stack and a class-AB EL34 push-pull power stage. The 80s rock/metal standard - tight, aggressive, harmonically rich.

Param	Range	Default	Unit
Gain	0 – 1	0.5	—
Bass	0 – 1	0.5	—
Mid	0 – 1	0.5	—
Treble	0 – 1	0.6	—
Master	0 – 1	0.6	—
Level	-24 – 6	-6	dB

AC30 1 input

Vox AC30 Top Boost-style amp: two cascaded Koren 12AX7 stages into a chimy top-boost tone stack (flat ~800 Hz mid, bright treble) and a class-A EL84 push-pull power stage that compresses and sags early - the jangly British chime. EL84 (lower max current) + class-A bias distinguish it from the EL34 Marshalls.

Param	Range	Default	Unit
Gain	0 – 1	0.5	—
Bass	0 – 1	0.5	—
Mid	0 – 1	0.5	—
Treble	0 – 1	0.6	—
Master	0 – 1	0.6	—
Level	-24 – 6	-6	dB

Modulation

198 modules

SynthLFO 0 inputs

Low-frequency oscillator (sine/tri/saw/square) modulation source. Phase offsets the start point; Smooth slews the output for a softer shape.

Param	Range	Default	Unit
Rate	0.01 – 20	1	Hz
Shape	0 – 3	0	—
Depth	0 – 1	1	—
Smooth	0 – 1	0	—
Phase	0 – 1	0	—
Sync	0 – 8	0	—

SynthSampleHold 1 input

Sample-and-hold of the input at a clocked rate, with glide.

Param	Range	Default	Unit
Rate	0.1 – 200	8	Hz
Glide	0 – 1	0	—

StepLFO 0 inputs

Step LFO: descending stair or per-step random sample-and-hold, smoothed.

Param	Range	Default	Unit
Rate	0.01 – 40	4	Hz
Steps	2 – 16	8	—
Shape	0 – 1	1	—
Smooth	0 – 1	0.2	—
Sync	0 – 8	0	—

NoiseLFO 0 inputs

Noise modulation source (white/pink/brown) with optional sample-and-hold.

Param	Range	Default	Unit
Rate	0 – 200	0	Hz
Color	0 – 2	1	—
Depth	0 – 1	1	—
Sync	0 – 8	0	—

SwitcherLFO 0 inputs

Massive X morphing LFO: runs one phase at Rate through a bank of 8 bipolar shapes (sine, triangle, saw up/down, square, narrow pulse, s-curve, double sine), crossfading between adjacent shapes and mapping the result to a 0 to 1 output. Rate sets the cycle speed, Shape selects and morphs continuously across the bank, and Phase offsets the read position. Sweeping Shape gives a continuous waveform-morph modulator rather than a fixed LFO shape.

Param	Range	Default	Unit
Rate	0.01 – 20	1	Hz
Shape	0 – 1	0	—
Phase	0 – 1	0	—

DAHDSR 1 input

Six-stage modulation envelope driven by the gate at in0: on gate-on it runs Delay, Attack, Hold, Decay then holds at Sustain, and on gate-off it enters Release, outputting a 0 to 1 level. Delay/Attack/Hold/Decay/Release are times in seconds and Sustain is the held level. The extra Delay and Hold stages over a plain ADSR let it shape slow swells or gated bursts as a modulation source.

Param	Range	Default	Unit
Delay	0 – 4	0	s
Attack	0 – 5	0.01	s
Hold	0 – 4	0	s
Decay	0 – 5	0.2	s
Sustain	0 – 1	0.7	—
Release	0 – 5	0.3	s

Glide 2 inputs

Portamento: one-pole smooths the pitch/control signal at in0 toward each new value over Time, gated by in1. Time sets the glide duration in seconds (zero snaps instantly), and Mode chooses Always (glide on every note change) or Legato (snap on an isolated note, glide only between overlapping notes). Drop it between a note source and an oscillator for per-layer portamento.

Param	Range	Default	Unit
Time	0 – 2	0.1	s
Mode	0 – 1	0	—

PathLFO 0 inputs

Drawable XY-path LFO: a Rate-driven phase traverses the loop of 2D waypoints set in node config and reads back one coordinate as a bipolar -1 to 1 output, scaled by Level. Rate sets the traversal speed, Axis chooses whether the X or Y coordinate drives the output, and Level scales the result. Pair two nodes set to X and Y on the same drawn path to modulate two targets in a coordinated 2D motion.

Param	Range	Default	Unit
Rate	0.01 – 50	1	Hz
Axis	0 – 1	0	—
Level	0 – 1	1	—

MSEG 1 input

Multi-segment envelope generator: a gate edge resets a phase that ramps from 0 to 1 over Rate seconds, and the output linearly interpolates between user-defined 'time,level' breakpoints parsed from the node config. Rate sets the total run time of the shape, Loop makes the phase wrap and repeat instead of holding the last level, Level scales the final output, and Sync (1/1..1/16T) locks the cycle length to host tempo instead of the free Rate seconds. Breakpoints come from a 't,l;t,l;...' config string (defaulting to a

fast attack and gentle decay when fewer than two are given), giving a flexible Serum/Vital-style modulation source for arbitrary multi-stage curves.

Param	Range	Default	Unit
Rate	0.05 – 8	1	s
Loop	0 – 1	0	—
Level	0 – 1	1	—
Sync	0 – 8	0	—

Performer 1 input

Performer (Massive X-style step modulator): a tempo-synced step sequencer that runs a drawn pattern of up to 64 step levels (up to ~8 bars) locked to the host tempo. Draw the pattern in the step-bar editor; Sync sets the per-step note division (1/1..1/16T), Glide slews between steps (hard steps to a smooth contour), Level scales the output, and a gate on in0 restarts the pattern from step 1. Patch its output into cutoff, pitch, level or any mod input for long, evolving rhythmic movement that locks to the transport.

Param	Range	Default	Unit
Sync	1 – 8	4	—
Glide	0 – 1	0	—
Level	0 – 1	1	—

Barberpole 1 input

Infinite (barberpole) phaser: two 4-stage allpass banks a half-cycle apart, each amplitude-windowed so one fades in as the other wraps, giving an endless rising or falling sweep. Rate sets the sweep speed (Hz), Depth scales the allpass coefficient swing, Dir flips between rising and falling, and Mix blends wet against dry. A Shepard-tone style effect that seems to climb or descend forever.

Param	Range	Default	Unit
Rate	0.02 – 4	0.3	Hz
Depth	0 – 1	0.7	—
Dir	0 – 1	0	—
Mix	0 – 1	0.5	—

PeakPhaser 1 input

A 6-stage allpass phaser whose sweep is driven by a peak envelope follower on the input, so transients trigger the filter motion instead of an LFO. Soften sets the envelope release time, Feedback sets the allpass regeneration (up to 0.9), Amount scales how far each peak sweeps the phaser, and Mix blends wet against dry. Part of the Traktor Pulse family, it pumps the phaser in time with the signal's own dynamics.

Param	Range	Default	Unit
Soften	0 – 1	0.5	—
Feedback	0 – 0.9	0.5	—
Amount	0 – 1	0.7	—
Mix	0 – 1	0.5	—

TriceraChorus 1 input

Tri-voice chorus with a micro-detune swirl: three LFO-modulated delay taps at spread phases plus a slow pitch-detuned voice for a rich animated widening.

Param	Range	Default	Unit
Rate	0.05 – 6	0.5	Hz
Depth	0 – 1	0.6	—
Detune	0 – 1	0.3	—
Mix	0 – 1	0.5	—

DimensionD 1 input

Roland Dimension D-style BBD chorus: two anti-phase bucket-brigade lines (compander + clock-tracked reconstruction filter) for a wide, 'motionless' thickening with no audible wobble.

Param	Range	Default	Unit
Mode	1 · 2 · 3 · 4	—	—
Mix	0 – 1	0.5	—

CE1 1 input

Boss CE-1-style BBD chorus/vibrato: a triangle-LFO chorus or a wet-only vibrato through a companded bucket-brigade line, with preamp coloration.

Param	Range	Default	Unit
Mode	Chorus · Vibrato	—	—
Rate	0.1 – 8	1	Hz
Depth	0 – 1	0.5	—
Mix	0 – 1	0.5	—

Solina 1 input

Solina/string-machine-style ensemble: three BBD taps on 120-degree-phased dual LFOs (slow + fast) through one companded bucket-brigade line for a lush, moving multi-voice swirl.

Param	Range	Default	Unit
Slow	0.1 – 2	0.6	Hz
Fast	1 – 12	6	Hz
Depth	0 – 1	0.6	—
Mix	0 – 1	0.5	—

ElectricMistress 1 input

EHX Electric Mistress-style flanger: a swept companded bucket-brigade line forms moving comb notches; Filter Matrix freezes the sweep for a fixed metallic tone.

Param	Range	Default	Unit
Rate	0.05 – 8	0.4	Hz
Range	0 – 1	0.5	—
Color	0 – 1	0.4	—
Matrix	Sweep · Freeze	—	—

TriStereo 1 input

Dytronics Tri-Stereo Chorus-style ensemble: three bucket-brigade voices (componder + clock-tracked reconstruction) at staggered rates for a lush, wide three-dimensional chorus.

Param	Range	Default	Unit
Rate	0.1 – 3	0.5	Hz
Depth	0 – 1	0.6	—
Mix	0 – 1	0.5	—

MXRFlanger 1 input

MXR Flanger-style BBD flanger: a companded bucket-brigade line swept for a deep, metallic comb with line-level regeneration for jet-plane sweeps.

Param	Range	Default	Unit
Rate	0.05 – 6	0.3	Hz
Depth	0 – 1	0.7	—
Regen	0 – 0.9	0.4	—
Mix	0 – 1	0.5	—

Phase90 1 input

MXR Phase 90-style 4-stage phaser: four JFET all-pass stages (bias-dependent frequency shift + transfer-curve saturation) give two evenly-tracking notches; Script (clean) or Block (resonant feedback).

Param	Range	Default	Unit
Rate	0.05 – 5	0.5	Hz
Block	Script · Block		—
Mix	0 – 1	0.5	—

SmallStone 1 input

EHX Small Stone-style 4-stage OTA phaser: nonlinear OTA all-pass stages; the Color switch adds feedback, drives the OTAs harder and widens the notch sweep for a more vocal, resonant swirl.

Param	Range	Default	Unit
Rate	0.05 – 5	0.4	Hz
Color	Off · On		—
Mix	0 – 1	0.5	—

UniVibe 1 input

Uni-Vibe-style 4-stage phaser: staggered all-pass frequencies driven by an asymmetric lamp-filament LFO through a power-law photocell I-V curve, with nonlinear stages - the watery, throbbing swirl. Chorus or Vibrato.

Param	Range	Default	Unit
Rate	0.05 – 8	2	Hz
Depth	0 – 1	0.7	—
Mode	Chorus · Vibrato		—
Mix	0 – 1	0.5	—

OptoTrem 1 input

Fender optical tremolo: a lamp drives a photocell whose asymmetric lag (cools slower than it warms) and sublinear I-V curve round the throb; Shape sets the cell response speed (smooth to choppy).

Param	Range	Default	Unit
Rate	0.5 – 12	5	Hz
Depth	0 – 1	0.6	—
Shape	0 – 1	0.7	—

BiasTrem 1 input

Tube bias tremolo: the LFO shifts the tube grid bias (operating point), so the throb modulates both amplitude and 2nd-harmonic content - a richer, harmonically-moving wobble than plain amplitude tremolo.

Param	Range	Default	Unit
Rate	0.5 – 10	4	Hz
Depth	0 – 1	0.5	—
Bias	0 – 1	0.3	—

FuncGen 0 inputs

H3000-style function generator: a modulation source with 19 selectable waveshapes plus random sample-and-hold, smooth random and noise (no audio input).

Param	Range	Default
Rate	0.01 – 30	1
Shape	Sine · Triangle · Saw Up · Saw Down · Square · Pulse · Sine2 · Rectified · Half Sine · Exp Up · Exp Down · Log · Trapezoid · Staircase · Random	
Depth	0 – 1	1
Phase	0 – 1	0
Width	0.05 – 0.95	0.5
Bias	-1 – 1	0

BiPhase 1 input

Mu-Tron Bi-Phase dual phaser: TWO complete 6-stage OTA phasers with independent LFO rates (RateA / RateB), whose two moving notch combs beat against each other for a swirling, three-dimensional sweep no single-sweep phaser (Phase 90, Small Stone) can reach. Depth sets the sweep width, Feedback the resonant emphasis of the notches, Mix the wet blend. Six 1st-order all-pass stages per side.

Param	Range	Default	Unit
RateA	0.02 – 8	0.3	Hz
RateB	0.02 – 8	0.5	Hz
Depth	0 – 1	0.7	—
Feedback	0 – 0.9	0.3	—
Mix	0 – 1	0.5	—

BodeRing 1 input

Bode / Moog 6401-style ring modulator: the input multiplied by an internal tunable sine carrier through a diode-ring nonlinearity, giving the metallic sum-and-difference sidebands of the classic sci-fi/bell sound. Freq tunes the carrier, Drive sets the diode-bridge bite, Mix blends modulated against dry.

Param	Range	Default	Unit
Freq	1 – 4000	220	Hz
Drive	0 – 1	0.3	—
Mix	0 – 1	1	—

Random inputs

Self-contained random modulation source: latches a new random value at Rate (internal sample-and-hold of a noise source) with Smooth slewing between steps - from hard random steps to smooth wandering drift. Depth scales the swing and Bias offsets the centre. Needs no input; patch its output into any mod input for evolving, never-repeating movement.

Param	Range	Default	Unit
Rate	0.01 – 50	2	Hz
Smooth	0 – 1	0	—
Depth	0 – 1	1	—
Bias	-1 – 1	0	—

MIDI CC inputs

MIDI continuous-controller mod source: outputs the live value (0..1) of MIDI CC number CC, so any hardware knob, fader, expression or breath controller becomes a modulation source - patch its output into any parameter's mod input. Smooth slews fast controller jumps, Depth scales the range and Bias offsets the centre. Common numbers: 1 mod wheel, 2 breath, 7 volume, 11 expression, 64 sustain, 74 brightness.

Param	Range	Default	Unit
CC	0 – 127	1	#
Smooth	0 – 1	0	—
Depth	0 – 1	1	—
Bias	-1 – 1	0	—

ThruZero input

Through-zero flanger: two delay taps sweep symmetrically about a short centre delay - one lengthening while the other shortens - so as the LFO crosses zero the taps pass through each other, producing the deep notch that sweeps all the way down to DC and momentarily cancels, the unmistakable tape-flanger 'through-zero' sound a normal flanger (with its fixed minimum delay) can't make. Rate sets the sweep speed, Depth the excursion, Feedback the resonance, and Mix the wet blend.

Param	Range	Default	Unit
Rate	0.01 – 8	0.3	Hz
Depth	0 – 5	3	ms
Feedback	0 – 0.95	0.3	—
Mix	0 – 1	0.5	—

LFO inputs

Low-frequency control generator with sine, triangle, saw and pulse shapes (Shape), free-running at Rate. Depth scales the swing, Bias offsets the centre, Phase shifts the start point, and Width sets the pulse duty cycle. Patch its output into any parameter's mod input for vibrato, tremolo, filter sweeps and rhythmic movement.

Param	Range	Default	Unit
Rate	0.01 – 100	1	Hz
Shape	0 – 3	0	—

Param	Range	Default	Unit
Depth	0 – 1	1	—
Phase	0 – 1	0	—
Bias	-1 – 1	0	—
Width	0.05 – 0.95	0.5	—

RingMod 2 inputs

Ring and amplitude modulator that multiplies the input by a carrier on in 2 (plus Bias). Mode morphs from true ring modulation (bipolar carrier, inharmonic metallic sum/difference tones) to amplitude modulation (unipolar, keeps the original plus sidebands), and Amount sets dry/wet. An audio-rate carrier yields clangorous bell tones; a low-rate carrier becomes tremolo.

Param	Range	Default	Unit
Amount	0 – 1	1	—
Mode	0 – 1	0	—
Bias	-1 – 1	0	—

PhaseMod 2 inputs

FM / phase-modulation processor: phase-modulates the in0 carrier with the in1 modulator (stack any number of sources into either input) to make classic FM sidebands - the modular way to FM oscillators together without per-oscillator routing. Patch one oscillator into in0 and another into in1 for two-operator FM; stack more cables on in1 to FM with several sources at once. Index sets the modulation depth, Feedback routes the output back into the modulator for harmonic edge toward noise, and Mix blends against the dry carrier. Distinct from the self-contained FM oscillator: this FM's whatever audio you feed it.

Param	Range	Default	Unit
Index	0 – 1	0.3	—
Feedback	0 – 0.95	0	—
Mix	0 – 1	1	—

Clock 0 inputs

Tempo-synced clock generator: emits a gate-pulse train locked to the host tempo (or a free Hz Rate). Sync picks the note division (0=free Hz, 1=1/1 .. 8=1/16T), Rate is the free-running frequency when Sync is 0, and Width sets the pulse length as a fraction of the step. Patch its output into envelopes, sample-and-holds, ClockDiv or any gate input to drive rhythmic movement from the host transport.

Param	Range	Default	Unit
Rate	0.1 – 40	2	Hz
Width	0.05 – 0.95	0.5	—
Sync	0 – 8	3	—

Euclid 0 inputs

Euclidean rhythm generator: distributes Pulses hits as evenly as possible across Steps positions (the algorithm behind countless drum patterns) and clocks them at the tempo-synced step Rate. Steps is the pattern length, Pulses the number of hits, Rotate shifts the pattern start, Rate the per-step note division (1=1/1 .. 8=1/16T), and Width the gate length. Outputs a gate on each active step - feed a drum/pluck envelope for generative rhythm that locks to the host.

Param	Range	Default	Unit
Steps	1 – 32	16	—
Pulses	0 – 32	4	—

Param	Range	Default	Unit
Rotate	0 – 31	0	—
Rate	1 – 8	5	—
Width	0.05 – 0.95	0.5	—

Ratchet 1 input

Ratchet / retrigger: while the in0 gate is high, replaces the held gate with a fast burst of Count sub-gates at the tempo-synced Rate - the classic 'stutter/roll' on a step. Count sets how many retriggers fire per gate, Rate the sub-division ($1=1/1$.. $8=1/16T$), and Width each sub-gate's length. Patch a clock or sequencer gate into in0 and the ratcheted gate out into an envelope for rolls and rhythmic fills.

Param	Range	Default	Unit
Count	1 – 8	2	—
Rate	1 – 8	6	—
Width	0.05 – 0.95	0.5	—

Turing 1 input

Turing-machine sequencer (looping shift register): on each rising edge of the in0 clock it shifts an internal register of step values and feeds the end back to the front - but with probability set by Lock it injects a fresh random value instead, so Lock fully up gives a locked repeating loop, fully down a new random sequence each pass, and in between a slowly mutating pattern. Length sets the loop length (2-16 steps), Lock the lock/mutation amount, Range the CV span, and Quantize snaps each step to a chromatic grid. The classic generative Eurorack CV source - patch its output into an oscillator pitch or any mod input.

Param	Range	Default	Unit
Length	2 – 16	8	—
Lock	0 – 1	0.5	—
Range	0 – 1	1	—
Quantize	0 – 1	0	—

Grids 0 inputs

Topographic drum trigger generator (Mutable Instruments Grids): walks a 32-step groove map and fires a gate when the selected Voice's pattern density at the current step beats the Density threshold, so turning Density up adds hits and down thins them out. Map morphs the rhythm between two feels, Chaos sprays in random hits/drops, and Voice picks Kick, Snare or Hat (run three Grids off the same patch for a full kit). Self-clocks at Rate. Distinct from Euclid's evenly-spread Euclidean rhythms - these are curated grooves.

Param	Range	Default	Unit
Rate	0.1 – 100	8	Hz
Map	0 – 1	0.5	—
Density	0 – 1	0.5	—
Chaos	0 – 1	0	—
Voice	Kick · Snare · Hat		—

Marbles 0 inputs

Random voltage source with a loop lock (Mutable Instruments Marbles): on each internal Rate step it generates a new random value around Bias with a width set by Spread, but Deja Vu blends in replay of a stored loop - 1 = a locked, repeating random melody of Length steps, 0 = fresh randomness every step. A stepped CV source between pure chance and a fixed sequence; quantize the output for pitch. Distinct from Random (no loop memory) and Turing (binary shift register, not continuous CV).

Param	Range	Default	Unit
Rate	0.1 – 100	6	Hz
Bias	0 – 1	0.5	—
Spread	0 – 1	0.5	—
DejaVu	0 – 1	0.5	—
Length	1 – 16	8	—

Wobblebug 0 inputs

Source of uncertainty (Make Noise Wobblebug / Buchla 266): an irregular internal clock latches a new random value each tick; Smooth blends from hard stepped voltages toward a smoothly wandering output, and Wobble rings each step with a decaying sinusoid for the burbling Buchla character. A self-running random modulation source - no input needed. Distinct from Random (steady clock, no wobble) and Chaos/LorenzLFO (deterministic attractors).

Param	Range	Default	Unit
Rate	0.05 – 60	4	Hz
Smooth	0 – 1	0.4	—
Wobble	0 – 1	0.3	—

Maths 1 input

Function generator (Make Noise Maths / Buchla 281 / Serge DUSG): a rise/fall slope with shapeable curves. Rise and Fall set the up/down times; Curve bends them from logarithmic through linear to exponential. Mode: Cycle self-runs as a variable-shape LFO; Trig fires a one-shot AD envelope on a gate at in0; Gate rises while in0 is held and falls when released (AR). The CV building block behind envelopes, slope LFOs and generative Krell patches. Patch a Clock/Euclid into in0 to trigger it.

Param	Range	Default	Unit
Rise	0.001 – 5	0.05	s
Fall	0.001 – 10	0.4	s
Curve	0 – 1	0.5	—
Mode	Cycle · Trig · Gate		—

Tides 0 inputs

Tidal modulator (Mutable Instruments Tides): a slope source whose shape morphs continuously. Rate sets the speed (LFO into audio range); Slope skews the ramp from a fast-rise/slow-fall sawtooth through a symmetric triangle to slow-rise/fast-fall; Smooth rounds the contour from a sharp triangle toward a smooth sine swell. Use it as an evolving LFO, an envelope, or a raw oscillator. Distinct from the stepped MSEG and the fixed LFO shapes.

Param	Range	Default	Unit
Rate	0.01 – 4000	1	Hz
Slope	0 – 1	0.5	—
Smooth	0 – 1	0.5	—

QuadLFO 0 inputs

Quadrature LFO (Doepfer A-143-9): one rate, four phase-locked outputs 90 degrees apart. Each instance reads the shared transport clock, so several QuadLFOs at the same Rate stay phase-coherent and Phase (0/90/180/270) picks which quadrature tap this one outputs - patch four for circular/3D panning, stereo movement or evolving cross-modulation. Shape selects sine, triangle, saw or square. Distinct from the free-running LFO (no fixed phase relationship between instances).

Param	Range	Default	Unit
Rate	0.01 – 40	1	Hz
Phase	0 · 90 · 180 · 270		—
Shape	Sine · Tri · Saw · Square		—

PLL 1 input

Phase-locked loop (Doepfer A-196): an internal square-wave VCO chases the frequency of the in0 input through an XOR phase comparator and a loop filter, locking onto it (or a multiple) - the lock is never perfect, so it warbles, glitches and screams in the classic PLL way. Base sets the VCO centre, Lock the loop tightness (low = sloppy, unstable, vocal; high = quick, tight tracking), and Range how far it can pull (octaves). Feed it audio or a clock for frequency multiplication, robot voices and chaotic tracking tones.

Param	Range	Default	Unit
Base	20 – 4000	220	Hz
Lock	0 – 1	0.4	—
Range	1 – 5	3	oct

Bloom 0 inputs

Fractal sequencer (Qu-Bit Bloom): grows a melody from a short Trunk seed sequence by recursively grafting variations onto it. Length sets the trunk length; Branch adds fractal sub-sequences derived from the trunk (0 = play the plain trunk, up = ever more elaborate variations); Path chooses which route through the recursive tree is traversed; Scale sets the output range. Self-clocks at Rate. A generative stepped-CV source - quantize for pitch. Distinct from Turing (shift register) and StepSeq (fixed).

Param	Range	Default	Unit
Rate	0.1 – 100	8	Hz
Length	2 – 16	8	—
Branch	0 – 1	0.3	—
Path	0 – 1	0	—
Scale	0 – 1	1	—

Frames 4 inputs

Keyframe morphing mixer (Mutable Instruments Frames): blends its four inputs (A-D) by sweeping one Frame knob through a chain of keyframes, crossfading from A toward D as Frame moves 0 to 1 - the modular take on animation keyframing. Frame sets the position and Anim, when above zero, animates it automatically at that rate (an evolving 4-source morph). Patch four oscillators or modulators in for smooth timbral or CV morphs. Distinct from the static Mixer and the XY-plane Vector.

Param	Range	Default	Unit
Frame	0 – 1	0	—
Anim	0 – 10	0	Hz

Stages 1 input

Segment generator (Mutable Instruments Stages): builds a function from a chain of Segs segments, each ramping to a new level over Time. Shape morphs every segment from a hard step, through a linear ramp, to a smooth S-curve; Loop makes the chain cycle as an evolving LFO, or one-shot it from a gate at in0 as a complex envelope. The flexible CV building block between an envelope, an LFO and a slow sequencer. Distinct from the breakpoint MSEG.

Param	Range	Default	Unit
Rate	0.05 – 60	2	Hz
Segs	2 – 8	4	—

Param	Range	Default	Unit
Shape	0 – 1	0.5	—
Loop	OneShot · Loop		—

JustFriends 0 inputs

Manifold function generator (Mannequins Just Friends): one shape, six harmonically-related copies summed into a rich evolving wave. Time sets the base rate (slow LFO up to audio pitch); Intone spreads the six harmonics from a tight integer series toward stretched, detuned ratios; Curve bends every slope from a linear ramp toward a smooth sine; Level trims output. A function generator and harmonic oscillator in one - drones, organ-ish tones and undulating modulation. Distinct from the partial-summing Additive (pure sines, spectral tilt).

Param	Range	Default	Unit
Time	0.05 – 2000	4	Hz
Intone	0 – 1	0	—
Curve	0 – 1	0.5	—
Level	0 – 1	0.7	—

SampleHold 1 input

Clocked sample-and-hold that latches the input value at Rate and holds it until the next tick, producing stepped random or quantized control. Glide slews between steps for a smoother, portamento-like contour. Feed it noise for classic random sample-and-hold modulation, or an audio source to quantize it in time.

Param	Range	Default	Unit
Rate	0.1 – 200	8	Hz
Glide	0 – 1	0	—

Slew 1 input

Slew-rate limiter that caps how fast the signal can rise and fall, with independent Rise and Fall rates. Use it as portamento/glide on pitch CV, a smoother on stepped modulation, or a lag to de-zipper control changes. Asymmetric Rise/Fall gives simple envelope-like attack and decay shaping.

Param	Range	Default	Unit
Rise	0 – 1	0.5	—
Fall	0 – 1	0.5	—

Chaos 0 inputs

Chaos modulator driven by the logistic map ($x = r*x*(1-x)$): at each Rate tick it iterates the map, and Chaos raises r from the periodic into the chaotic regime for unpredictable-but-deterministic motion. Smooth slews between values for a flowing control source rather than hard steps. Use it for evolving, never-quite-repeating modulation.

Param	Range	Default	Unit
Rate	0.1 – 500	20	Hz
Chaos	0 – 1	0.8	—
Smooth	0 – 1	0.5	—

Phaser 1 input

All-pass cascade phaser: an LFO sweeps 2-8 all-pass stages (Stages) to move a series of notches through the spectrum, with Feedback sharpening their resonance. Depth sets the sweep range and Mix the dry/wet blend. More stages add more notches for a deeper, more vocal swirl.

Param	Range	Default	Unit
Rate	0.05 – 8	0.4	Hz
Depth	0 – 1	0.7	—
Feedback	0 – 0.9	0.3	—
Stages	2 – 8	4	—
Mix	0 – 1	0.5	—

Tremolo 0 inputs

Amplitude tremolo: an LFO at Rate modulates the signal level by Depth, switchable between a smooth sine and a choppy square (Shape). Classic for rhythmic volume pulsing, surf-guitar wobble and rotary-style throb. At full Depth with the square shape it becomes a hard on/off gate.

Param	Range	Default	Unit
Rate	0.05 – 20	5	Hz
Depth	0 – 1	0.5	—
Shape	0 – 1	0	—

Dropout 1 input

Random rhythmic muting: a clock at Rate decides per cycle whether to drop the signal to silence (Amount sets the drop probability), smoothed by Smooth to avoid clicks. Emulates a bad cable, dying battery or stuttery transmission. Subtle for movement, extreme for glitch and IDM textures.

Param	Range	Default	Unit
Rate	0.1 – 30	4	Hz
Amount	0 – 1	0.3	—
Smooth	0 – 1	0.5	—

TranceGate 1 input

Rhythmic hard gate: an LFO at Rate chops the signal on and off with a Duty cycle, smoothed by Smooth and scaled by Depth. Modulate Rate to lock it to tempo for the classic trance/EDM stutter. See StepSeq for arbitrary per-step patterns rather than a fixed duty.

Param	Range	Default	Unit
Rate	0.1 – 30	8	Hz
Duty	0.05 – 0.95	0.5	—
Smooth	0 – 1	0.2	—
Depth	0 – 1	1	—

Vector 4 inputs

Bilinear XY vector mixer: blends four inputs across an X/Y plane (in 1/in 2 the top corners, in 3/in 4 the bottom), with X and Y setting the morph position. Classic vector synthesis (Prophet VS / Wavestation) - feed it four oscillators or sources and automate X/Y to morph between them.

Param	Range	Default	Unit
X	0 – 1	0.5	—
Y	0 – 1	0.5	—

Scan 4 inputs

Scanning four-source mixer: a read position sweeps across in 1..in 4 and crossfades between adjacent sources, wrapping 4 back to 1 for a continuous loop. Pos sets the position by hand; Rate above zero auto-scans the position and loops it; Smooth rounds the crossfade from linear to an S-curve. Feed it four oscillators or signals and scan or automate to morph between them - the sequential companion to the bilinear Vector mixer.

Param	Range	Default	Unit
Pos	0 – 1	0	—
Rate	0 – 30	0	Hz
Smooth	0 – 1	1	—

StepSeq 0 inputs

Step sequencer modulation source: cycles through up to 6 step levels (S1-S6) at Rate steps per second, outputting the held value of the current step. Use it as a rhythmic mod source for cutoff, pitch or level, or to chop gain into patterns. Modulate Rate to lock to tempo; see TranceGate for a simple on/off duty gate.

Param	Range	Default	Unit
Rate	0.1 – 30	4	Hz
Steps	1 – 6	4	—
S1	0 – 1	1	—
S2	0 – 1	0.5	—
S3	0 – 1	0.75	—
S4	0 – 1	0.25	—
S5	0 – 1	0.6	—
S6	0 – 1	0.4	—

Stepper 0 inputs

Tempo-synced step-sequencer modulator: steps through S1..S5 levels locked to the host tempo, with Glide smoothing the jumps from hard sample-and-hold steps to a slewed contour. Sync sets the per-step note division (1/1..1/16T), Steps how many levels are used, and Glide the slew time. Per-voice (each instance runs independently); patch it into cutoff, pitch or level for rhythmic, tempo-locked movement. Distinct from StepSeq, which free-runs in Hz with no glide or sync.

Param	Range	Default	Unit
Sync	1 – 8	4	—
Steps	1 – 5	4	—
Glide	0 – 1	0	—
S1	0 – 1	1	—
S2	0 – 1	0.4	—
S3	0 – 1	0.7	—
S4	0 – 1	0.2	—
S5	0 – 1	0.55	—

KeyTrack 1 input

Keyboard-tracking mod source: turns the note pitch into a centered, scalable modulation - the classic filter keytrack. Patch the Note source into in0; the output is (note - Center) x Amount around the midpoint, so notes above Center push the modulation up and

notes below push it down (Amount 1 = full 1:1 tracking across the keyboard, 0.5 = half, negative inverts). Patch the output into a filter cutoff (or any param) so the timbre follows the keyboard. Center sets the pivot note (0-127), Amount the bipolar tracking depth. Unlike the raw Note source this is centered and scalable, so the modulation is zero at the pivot.

Param	Range	Default	Unit
Center	0 – 127	60	—
Amount	-2 – 2	1	—

Drift 0 inputs

Slow smoothed random-walk modulator centred on 0.5, emulating vintage analog drift. Rate sets how often it picks a new target and Depth the deviation. Patch a little into pitch, cutoff or level to take the static, too-perfect edge off a patch.

Param	Range	Default	Unit
Rate	0.05 – 5	0.5	Hz
Depth	0 – 1	0.5	—

Allpass 1 input

All-pass diffuser: a cascade of 1-8 first-order all-pass sections (Stages) that rotate phase while leaving magnitude flat, set by Coef. A building block for reverb diffusion, phasers and stereo de-correlation rather than an audible filter on its own. More stages and higher Coef smear transients harder.

Param	Range	Default	Unit
Coef	-0.95 – 0.95	0.7	—
Stages	1 – 8	4	—
Mix	0 – 1	1	—

LorenzLFO 0 inputs

Lorenz strange-attractor chaos modulator (Serum-style): integrates the Lorenz system and outputs one Axis (X/Y/Z), normalized and scaled by Depth. Produces smooth, organic, never-repeating motion - more flowing than stepped random. Great for evolving cutoff, pitch or pan drift.

Param	Range	Default	Unit
Rate	0.01 – 50	2	Hz
Axis	0 – 2	0	—
Depth	0 – 1	1	—

Combine 2 inputs

Combines two modulation or audio inputs (Pigments-style): Mode selects sum, ring (multiply), max, min, difference or sign-XOR, scaled by Mix. A math/logic node for the mod matrix - build complex modulators from simpler ones, or use it as a flexible ring/AM stage. Complements the Expr script node for inline math.

Param	Range	Default	Unit
Mode	0 – 5	0	—
Mix	0 – 1	1	—

TZFlanger 1 input

Through-zero flanger that reads two short delay taps swept in opposite directions by a cosine LFO, so the offsets cross and produce a deep tape-style null. Rate sets the LFO speed, Depth scales the sweep range, Feedback (up to 0.95) regenerates one tap for sharper resonance, and Mix sets dry/wet. The counter-swept design gives the dramatic zero-crossing whoosh that fixed flangers cannot reach.

Param	Range	Default	Unit
Rate	0.01 – 8	0.3	Hz
Depth	0 – 1	0.6	—
Feedback	0 – 0.95	0.4	—
Mix	0 – 1	0.5	—

Vibe 1 input

Photocell-style vibe that runs the signal through four staggered first-order all-pass stages whose coefficients are swept by a cosine LFO, each stage offset slightly to mimic mismatched optical cells. Rate sets the swirl speed, Depth scales how far the all-pass coefficients move, and Mix blends the phase-shifted output against the dry signal. The uneven staging gives the watery, throbbing swirl associated with classic vibe pedals.

Param	Range	Default	Unit
Rate	0.05 – 8	1	Hz
Depth	0 – 1	0.7	—
Mix	0 – 1	0.5	—

HarmTremolo 1 input

Harmonic tremolo that splits the signal into low and high bands with a one-pole low-pass and amplitude-modulates them in opposition with a sine LFO, so as the lows swell the highs dip and back again. Rate sets the LFO speed, Depth sets the modulation intensity, Freq places the band split, and Mix sets dry/wet. The opposing band motion gives a richer, more vocal sweep than amplitude tremolo.

Param	Range	Default	Unit
Rate	0.1 – 12	4	Hz
Depth	0 – 1	0.6	—
Freq	200 – 3000	800	Hz
Mix	0 – 1	1	—

WowFlutter 1 input

Tape pitch instability emulated by a short delay line (centred near 5 ms) whose read time is modulated by two sine LFOs, a slow ~0.6 Hz wow and a fast ~7 Hz flutter, producing continuous pitch drift. Wow scales the slow drift depth and Flutter scales the fast wobble, while Mix blends the warped signal against the dry input. Use low amounts for subtle tape warmth and higher settings for seasick, detuned movement.

Param	Range	Default	Unit
Wow	0 – 1	0.4	—
Flutter	0 – 1	0.3	—
Mix	0 – 1	1	—

Swarm 0 inputs

Flocking modulation source (a first as a synth block): a swarm of six agents drift under boids rules - steering toward the flock centre (Cohesion), away from crowding neighbours (Separation), toward the average heading, plus a little wander - and the output

is the swarm's collective centre as a slowly evolving, organic CV. Rate sets the motion speed. Unlike an LFO (a fixed shape) or the memoryless random of Marbles/Wobblebug, the swarm self-organises: it flocks, splits and regroups, moving smoothly but never repeating. Needs no input.

Param	Range	Default	Unit
Rate	0 – 1	0.4	—
Cohesion	0 – 1	0.5	—
Separation	0 – 1	0.5	—

DoublePendulum 0 inputs

Double-pendulum chaos source (a first as a synth block): integrates the real equations of motion of two jointed pendulums - the textbook deterministic-chaos system - and outputs the lower bob's swing as a smooth, never-repeating CV. Tiny differences in motion explode into wholly different trajectories, so it wanders organically forever without an LFO's repetition or a noise source's harshness. Rate sets the simulation speed; the output stays bounded in +/-1 however wildly the arms tumble. A different, smoother chaos than the logistic Chaos or Lorenz blocks.

Param	Range	Default	Unit
Rate	0.05 – 4	1	—

Chua 0 inputs

Chua's-circuit chaos source (a first as a synth block): integrates the dimensionless Chua equations - the simplest electronic circuit that is provably chaotic, a real analog oscillator built from an op-amp, two capacitors, an inductor and the piecewise-linear 'Chua diode'. The trajectory winds around two attracting loops and flips unpredictably between them (the famous double-scroll), giving a smooth bipolar CV that meanders forever without repeating. Rate sets the integration speed. A circuit-born chaos, smoother than a map and distinct from the mechanical DoublePendulum or the Lorenz LFO.

Param	Range	Default	Unit
Rate	0.05 – 4	1	—

Duffing 0 inputs

Driven-Duffing chaos source (a first as a synth block): integrates the forced double-well Duffing oscillator $x'' + d*x' - x + x^3 = Drive*cos(w t)$ - a mass in a twin-valley potential shaken by a periodic drive. At chaotic settings it jumps erratically between the two wells, never locking to the drive's period, yielding a smooth bipolar CV with a rhythmic pulse running under the chaos. Rate sets the speed, Drive the forcing amplitude (raise it to push from periodic motion into chaos). A forced-oscillator chaos distinct from the autonomous Chua or DoublePendulum.

Param	Range	Default	Unit
Rate	0.05 – 4	1	—
Drive	0 – 0.6	0.37	—

Standard 0 inputs

Chirikov standard-map source (a first as a synth block): the kicked-rotor map $p <- p + K*sin(theta)$, $theta <- theta + p$ on a torus - the textbook model of Hamiltonian chaos and the onset of turbulence. Below the critical kick the motion is smooth and quasi-periodic; above it (raise K) the phase space dissolves into a chaotic sea. It free-runs at Rate, stepping the map and outputting $sin(theta)$ as a bipolar CV - regular and rolling at low K, wildly scattered at high K. A momentum-kicked chaos distinct from the dissipative attractors.

Param	Range	Default	Unit
Rate	0.1 – 50	8	Hz
K	0.5 – 6	2	—

Gingerbread inputs

Gingerbreadman-map source (a first as a synth block): iterates the piecewise-linear chaotic map $x \leftarrow 1 - y + |x|$, $y \leftarrow x$ - an area-preserving map whose orbits fill a strange six-sided 'gingerbread man' region of interleaved chaotic seas and stable islands. It free-runs at Rate, stepping the map and emitting x as a bipolar stepped CV that hops around the figure forever without repeating. Built from only an absolute value and a subtraction, it is a different, more angular chaos than the trig-based attractors.

Param	Range	Default	Unit
Rate	0.1 – 50	8	Hz

Clifford inputs

Clifford-attractor source (a first as a synth block): iterates the strange-attractor map $x \leftarrow \sin(A*y) + \cos(A*x)$, $y \leftarrow \sin(1.6*x) + 0.7*\cos(1.6*y)$, whose orbit is forever drawn toward a delicate folded-ribbon fractal. It free-runs at Rate, stepping the map and outputting x as a smooth bipolar CV that traces the attractor's swirls. A sets the warp of the figure - small changes redraw the whole attractor - giving a rich family of organic, never-repeating motions distinct from the scrolls of Chua or the kicks of the standard map.

Param	Range	Default	Unit
Rate	0.1 – 50	8	Hz
A	-2 – -0.5	-1.4	—

CircleMap inputs

Sine-circle-map source (a first as a synth block): iterates $\theta \leftarrow \theta + \Omega - (K/2\pi) \sin(2\pi \theta)$ on a circle - the model of an oscillator being kicked once per turn, and the textbook route to MODE-LOCKING. For bands of settings (the Arnold tongues) it locks onto a rational rhythm and repeats exactly; between them it runs quasi-periodically or, at high K, chaotically - so sweeping Omega walks a devil's-staircase of locked and unlocked ratios. It free-runs at Rate, outputting the angle as a 0..1 CV. K sets the kick strength, Omega the bare turn rate. A locking oscillator distinct from the dissipative attractors.

Param	Range	Default	Unit
Rate	0.1 – 50	8	Hz
K	0 – 6	4	—
Omega	0 – 1	0.33	—

GaussMap inputs

Gauss-map (mouse-map) source (a first as a synth block): iterates $x \leftarrow \exp(-\text{Alpha } x^2) + \text{Beta}$, a bell-shaped map named for the Gaussian in it. Sweeping Beta marches it through a remarkable cascade - fixed points, period-doubling, and period-ADDING windows where the cycle length grows by one - drawing the 'mouse' shape in its bifurcation diagram. It free-runs at Rate, stepping the map and outputting x as a bipolar CV that ranges from steady to multi-step periodic to chaotic as Alpha and Beta move. A bell-curve map distinct from the parabolic logistic and tent maps.

Param	Range	Default	Unit
Rate	0.1 – 50	8	Hz
Alpha	4 – 8	6.2	—
Beta	-0.7 – 0.3	-0.5	—

Cusp inputs

Cusp-map source (a first as a synth block): iterates $x \leftarrow 1 - 2\sqrt{|x|}$, a map with a sharp cusp at the origin instead of a smooth parabola. The square-root gives it an infinitely steep peak, so it stretches and folds the line more violently near zero than the polynomial maps, producing fully-developed chaos with a characteristic spiky density. It free-runs at Rate, stepping the map and outputting x as a bipolar CV that hops sharply across $[-1, 1]$. A non-smooth chaos distinct from the parabolic logistic and the piecewise tent.

Param	Range	Default	Unit
Rate	0.1 – 50	8	Hz

SineMap inputs

Sine-map source (a first as a synth block): iterates $x \leftarrow R\sin(\pi x)$, the trigonometric twin of the logistic map - a single smooth hump that, as R approaches 1, period-doubles its way into chaos by the very same Feigenbaum route, which is why it is a favourite engine for chaotic encryption. It free-runs at Rate, stepping the map and outputting x as a 0..1 CV. R sets the position on the road to chaos: low and it settles, near 1 and it never repeats. A sinusoidal unimodal map distinct from the polynomial logistic. Needs no input.

Param	Range	Default	Unit
Rate	0.1 – 50	8	Hz
R	0.7 – 1	0.99	—

Cubic inputs

Cubic-map source (a first as a synth block): iterates $x \leftarrow A^3x - x^3$, an ODD-symmetric map (it treats positive and negative x as mirror images) unlike the lopsided logistic. That symmetry gives it a distinctive bifurcation diagram with two mirrored chaotic bands that can suddenly merge, and orbits that swing through both signs. It free-runs at Rate, stepping the map and outputting x as a bipolar CV. A sets the gain on the road to chaos. A symmetric cubic chaos distinct from the asymmetric quadratic maps. Needs no input.

Param	Range	Default	Unit
Rate	0.1 – 50	8	Hz
A	2 – 3	2.59	—

De Jong inputs

De-Jong attractor source (a first as a synth block): iterates Peter de Jong's map $x \leftarrow \sin(Ay) - \cos(Bx)$, $y \leftarrow \sin(Cx) - \cos(Dy)$, whose orbit weaves into a dense web of looping filaments - one of the most varied strange-attractor families, redrawing completely with the smallest change of constants. It free-runs at Rate, stepping the map and outputting x as a smooth bipolar CV that wanders the web. A warps the figure. A four-sine attractor distinct from the two-sine Clifford. Needs no input.

Param	Range	Default	Unit
Rate	0.1 – 50	8	Hz
A	1 – 1.8	1.4	—

Tinkerbell inputs

Tinkerbell-map source (a first as a synth block): iterates $x \leftarrow x^2 - y^2 + Ax + By$, $y \leftarrow 2xy + Cx + Dy$ - a quadratic map of the plane whose orbit traces a comma-shaped fractal said to resemble Tinkerbell's flight path. Its mix of squaring and cross-coupling folds the trajectory into a tight curl with delicate internal banding. It free-runs at Rate, stepping the map and outputting x as a bipolar CV. A quadratic-plane attractor distinct from the trig-based De Jong and Clifford. Needs no input.

Param	Range	Default	Unit
Rate	0.1 – 50	8	Hz

GumowskiMira inputs

Gumowski-Mira attractor source (a first as a synth block): iterates a map built around the recursive nonlinearity $f(x) = A*x + 2(1-A)x^2/(1+x^2)$, devised at CERN to model the chaotic drift of particles in an accelerator beam. It is famous for the astonishing variety of shapes it draws - rings, swirls, lattices and creatures - from tiny parameter changes. It free-runs at Rate, stepping the map and outputting x as a bipolar CV. A particle-physics attractor distinct from the trigonometric and quadratic maps. Needs no input.

Param	Range	Default	Unit
Rate	0.1 – 50	8	Hz

Lozi inputs

Lozi-map source (a first as a synth block): iterates $x <- 1 - A*|x| + y$, $y <- B*x$ - the piecewise-linear cousin of the Henon map, with an absolute value where Henon has a square. That sharp corner gives it a strange attractor made of straight-line segments rather than smooth curves, and one of the few chaotic maps proven rigorously to be chaotic. It free-runs at Rate, stepping the map and outputting x as a bipolar CV. A sets the fold sharpness. An angular, provably-chaotic attractor distinct from the smooth Henon. Needs no input.

Param	Range	Default	Unit
Rate	0.1 – 50	8	Hz
A	1.4 – 1.8	1.7	—

Svensson inputs

Svensson attractor source (a first as a synth block): iterates Johnny Svensson's map $x <- D*\sin(A*x) - \sin(B*y)$, $y <- C*\cos(A*x) + \cos(B*y)$, a sine-and-cosine attractor that draws tight rosette and starburst figures with sharp radial symmetry. The mix of large and small coefficients pulls the orbit into spiked petals quite unlike the woven webs of De Jong. It free-runs at Rate, stepping the map and outputting x as a bipolar CV. A radial-symmetry attractor distinct from the other trigonometric maps. Needs no input.

Param	Range	Default	Unit
Rate	0.1 – 50	8	Hz

Thomas inputs

Thomas cyclically-symmetric attractor (a first as a synth block): integrates $x' = \sin(y) - b*x$ and its two cyclic rotations - a flow built entirely from sines, completely symmetric under swapping $x \rightarrow y \rightarrow z$. As the damping b is lowered the trajectory unwinds from a simple loop into a labyrinthine chaotic walk that drifts through space like a particle in a turbulent crystal lattice. It free-runs at Rate, outputting x as a smooth, gently-wandering bipolar CV. A trig-flow chaos distinct from the polynomial Lorenz or Chua systems.

Param	Range	Default	Unit
Rate	0.05 – 4	1	—

Halvorsen inputs

Halvorsen attractor (a first as a synth block): integrates the cyclically-symmetric system $x' = -a*x - 4y - 4z - y^2$ (and its $x \rightarrow y \rightarrow z$ rotations), whose quadratic cross-terms wind the trajectory into a tight three-armed spiral knot. It free-runs at Rate, outputting x as a smooth bipolar CV that loops and lurches around the attractor's arms. A denser, more wound-up chaos than the open Lorenz butterfly, distinct from the trig-based Thomas. Internally 3 small Euler sub-steps keep the stiff dynamics stable.

Param	Range	Default	Unit
Rate	0.05 – 4	1	—

Aizawa inputs

Aizawa attractor (a first as a synth block): integrates a six-parameter forced system whose trajectory wraps around a sphere while a hole opens through its top axis, producing a shape unlike any other strange attractor - a knotted ball with a drilled core. It free-runs at Rate, outputting x as a smooth bipolar CV that orbits the sphere and dips through the hole, giving rich circular motion with sudden axial excursions. A visually and sonically distinct chaos from the butterfly and spiral attractors.

Param	Range	Default	Unit
Rate	0.05 – 4	1	—

Qi inputs

Qi-Chen four-wing attractor (a first as a synth block): integrates $x' = a*(y - x) + y*z$, $y' = c*x - y - x*z$, $z' = x*y - b*z$, a system famous for growing FOUR wings instead of the usual two, the trajectory hopping chaotically between all four lobes. The extra quadratic cross-terms give it very wide, energetic swings. It free-runs at Rate, outputting x as a smooth bipolar CV with large excursions. A four-wing chaos distinct from the two-wing Lorenz and Chen attractors.

Param	Range	Default	Unit
Rate	0.05 – 4	1	—

ChenLee inputs

Chen-Lee attractor (a first as a synth block): integrates $x' = a*x - y*z$, $y' = b*y + x*z$, $z' = c*z + x*y/3$, a model of a rigid body rotating under feedback torque, derived from Euler's equations for a gyroscope. Its trajectory traces a pair of stacked, twisted rings (a double-scroll that loops rather than wings), wandering chaotically between them. It free-runs at Rate, outputting x as a smooth bipolar CV with wide swings. A rotational-dynamics chaos distinct from the Lorenz and Rossler families.

Param	Range	Default	Unit
Rate	0.05 – 4	1	—

ShimizuMorioka inputs

Shimizu-Morioka attractor (a first as a synth block): integrates $x' = y$, $y' = x - a*y - x*z$, $z' = -b*z + x^2$, a simplified model of the Lorenz system near the onset of chaos. It shows a symmetric butterfly that is born through a sequence of bifurcations as the parameters change, a clean laboratory for studying how order gives way to chaos. It free-runs at Rate, outputting x as a smooth bipolar CV that flips between the two wings. A reduced-Lorenz chaos distinct from the full Lorenz attractor.

Param	Range	Default	Unit
Rate	0.05 – 4	1	—

WangSun inputs

Wang-Sun attractor (a first as a synth block): integrates a four-term quadratic system $x' = a*x + c*y*z$, $y' = b*x + d*y - x*z$, $z' = e*z + f*x*y$ with an unusual mix of small and large coefficients, one of the gallery of distinct chaotic flows catalogued in the search for new attractor shapes. Its orbit folds into a lopsided, asymmetric scroll. It free-runs at Rate, outputting x as a smooth bipolar CV. A custom-coefficient chaos distinct from the named classical attractors.

Param	Range	Default	Unit
Rate	0.05 – 4	1	—

Rucklidge inputs

Rucklidge attractor (a first as a synth block): integrates $x' = -k*x + a*y - y*z$, $y' = x$, $z' = -z + y^2$, a model of convection in a fluid layer subject to a magnetic field (double-diffusive convection). It produces a symmetric two-lobed attractor as the rolls of fluid reverse chaotically. It free-runs at Rate, outputting x as a smooth bipolar CV with broad excursions. A convective-magnetic chaos distinct from the rigid-body and reduced-Lorenz flows.

Param	Range	Default	Unit
Rate	0.05 – 4	1	—

Yu inputs

Yu-Wang attractor (a first as a synth block): integrates $x' = a(y - x)$, $y' = b*x - c*x*z$, $z' = \exp(x*y) - d*z$ - unusual for its EXPONENTIAL term, which makes the z dynamics snap rather than curve. The orbit forms a tight two-scroll butterfly with sharp, fast reinjections. It free-runs at Rate, outputting x as a smooth bipolar CV. An exponential-nonlinearity chaos distinct from the purely polynomial attractors.

Param	Range	Default	Unit
Rate	0.05 – 4	1	—

ThreeScroll inputs

Three-scroll attractor (a first as a synth block): integrates the Pan-Xu-Zhou system $x' = a(y - x) + d*x*z$, $y' = c*y - x*z$, $z' = b*z + x*y - e*x*x$, a unified chaotic flow whose trajectory weaves through THREE separate scroll centres instead of the usual two, hopping between all three lobes. It free-runs at Rate, outputting x as a smooth bipolar CV with wide swings. A three-lobe chaos distinct from the two-wing Lorenz and four-wing Qi attractors.

Param	Range	Default	Unit
Rate	0.05 – 4	1	—

Lorenz84 inputs

Lorenz-84 attractor (a first as a synth block): integrates Edward Lorenz's low-order model of the GLOBAL ATMOSPHERIC circulation, $x' = -y^2 - z^2 - a*x + a*F$, $y' = x*y - b*x*z - y + G$, $z' = b*x*y + x*z - z$, where x is the strength of the westerly wind current and y, z are a travelling weather wave. It chaotically switches between calm and stormy regimes, a toy climate that never repeats. It free-runs at Rate, outputting x as a smooth bipolar CV. A climate-model chaos distinct from the original Lorenz convection attractor.

Param	Range	Default	Unit
Rate	0.05 – 4	1	—

MooreSpiegel inputs

Moore-Spiegel oscillator (a first as a synth block): integrates the jerk system $x' = y$, $y' = z$, $z' = -z - (T - R + R*x*x)*y - T*x$, devised to model the aperiodic flickering of a parcel of gas in a star's atmosphere (thermo-mechanical oscillations under radiation). The cubic stiffness term makes it burst between small and large oscillations chaotically. It free-runs at Rate, outputting x as a smooth bipolar CV. An astrophysical jerk-chaos distinct from the rotational and convective attractors.

Param	Range	Default	Unit
Rate	0.05 – 4	1	—

Dequan inputs

Dequan-Li attractor (a first as a synth block): integrates a five-term unified chaotic system $x' = a(y - x) + d*x*z$, $y' = k*x + f*y - x*z$, $z' = c*z + x*y - e*x*x$ with large coefficients that drive very energetic, wide-ranging orbits forming a dense double scroll. It free-runs at Rate, outputting x as a smooth bipolar CV with broad excursions. A high-energy unified chaos distinct from the gentler classical attractors.

Param	Range	Default	Unit
Rate	0.05 – 4	1	—

SprottJ inputs

Sprott-J attractor (a first as a synth block): integrates $x' = 2z$, $y' = -2y + z$, $z' = -x + y + y^2$ - one of the 19 algebraically SIMPLEST chaotic flows that Julien Sprott found by computer search, each with the bare minimum of terms needed for chaos. Case J folds a single quadratic term into an otherwise linear system. It free-runs at Rate, outputting x as a smooth bipolar CV. A minimal-quadratic chaos distinct from the elaborate named attractors.

Param	Range	Default	Unit
Rate	0.05 – 4	1	—

SprottK inputs

Sprott-K attractor (a first as a synth block): integrates $x' = x*y - z$, $y' = x - y$, $z' = x + 0.3*z$ - another of Sprott's minimal chaotic systems, this one driven by the single product term $x*y$. Despite having only one nonlinearity it produces a fully chaotic orbit. It free-runs at Rate, outputting x as a smooth bipolar CV. A single-product chaos distinct from the other minimal Sprott cases.

Param	Range	Default	Unit
Rate	0.05 – 4	1	—

SprottN inputs

Sprott-N attractor (a first as a synth block): integrates $x' = -2y$, $y' = x + z^2$, $z' = 1 + y - 2z$ - a minimal chaotic flow whose only nonlinearity is the squared z term. It traces a folded sheet that the orbit wanders chaotically. It free-runs at Rate, outputting x as a smooth bipolar CV. A squared-term minimal chaos distinct from the product-driven Sprott cases.

Param	Range	Default	Unit
Rate	0.05 – 4	1	—

SprottR inputs

Sprott-R attractor (a first as a synth block): integrates $x' = 0.9 - y$, $y' = 0.4 + z$, $z' = x*y - z$ - a minimal chaotic system with constant forcing terms and a single product nonlinearity. The constants push the orbit steadily while the $x*y$ coupling folds it into chaos. It free-runs at Rate, outputting x as a smooth bipolar CV. A forced minimal chaos distinct from the unforced Sprott cases.

Param	Range	Default	Unit
Rate	0.05 – 4	1	—

FinanceChaos 0 inputs

Financial-chaos attractor (a first as a synth block): integrates a model of a chaotic macroeconomy $x' = z + (y - a)x$, $y' = 1 - b*y - x^2$, $z' = -x - c*z$, where x is the interest rate, y the investment demand and z the price index. It shows how a deterministic economy with no outside shocks can still wander unpredictably between booms and busts. It free-runs at Rate, outputting x as a smooth bipolar CV. An econophysics chaos distinct from the mechanical and climate attractors.

Param	Range	Default	Unit
Rate	0.05 – 4	1	—

NoseHoover 0 inputs

Nose-Hoover oscillator (a first as a synth block): integrates the thermostatted system $x' = y$, $y' = -x + y*z$, $z' = a - y^2$ - the equations a physicist uses to hold a simulated molecule at constant temperature, which famously refuse to settle and instead wander chaotically. Unlike the dissipative attractors it is volume-CONSERVING, so the trajectory never collapses onto a thin sheet but fills a fat tube of phase space, giving a fuller, more even chaotic motion. It free-runs at Rate, outputting x as a smooth bipolar CV. A conservative chaos distinct from the dissipative attractors.

Param	Range	Default	Unit
Rate	0.05 – 4	1	—

Rikitake 0 inputs

Rikitake-dynamo attractor (a first as a synth block): integrates the two-disc Rikitake dynamo $x' = -\mu*x + z*y$, $y' = -\mu*y + (z - a)*x$, $z' = 1 - x*y$ - a model of the Earth's geomagnetic field built from two coupled current discs. Its chaos is the textbook explanation for why the planet's magnetic poles REVERSE at irregular intervals: the trajectory orbits one polarity for a while then flips to the other, unpredictably. It free-runs at Rate, outputting x as a smooth bipolar CV with long stretches in one sign broken by sudden reversals - a switching chaos distinct from the steadily-wandering attractors.

Param	Range	Default	Unit
Rate	0.05 – 4	1	—

Chen 0 inputs

Chen attractor (a first as a synth block): integrates Chen's system $x' = a(y-x)$, $y' = (c-a)x - xz + cy$, $z' = xy - bz$ - a close relative of the Lorenz butterfly discovered by feedback control, but with a more tightly wound double scroll whose two lobes are pulled closer and spun faster. It free-runs at Rate, outputting x as a smooth bipolar CV that loops within one lobe then flips chaotically to the other. A wound-up cousin of Lorenz distinct from the Rossler and Thomas flows.

Param	Range	Default	Unit
Rate	0.05 – 4	1	—

Lu 0 inputs

Lu (Lu-Chen) attractor (a first as a synth block): integrates $x' = a(y-x)$, $y' = -xz + cy$, $z' = xy - bz$ - the system that forms the missing BRIDGE between the Lorenz and Chen attractors, morphing continuously from one to the other as its parameter changes. Its double scroll sits between the two, with its own balance of slow orbiting and sudden lobe-switching. It free-runs at Rate, outputting x as a smooth bipolar CV. A transitional double-scroll chaos distinct from both Lorenz and Chen.

Param	Range	Default	Unit
Rate	0.05 – 4	1	—

BurkeShaw inputs

Burke-Shaw attractor (a first as a synth block): integrates $x' = -s(x+y)$, $y' = -y - s*x*z$, $z' = s*x*y + V$ - a tightly-coiled variant of the Lorenz system that winds into a dense, near-toroidal spiral rather than two separate wings, spinning rapidly around a doughnut of trajectories. It free-runs at Rate, outputting x as a smooth, fast-cycling bipolar CV with a strong underlying rotation. A coiled, rotation-dominated chaos distinct from the wing-flipping Lorenz and Chen.

Param	Range	Default	Unit
Rate	0.05 – 4	1	—

NewtonLeipnik inputs

Newton-Leipnik attractor (a first as a synth block): integrates a rigid-body rotation system (from the Euler equations of a tumbling object with feedback) that has the rare property of TWO coexisting strange attractors - depending on where it starts, the trajectory settles onto one of two different chaotic shapes. It free-runs at Rate, outputting x as a smooth bipolar CV that winds through a delicate folded-ribbon attractor. A two-attractor chaos distinct from the single-attractor Lorenz family.

Param	Range	Default	Unit
Rate	0.05 – 4	1	—

FourWing inputs

Four-wing attractor (a first as a synth block): integrates $x' = a*x + y*z$, $y' = b*x + c*y - x*z$, $z' = -z - x*y$, a system whose trajectory visits FOUR separate wings instead of the usual two, wandering chaotically between all four quadrants in a butterfly with double the lobes. It free-runs at Rate, outputting x as a smooth bipolar CV that jumps unpredictably between the four wings. A four-lobed chaos distinct from the two-winged Lorenz, Chen and Lu attractors.

Param	Range	Default	Unit
Rate	0.05 – 4	1	—

SprottB inputs

Sprott-B attractor (a first as a synth block): integrates $x' = y*z$, $y' = x - y$, $z' = 1 - x*y$, one of the algebraically SIMPLEST possible chaotic systems - J.C. Sprott found, by computer search, the minimal sets of terms that still produce chaos, and this is among them. Despite having only five terms it folds a genuine strange attractor. It free-runs at Rate, outputting x as a smooth bipolar CV. A minimal-form chaos distinct from the larger Lorenz and Chen systems. Needs no input.

Param	Range	Default	Unit
Rate	0.05 – 4	1	—

SprottC inputs

Sprott-C attractor (a first as a synth block): integrates $x' = y*z$, $y' = x - y$, $z' = 1 - x^2$, a sibling of the Sprott-B system with the cross-term $x*y$ swapped for the square x^2 - another of Sprott's minimal five-term chaotic flows. The small change reshapes the attractor into its own distinct double-lobed fold. It free-runs at Rate, outputting x as a smooth bipolar CV. A minimal quadratic chaos distinct from Sprott-B and the classical attractors. Needs no input.

Param	Range	Default	Unit
Rate	0.05 – 4	1	—

SprottE inputs

Sprott-E attractor (a first as a synth block): integrates $x' = y^2z$, $y' = x^2 - y$, $z' = 1 - 4x$, another of Sprott's algebraically-minimal chaotic systems, with a square nonlinearity in the y equation and a linear ramp in z. Its orbit winds through a compact, tightly-folded strange attractor. It free-runs at Rate, outputting x as a smooth bipolar CV. A minimal-form chaos distinct from the Sprott-B and Sprott-C variants. Needs no input.

Param	Range	Default	Unit
Rate	0.05 – 4	1	—

SprottG inputs

Sprott-G attractor (a first as a synth block): integrates $x' = 0.4x + z$, $y' = xz - y$, $z' = -x + y$, a minimal chaotic flow with a small linear self-feedback that lets it spiral outward before the nonlinear term folds it back. Its attractor is a delicate scrolled spiral. It free-runs at Rate, outputting x as a smooth bipolar CV that winds and folds. A spiral minimal chaos distinct from the other Sprott systems. Needs no input.

Param	Range	Default	Unit
Rate	0.05 – 4	1	—

Bouali inputs

Bouali attractor (a first as a synth block): integrates a three-variable system devised to model fluctuating economic and predator-prey-like quantities - $x' = x(4 - y) + 0.3z$, $y' = -y(1 - x^2)$, $z' = -x(1.5 - 0.05z) - 0.05z$ - whose trajectory winds onto a broad, layered, sheet-like strange attractor. It free-runs at Rate, outputting x as a smooth bipolar CV with a strong slow sway under faster folds. A sheet-like chaos distinct from the scroll attractors. Needs no input.

Param	Range	Default	Unit
Rate	0.05 – 4	1	—

Hopalong inputs

Hopalong-map source (a first as a synth block): iterates Barry Martin's 'hopalong' map $x \leftarrow y - \text{sign}(x)\sqrt{|x|}$, $y \leftarrow A - x$, which hurls the orbit back and forth across the origin, the point hopping along an intricate web of nested arcs and swirls. The square-root fold keeps it from running away while letting it explore a wide, lacy attractor. It free-runs at Rate, stepping the map and outputting x as a bipolar CV that leaps across the figure. An orbital, root-folded chaos distinct from the trigonometric and polynomial maps.

Param	Range	Default	Unit
Rate	0.1 – 50	8	Hz

Bedhead inputs

Bedhead-map source (a first as a synth block): iterates the trigonometric map $x \leftarrow \sin(x^2y/b)y + \cos(ax - y)$, $y \leftarrow x + \sin(y)/b$, one of the 'bedhead' attractors whose orbit tangles into a soft, hair-like snarl of overlapping loops. The product term x^2y inside the sine couples the two coordinates nonlinearly, giving a dense, organic, woven motion. It free-runs at Rate, outputting x as a bipolar CV. A coupled-trig chaos distinct from the additive De Jong and Clifford maps.

Param	Range	Default	Unit
Rate	0.1 – 50	8	Hz

FractalDream ▯ inputs

Fractal-dream-map source (a first as a synth block): iterates Clifford Pickover's 'fractal dream' map $x \leftarrow \sin(B*y) + C*\sin(B*x)$, $y \leftarrow \sin(A*x) + D*\sin(A*y)$, a four-term sine system whose orbit settles into soft, dreamlike clouds of nested filaments. It free-runs at Rate, stepping the map and outputting x as a smooth bipolar CV that drifts through the dream. A four-sine attractor distinct from the De Jong and Svensson maps. Needs no input.

Param	Range	Default	Unit
Rate	0.1 – 50	8	Hz

Martin ▯ inputs

Martin-map source (a first as a synth block): iterates the sine variant of Barry Martin's map $x \leftarrow y - \sin(x)$, $y \leftarrow A - x$, whose orbit hops across a broad, wavy lattice of interleaved bands rather than a compact attractor. The single sine fold gives a wide, drifting, quasi-tiled motion that roams far before folding back. It free-runs at Rate, outputting x as a bipolar CV. A wide-ranging, lattice-like chaos distinct from the compact Hopalong and the trig attractors.

Param	Range	Default	Unit
Rate	0.1 – 50	8	Hz

Pickover ▯ inputs

Pickover attractor source (a first as a synth block): iterates Clifford Pickover's 3-D map $x \leftarrow \sin(a*y) - z*\cos(b*x)$, $y \leftarrow z*\sin(c*x) - \cos(d*y)$, $z \leftarrow \sin(x)$, whose orbit is drawn into a delicate, three-dimensional thread-like sculpture of looping filaments. The z-coupling pulls the motion out of the plane, giving a richer wander than the 2-D maps. It free-runs at Rate, outputting x as a smooth bipolar CV. A 3-D threaded attractor distinct from the planar De Jong, Clifford and Svensson maps.

Param	Range	Default	Unit
Rate	0.1 – 50	8	Hz

Julia ▯ inputs

Julia-set source (a first as a synth block): iterates the complex map $z \leftarrow z^2 + c$, the equation behind the Julia fractals - a point in the complex plane is squared and shifted again and again. For points inside the set it wanders forever in an intricate bounded orbit; when one escapes to infinity the source reseeds near the centre and sets off again, so the output is a restless, fractal-flavoured CV. CReal and CImag pick which Julia set (its shape and dynamics) to explore. It free-runs at Rate, outputting the real part. A complex-dynamics source unlike any real-valued chaos.

Param	Range	Default	Unit
Rate	0.1 – 50	8	Hz
CReal	-1 – 0.5	-0.8	—
CImag	-1 – 1	0.156	—

BurningShip ▯ inputs

Burning-Ship source (a first as a synth block): iterates $z \leftarrow (|\operatorname{Re} z| + i|\operatorname{Im} z|)^2 + c$, the Burning Ship fractal's map, where taking the ABSOLUTE VALUE of each part before squaring breaks the usual symmetry and produces the jagged, flame-and-mast shape it is named for. The folding makes its bounded orbits sharper and more angular than a Julia set; escapes reseed near the centre. CReal and CImag pick the region. It free-runs at Rate, outputting the real part. An absolute-value complex map distinct from the Julia and Tricorn sources.

Param	Range	Default	Unit
Rate	0.1 – 50	8	Hz
CReal	-2 – 0	-1.8	—
CImag	-1 – 1	-0.08	—

Tricorn inputs

Tricorn (Mandelbar) source (a first as a synth block): iterates $z \leftarrow \text{conj}(z)^2 + c$, the same squaring as a Julia set but on the CONJUGATE of z (flipping the sign of its imaginary part each step). That conjugation gives the fractal a three-cornered symmetry and makes its boundary fractal in a different way, so the bounded orbits twist with a distinctive three-fold pull; escapes reseed near the centre. CReal and CImag pick the region. It free-runs at Rate, outputting the real part. A conjugate complex map distinct from the Julia and Burning Ship sources.

Param	Range	Default	Unit
Rate	0.1 – 50	8	Hz
CReal	-1 – 0.5	-0.5	—
CImag	-1 – 1	0.5	—

Phoenix inputs

Phoenix-set source (a first as a synth block): iterates $z \leftarrow z^2 + c + p \cdot z_{\text{prev}}$, a complex map with MEMORY - the next value depends not only on the current z but on the PREVIOUS one too, the extra feedback that grows the curling, bird-like Phoenix fractal. That one-step memory makes its bounded orbits curl and spiral more than a plain Julia set; escapes reseed near the centre. It free-runs at Rate, outputting the real part. A memory-fed complex map distinct from the memoryless Julia, Tricorn and Burning Ship.

Param	Range	Default	Unit
Rate	0.1 – 50	8	Hz
CReal	0 – 1	0.567	—
P	-1 – 0	-0.5	—

Newton inputs

Newton-fractal source (a first as a synth block): runs Newton's root-finding method on the equation $z^3 = 1$ in the complex plane - $z \leftarrow z - (z^3 - 1)/(3z^2)$ - which always homes in on one of the three cube roots of unity, but WHICH one depends with fractal sensitivity on where it starts. The source iterates until it locks onto a root, then reseeds from a fresh random point, so the output jumps between the three roots' values with a swirling transient before each lock. It free-runs at Rate, outputting the real part. A root-finding chaos distinct from the squaring fractals.

Param	Range	Default	Unit
Rate	0.1 – 50	8	Hz

Firefly inputs

Firefly-synchronisation source (a first as a synth block): models a swarm of twelve fireflies, each charging toward its flash; when one fires it nudges all the others a little closer to firing too. Through this pulse coupling they spontaneously fall into rhythm - the real mechanism behind fields of fireflies flashing in unison. The output is the swarm's phase COHERENCE (0 = scattered, 1 = all flashing together), which rises as they sync and breaks up again, breathing in and out. Couple sets how strongly each flash pulls the others. A self-organising sync source distinct from the chaos oscillators. Needs no input.

Param	Range	Default	Unit
Rate	0.1 – 50	12	Hz
Couple	0 – 0.4	0.2	—

Vicsek 0 inputs

Vicsek-flocking source (a first as a synth block): models twelve self-propelled particles on a ring, each steering its heading toward the AVERAGE direction of its neighbours plus a dash of random Noise - the minimal model of how birds flock, fish school and bacteria swarm. At low noise they all line up and move as one (high order); raise the noise past a critical point and the flock shatters into disorder. The output is the order parameter (how aligned the flock is), which hovers and flickers near the flocking transition. A collective-motion source distinct from the single-body chaos. Needs no input.

Param	Range	Default	Unit
Rate	0.1 – 50	12	Hz
Noise	0 – 1.5	0.3	—

Deffuant 0 inputs

Deffuant opinion-dynamics source (a first as a synth block): models twelve people each holding an opinion between 0 and 1; at each step two meet at random and, only if their views are already within Tolerance of each other, they compromise and move closer - if they are too far apart they simply ignore one another. This 'bounded confidence' makes the crowd settle into one consensus (narrow Tolerance gives several rival camps), occasionally shaken up when someone changes their mind. The output is the SPREAD of opinion (wide = divided, near zero = agreed), drifting as clusters form and dissolve. A social-dynamics source distinct from the physical models. Needs no input.

Param	Range	Default	Unit
Rate	0.1 – 50	12	Hz
Tolerance	0.05 – 0.5	0.3	—

Galam 0 inputs

Galam majority-rule source (a first as a synth block): models twelve people on a ring each holding a yes/no opinion; at each step a small group is picked and EVERYONE in it adopts the local MAJORITY view - the mechanism Galam used to model how rumours, fashions and political opinions sweep through a crowd toward unanimity. A trickle of independent mind-changing keeps it from freezing forever. The output is the fraction holding 'yes', which lurches toward total consensus (all-yes or all-no) and then gets knocked back. A majority-driven opinion source distinct from the pairwise Deffuant model. Needs no input.

Param	Range	Default	Unit
Rate	0.1 – 50	12	Hz
Flip	0 – 0.05	0.01	—

Schelling 0 inputs

Schelling-segregation source (a first as a synth block): models twelve agents of two kinds on a ring; an agent is content only if enough of its neighbours are of its OWN kind, and a discontented one swaps places with a random other - Schelling's Nobel-winning demonstration that even a mild preference for similar neighbours drives a population into sharply segregated blocks. A little random relocation keeps it stirring. The output is the segregation index (how clumped the two kinds are), climbing as the blocks form. A self-sorting source distinct from the opinion models. Needs no input.

Param	Range	Default	Unit
Rate	0.1 – 50	12	Hz
Stir	0 – 0.05	0.01	—

SIR 0 inputs

SIR epidemic source (a first as a synth block): integrates the textbook Susceptible-Infected-Recovered model - the equations Kermack and McKendrick wrote in 1927 that underlie every epidemic forecast. A pathogen burns through the susceptible population

in a wave that peaks and fades as the recovered build up herd immunity; births slowly refill the susceptible pool and a seasonal swing in contagion keeps the waves coming. The output is the infected fraction, a rising-and-falling epidemic curve. R_0 sets the basic reproduction number (how contagious), Season the seasonal swing. A disease-dynamics source distinct from the chemical and predator-prey oscillators. Needs no input.

Param	Range	Default	Unit
Rate	0.1 – 4	1	—
R_0	1.5 – 8	4	—
Season	0 – 0.4	0.15	—

SIRS inputs

SIRS epidemic source (a first as a synth block): the Susceptible-Infected-Recovered-Susceptible model, where immunity WANES - the recovered slowly become susceptible again and can be reinfected, so instead of a single epidemic the disease settles into recurring waves, the dynamics behind seasonal flu and the common cold. The output is the infected fraction, oscillating endlessly as immunity builds and fades. R_0 sets the contagiousness, Season the seasonal forcing. A waning-immunity epidemic distinct from the one-shot SIR. Needs no input.

Param	Range	Default	Unit
Rate	0.1 – 4	1	—
R_0	1.5 – 8	4	—
Season	0 – 0.4	0.15	—

SIS inputs

SIS epidemic source (a first as a synth block): the Susceptible-Infected-Susceptible model, where there is NO immunity at all - the infected recover straight back to susceptible and can immediately catch it again, the model for diseases like the common cold or many bacterial infections that confer no lasting protection. Above its threshold the disease never dies out but stays endemic at a high steady level, breathing up and down with the seasons. The output is the infected fraction, hovering high. R_0 sets the contagiousness, Season the swing. A no-immunity epidemic distinct from the SIR and SIRS. Needs no input.

Param	Range	Default	Unit
Rate	0.1 – 4	1	—
R_0	1.5 – 8	4	—
Season	0 – 0.4	0.15	—

SEIR inputs

SEIR epidemic source (a first as a synth block): the Susceptible-Exposed-Infected-Recovered model, which adds a LATENT period - the newly-infected are Exposed but not yet contagious for a while before they become Infectious, the more realistic model used for measles and COVID. That incubation delay slows and stretches the epidemic wave and makes its peaks lag behind the cause. The output is the infected fraction, a delayed rising-and-falling curve. R_0 sets the contagiousness, Season the seasonal swing. A latent-period epidemic distinct from the instantaneous SIR. Needs no input.

Param	Range	Default	Unit
Rate	0.1 – 4	1	—
R_0	1.5 – 8	4	—
Season	0 – 0.4	0.15	—

SEIRS inputs

SEIRS epidemic source (a first as a synth block): the full Susceptible-Exposed-Infected-Recovered-Susceptible cycle, combining BOTH a latent incubation period AND waning immunity - the most complete of the classic compartment models, used for endemic

seasonal diseases. The exposed delay and the loss of immunity together produce rich, sustained, sometimes multi-year oscillation cycles in the infection level. The output is the infected fraction, oscillating with a layered rhythm. R_0 sets the contagiousness, Season the seasonal forcing. The full-cycle epidemic distinct from the simpler SIR, SIRS and SEIR. Needs no input.

Param	Range	Default	Unit
Rate	0.1 – 4	1	—
R_0	1.5 – 8	4	—
Season	0 – 0.4	0.15	—

ElFarol inputs

El-Farol-bar source (a first as a synth block): models Brian Arthur's famous bar problem - twelve regulars each decide whether to go out, wanting to go only if the bar will NOT be too crowded. Since everyone is second-guessing everyone else, no fixed rule works, yet through trial-and-error adjustment the crowd self-organises so attendance hovers right around Capacity, never settling. The output is the attendance, fluctuating around the comfort threshold - the founding model of bounded-rationality economics and inductive reasoning. A self-defeating-prediction source distinct from the consensus-seeking opinion models. Needs no input.

Param	Range	Default	Unit
Rate	0.1 – 50	12	Hz
Capacity	0.3 – 0.7	0.6	—

Cobweb inputs

Cobweb-market source (a first as a synth block): models the boom-bust price cycle of a market where producers decide TODAY'S supply based on YESTERDAY'S price - the cobweb theorem behind hog and crop cycles. High prices trigger overproduction, which crashes prices, which triggers underproduction, and so on. With a realistic exponential (Ricker) supply response the price never settles: it oscillates and, when producers over-react (high R), tips into genuine chaos. The output is the price. R sets how strongly supply reacts. A lagged-market chaos distinct from the smooth predator-prey and the logistic map. Needs no input.

Param	Range	Default	Unit
Rate	0.1 – 50	8	Hz
R	2 – 3.8	3	—

Bass inputs

Bass-diffusion source (a first as a synth block): the marketing model of how a new product spreads - a few INNOVATORS adopt on their own, then a wave of IMITATORS follows as they see others using it, giving the classic S-shaped adoption curve. Here the market saturates and resets, so the output is a repeating slow swell from zero to full adoption: a smooth, asymmetric ramp that starts slow, accelerates through the tipping point, then levels off. Innovate sets the spontaneous-adoption rate, Imitate the word-of-mouth strength. A technology-adoption curve distinct from the oscillating dynamical sources. Needs no input.

Param	Range	Default	Unit
Rate	0.1 – 50	4	Hz
Innovate	0.005 – 0.1	0.03	—
Imitate	0.1 – 0.6	0.4	—

Goodwin inputs

Goodwin growth-cycle source (a first as a synth block): Richard Goodwin's model of the capitalist business cycle, which borrows the predator-prey equations from ecology - here EMPLOYMENT plays the prey and the WORKERS' WAGE SHARE the predator. High employment lets workers win higher wages, which squeezes profits and investment, which raises unemployment, which weakens wages, which restores profits, and round it goes. The output is the employment rate, cycling endlessly like a boom-and-bust. A class-struggle oscillation, integrated symplectically so the cycle never winds down, distinct from the biological predator-prey. Needs no input.

Param	Range	Default	Unit
Rate	0.05 – 4	1	—

Kirman inputs

Kirman-herding source (a first as a synth block): Alan Kirman's 'ants' model of why crowds and markets swing between extremes - twelve agents each hold one of two views, and at each step one agent either switches by IMITATING a random other (herding) or changes its mind spontaneously. With strong herding the population does not settle in the middle but lurches as a herd, swinging to nearly all-one-way, lingering, then stampeding to the other - the bimodal switching behind fads, bank runs and market bubbles. The output is the fraction holding one view. Herd sets the imitation strength. A herding-instability source distinct from the consensus-seeking opinion models. Needs no input.

Param	Range	Default	Unit
Rate	0.1 – 50	12	Hz
Herd	0 – 1	0.4	—
Spontaneous	0 – 0.1	0.01	—

Excitable inputs

Excitable-medium source (a first as a synth block): a ring of sixteen FitzHugh-Nagumo cells coupled by diffusion, the model of an excitable tissue like heart muscle or a nerve sheet. A pulse, once lit, travels around the ring re-igniting each cell as it passes - a self-sustaining wave of excitation, the same mechanism behind a heartbeat and the spiral waves of a fibrillating heart. The output is one cell's voltage, spiking each time the wave sweeps by. Drive sets the excitability. A travelling-wave source distinct from the single-cell oscillators. Needs no input.

Param	Range	Default	Unit
Rate	0.05 – 4	1	—
Drive	0 – 1	0.5	—

KuramotoRing inputs

Kuramoto-ring source (a first as a synth block): sixteen phase oscillators arranged in a circle, each pulled toward the phase of its two NEIGHBOURS only (not the whole crowd as in the classic Kuramoto model). Local coupling lets waves of synchrony travel around the ring - bands of in-step oscillators sweeping past out-of-step ones, the metachronal waves seen in beating cilia and centipede legs. The output is one oscillator's value, beating as the synchrony waves roll by. Couple sets the pull toward neighbours. A spatially-coupled sync source distinct from the all-to-all chaos. Needs no input.

Param	Range	Default	Unit
Rate	0.05 – 8	2	—
Couple	0 – 2	0.5	—

CoupledPendula inputs

Coupled-pendulum-chain source (a first as a synth block): a ring of sixteen pendulums, each linked by a spring to its neighbours - the classic lecture-hall wave machine. Give one a push and the swing travels along the chain as a wave, reflecting and interfering with itself (this is the sine-Gordon system, famous for its solitons). With almost no friction the energy sloshes around the ring forever. The output is one pendulum's swing, rising and falling as the waves pass through it. A mechanical wave source distinct from the electronic and chemical oscillators. Needs no input.

Param	Range	Default	Unit
Rate	0.05 – 4	1	—
Coupling	0 – 1.5	0.5	—

LambdaOmega 0 inputs

Lambda-omega medium source (a first as a synth block): a ring of sixteen cells each running a little rotational oscillator (the lambda-omega reaction-diffusion system, the normal form for any chemical or biological medium near the onset of oscillation), coupled by diffusion. The cells phase-lock into travelling and spiral waves - the spinning patterns of the Belousov-Zhabotinsky reaction and slime-mould aggregation. The output is one cell's concentration, oscillating as the spiral arms sweep past. A pattern-forming oscillatory medium distinct from the excitable Excitable and the single-cell oscillators. Needs no input.

Param	Range	Default	Unit
Rate	0.05 – 4	1	—
Diffuse	0 – 1	0.4	—

Toda 0 inputs

Toda-lattice source (a first as a synth block): a ring of sixteen masses joined by springs whose force grows EXPONENTIALLY with compression - the Toda lattice, one of the rare nonlinear systems that is exactly solvable and supports SOLITONS, lumps of energy that travel through the chain and pass through each other unchanged. Kick it and solitons circulate the ring forever, colliding and re-emerging. The output is one mass's momentum, kicking each time a soliton passes through. An integrable-soliton source distinct from the chaotic lattices. Needs no input.

Param	Range	Default	Unit
Rate	0.05 – 4	1	—

Henon 2 inputs

Henon-map chaos sequencer (a first as a synth block): iterates the classic 2-D strange-attractor map $x' = 1 - A*x^2 + y$, $y' = B*x$ on every rising edge of the in0 clock, outputting x as a stepped CV that hops chaotically across the attractor. At the textbook $A=1.4$, $B=0.3$ it traces the famous folded Henon curve - deterministic yet never repeating, a sharper, more angular chaos than the smooth DoublePendulum or the logistic Chaos. A rising edge on in1 reseeds to the origin.

Param	Range	Default	Unit
A	1 – 1.4	1.4	—
B	0 – 0.3	0.3	—

Telegraph 0 inputs

Random-telegraph-signal source (a first as a synth block): the dichotomous noise of physics - a two-level signal that flips between 0 and 1 at random (memoryless) moments, the model for a trapped electron blinking or a bistable circuit jittering. Rate sets the mean number of flips per second; because the wait between flips is exponentially distributed the result is a random square wave with no fixed period, holding each level for an unpredictable time. A genuine stochastic gate, distinct from the random-but-clocked Sample-and-Hold sources. Needs no input.

Param	Range	Default	Unit
Rate	0.1 – 50	4	Hz

Poisson 0 inputs

Poisson-process trigger (a first as a synth block): fires a one-sample impulse at random, memoryless moments at an average Rate of events per second - the canonical model for radioactive decay clicks, photon arrivals or raindrops. The gaps between triggers follow an exponential distribution, so the pulses are genuinely random in time yet hold a steady long-run density. Patch it to gate or trigger anything that wants un-quantised, never-repeating randomness. Distinct from a regular Clock and from the Bernoulli gate (which only thins an existing clock). Needs no input.

Param	Range	Default	Unit
Rate	0.1 – 100	8	Hz

Hawkes ▯ inputs

Self-exciting (Hawkes) trigger (a first as a synth block): a point process where every event briefly RAISES the chance of another - so triggers arrive in bursts and clusters rather than evenly, the model for earthquake aftershocks, neuron avalanches and viral cascades. Rate sets the background event rate, Excite how strongly each firing kicks the intensity up, and Decay how fast that excitement fades. With low Excite it behaves like a plain Poisson source; raise it and the output clumps into flurries of activity separated by lulls. The branching is kept sub-critical so it can never run away. Needs no input.

Param	Range	Default	Unit
Rate	0.5 – 30	6	Hz
Excite	0 – 40	20	—
Decay	5 – 500	21	ms

Renewal ▯ inputs

Regularised (Erlang renewal) trigger (a first as a synth block): a point process that is MORE evenly spaced than random - it fires only every Regularity-th pulse of a faster internal Poisson source, so the wait between outputs is an Erlang sum of exponentials. Rate sets the output event rate. At Regularity 1 it is plain Poisson (fully random); raise it and the triggers tighten toward a steady metronome while keeping a little jitter - the sub-Poisson, anti-clustering opposite of Hawkes. The natural model for a neuron's refractory firing or a heartbeat's near-regular rhythm. Needs no input.

Param	Range	Default	Unit
Rate	0.1 – 100	8	Hz
Regularity	1 – 8	4	—

Ornstein ▯ inputs

Ornstein-Uhlenbeck source (a first as a synth block): the mean-reverting random process - Brownian motion on a spring, which wanders randomly but is always pulled back toward its centre, so it drifts and meanders without ever running off the way a free random walk does. It is the standard model for a noisy voltage settling, a particle in a fluid, or an interest rate. Pull sets how strongly it springs back to centre (higher = tighter, faster wobble) and Depth the noise amount. A smooth, naturally-bounded random CV distinct from the target-hopping Drift and the free Levy flight. Needs no input.

Param	Range	Default	Unit
Pull	0 – 0.02	0.002	—
Depth	0 – 0.1	0.02	—

VanDerPol ▯ inputs

Van-der-Pol relaxation oscillator (a first as a synth block): integrates $x'' - \mu(1 - x^2)x' + x = 0$, the classic self-sustaining oscillator from early radio valves and the heartbeat. Unlike a sine it has nonlinear damping that pumps energy in when the swing is small and bleeds it off when large, so it settles into a fixed-amplitude limit cycle. At low μ it is nearly sinusoidal; raise μ and it snaps into a sharp relaxation wave - slow charge, sudden flip - the sawtooth-like pulse of a blinking neon or a firing neuron. Rate sets the speed. A self-stabilising oscillator distinct from the chaotic attractors.

Param	Range	Default	Unit
Rate	0.05 – 4	1	—
μ	0.5 – 5	2	—

FitzHughNagumo 0 inputs

FitzHugh-Nagumo neuron (a first as a synth block): integrates the two-variable simplification of the Hodgkin-Huxley nerve-impulse equations, $v' = v - v^3/3 - w + \text{Drive}$, $w' = 0.08(v + 0.7 - 0.8w)$. It is the textbook EXCITABLE system: a fast voltage v that fires and a slow recovery w that resets it. With enough Drive it spikes over and over - a sharp upstroke, a refractory dip, a slow recharge - the canonical shape of a firing neuron. Rate sets the speed, Drive the injected current (turn it up to cross from silent into repetitive spiking). A spiking oscillator distinct from the smooth chaos sources.

Param	Range	Default	Unit
Rate	0.05 – 4	1	—
Drive	0 – 1	0.5	—

HindmarshRose 0 inputs

Hindmarsh-Rose bursting neuron (a first as a synth block): integrates a three-variable neuron model with a slow adaptation current that makes it BURST - it fires a rapid cluster of spikes, then falls quiet while the slow variable recovers, then bursts again, often chaotically so no two bursts match. This spike-then-rest-then-spike rhythm is exactly how many real neurons fire. The output is the membrane voltage: fast spiking riding on a slow up-and-down envelope. Rate sets the speed. A bursting chaos distinct from the steady spiking of FitzHugh-Nagumo and the smooth attractors.

Param	Range	Default	Unit
Rate	0.05 – 4	1	—

WilsonCowan 0 inputs

Wilson-Cowan neural-mass oscillator (a first as a synth block): models a whole population of brain cells, not one neuron - a pool of excitatory cells E that drives itself and an inhibitory pool I that reins it back, each passed through a sigmoid firing curve. The push-pull between excitation and delayed inhibition makes the population activity oscillate, the mechanism behind brain rhythms like the alpha and gamma waves on an EEG. The output is the excitatory activity, a smooth 0..1 rhythmic swell. Rate sets the speed and Drive the background input. A network-level oscillation distinct from the single-cell neuron models.

Param	Range	Default	Unit
Rate	0.05 – 4	1	—
Drive	0 – 1.5	0.5	—

Rulkov 0 inputs

Rulkov-map neuron (a first as a synth block): a two-variable MAP (not a differential equation) that reproduces neuron spiking and bursting with just $x' = \text{Alpha}/(1 + x^2) + y$ and a slow y - prized because it captures the rich firing of a real cell at a fraction of the cost. A fast x that spikes sits on a slow y that drifts it in and out of bursting, so raising Alpha moves it from silence through tonic spikes into chaotic bursts. It free-runs at Rate, stepping the map and outputting x as a bipolar CV. A discrete-map neuron distinct from the integrated Hindmarsh-Rose and FitzHugh-Nagumo models.

Param	Range	Default	Unit
Rate	0.1 – 50	8	Hz
Alpha	4 – 4.6	4.3	—

LotkaVolterra 0 inputs

Lotka-Volterra predator-prey oscillator (a first as a synth block): the founding equations of mathematical ecology - prey x breed freely but are eaten by predators y , whose numbers in turn rise and fall on the food supply. The two populations chase each other in a perpetual cycle: prey boom, predators follow and crash them, then starve and let the prey boom again. Rate sets the speed and Growth the prey's breeding rate. The output is the prey population as a smooth 0..1 swell, integrated symplectically so the cycle never winds down. An ecological oscillation distinct from the neuron and chaos sources.

Param	Range	Default	Unit
Rate	0.05 – 4	1	—
Growth	0.5 – 2	1.1	—

Brusselator 0 inputs

Brusselator chemical oscillator (a first as a synth block): the textbook model of an autocatalytic chemical clock, where a reactant feeds a loop that periodically builds up and consumes an intermediate, so the concentration pulses up and down on its own - the kind of self-organising rhythm seen in real oscillating reactions. Above the critical feed it locks into a stable limit cycle. Rate sets the speed and B the feed parameter (raise it past 2 to start the oscillation). The output is the concentration x. A chemistry-born oscillation distinct from the biological and chaotic sources.

Param	Range	Default	Unit
Rate	0.05 – 4	1	—
B	2 – 4	3	—

Sel'kov 0 inputs

Sel'kov glycolytic oscillator (a first as a synth block): the model of glycolysis - the pathway cells use to burn sugar - which does not proceed steadily but PULSES, the concentration of an intermediate rising and falling in a self-sustained rhythm (the glycolytic oscillations seen in yeast extract). Rate sets the speed and Feed the substrate supply. The output is the oscillating concentration. A metabolic oscillation distinct from the Brusselator chemistry and the neuron models.

Param	Range	Default	Unit
Rate	0.05 – 4	1	—
Feed	0.4 – 0.8	0.6	—

Repressilator 0 inputs

Repressilator gene oscillator (a first as a synth block): the first synthetic gene circuit built to oscillate - three genes wired in a ring where each represses the next (A silences B, B silences C, C silences A), so no state is stable and the three protein levels cycle round in turn, a genetic clock engineered into living bacteria. Rate sets the speed and Strength the repression (raise it to deepen the swing). The output is the first protein's level. A synthetic-biology oscillation distinct from the metabolic and neural models.

Param	Range	Default	Unit
Rate	0.05 – 4	1	—
Strength	4 – 16	10	—

Lengyel 0 inputs

Lengyel-Epstein chemical oscillator (a first as a synth block): the model of the CIMA reaction (chlorine-dioxide / iodine / malonic-acid), the experiment that first produced stationary Turing patterns in a real chemical and which, well stirred, oscillates in time instead. Two coupled concentrations chase each other into a stable limit cycle. Rate sets the speed and Feed the reactant supply. The output is the activator concentration. A pattern-forming chemistry distinct from the Brusselator and the biological oscillators.

Param	Range	Default	Unit
Rate	0.05 – 4	1	—
Feed	8 – 14	10	—

Perlin ▯ inputs

Perlin-noise source (a first as a synth block): smooth coherent gradient noise - the organic, flowing randomness used to generate natural terrain, clouds and fire in computer graphics. Unlike white noise or a sample-and-hold (which jump instantly), Perlin noise glides: nearby moments are always close in value, so the output meanders gently and naturally with no abrupt steps, set by a quintic-smoothed lattice of random gradients. Rate sets how fast it drifts through the noise field. The natural-movement modulator, distinct from the white/pink NoiseLFO and the stepped Random.

Param	Range	Default	Unit
Rate	0.05 – 20	1	Hz

fBm ▯ inputs

Fractal-Brownian-motion source (a first as a synth block): stacks several octaves of Perlin noise - each one twice as fast and half as loud as the last - into the rich, self-similar randomness of natural terrain, coastlines and mountain skylines. The low octaves give the slow overall drift while the high ones add fine detail, so the motion is organic at every timescale at once. Rate sets the base speed and Octaves how many layers (more = finer, rougher detail). A multi-scale coherent-noise modulator distinct from the single-octave Perlin. Needs no input.

Param	Range	Default	Unit
Rate	0.05 – 20	1	Hz
Octaves	1 – 6	4	—

Ridged ▯ inputs

Ridged-multifractal source (a first as a synth block): a fractal noise that takes each octave's absolute value and inverts it ($1 - |\text{noise}|$), so the smooth valleys of fBm become sharp upward RIDGES - the technique that carves mountain crests, canyon walls and lightning veins in procedural terrain. The output rides high and crinkles into sudden creases instead of rolling smoothly. Rate sets the base speed and Octaves the detail. A ridged, peaky coherent-noise modulator distinct from the rounded fBm and Billow. Needs no input.

Param	Range	Default	Unit
Rate	0.05 – 20	1	Hz
Octaves	1 – 6	4	—

Billow ▯ inputs

Billow-noise source (a first as a synth block): a fractal noise that takes the absolute value of each octave ($2|\text{noise}| - 1$), turning the gentle zero-crossings of fBm into rounded, puffy lobes - the look of cumulus clouds, rolling smoke and bubbling foam in procedural textures. The output swells up in soft billows with creased troughs where it folds back on itself, the rounded opposite of the sharp Ridged. Rate sets the base speed and Octaves the detail. A billowy coherent-noise modulator distinct from fBm and Ridged. Needs no input.

Param	Range	Default	Unit
Rate	0.05 – 20	1	Hz
Octaves	1 – 6	4	—

Worley ▯ inputs

Worley (cellular) noise source (a first as a synth block): scatters random feature points along the timeline and outputs the distance from the moving read-head to the NEAREST one - the cellular/Voronoi noise behind stone, scales, cracked mud and water-caustic textures. The value falls to zero each time the read-head passes a feature point and swells up in the gaps between them, giving a bumpy, organic ramp-and-dip motion quite unlike the smooth hills of Perlin. Rate sets how fast it travels between cells. A cellular coherent-noise modulator distinct from the gradient-noise family. Needs no input.

Param	Range	Default	Unit
Rate	0.05 – 20	1	Hz

Ehrenfest ▯ inputs

Ehrenfest-urn source (a first as a synth block): the textbook model of diffusion and the arrow of time - Balls are split between two boxes and at each step ONE random ball is moved to the other box. Whichever box has more balls is more likely to lose one, so the split is gently pulled back toward 50/50 and then jiggles around it forever. The output is the fraction in one box: a mean-reverting random wander that, unlike a free random walk, never strays far - a statistical-mechanics restoring force rather than the spring of Ornstein-Uhlenbeck. Rate sets the step speed. Needs no input.

Param	Range	Default	Unit
Rate	0.1 – 50	8	Hz
Balls	8 – 128	32	—

PolyaUrn ▯ inputs

Polya-urn source (a first as a synth block): the canonical 'rich get richer' process - an urn holds red and black balls; draw one at random, then put it back ALONG WITH another of the same colour, so every draw makes that colour likelier still. The proportion drifts and then locks onto a random value that is different every run (the path-dependent limit). Here the urn is periodically emptied and reseeded, so the output settles to one plateau, holds, then jumps and settles on a fresh one - a self-reinforcing sample-and-hold quite unlike uniform randomness. Rate sets the draw speed. Needs no input.

Param	Range	Default	Unit
Rate	0.1 – 50	8	Hz

Moran ▯ inputs

Moran-process source (a first as a synth block): the model of random genetic drift in a fixed population - Balls individuals are each one of two types; at every step one is chosen to reproduce and one (at random) dies, so the share of each type wanders purely by chance. A trickle of Mutation flips the occasional offspring, which keeps the share from ever getting permanently stuck at all-one-type. The output is that share: an unbiased random walk on a population, the discrete cousin of neutral evolution. Rate sets the generation speed. Needs no input.

Param	Range	Default	Unit
Rate	0.1 – 50	8	Hz
Balls	8 – 128	32	—
Mutation	0 – 0.2	0.05	—

GamblersRuin ▯ inputs

Gambler's-ruin source (a first as a synth block): a fortune that takes a coin-flip step up or down each turn between two walls - bankruptcy at the bottom, the jackpot at the top. The path is a pure random walk, but the moment it touches either wall the game restarts from the middle, so the output is a stream of random excursions: long meandering climbs and slides that always end in a sudden reset to centre. Rate sets the bet speed and Wall the playing-field size (wider walls = longer, rarer excursions). An absorbing-barrier walk distinct from the mean-reverting urns. Needs no input.

Param	Range	Default	Unit
Rate	0.1 – 50	8	Hz
Wall	8 – 128	32	—

BirthDeath inputs

Birth-death (queue) source (a first as a synth block): a population (or a queue of waiting jobs) where new arrivals appear at a steady Birth rate while each member already present has a Death chance of leaving, so the more there are the faster they go - a self-limiting balance that settles around Birth/Death. The output is the population, an ASYMMETRIC mean-reverting wander: it drifts up on a run of arrivals then thins out faster the higher it climbs, the Poisson-occupancy shape of a busy queue. Distinct from the symmetric Ehrenfest diffusion. Rate sets the event speed. Needs no input.

Param	Range	Default	Unit
Rate	0.1 – 50	8	Hz
Birth	1 – 10	4	—
Death	0.1 – 2	0.5	—

Rose inputs

Rose-curve (rhodonea) LFO (a first as a synth block): traces the petal curve $r = \cos(\text{Petals} * \text{angle})$ and outputs its horizontal sweep, a smooth modulation whose single cycle folds into a rosette of Petals lobes. The result is a repeating shape far richer than a sine - a flower of evenly-spaced swells and dips - whose symmetry is set directly by the petal count. Rate sets the speed. A geometric multi-lobe LFO distinct from the sine, ramp and random sources.

Param	Range	Default	Unit
Rate	0.01 – 20	1	Hz
Petals	2 – 12	5	—

Spirograph inputs

Spirograph (hypotrochoid) LFO (a first as a synth block): traces the looping curve a small circle draws as it rolls inside a larger one - the toy-spirograph pattern - and outputs its sweep, the sum of two circular motions whose speeds are set by Ratio. The result is an epicyclic LFO with a fast inner wobble riding on a slow outer swing, Depth balancing the two. A geometric two-rate modulation distinct from a plain sine or a synced pair. Rate sets the speed.

Param	Range	Default	Unit
Rate	0.01 – 20	1	Hz
Ratio	2 – 9	3	—
Depth	0 – 1	0.6	—

Epicycloid inputs

Epicycloid LFO (a first as a synth block): traces the curve a point on a small circle draws as it rolls around the OUTSIDE of a larger one - the path of a planet in a geocentric sky - and outputs its sweep. With Cusps points it forms a star-like loop of sharp, evenly-spaced cusps, so the LFO sweeps smoothly then snaps at each cusp, a spiky repeating contour. Rate sets the speed. A rolling-circle modulation distinct from the inner-rolling Spirograph.

Param	Range	Default	Unit
Rate	0.01 – 20	1	Hz
Cusps	2 – 10	4	—

Harmonograph inputs

Harmonograph LFO (a first as a synth block): models the Victorian drawing machine where two pendulums swinging at slightly-detuned frequencies trace a slowly-evolving figure. It sums two sinusoids at the FreqA and FreqB harmonics with a tiny detuning, so the shape drifts and precesses cycle after cycle, never quite repeating - an organic, breathing modulation rather than a fixed loop. Rate sets the base speed, FreqA and FreqB the two pendulum rates. A quasi-periodic curve distinct from the exactly-repeating Rose and Spirograph.

Param	Range	Default	Unit
Rate	0.01 – 20	1	Hz
FreqA	1 – 6	2	—
FreqB	1 – 6	3	—

Lissajous 0 inputs

Lissajous LFO (a first as a synth block): multiplies two sinusoids whose frequencies are in the ratio RatioX:RatioY - the pair that, plotted against each other, trace the classic Lissajous figures on an oscilloscope. The product is a complex, symmetric modulation that beats and nulls as the two rates cross, its character set entirely by the integer ratio. Simple ratios give clean repeating shapes, larger ones dense interference. Rate sets the speed. A two-tone interference LFO distinct from the rolling-circle curves.

Param	Range	Default	Unit
Rate	0.01 – 20	1	Hz
RatioX	1 – 8	3	—
RatioY	1 – 8	2	—

Cardioid 0 inputs

Cardioid LFO (a first as a synth block): traces the heart-shaped curve $r = 1 - \cos(\text{angle})$ and outputs its horizontal sweep, a strongly ASYMMETRIC modulation - a slow, lingering dwell near one extreme and a quick sweep through the other, the single-cusped 'heart' contour. Its lopsided shape makes it good for envelope-like swells that rise gently and fall fast (or vice versa). Rate sets the speed. A heart-curve LFO distinct from the symmetric sine and the multi-lobe Rose.

Param	Range	Default	Unit
Rate	0.01 – 20	1	Hz

Lemniscate 0 inputs

Lemniscate LFO (a first as a synth block): traces the figure-eight curve of Bernoulli ($r\text{-squared} = \cos(2*\text{angle})$) and outputs its sweep, which produces TWO smooth pulses per cycle separated by flat dead zones where the curve pinches to nothing at the crossing point. The result is a paired-pulse modulation - two lobes of motion, then rest, then two more - quite unlike a single sine swing. Rate sets the speed. A figure-eight LFO distinct from the single-loop curves.

Param	Range	Default	Unit
Rate	0.01 – 20	1	Hz

Astroid 0 inputs

Astroid LFO (a first as a synth block): traces the four-cusped star curve $x = \cos\text{-cubed}(\text{angle})$ and outputs it directly - a cosine with FLATTENED peaks and steepened zero-crossings, so it lingers at the extremes and snaps quickly through the middle, the inverse of a sine's rounded shape. A gentle, plateau-topped modulation good for slow holds with fast transitions. Rate sets the speed. A cubed-cosine star curve distinct from the sine and triangle shapes.

Param	Range	Default	Unit
Rate	0.01 – 20	1	Hz

Deltoid 0 inputs

Deltoid LFO (a first as a synth block): traces the three-cusped hypocycloid (a circle rolling inside one three times its size), outputting $(2*\cos(\text{angle}) + \cos(2*\text{angle})) / 3$ - a smooth modulation with a strong main swell and a smaller secondary bump, the triangular

cousin of the astroid. Its three-fold asymmetry gives a lopsided, gently galloping motion. Rate sets the speed. A three-cusp curve distinct from the four-cusp astroid and the rolling-circle epicycloid.

Param	Range	Default	Unit
Rate	0.01 – 20	1	Hz

Superformula 0 inputs

Superformula LFO (a first as a synth block): Johan Gielis's single equation that draws a vast family of natural shapes - circles, polygons, stars, flowers, leaves - from just two knobs, by setting the radius from $|\cos(\text{Sym} \cdot t/4)|$ and $|\sin(\text{Sym} \cdot t/4)|$ raised to a power. Symmetry sets how many lobes (the polygon/star order) and Shape how sharp or rounded they are, so sweeping them MORPHS the LFO contour continuously from a smooth blob to a spiky star. The most shape-versatile single LFO, distinct from every fixed-curve source. Rate sets the speed.

Param	Range	Default	Unit
Rate	0.01 – 20	1	Hz
Symmetry	2 – 12	6	—
Shape	0.3 – 3	1	—

Rossler 0 inputs

Rossler-attractor chaos source (a first as a synth block): integrates the three Rossler equations $dx = -y - z$, $dy = x + a \cdot y$, $dz = b + z \cdot (x - c)$ - a smooth deterministic-chaos system that traces a spiralling band with an occasional fold. The output is x as a slowly wandering CV: gentler and more periodic-looking than Lorenz, with long quasi-cycles broken by sudden excursions. Rate sets the integration speed. A continuous chaos with a different character from the angular Henon or the tumbling DoublePendulum; needs no input.

Param	Range	Default	Unit
Rate	0.05 – 4	1	—

Kuramoto 0 inputs

Kuramoto synchronisation source (a first as a synth block): eight phase oscillators, each at its own natural frequency, are nudged toward each other by Coupling - the model of how fireflies, neurons and metronomes spontaneously fall into step. The output is their combined mean field: with low Coupling the eight detuned waves smear into a restless wobble, and as Coupling rises they lock into one coherent oscillation that swells in amplitude. Rate sets the centre frequency and Spread the detuning. A self-organising motion no LFO or noise source produces. Needs no input.

Param	Range	Default	Unit
Rate	0.05 – 8	1	Hz
Spread	0 – 1	0.3	—
Coupling	0 – 4	1	—

Orbit 0 inputs

Kepler-orbit LFO (a first as a synth block): a body sweeps an elliptical orbit under gravity, so unlike a sine it lingers slowly at the far point (aphelion) then whips quickly through the near point (perihelion) - the asymmetric speed of a real planet, set by Eccentricity. It solves Kepler's equation each sample for the eccentric anomaly and outputs its cosine. At zero eccentricity it is a pure cosine; cranked up it becomes a lopsided swing that hangs and then lunges. Rate sets the orbital period. A gravitationally-shaped modulation curve, needs no input.

Param	Range	Default	Unit
Rate	0.01 – 20	1	Hz

Param	Range	Default	Unit
Eccentricity	0 – 0.9	0.5	—

Ising inputs

Ising-model magnetisation source (a first as a synth block): a 4x4 lattice of up/down spins evolves by the Metropolis rule - a spin flips if it lowers the energy, or randomly with a chance set by Temperature - the textbook model of a magnet. The output is the net magnetisation: cold, the spins align into one domain and it drifts toward +/-1; hot, they tumble and it jitters around zero; near the critical temperature it swings in large, slow, self-similar fluctuations. A statistical-physics modulation source. Needs no input.

Param	Range	Default	Unit
Temperature	0.1 – 5	2.27	—

ForestFire inputs

Forest-fire CA (a first as a synth block): a 4x4 grid of cells that are empty, growing a tree, or burning. Each sample a tree sprouts on an empty cell with chance Growth, a stray tree catches fire by Lightning, and fire spreads to neighbouring trees then burns out - the Drossel-Schwabl model of self-organised criticality. The output is the burning density: quiet while the forest regrows, then sudden sweeping fire-fronts, with the avalanche statistics of real wildfires. A different criticality than the Sandpile. Needs no input.

Param	Range	Default	Unit
Growth	0 – 0.05	0.01	—
Lightning	0 – 0.01	0.001	—

Levy inputs

Levy-flight modulation source (a first as a synth block): a random walk whose step sizes are drawn from a heavy-tailed (Cauchy) distribution, so it mostly creeps in tiny steps but every so often makes a sudden enormous leap across the range - the foraging pattern of albatrosses and the maths of anomalous diffusion. Step sets the typical jump scale and Rate how often it moves; the walk reflects off +/-1 to stay in range. Where a smoothed-noise LFO wanders gently, Levy lurks then pounces - clustered stillness punctuated by leaps. Needs no input.

Param	Range	Default	Unit
Step	0.001 – 0.2	0.02	—
Rate	0 – 1	0.5	—

NBody inputs

Gravitational N-body source (a first as a synth block): three point masses pull on one another by Newton's inverse-square law inside a box with reflecting walls, tracing the famously unsolvable three-body motion - bodies swing past, sling-shot and tumble in an endless gravitational dance. Gravity sets the attraction strength and Rate the simulation speed; the output follows the first body's horizontal position. A softened-gravity, wall-bounded chaos with its own swooping character, distinct from the pendulum or the strange-attractor maps. Needs no input.

Param	Range	Default	Unit
Gravity	0.05 – 2	0.5	—
Rate	0.05 – 4	1	—

Voter inputs

Voter-model consensus source (a first as a synth block): a 4x4 grid of cells each holding one of two opinions; each step a random cell simply copies a random neighbour, the simplest model of imitation, gossip and herd behaviour. Domains coarsen and fight along their borders until - on a finite grid - one opinion wins and the whole field falls into consensus, at which point it freezes. The output is the net opinion, a CV that wanders then locks. Unlike the Ising model there is no temperature, only blind copying. Needs no input.

Param	Range	Default	Unit
Rate	1 – 8	4	—

RandomWalk inputs

Random-walk source (a first as a synth block): the classic drunkard's walk - each step nudges the value by a random amount, so it wanders with no memory of where it has been, the very model of Brownian diffusion. It free-runs at Rate; reflecting walls at the rails keep it bouncing inside the range forever. Step sets how far each move can carry it. A memoryless diffusion distinct from the mean-pulled Ornstein-Uhlenbeck and the heavy-tailed Levy flight. Needs no input.

Param	Range	Default	Unit
Rate	0.1 – 50	8	Hz
Step	0 – 1	0.1	—

OrnsteinUhlenbeck inputs

Ornstein-Uhlenbeck source (a first as a synth block): a random walk on an elastic leash - noise kicks it around but a restoring pull always drags it back toward the centre, the textbook mean-reverting process used to model everything from particle velocities to interest rates. The result is smooth, organic, band-limited drift (Brownian motion's well-behaved cousin) with no sharp jumps. Revert sets the strength of the pull, Sigma the noise. A mean-reverting drift distinct from the free random walk. Needs no input.

Param	Range	Default	Unit
Rate	0.1 – 200	30	Hz
Revert	0 – 1	0.05	—
Sigma	0 – 1	0.12	—

LevyFlight inputs

Levy-flight source (a first as a synth block): a random walk whose step sizes are drawn from a heavy-tailed Cauchy distribution, so most moves are tiny but every so often it takes a wild, far-reaching leap - the search pattern of foraging animals and the signature of anomalous, super-diffusive transport. The value mostly hovers, then teleports across the range and resumes. Jump scales the leaps; the rails wrap so a big jump reappears on the far side. A heavy-tailed flight distinct from the Gaussian random walk. Needs no input.

Param	Range	Default	Unit
Rate	0.1 – 50	6	Hz
Jump	0 – 0.2	0.02	—

TelegraphProcess inputs

Random-telegraph source (a first as a synth block): a two-state signal that sits at one rail and then randomly flips to the other, the dichotomous noise that models a flickering switch, burst noise in electronics, and ion channels snapping open and shut. The hold times are exponentially distributed, so it is a stepped square wave with utterly unpredictable timing. Switch sets the flip probability per step (the average rate of jumps). A two-level jump process distinct from the continuous random walks. Needs no input.

Param	Range	Default	Unit
Rate	0.1 – 200	40	Hz
Switch	0 – 1	0.05	—

BrownianBridge 0 inputs

Brownian-bridge source (a first as a synth block): a random walk that is tied down at both ends - it wanders freely in the middle but is mathematically guaranteed to arrive back at zero exactly at the end of each cycle, then starts a fresh bridge. This is the conditioned Brownian motion used to interpolate between fixed points and to model paths with a known destination. Length sets the cycle in steps. A pinned, self-resetting drift distinct from the open random walk. Needs no input.

Param	Range	Default	Unit
Rate	0.1 – 200	40	Hz
Length	4 – 256	64	—

CML 0 inputs

Coupled-map-lattice source (a first as a synth block): eight logistic maps sit in a ring, each iterating its own chaos while a Coupling term shares a fraction of each cell with its neighbours. The competition between local chaos and diffusive smoothing produces spatiotemporal chaos - travelling fronts, frozen-random patches and turbulent bursts that no single attractor shows. The output is the lattice average, a richly textured wandering CV. R sets each map's chaos and Coupling the diffusion. A whole field of coupled chaos rather than one oscillator. Needs no input.

Param	Range	Default	Unit
R	3.6 – 4	3.9	—
Coupling	0 – 0.5	0.1	—

Dimension 1 input

Dimension-D-style BBD chorus: a slow ~0.5 Hz LFO modulates a short delay around 8 ms, with the right channel running anti-phase, and the wet copy is blended with the dry at a fixed ratio. Depth scales how far the delay time is swept. Out picks L or R for the per-channel modulation phase, giving the lush stereo movement of the classic unit rather than a deep vibrato.

Param	Range	Default	Unit
Depth	0 – 1	0.5	—
Out	0 – 1	0	—

Gate 1 input

Step-sequenced tremolo gate that reads a per-step on/off pattern from a 16-bit mask, advancing through the steps on a sample-clock phase for tempo coherence. Rate sets the cycle frequency, Pattern is the bitmask of active steps, Steps sets how many steps are used, and Smooth softens the transitions between open and closed. Mix sets how deeply the gate cuts the signal, from subtle pulsing to full rhythmic chopping.

Param	Range	Default	Unit
Rate	0.25 – 8	2	Hz
Pattern	0 – 65535	43690	—
Steps	1 – 16	8	—
Smooth	0 – 1	0.3	—
Mix	0 – 1	1	—

Undulator 1 input

Modulated short delay combined with an envelope swell and a tremolo running at twice the LFO rate for H3000-style undulation. Rate sets the LFO speed, Depth sets how far the delay time is swept for pitch-wobble and chorus, and Swell controls the tremolo depth driven from a slow input follower. Mix blends the moving, detuned voice against the dry signal for shimmering, breathing modulation.

Param	Range	Default	Unit
Rate	0.05 – 5	0.5	Hz
Depth	0 – 1	0.5	—
Swell	0 – 1	0.5	—
Mix	0 – 1	0.5	—

Delay

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FreqEcho 1 input

Frequency-shifted echo: a delay line whose feedback is single-sideband frequency-shifted via a polyphase Hilbert pair, so every repeat slides up or down in Hz rather than in pitch. Time sets the delay (1 to 1000 ms), Shift sets the linear frequency offset (-500 to +500 Hz), Feedback (0 to 0.95) sets repeat count, and Mix blends dry/wet. Because the shift is inharmonic, the tail detunes and smears with each pass for metallic, ever-evolving echoes.

Param	Range	Default	Unit
Time	1 – 1000	250	ms
Shift	-500 – 500	80	Hz
Feedback	0 – 0.95	0.6	—
Mix	0 – 1	0.4	—

Multitap 1 input

Multi-tap delay: the input is written into a circular buffer and read back at several time-spaced taps that are summed, averaged, and fed back into the buffer. Time sets the base delay in ms, Taps chooses 1 to 6 read points, Feedback recirculates the averaged taps for repeats, Spread pulls the taps toward the longer end for uneven echo timing, and Mix crossfades dry input against the wet sum. Low Spread gives evenly stacked echoes while high Spread spaces them out, and higher Feedback builds longer rhythmic delay trails.

Param	Range	Default	Unit
Time	1 – 1000	250	ms
Taps	1 – 6	3	—
Feedback	0 – 0.9	0.4	—
Spread	0 – 1	0.5	—
Mix	0 – 1	0.4	—

Bouncer 2 inputs

A feedback delay whose delay time continually shrinks, so the echoes speed up toward a stutter like a bouncing ball; a gate on in1 re-arms the delay back to its start time. Time sets the initial delay (20..600 ms), Accel sets how fast the time shrinks, Feedback sets the echo regeneration (up to 0.95), and Mix blends wet against dry. Higher Accel collapses the repeats into a rapid buzz, while a gate retrigger resets the bounce.

Param	Range	Default	Unit
Time	20 – 600	250	ms
Accel	0 – 1	0.5	—
Feedback	0 – 0.95	0.7	—
Mix	0 – 1	0.5	—

Beatmasher 2 inputs

Beat-repeat: while the gate on in1 is held it records a slice of audio on the first pass then loops that captured slice; releasing the gate passes the dry signal through. Length sets the size of the captured/looped slice (20..1000 ms) and Mix blends the looped slice against the dry input. Hold the gate to freeze and stutter a fragment, release to drop back to live audio.

Param	Range	Default	Unit
Length	20 – 1000	250	ms
Mix	0 – 1	1	—

BeatSlicer 2 inputs

Captures a buffer while the gate on in1 is held, divides it into 8 slices, and replays them in one of several reorder patterns; releasing the gate passes dry audio through. Length sets the buffer/loop size (100..2000 ms), Pattern selects the slice reordering (0..4, e.g. straight, swapped pairs, doubled, reversed, interleaved), and Mix blends the sliced output against dry. Hold the gate to scramble a phrase into a rhythmic rearrangement.

Param	Range	Default	Unit
Length	100 – 2000	500	ms
Pattern	0 – 4	1	—
Mix	0 – 1	1	—

SliceShuffler 1 input

Continuous buffer-shuffler (Max-for-Live Buffer-Shuffler style): records the incoming audio one bar at a time and, while recording the next bar, plays the last bar back as Steps slices re-ordered by a Seed-driven permutation, with a fraction of slices (Reverse) played backwards and a 16-step on/off Gate mask muting chosen slots. Length sets the bar in milliseconds, Steps the slice count (4/8/16), Seed the rearrangement (every value a different reproducible shuffle), and Mix blends against dry. Unlike BeatSlicer's five fixed patterns this is a fully addressable per-slice remap that runs without a trigger.

Param	Range	Default	Unit
Length	50 – 2000	500	ms
Steps	4 · 8 · 16		—
Seed	0 – 255	0	—
Reverse	0 – 1	0	—
Gate	0 – 65535	65535	—
Mix	0 – 1	1	—

ReverseGrain 2 inputs

Captures a buffer while the gate on in1 is held, then plays it backward at a pitch-shifted read rate for a reversed, stuttered texture; releasing the gate passes dry audio through. Length sets the captured grain size (50..1500 ms), Pitch transposes the reverse playback rate (+/-12 semitones), and Mix blends the reversed grain against dry. Hold the gate to suck a phrase into a backward swell.

Param	Range	Default	Unit
Length	50 – 1500	400	ms
Pitch	-12 – 12	0	st
Mix	0 – 1	1	—

PeakFlanger 1 input

A comb flanger whose delay time is swept by a peak envelope follower on the input, so transients drive the flange motion (delay ranges ~0.5..9.5 ms). Soften sets the envelope release time, Feedback sets the comb regeneration (+/-0.95, negative inverts the comb), Amount scales how far each peak sweeps the delay, and Mix blends wet against dry. The Traktor Flanger Pulse character: the jet sweep follows the music's own dynamics rather than a fixed LFO.

Param	Range	Default	Unit
Soften	0 – 1	0.5	—

Param	Range	Default	Unit
Feedback	-0.95 – 0.95	0.5	—
Amount	0 – 1	0.7	—
Mix	0 – 1	0.5	—

SpaceEcho 1 input

Roland Space Echo-style tape delay: wow+flutter-modulated tape transport with magnetic hysteresis (asymmetric saturation + remanence) and tone-shaped head gap loss in the feedback repeats.

Param	Range	Default	Unit
Time	50 – 700	300	ms
Feedback	0 – 0.95	0.4	—
Wow	0 – 1	0.2	—
Tone	1000 – 12000	5000	Hz
Mix	0 – 1	0.4	—

Echorec 1 input

Binson Echorec-style magnetic drum delay: several fixed playback heads give rhythmic multi-tap echoes with rotating-drum wobble and magnetic hysteresis in the feedback.

Param	Range	Default	Unit
Time	40 – 350	150	ms
Heads	1 · 1-2 · 1-2-3 · All	—	—
Feedback	0 – 0.9	0.4	—
Mix	0 – 1	0.4	—

MemoryMan 1 input

EHX Memory Man-style analog BBD delay: a companded bucket-brigade line whose clock-tracked reconstruction filter darkens each repeat with delay length, gently chorused, with line-level regeneration.

Param	Range	Default	Unit
Time	30 – 550	250	ms
Feedback	0 – 0.9	0.35	—
Chorus	0 – 1	0.3	—
Mix	0 – 1	0.4	—

SDD3000 1 input

Korg SDD-3000-style delay: a characterful input preamp into a modeled converter (ADC anti-alias + 13-bit quantization) - clean and bright with subtle digital grain.

Param	Range	Default	Unit
Time	20 – 900	350	ms
Feedback	0 – 0.9	0.35	—
Tone	2000 – 16000	9000	Hz
Mix	0 – 1	0.4	—

TimeCube 1 input

Cooper Time Cube-style delay: a short, dark acoustic-tube delay (sound piped through a garden hose between transducers) for tight doubling and slapback. Time sets the short delay, Feedback the repeats and Mix the blend. Naturally band-limited and woody - characterful on vocals and snare, not a clean digital echo.

Param	Range	Default	Unit
Time	5 – 80	30	ms
Feedback	0 – 0.6	0.2	—
Mix	0 – 1	0.4	—

Echoplex 1 input

Maestro Echoplex EP-34-style tape echo: warm, wow+flutter-modulated repeats with magnetic hysteresis (asymmetric saturation + remanence) and head gap-loss HF roll-off.

Param	Range	Default	Unit
Time	80 – 650	300	ms
Feedback	0 – 0.9	0.4	—
Wow	0 – 1	0.3	—
Mix	0 – 1	0.4	—

Delay 3 inputs

Feedback delay line (1-8000 ms) with a damping low-pass in the feedback path that progressively darkens each repeat. Time is modulated by in 3 (scaled by Mod) for chorus and flange-style movement, Feedback sets regeneration up to 0.99, and Mix sets dry/wet. Damp at 0 gives bright digital echoes; higher values emulate tape/BBD high-frequency loss.

Param	Range	Default	Unit
Time	1 – 8000	250	ms
Feedback	0 – 0.99	0.35	—
Mod	0 – 500	0	ms
Damp	0 – 1	0	—
Mix	0 – 1	1	—

AlignDelay 1 input

Alignment delay (Ableton Align Delay-style): a clean dry delay used to time-align tracks or push a signal later by a precise amount - no feedback, no filtering, no colour. Time sets the delay in milliseconds; leave Mix fully wet for pure alignment, or blend for a simple slap double.

Param	Range	Default	Unit
Time	0 – 100	0	ms
Mix	0 – 1	1	—

FilterDelay 1 input

Filtered multi-tap delay (Ableton Filter Delay-style): three delay taps at spreading times, each band-passed at Color before feedback, so the echoes are tuned and coloured rather than full-range. Time sets the first tap, Spread the spacing of the other two, Color the band-pass centre, Feedback the regeneration, and Mix the wet blend. (Mono sum of the three taps; the device's L/C/R panning needs a stereo rack.)

Param	Range	Default	Unit
Time	1 – 8000	250	ms
Spread	0 – 1	0.5	—
Color	100 – 6000	1200	Hz
Feedback	0 – 0.9	0.4	—
Mix	0 – 1	0.5	—

Rainmaker 1 input

Rhythmic tapped delay (Intellijel / Cylonix Rainmaker): spreads up to Taps echoes across the delay Time in a decaying rhythm, then runs the result through a tuned Comb resonator for ringing, pitched repeats - rhythmic delays, resonant reverbs and string-like tones in one. Time sets the span, Taps the number of echoes, Feedback the regeneration, Comb the resonator pitch/amount, and Mix the wet blend. A 16-tap rhythm engine, distinct from the fixed-tap Multitap.

Param	Range	Default	Unit
Time	20 – 8000	400	ms
Taps	1 – 16	8	—
Feedback	0 – 0.9	0.3	—
Comb	0 – 1	0.3	—
Mix	0 – 1	0.5	—

Morphagene 1 input

Tape-splice granular (Make Noise / Soundhack Morphagene): continuously records the in0 input onto a virtual tape, then plays back a Gene (grain) from it. Slide scans the playback position along the reel, Vari sets the varispeed (pitch + direction, reverse below centre), Gene the grain length, Morph the grain overlap/smeared, and Mix the blend. Scrub, stretch, reverse and granulate live audio - distinct from the fixed-loop Looper and the cloud-style granular blocks.

Param	Range	Default	Unit
Gene	0.02 – 0.5	0.15	s
Vari	-1 – 1	1	—
Slide	0 – 1	0	—
Morph	0 – 1	0.5	—
Mix	0 – 1	0.5	—

Comb 1 input

Tuned comb resonator: a short feedback delay set by Freq (20-2000 Hz) with a damping low-pass, which doubles as a Karplus-Strong plucked string when excited by a transient. Feedback sets resonance and sustain while Damp darkens the decay. High feedback rings at the tuned pitch; pair it with noise or an impulse for string and membrane tones.

Param	Range	Default	Unit
Freq	20 – 2000	220	Hz
Feedback	0 – 0.99	0.9	—
Damp	0 – 1	0.3	—
Mix	0 – 1	1	—

Chorus 1 input

Single-voice chorus: an LFO (Rate, Depth) modulates a short base Delay (1-30 ms) and the detuned copy is blended back via Mix to thicken and widen the source. Short delays with light depth give subtle doubling; longer and deeper settings approach vibrato and flange. See Ensemble for a three-voice string-machine variant.

Param	Range	Default	Unit
Rate	0.05 – 8	0.8	Hz
Depth	0 – 1	0.5	—
Delay	1 – 30	12	ms
Mix	0 – 1	0.5	—

Flanger 1 input

Swept-comb flanger: an LFO sweeps a very short delay (0.5-10 ms) mixed with the dry signal to create moving comb notches, with bipolar Feedback resonating the peaks (negative values invert for a hollow tone). Depth sets the sweep width and Mix the blend. Short delays with high feedback give the classic metallic jet-whoosh.

Param	Range	Default	Unit
Rate	0.05 – 8	0.3	Hz
Depth	0 – 1	0.7	—
Feedback	-0.95 – 0.95	0.3	—
Delay	0.5 – 10	2	ms
Mix	0 – 1	0.5	—

Vibrato 1 input

Pitch vibrato: an LFO modulates a short delay's read position to bend pitch, output fully wet (no dry blend) for true vibrato rather than chorus. Rate sets the wobble speed and Depth its pitch range. Gentle settings add expression; deep settings give a seasick warble.

Param	Range	Default	Unit
Rate	0.1 – 12	5	Hz
Depth	0 – 1	0.3	—
Delay	1 – 20	5	ms

Stutter 1 input

Auto-rhythmic buffer repeat: at each Rate cycle it snapshots a Grab-length window and loops it for the cycle, creating stutters and beat-repeat artefacts, with Mix blending against the live signal. Short grabs at high rates give granular buzz; longer grabs repeat recognizable slices. See Beatmasher and BeatSlicer for tempo-locked glitching.

Param	Range	Default	Unit
Rate	0.5 – 16	4	Hz
Grab	10 – 250	60	ms
Mix	0 – 1	1	—

TapeDelay 1 input

Vintage tape echo: a ~1.5 Hz wow modulates the delay time while a Tone low-pass and tanh tape Saturation in the feedback path darken and thicken each repeat. Feedback sets regeneration and Mix the blend. Unlike the clean Delay, repeats degrade and warble musically - raise Wow and Sat for worn-tape character.

Param	Range	Default	Unit
Time	1 – 8000	300	ms
Feedback	0 – 0.99	0.45	—
Wow	0 – 1	0.2	—
Tone	500 – 16000	5000	Hz

Param	Range	Default	Unit
Sat	0 – 1	0.3	—
Mix	0 – 1	0.4	—

Ensemble 1 input

Three-voice string-machine chorus: one LFO modulates three delay taps at evenly spread phases, summing into a lush, wide thickening richer than the single-voice Chorus. Rate and Depth set the swirl and Mix the blend. Classic for pads, strings and synth-brass.

Param	Range	Default	Unit
Rate	0.05 – 4	0.6	Hz
Depth	0 – 1	0.6	—
Mix	0 – 1	0.5	—

Rotary 1 input

Leslie rotary speaker: combines a doppler pitch wobble (modulated short delay) with a phase-offset amplitude tremolo so the tone swirls as the virtual horn spins. Speed sets the rotation rate and Depth its intensity. Iconic on organ, electric piano and guitar - automate Speed for the slow-to-fast chorale ramp.

Param	Range	Default	Unit
Speed	0.5 – 8	6	Hz
Depth	0 – 1	0.5	—
Mix	0 – 1	0.5	—

Reverse 1 input

Reverse delay: snapshots a Window-length slice of input and plays it backward on a loop, a glitchy pre-echo/reverse texture, blended via Mix. Short windows give fluttering grains; longer windows reverse recognizable phrases. See ReverseGrain and ReverseVerb for related effects.

Param	Range	Default	Unit
Window	20 – 500	120	ms
Mix	0 – 1	1	—

BBDDelay 1 input

Bucket-brigade style echo whose delayed tap is passed through a one-pole low-pass and a tanh saturation in the feedback path, so each repeat grows darker and grittier. Time sets the delay length (1-8000 ms), Feedback sets regeneration up to 0.9, Tone sets the high-frequency rolloff of the repeats, and Mix blends the filtered echo with the dry signal. Lower Tone yields the murky, decaying character of analog BBD delays.

Param	Range	Default	Unit
Time	1 – 8000	250	ms
Feedback	0 – 0.9	0.4	—
Tone	0 – 1	0.5	—
Mix	0 – 1	0.5	—

ReverseDelay 1 input

Captures the input into a delay buffer and reads it back newest-to-oldest within a repeating window, so each chunk plays in reverse. Time sets the window length (50-8000 ms) that defines how much audio is flipped per cycle, and Mix blends the reversed playback against the dry signal. Useful for backward-tape swells and disorienting rhythmic stutters.

Param	Range	Default	Unit
Time	50 – 8000	400	ms
Mix	0 – 1	0.5	—

DuckDelay 1 input

Feedback delay line whose wet output is attenuated by an envelope follower tracking the dry signal, so repeats drop out while you play and swell up in the gaps. Time sets the delay length (1-8000 ms), Feedback sets regeneration up to 0.9, Duck sets how hard the dry level suppresses the echoes, and Mix blends wet and dry. Keeps echoes from cluttering busy passages while letting them bloom in the silences.

Param	Range	Default	Unit
Time	1 – 8000	300	ms
Feedback	0 – 0.9	0.4	—
Duck	0 – 1	0.7	—
Mix	0 – 1	0.5	—

GrainDelay 1 input

Granular feedback delay that reads overlapping windowed grains from the delay line, pitch-shifting them and scattering their read positions. Time sets the base tap (10-8000 ms), Pitch transposes the grains +/-12 semitones, and Scatter randomizes each grain's offset for a smeared, diffuse cloud. Mix sets dry/wet; the shifted grains are fed back at reduced gain, so repeats drift in pitch and position.

Param	Range	Default	Unit
Time	10 – 8000	300	ms
Pitch	-12 – 12	0	st
Scatter	0 – 1	0.4	—
Mix	0 – 1	0.5	—

DiffuseDelay 1 input

Feedback delay whose tap is smeared through a 4-stage all-pass diffusion chain, pushing echoes toward reverb. Time sets the delay (1-1000 ms), Feedback sets regeneration up to 0.9, and Diffuse sets the all-pass coefficient that controls how much each repeat is blurred. Mix sets dry/wet; low Diffuse gives clean echoes, high Diffuse dissolves them into a diffuse wash.

Param	Range	Default	Unit
Time	1 – 1000	300	ms
Feedback	0 – 0.9	0.4	—
Diffuse	0 – 0.9	0.6	—
Mix	0 – 1	0.5	—

Looper 1 input

Circulating-buffer looper that re-reads a fixed-length loop and either overdubs new input onto it or recirculates it unchanged. Length sets the loop time (100-2000 ms) and Record switches between overdub (input mixed with the looped buffer at half gain) and pure recirculation when off. Mix sets dry/wet; leave Record on to layer takes, turn it off to freeze and play back the captured loop.

Param	Range	Default	Unit
Length	100 – 2000	1000	ms
Record	0 – 1	1	—
Mix	0 – 1	0.5	—

Shuffler 1 input

Beat-shuffler that records the input and, at each segment boundary, may jump its playback to one of a few earlier segment offsets. Segment sets the slice length (50-500 ms) and Chance sets the probability that a new boundary triggers a jump to a randomly chosen prior offset. Mix sets dry/wet; low Chance plays mostly straight, high Chance stutters and rearranges the beat.

Param	Range	Default	Unit
Segment	50 – 500	200	ms
Chance	0 – 1	0.5	—
Mix	0 – 1	0.5	—

Scratch 1 input

Turntable-scratch effect where a raised-cosine LFO sweeps the delay read head back and forth through recent input. Rate sets the sweep speed (0.1-8 Hz) and Depth sets how far back the head travels (1-200 ms). Mix sets dry/wet; faster Rate and deeper Depth give more pronounced vinyl-scratch wobble.

Param	Range	Default	Unit
Rate	0.1 – 8	2	Hz
Depth	1 – 200	80	ms
Mix	0 – 1	0.5	—

PingPong 1 input

Ping-pong delay for a mono source: a single line is read at time T or 2T so the two outputs hear staggered taps that appear to bounce between speakers, with regeneration fed back through the longer tap. Time sets the base delay, Feedback the number of repeats up to 0.9, and Mix dry/wet. Out picks L or R, where R hears T and L hears 2T to create the alternating bounce.

Param	Range	Default	Unit
Time	1 – 1000	300	ms
Feedback	0 – 0.9	0.45	—
Out	0 – 1	0	—
Mix	0 – 1	0.5	—

Glitch 1 input

Probabilistic buffer glitcher that records the input and, at each slice boundary, randomly jumps to a captured slice and may play it forward or reversed. Slice sets the segment length in milliseconds and Chance sets the probability that a given slice is repeated or reversed rather than passed through cleanly. Mix blends the glitched playback against the dry signal for stutter, repeat and reverse artifacts.

Param	Range	Default	Unit
Slice	50 – 500	150	ms
Chance	0 – 1	0.4	—
Mix	0 – 1	1	—

Reverb

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Convolution 1 input

Convolution reverb: input is buffered into blocks and convolved with an impulse response via partitioned FFT, then streamed back with block latency. The IR comes from the node config (an audio file path loads that real IR, otherwise a decay time in seconds generates a decaying-noise tail); PreDelay offsets the wet by 0 to 200 ms, Tone is a one-pole low-pass on the wet, and Mix blends dry/wet. Use it to place sounds in real spaces or to apply any captured IR.

Param	Range	Default	Unit
Mix	0 – 1	0.3	—
PreDelay	0 – 200	0	ms
Tone	0 – 1	1	—

Plate 1 input

Plate reverb: the input runs through four series allpass diffusers and then into four parallel damped feedback combs that are summed for the wet tail. Decay sets the comb feedback (0 to 0.95), Damp is a one-pole low-pass inside the feedback that darkens later reflections, Size scales the delay-line lengths, and Mix blends dry/wet. Dense, smooth metallic decay suited to plates and tight ambiences.

Param	Range	Default	Unit
Size	0 – 1	0.6	—
Decay	0 – 0.95	0.7	—
Damp	0 – 1	0.4	—
Mix	0 – 1	0.3	—

Shimmer 1 input

Shimmer reverb: a delay line whose feedback path is pitch-shifted by a two-window granular shifter so repeats climb in pitch each pass. Shift sets the transposition in semitones (-12 to +24), Feedback (0 to 0.95) controls how long the tail sustains, Size sets the delay time in ms, Damp low-passes the feedback, and Mix blends dry/wet. With Shift at +12 it builds the classic ethereal rising-octave wash.

Param	Range	Default	Unit
Size	50 – 800	300	ms
Shift	-12 – 24	12	st
Feedback	0 – 0.95	0.7	—
Damp	0 – 1	0.4	—
Mix	0 – 1	0.4	—

Iceverb 1 input

Four lowpass-damped comb filters form the reverb body, then a self-oscillating resonant state-variable filter is applied to the tail for a cold, ringing sheen. Size sets the comb feedback/decay (up to 0.95), Color sets the resonant filter frequency (~100 Hz..14 kHz), Icing sets its resonance toward self-oscillation, and Mix blends wet against dry. High Icing makes the tail sing into a metallic, icy ring.

Param	Range	Default	Unit
Size	0 – 0.95	0.7	—
Color	0 – 1	0.5	—
Icing	0.5 – 14	8	—
Mix	0 – 1	0.35	—

ModEchoVerb 1 input

Fused reverb and modulated delay: the input feeds a wow-modulated feedback delay and a short comb reverb tank at once, blending echo and space in one block.

Param	Range	Default	Unit
Size	0 – 0.95	0.6	—
Time	20 – 1000	300	ms
Feedback	0 – 0.9	0.4	—
ModRate	0.05 – 5	0.4	Hz
Mix	0 – 1	0.4	—

EMT140 1 input

EMT 140-style plate reverb: a dispersive modulated all-pass tank (the plate shimmer where HF and LF propagate at different speeds), instant dense diffusion, bright sheen, HF decaying faster than the low tail.

Param	Range	Default	Unit
Decay	0 – 1	0.6	—
Tone	0 – 1	0.4	—
PreDelay	0 – 200	0	ms
Mix	0 – 1	0.3	—

Chamber 1 input

Chamber reverb: higher modal density than a plate with a diffuse onset and warm, non-metallic tail - between a plate and a room.

Param	Range	Default	Unit
Decay	0 – 1	0.6	—
Tone	0 – 1	0.6	—
PreDelay	0 – 200	10	ms
Mix	0 – 1	0.3	—

BX20 1 input

AKG BX-20-style torsional-spring reverb: a dispersive modulated all-pass tank models the spring chirp smeared smooth - long, dark, no boing.

Param	Range	Default	Unit
Decay	0 – 1	0.7	—
Tone	0 – 1	0.7	—
Mix	0 – 1	0.3	—

Lex224 1 input

Lexicon 224-style digital hall: a long, lush, slowly-modulated reverb tail (chorusing avoids metallic ringing) with bright, dense diffusion.

Param	Range	Default	Unit
Decay	0 – 1	0.7	—
Tone	0 – 1	0.5	—
PreDelay	0 – 200	20	ms
Mix	0 – 1	0.3	—

Lex480 1 input

Lexicon 480L-style digital reverb: a brighter, denser random-hall/plate with a modulated tail and fast build-up.

Param	Range	Default	Unit
Decay	0 – 1	0.65	—
Tone	0 – 1	0.6	—
PreDelay	0 – 200	10	ms
Mix	0 – 1	0.3	—

EMT250 1 input

EMT 250-style early digital reverb: a dense tank with first-generation converter quantization grain - bright and unmistakably early-digital.

Param	Range	Default	Unit
Decay	0 – 1	0.55	—
Tone	0 – 1	0.55	—
PreDelay	0 – 150	8	ms
Mix	0 – 1	0.3	—

RMX16 1 input

AMS RMX16-style digital reverb (Nonlin program): a dense, gated tail that holds then cuts off instead of decaying naturally - the classic 80s gated sound.

Param	Range	Default	Unit
Density	0 – 1	0.6	—
Hold	20 – 400	150	ms
Tone	0 – 1	0.5	—
Mix	0 – 1	0.3	—

SpringReverb 1 input

Spring-tank reverb: a feedback delay whose loop runs through a chain of first-order all-pass sections, giving the frequency-dependent (dispersive) delay that produces a real spring's characteristic 'boing' and metallic chirp - distinct from the dense diffusion of plate or hall reverbs. Time sets the tank length, Decay the tail, Dispersion the all-pass chirp amount, Damp the high-frequency loss, and Mix the wet/dry blend.

Param	Range	Default	Unit
Time	10 – 200	60	ms

Param	Range	Default	Unit
Decay	0 – 0.95	0.7	—
Dispersion	0 – 0.9	0.6	—
Damp	0 – 1	0.4	—
Mix	0 – 1	0.4	—

Plonk 1 input

Modal resonator (Rings/Plonk-style): excites a bank of five tuned band-pass resonators with the input, imposing the ringing modes of a struck object - bar, plate, bell, drumhead - onto any signal. Tune sets the fundamental, Structure morphs the partials from a harmonic stack toward inharmonic metallic ratios, Decay sets the ring time (resonator Q), and Mix the wet blend. Feed it drums, noise or a synth for plucked, struck and resonated tones. Distinct from the single-mode Resonator and the self-struck Modal oscillator.

Param	Range	Default	Unit
Tune	40 – 2000	220	Hz
Structure	0 – 1	0.5	—
Decay	1 – 40	12	—
Mix	0 – 1	0.5	—

Diffuser 1 input

Allpass diffuser: a cascade of four Schroeder all-pass sections with prime-spaced delays smears transients into a short, dense burst without adding a reverb tail or colouring the magnitude spectrum - the diffusion stage that lives inside reverbs, exposed on its own to thicken percussion, blur clicks, or pre-soften a signal before further processing. Amount sets the all-pass coefficient (smear density), Size scales the delay times, and Mix blends against the dry signal.

Param	Range	Default	Unit
Amount	0 – 0.85	0.6	—
Size	0.2 – 2	1	—
Mix	0 – 1	0.5	—

Aetherizer 1 input

Grain cloud (Absynth Aetherizer-style): continuously records the input and sprays a swarm of overlapping Hann-windowed grains read back from random recent positions, each at a randomized pitch, so a held sound dissolves into an evolving shimmering cloud. Size sets grain length, Density the grains per second, Pitch the centre transpose, Spread the random pitch scatter, Scatter how far back grains are picked, and Mix the wet amount. A texture/sound-design effect distinct from the granular delay and granular reverb.

Param	Range	Default	Unit
Size	20 – 200	80	ms
Density	5 – 120	40	Hz
Pitch	-12 – 12	0	st
Spread	0 – 12	3	st
Scatter	20 – 1000	300	ms
Mix	0 – 1	0.5	—

Sympathetic 1 input

Sympathetic-string resonator bank: four high-Q tuned resonators (like the drone strings of a sitar or the undamped strings of a piano) ring in sympathy with the input, voiced as a chord above the Tune root. Tune sets the root pitch, Decay how long the strings ring (resonator Q), Spread the chord voicing, and Mix the resonated blend. Feed it a plucked or percussive source for shimmering resonant tails - distinct from reverb and from a single modal resonator.

Param	Range	Default	Unit
Tune	40 – 1000	220	Hz
Decay	0 – 1	0.7	—
Spread	0 – 1	0.5	—
Mix	0 – 1	0.5	—

ErbeVerb 1 input

Modeless morphing reverb (Make Noise / Soundhack Erbe-Verb): a continuously-variable feedback-delay-network reverb with no presets - dial Size and Decay anywhere from a tight room through a cavernous hall, and Tilt from dark to bright. Mod animates the delay network for chorused, evolving and unreal spaces, and Mix blends wet against dry. One reverb that morphs smoothly across the whole space rather than switching between fixed types.

Param	Range	Default	Unit
Size	0 – 1	0.5	—
Decay	0 – 1	0.6	—
Tilt	0 – 1	0.5	—
Mod	0 – 1	0.2	—
Mix	0 – 1	0.4	—

Reverb 1 input

Algorithmic room reverb (Freeverb-style comb/all-pass network) with Size scaling the decay time and Damp rolling off the highs in the tail. Width sets the stereo spread and Mix the dry/wet blend. Large Size with low Damp gives long bright halls; high Damp yields short dark rooms - see Massive, Plate, Spring and Shimmer for character-specific spaces.

Param	Range	Default	Unit
Size	0 – 1	0.5	—
Damp	0 – 1	0.5	—
Width	0 – 1	1	—
Mix	0 – 1	0.3	—

Massive 1 input

Feedback-delay-network reverb (Valhalla Supermassive style): four modulated delay lines mixed through a unitary Hadamard matrix give huge, smooth, near-infinite tails. Size scales delay length up to multi-second, Decay sets feedback, Damp absorbs highs in the loop, and Mod choruses the lines for shimmer. Mode picks Lush / Huge / Cosmos voicings, from tight to cathedral to cosmic.

Param	Range	Default	Unit
Size	0 – 1	0.6	—
Decay	0 – 1	0.85	—
Damp	0 – 1	0.4	—
Mod	0 – 1	0.3	—
Mode	0 – 2	1	—
Mix	0 – 1	0.4	—

Spring 1 input

Spring-reverb approximation: a short (~30 ms) delay loop diffused through an all-pass cascade for the characteristic dispersive boing, with a Tone low-pass in the feedback. Decay sets the tail length and Mix the blend. The classic guitar-amp and dub reverb - splashy on transients, drippy at high Decay.

Param	Range	Default	Unit
Decay	0 – 0.95	0.6	—
Tone	500 – 8000	3000	Hz
Mix	0 – 1	0.3	—

Ascend 1 input

Pitch-rising shimmer reverb: a long delay whose feedback is pitch-shifted up an octave (granular two-tap) and damped, so the tail ascends and blooms into ethereal pads. Size sets the delay length, Shimmer the octave-up feedback amount and Damp the tail brightness. Lush on pads, vocals and ambient guitar - see Crystals for reverse+pitch and Gravity for huge inverted tails.

Param	Range	Default	Unit
Size	0 – 1	0.6	—
Shimmer	0 – 0.95	0.5	—
Damp	0 – 1	0.4	—
Mix	0 – 1	0.3	—

GatedVerb 1 input

Reverb whose wet tail is muted by a fast envelope-driven gate, so the reverb only passes while the dry input stays above a threshold. Size sets the room length and Damp the high-frequency decay, Threshold sets the level below which the tail is cut, and Mix blends the gated reverb against the dry signal. Reproduces the abrupt, chopped-off reverb of 80s drum production.

Param	Range	Default	Unit
Size	0 – 1	0.6	—
Damp	0 – 1	0.4	—
Threshold	-60 – 0	-30	dB
Mix	0 – 1	0.3	—

FreezeVerb 1 input

Reverb that can latch into an infinite-hold freeze mode, sustaining the current tail indefinitely instead of decaying. Size sets the room length and Damp the high-frequency decay, Freeze above its midpoint engages the infinite hold, and Mix blends the reverb against the dry signal. Use the freeze to build sustained drones and pads from a single moment of input.

Param	Range	Default	Unit
Size	0 – 1	0.7	—
Damp	0 – 1	0.3	—
Freeze	0 – 1	0	—
Mix	0 – 1	0.4	—

DuckVerb 1 input

Reverb whose wet signal is attenuated by an envelope follower tracking the dry input, so the tail dips while you play and swells back in the gaps. Size sets the room length and Damp the high-frequency decay, Duck sets how strongly the dry level suppresses the wet, and Mix blends the ducked reverb against the dry signal. Keeps the reverb from washing out a busy performance while preserving the tail between phrases.

Param	Range	Default	Unit
Size	0 – 1	0.6	—
Damp	0 – 1	0.4	—
Duck	0 – 1	0.7	—

Param	Range	Default	Unit
Mix	0 – 1	0.3	—

ModVerb 1 input

Reverb whose room size is continuously modulated by a sine LFO, detuning the tail for a chorused, shimmering character. Size sets the base room length and Damp the high-frequency decay, Rate sets the LFO speed (0.05-5 Hz) and Depth how far it sweeps the size, and Mix blends the moving reverb against the dry signal. Adds motion and width to otherwise static reverb tails.

Param	Range	Default	Unit
Size	0 – 1	0.6	—
Damp	0 – 1	0.4	—
Rate	0.05 – 5	0.5	Hz
Depth	0 – 1	0.3	—
Mix	0 – 1	0.3	—

ReverseVerb 1 input

Reverb whose wet level swells up from silence after each detected onset, fading the tail in rather than out for a backward, blooming effect. Size sets the room length and Damp the high-frequency decay, Swell sets the fade-in time (1-500 ms) restarted on every new transient, and Mix blends the swelling reverb against the dry signal. Produces the rising, sucked-in quality of reversed reverb.

Param	Range	Default	Unit
Size	0 – 1	0.6	—
Damp	0 – 1	0.4	—
Swell	1 – 500	150	ms
Mix	0 – 1	0.4	—

Gravity 1 input

Oversized 4-line feedback delay network (FDN) reverb, each line damped by a low-pass and length-modulated by a shared LFO. Size stretches the base delay times up to 9x, Gravity raises feedback toward unity for near-infinite tails, Mod sets the LFO modulation depth, and Freeze pins feedback to 1.0 while muting new input. Mix blends the Hadamard-mixed wet sum against the dry; pushed hard it yields huge, slowly-evolving, freezable spaces.

Param	Range	Default	Unit
Size	0 – 1	0.8	—
Gravity	0 – 1	0.9	—
Mod	0 – 1	0.4	—
Freeze	0 – 1	0	—
Mix	0 – 1	0.4	—

NonlinVerb 1 input

Nonlinear 4-line FDN reverb whose wet tail is gated by an envelope follower: a transient above the running level retriggers a hold, then the gain holds flat before cutting off. Size scales the short base delay times up to 5x, while Hold sets how long (10-500 ms) the tail sustains at full level before it decays away. Mix sets dry/wet; the held-then-cut behaviour gives gated, non-natural reverb bursts rather than a smooth decay.

Param	Range	Default	Unit
Size	0 – 1	0.5	—
Hold	10 – 500	150	ms

Param	Range	Default	Unit
Mix	0 – 1	0.4	—

Crystals 1 input

Shimmer reverb: a 4-line FDN whose output is pitch-shifted an octave up and re-injected into the network, building rising crystalline overtones. Size scales the base delay times up to 6x, Shimmer sets how much octave-up signal feeds back in, and Damp sets the per-line low-pass that darkens repeats. Mix blends the wet sum against the dry; higher Shimmer ascends further, higher Damp tames the brightness.

Param	Range	Default	Unit
Size	0 – 1	0.6	—
Shimmer	0 – 1	0.5	—
Damp	0 – 1	0.4	—
Mix	0 – 1	0.4	—

GrainVerb 1 input

Granular reverb that windows the input with a moving Hann grain and injects it into a 4-line feedback delay network for diffusion. Size scales the four delay lengths up to 6x for longer tails, Grain sets the grain rate in milliseconds, and Damp applies a low-pass in each feedback line to darken successive reflections. Mix blends the diffused tail against the dry input for a smeared, granular ambience.

Param	Range	Default	Unit
Size	0 – 1	0.6	—
Grain	5 – 100	30	ms
Damp	0 – 1	0.4	—
Mix	0 – 1	0.3	—

Freezer 1 input

Audio freezer that continuously buffers the input and, when Freeze is engaged, loops the most recently captured window indefinitely. Size sets the length of the looped window in milliseconds, trading a short stuttering grain against a longer evolving pad, and Mix blends the frozen loop against the dry signal. With Freeze off it passes the input through while keeping the buffer primed for an instant capture.

Param	Range	Default	Unit
Freeze	0 – 1	0	—
Size	20 – 500	200	ms
Mix	0 – 1	1	—

MangledVerb 1 input

Reverb whose fully wet output is fed through a tanh saturation stage for gritty, broken ambience. Size sets the room size and tail length, Damp rolls off high frequencies in the decay, and Drive pushes the reverb output harder into the nonlinearity for increasing crush. Mix blends the distorted tail against the dry signal, ranging from a lightly dirtied space to a heavily mangled wash.

Param	Range	Default	Unit
Size	0 – 1	0.6	—
Damp	0 – 1	0.4	—
Drive	0 – 1	0.5	—
Mix	0 – 1	0.3	—

Dynamics

35 modules

Omnipressor 1 input

Extreme dynamics on one knob: Function sweeps from gating expansion through limiting compression to full dynamic reversal, where loud passages turn quiet and quiet turns loud.

Param	Range	Default	Unit
Function	-1 – 3	1	—
Threshold	-48 – 0	-18	dB
Attack	0.1 – 100	3	ms
Release	5 – 800	120	ms
Gain	-12 – 24	0	dB

VariMu 1 input

Fairchild/Manley-style tube vari-mu compressor: program-dependent ratio rises with level, and a remote-cutoff tube transfer whose genuine 2nd-harmonic warmth grows as gain reduction biases the tube toward cutoff.

Param	Range	Default	Unit
Threshold	-48 – 0	-18	dB
Ratio	1.5 – 8	3	—
Attack	1 – 100	20	ms
Release	50 – 2000	400	ms
Makeup	-6 – 24	0	dB

Opto 1 input

LA-2A-style optical compressor: a T4 photocell whose photoresistor gain law gives a soft knee and a ratio that rises with level, plus program-dependent release (recovers slower after sustained compression).

Param	Range	Default	Unit
Threshold	-48 – 0	-20	dB
Ratio	2 – 6	3	—
Attack	1 – 50	10	ms
Release	50 – 3000	600	ms
Makeup	-6 – 24	0	dB

FET 1 input

1176-style FET compressor: ultra-fast attack, fixed high ratios, and an asymmetric FET gain cell whose harmonic bite grows with gain reduction (max ratio = all-buttons slam).

Param	Range	Default	Unit
Threshold	-48 – 0	-16	dB
Ratio	4 – 20	8	—

Param	Range	Default	Unit
Attack	0.02 – 5	0.5	ms
Release	20 – 1200	200	ms
Makeup	-6 – 24	0	dB

SSL Bus 1 input

SSL bus-style VCA glue compressor: a dual-time-constant program-dependent auto-release for pumping-free glue, through a dB-linear VCA cell with control-port slew and gain-cell THD that grows with gain reduction.

Param	Range	Default	Unit
Threshold	-40 – 0	-14	dB
Ratio	2 – 10	4	—
Attack	0.1 – 30	10	ms
Release	0.1 – 1.2	0.3	s
Makeup	-6 – 24	0	dB

Dbx160 1 input

dbx 160-style VCA compressor: true-RMS detection and an 'over-easy' soft knee through a dB-linear VCA cell with control-port slew and gain-cell THD; punchy and musical.

Param	Range	Default	Unit
Threshold	-40 – 0	-18	dB
Ratio	1 – 20	4	—
Knee	0 – 20	12	dB
Makeup	-6 – 24	0	dB

Distressor 1 input

Distressor-style aggressive compressor: a FET gain cell whose bite grows with gain reduction, switchable harmonic distortion (Dist2 = genuine 2nd, Dist3 = 3rd) and a British-mode all-buttons slam.

Param	Range	Default	Unit
Threshold	-40 – 0	-16	dB
Ratio	1 – 20	6	—
Attack	0.05 – 5	0.5	ms
Release	20 – 1200	150	ms
Dist	Clean · Dist2 · Dist3	—	—
British	Off · On	—	—

LA3A 1 input

Teletronix LA-3A-style optical compressor: the LA-2A photocell gain law but a fast-attack, brighter solid-state amp - punchier optical leveling with program-dependent release.

Param	Range	Default	Unit
Threshold	-40 – 0	-18	dB
Ratio	2 – 8	4	—
Makeup	-6 – 24	0	dB
Mix	0 – 1	1	—

API2500 1 input

API 2500-style VCA bus compressor: a Thrust sidechain tilt with Old (feedback) or New (feedforward) detection, through a dB-linear VCA cell with control-port slew and gain-cell THD.

Param	Range	Default	Unit
Threshold	-40 – 0	-14	dB
Ratio	1.5 – 10	4	—
Attack	0.1 – 30	10	ms
Release	50 – 2000	300	ms
Thrust	Off · Med · Loud	—	—
Mode	Old · New	—	—

StaLevel 1 input

Gates Sta-Level-style tube vari-mu limiter: a very slow program-dependent auto-release and gentle ~3:1 leveling through a remote-cutoff tube cell whose 2nd-harmonic warmth (Warmth) grows with gain reduction.

Param	Range	Default	Unit
Threshold	-40 – 0	-16	dB
Release	0 – 1	0.5	—
Makeup	-6 – 24	0	dB
Warmth	0 – 1	0.3	—

Neve33609 1 input

Neve 33609-style diode-bridge bus compressor: smooth glue through a diode-bridge gain cell with control-port slew, a dual-time-constant release and transformer saturation.

Param	Range	Default	Unit
Threshold	-40 – 0	-16	dB
Ratio	1.5 – 6	3	—
Attack	0.1 – 30	8	ms
Release	50 – 2000	400	ms
Makeup	-6 – 24	0	dB

TubeTech 1 input

Tube-Tech CL 1B-style opto compressor: a photocell gain law (soft knee, rising ratio) with program-dependent release and tube makeup warmth.

Param	Range	Default	Unit
Threshold	-40 – 0	-18	dB
Ratio	1.5 – 10	4	—
Attack	5 – 300	50	ms
Release	100 – 5000	1000	ms
Makeup	-6 – 24	0	dB

ShadowHills 1 input

Shadow Hills Mastering-style compressor: a photocell opto stage (soft-knee, rising-ratio law, program-dependent release) into a discrete fixed-ratio VCA stage, with a transformer-flavour select (nickel/iron/steel).

Param	Range	Default	Unit
Threshold	-40 – 0	-16	dB
Opto Ratio	1.5 – 4	2	—
VCA Ratio	2 – 8	3	—
Release	100 – 2000	500	ms
Xfmr	Nickel · Iron · Steel		—
Makeup	-6 – 24	0	dB

Summit 1 input

Summit TLA-100-style tube/opto leveler: photocell gain law (soft knee, rising ratio) with program-dependent release, transformer warmth and tube makeup.

Param	Range	Default	Unit
Threshold	-40 – 0	-18	dB
Ratio	2 – 10	4	—
Attack	1 – 100	20	ms
Release	100 – 3000	600	ms
Makeup	-6 – 24	0	dB

SCComp 2 inputs

Sidechain compressor: compresses the signal (in 0) by the level of a key input (in 1) through a dB-linear VCA cell with control-port slew and gain-cell THD - cable the host sidechain (SC L/R) to the key for ducking/pumping.

Param	Range	Default	Unit
Threshold	-48 – 0	-18	dB
Ratio	1 – 20	4	—
Attack	0.1 – 100	5	ms
Release	1 – 500	100	ms
Makeup	-6 – 24	0	dB

DynaComp 1 input

MXR Dyna Comp-style OTA pedal compressor (CA3080): a full-wave-rectified envelope drives a high-ratio OTA gain cell for the squishy, sustaining 'country squeeze' - fast clamp, slow bloom, and the signature high-end softening. Sensitivity sets how hard it squeezes, Attack the detector speed, Output the makeup.

Param	Range	Default	Unit
Sensitivity	0 – 1	0.6	—
Attack	1 – 50	6	ms
Output	-6 – 24	8	dB

LA2A 1 input

Teletronix LA-2A-style opto leveling amplifier: a T4 electro-optical cell compresses slowly and smoothly with the cell's two-stage, program-dependent release - fast for the first part of recovery, then a long gentle tail - so it sits musically without surgical control. Peak Reduction sets how hard it leans, Gain the makeup, Emphasis tilts the detector toward highs (de-essing).

Param	Range	Default	Unit
Peak Reduction	0 – 1	0.5	—
Gain	-6 – 30	8	dB

Param	Range	Default	Unit
Emphasis	0 – 1	0.3	—

FATSO 1 input

Empirical Labs FATSO-style warmth/comp: a soft-knee compressor into a tape-style saturation stage (a tilting HF soft-clip) that rounds transients and adds harmonic 'warmth', the way the hardware glues and thickens a mix. Comp sets the squeeze, Warmth the saturation drive, Output the makeup.

Param	Range	Default	Unit
Comp	0 – 1	0.4	—
Warmth	0 – 1	0.3	—
Output	-6 – 18	4	dB

Drawmer 1 input

Drawmer DS201-style noise gate: opens fast above Threshold and closes after Release, with hysteresis so it doesn't chatter on the edge. Range sets how far it ducks when closed (hard gate vs gentle expander), Attack the open speed - the studio-standard gate for tightening drums and killing hum between phrases.

Param	Range	Default	Unit
Threshold	-60 – 0	-30	dB
Range	-80 – 0	-60	dB
Attack	0.1 – 50	1	ms
Release	5 – 1000	120	ms

EnvFollow 1 input

Envelope follower: tracks the rectified amplitude of the input with separate Attack and Release times, outputting a smooth control signal that rises quickly on transients and falls slowly back - the standard way to turn audio into a modulation source. Patch its output into any parameter's mod input so a filter, VCA or effect responds to playing dynamics; Gain scales the result. Unlike the duckers (which bury the follower inside one effect) this exposes the envelope itself.

Param	Range	Default	Unit
Attack	0.1 – 200	5	ms
Release	1 – 1000	100	ms
Gain	0 – 8	1	—

Envelope 1 input

Envelope follower that rectifies the input and smooths it with independent Attack and Release times into a control signal. Gain scales the output for driving a filter cutoff, VCA or any mod input (auto-wah, ducking, dynamics). Fast attack tracks transients; a longer release gives a smoother contour.

Param	Range	Default	Unit
Attack	0.1 – 200	5	ms
Release	1 – 1000	80	ms
Gain	0 – 8	1	—

Compressor 1 input

Peak-following compressor: tracks the input envelope with Attack/Release ballistics and, above Threshold, reduces gain by the set Ratio, with Makeup to restore level. Fast attack tames transients while slow release glues sustained material. A clean general-purpose dynamics stage - see the analog pack (FET, Opto, VariMu) for coloured compression.

Param	Range	Default	Unit
Threshold	-60 – 0	-18	dB
Ratio	1 – 20	4	—
Attack	0.1 – 100	5	ms
Release	1 – 500	80	ms
Makeup	-12 – 24	0	dB

Gate 1 input

Noise gate: when the detected level falls below Threshold it attenuates the signal toward Range (the floor), opening and closing with Attack/Release ballistics. Range at -80 dB fully mutes between events; smaller values leave a residual for gentler ducking. Use it to silence hiss and bleed between phrases or for rhythmic gating.

Param	Range	Default	Unit
Threshold	-80 – 0	-40	dB
Attack	0.1 – 50	1	ms
Release	1 – 500	100	ms
Range	-80 – 0	-80	dB

Limiter 1 input

Brick-wall peak limiter: instantly clamps gain so peaks never exceed Threshold, then releases smoothly back to unity over Release, with Output trimming the post-limit level. Set Threshold as the ceiling and drive the input for transparent loudness. See Maximizer for loudness-optimized limiting.

Param	Range	Default	Unit
Threshold	-24 – 0	-3	dB
Release	1 – 500	50	ms
Output	-12 – 12	0	dB

Transient 1 input

Transient shaper: a fast vs slow envelope follower separates the attack (fast > slow) from the sustain (fast < slow), letting Attack and Sustain each be boosted or cut independently, with Output trimming the result. Add punch to drums, or tighten/lengthen sustain - all independent of input level, unlike a compressor.

Param	Range	Default	Unit
Attack	-1 – 1	0.3	—
Sustain	-1 – 1	0	—
Output	-12 – 12	0	dB

Expander 1 input

Downward expander: attenuates signal that falls below Threshold by the set Ratio (the inverse of a compressor), tightening quiet passages and reducing bleed and hiss between notes. Attack/Release set the ballistics. Gentler than a gate - it fades low-level material rather than hard-muting it.

Param	Range	Default	Unit
Threshold	-60 – 0	-40	dB
Ratio	1 – 10	2	—
Attack	0.1 – 100	5	ms
Release	1 – 500	100	ms

MultiComp 1 input

Three-band compressor: crossovers split the signal into low, mid and high bands (Low/Mid and Mid/High), each compressed independently against a shared Threshold and Ratio with fast-attack/slow-release ballistics, then summed with Makeup. Tame a busy mix's bands separately - control boomy lows without dulling the highs. See the SSL/dbx analog models for coloured bus compression.

Param	Range	Default	Unit
Low/Mid	100 – 1000	300	Hz
Mid/High	1000 – 12000	3000	Hz
Threshold	-48 – 0	-18	dB
Ratio	1 – 20	4	—
Makeup	-12 – 24	0	dB

DeEsser 1 input

Tames sibilance by splitting off a high band with a one-pole low-pass, tracking its level with a fast-attack envelope, and applying gain reduction to that band only when it crosses the threshold before recombining it with the lows. Freq sets the sibilance band, Threshold sets where reduction begins, Amount scales how hard the high band is pulled down, and Mix sets dry/wet. Because only the highs are compressed, it removes harsh esses without dulling the rest of the signal.

Param	Range	Default	Unit
Freq	2000 – 12000	6000	Hz
Threshold	-60 – 0	-24	dB
Amount	0 – 1	0.7	—
Mix	0 – 1	1	—

Ducker 2 inputs

Sidechain ducker that follows the level of the key input (in 2) with a release-smoothed envelope and pulls down the main signal whenever that key exceeds the threshold. Threshold sets the key level that triggers ducking, Amount sets how far the main signal drops, and Release sets how quickly it recovers once the key falls back. Use it for music-under-voice ducking or rhythmic pumping driven by a separate trigger.

Param	Range	Default	Unit
Threshold	-60 – 0	-30	dB
Amount	0 – 1	0.8	—
Release	10 – 1000	200	ms

Maximizer 1 input

Loudness maximizer that applies a gain boost, tracks the peak envelope, and divides back any sample exceeding the ceiling before hard-clamping to it, acting as a brick-wall limiter. Boost sets how hard the input is driven up and Ceiling sets the absolute output peak in dB. Push Boost while keeping Ceiling near 0 dB to raise apparent loudness without overshooting the output.

Param	Range	Default	Unit
Boost	0 – 24	6	dB
Ceiling	-12 – 0	-0.3	dB

UpComp 1 input

Upward compressor that tracks the input envelope and, when it sits below the threshold, applies makeup gain proportional to how far below it is and to the ratio, lifting quiet passages toward the threshold. Threshold sets the level below which boosting begins, Ratio sets how aggressively the deficit is filled, Makeup adds a fixed output gain, and Mix sets dry/wet. Use it to raise low-level detail and tighten dynamic range from the bottom up rather than taming peaks.

Param	Range	Default	Unit
Threshold	-60 – 0	-30	dB
Ratio	1 – 8	2	—
Makeup	0 – 24	0	dB
Mix	0 – 1	1	—

DynEQ 1 input

Dynamic EQ band: an SVF band-pass at Freq/Q tracks an envelope follower, and when that band's energy crosses Threshold the band is subtracted from the signal in proportion to Amount. Freq and Q place and narrow the band, Threshold sets where reduction begins, and Mix blends the result against dry. It only acts when the band gets loud, so it tames resonances and harshness without static EQ cuts.

Param	Range	Default	Unit
Freq	100 – 12000	2000	Hz
Q	0.5 – 8	2	—
Threshold	-60 – 0	-24	dB
Amount	0 – 1	0.7	—
Mix	0 – 1	1	—

MBDynamics 1 input

Multiband dynamics processor that splits the input into low, mid and high bands (crossovers near 250 Hz and 2.5 kHz) and applies independent compression and gating to each. Threshold sets the level above which a band is compressed at Ratio, while Gate sets the level below which a band is attenuated downward. Mix blends the processed sum against the dry signal, useful for tightening or de-noising specific frequency regions at once.

Param	Range	Default	Unit
Threshold	-60 – 0	-24	dB
Ratio	1 – 20	3	—
Gate	-80 – -20	-60	dB
Mix	0 – 1	1	—

Pump 1 input

Tempo-synced sidechain-style volume pump that ducks the input each cycle and recovers using a sample-clock phase for coherent timing. Rate sets the duck frequency in Hz, Depth sets how far the level drops, and Curve shapes the recovery ramp from snappy to gradual. Mix blends the pumped signal against the dry input for the classic EDM ducking effect without an external sidechain.

Param	Range	Default	Unit
Rate	0.1 – 8	2	Hz

Param	Range	Default	Unit
Depth	0 – 1	0.8	—
Curve	0.2 – 4	1.5	—
Mix	0 – 1	1	—

Squash 1 input

Aggressive parallel smash: a fast peak-following envelope drives heavy gain reduction above a fixed threshold, and the gain-reduced signal is pushed through a tanh saturator for crushed, distorted dynamics. Amount steepens the gain-reduction curve (harder squash), Tone scales the saturated output level, and Mix blends the smashed signal in parallel with the dry input. Useful for adding punch and grit to drums and bus material.

Param	Range	Default	Unit
Amount	0 – 1	0.7	—
Tone	0 – 1	0.5	—
Mix	0 – 1	0.5	—

Pitch

15 modules

SynthGlide 1 input

Portamento: exponential glide of the note pitch.

Param	Range	Default	Unit
Time	0 – 1000	80	ms

FreqShift 1 input

Bode-style single-sideband frequency shifter: splits the input at in0 into in-phase and 90-degree branches via a polyphase IIR Hilbert allpass pair, then heterodynes against an internal carrier to shift every partial by a fixed Hz amount. Shift sets the offset (-1000 to +1000 Hz) and Mix blends the shifted signal against the dry input. Because the shift is in Hz rather than ratio, harmonic relationships break, giving inharmonic, metallic, clangorous tones.

Param	Range	Default	Unit
Shift	-1000 – 1000	100	Hz
Mix	0 – 1	1	—

Shifter 1 input

Pitch / frequency / ring shifter in one (Ableton Shifter-style). Mode switches between granular Pitch shift (transpose by Coarse semitones + Fine cents, length preserved), Bode Frequency shift (Coarse/Fine reinterpreted as Hz via a polyphase-IIR Hilbert SSB shifter, breaking harmonics into inharmonic/metallic tones), and Ring modulation (multiply by an internal sine carrier at the Coarse/Fine frequency). Window sets the pitch-mode grain size; Mix blends dry/wet.

Param	Range	Default	Unit
Mode	Pitch · Frequency · Ring		—
Coarse	-48 – 48	7	st
Fine	-100 – 100	0	ct
Window	20 – 120	60	ms
Mix	0 – 1	1	—

Spinback 2 inputs

Turntable brake: a gate on in1 starts reading the captured buffer at steadily falling speed, so playback drops in pitch and slows to a dead stop like braking a record. Time sets how long the brake takes to reach a halt (50..3000 ms) and Mix blends the braked signal against the dry input. Releasing the gate stops braking and returns to live audio.

Param	Range	Default	Unit
Time	50 – 3000	600	ms
Mix	0 – 1	1	—

H910 1 input

Harmonizer: a pitch shifter feeding a feedback delay, so each repeat is pitched again into a rising or falling cascade (the classic H910 fused pitch+delay).

Param	Range	Default	Unit
Pitch	-12 – 12	7	st
Delay	1 – 1000	200	ms
Feedback	0 – 0.9	0.4	—
Mix	0 – 1	0.5	—

Octavox 1 input

A cloud of four pitch-shifted voices at diatonic intervals, each on its own delay tap, summed into a shimmering harmonized pitch-delay (Octavox/Quadravox class).

Param	Range	Default	Unit
Spread	10 – 400	120	ms
Key	0 – 2	0	—
Feedback	0 – 0.85	0.3	—
Level	0 – 1	0.6	—
Mix	0 – 1	0.5	—

H3000 1 input

Eventide H3000-style micro-pitch doubler: two slightly detuned, short-delayed voices thicken the signal into a wide, lush double-track.

Param	Range	Default	Unit
Detune	0 – 30	12	ct
Delay	1 – 50	18	ms
Width	0 – 1	0.7	—
Mix	0 – 1	0.5	—

PitchShift 1 input

Granular pitch shifter using two overlapping, cross-faded grains (Window 20-120 ms) to transpose +/-24 semitones without changing length. Feedback re-injects the shifted output for cascading and shimmer effects, and Mix sets dry/wet. Shorter windows tighten transients but add warble; longer windows smooth the tone at the cost of latency.

Param	Range	Default	Unit
Pitch	-24 – 24	7	st
Mix	0 – 1	0.5	—
Feedback	0 – 0.9	0	—
Window	20 – 120	60	ms

Octaver 1 input

Sub-octave generator: a flip-flop toggled at each rising zero crossing produces a half-rate square tracked by the input envelope, low-passed by Tone for warmth and mixed under the Dry signal at Sub. Monophonic, analog-pedal style - best on clean single notes and bass. It glitches on chords, as expected of the technique.

Param	Range	Default	Unit
Sub	0 – 1	0.7	—
Dry	0 – 1	1	—
Tone	200 – 8000	2000	Hz

Unison 1 input

Unison thickener: turns a mono input into up to 16 pitch-detuned voices (the supersaw sound) using overlapping ramped-delay pitch shifters at symmetric cent offsets, so the centre stays at pitch and the outer voices spread apart. Detune sets the spread, Blend the outer-voice level and Mix dry/wet. Wide and lush on synths, pads and backing vocals.

Param	Range	Default	Unit
Voices	1 – 16	4	—
Detune	0 – 50	18	ct
Blend	0 – 1	0.7	—
Mix	0 – 1	1	—

MicroShift 1 input

Micro-pitch doubler that generates two delay-line voices detuned slightly up and slightly down, then sums them with the dry signal to widen the source. Detune sets the symmetric pitch offset (0-50 cents) and Delay sets the base tap time (0-40 ms) before the shifted voices. Mix sets dry/wet; small Detune thickens, larger Detune broadens into a chorus-like spread.

Param	Range	Default	Unit
Detune	0 – 50	12	cents
Delay	0 – 40	12	ms
Mix	0 – 1	0.5	—

Harmonizer 1 input

Single-voice harmonizer that pitch-shifts the input by a chosen musical interval using overlapping windowed delay-line reads, then adds it back to the dry signal. Interval sets the transposition (+/-24 semitones), Detune adds up to 30 cents of offset for a less rigid blend, and Level sets the harmony voice gain. Mix scales the added voice into the output; useful for parallel harmonies and octave doubling.

Param	Range	Default	Unit
Interval	-24 – 24	7	st
Detune	0 – 30	0	cents
Level	0 – 1	0.7	—
Mix	0 – 1	1	—

Pitch2Midi 1 input

Monophonic pitch tracker that measures the period between rising zero-crossings, converts it to a MIDI note number, and emits it as a control value. Glide smooths the tracked pitch over time for portamento, while Out switches the output between the normalized pitch CV and a simple gate that goes high whenever the input is active. Intended to drive other modules from a played signal rather than to process audio.

Param	Range	Default	Unit
Glide	0 – 0.99	0.3	—
Out	0 – 1	0	—

PitchCorrect 1 input

Rough auto-tune that detects the input period via zero-crossings, finds the nearest semitone, and applies a crossfaded two-tap windowed pitch shift to pull the note onto pitch. Amount scales how far the signal is moved toward the target semitone, from no correction to full snapping, and Mix blends the corrected voice against the dry input. Best on clean monophonic sources; artifacts increase with stronger correction.

Param	Range	Default	Unit
Amount	0 – 1	0.8	—
Mix	0 – 1	1	—

TimeStretch 1 input

Granular time-smear that writes the input into a delay line and reads it back with two overlapping Hann-windowed grains advancing slower than real time. Stretch sets how slowly the read position moves (lower values smear the sound out further) and Grain sets the grain length in milliseconds, trading smoothness against fluttering artifacts. Mix blends the stretched texture against the dry signal for frozen, drifting timbres.

Param	Range	Default	Unit
Stretch	0.25 – 1	0.5	—
Grain	20 – 200	80	ms
Mix	0 – 1	1	—

Utility

92 modules

VCA 2 inputs

Voltage-controlled amplifier: multiplies the signal by the control input.

SynthGain 1 input

Gain trim in decibels.

Param	Range	Default	Unit
Gain	-60 – 24	0	dB

SynthMixer 2 inputs

Sums two inputs with independent levels.

Param	Range	Default	Unit
In 1	0 – 1.5	1	—
In 2	0 – 1.5	1	—

Scaler 1 input

CV utility: scale, offset and bipolar curve-bend a control signal.

Param	Range	Default	Unit
Scale	-2 – 2	1	—
Offset	-1 – 1	0	—
Curve	-1 – 1	0	—

Subtract 3 inputs

Difference node: in0 minus in1 minus in2, plus a DC Offset. Feed two signals to get their difference (error/side-chain CV, mid-from-sum, delta detectors) or leave in2 unpatched for a simple two-input subtract. Unconnected inputs contribute 0.

Param	Range	Default	Unit
Offset	-2 – 2	0	—

Multiply 2 inputs

Two-quadrant multiplier / VCA: in0 times in1 times Scale. With an audio-rate in1 it is ring modulation; with a control-rate in1 it is a clean amplitude VCA. Both inputs must be patched - an unconnected input is 0, which mutes the output (the multiplicative identity is not assumed).

Param	Range	Default	Unit
Scale	-4 – 4	1	—

Divide 2 inputs

Safe divider: in0 divided by (in1 + Denom), with the result limited to +/-Limit so a near-zero denominator can never blow up. Denom both seeds the divisor when in1 is unpatched and biases it when it is, and the magnitude of the divisor is floored at a tiny epsilon. Useful for normalising, ratio CV, and reciprocal-style shaping.

Param	Range	Default	Unit
Denom	-4 – 4	1	—
Limit	1 – 64	16	—

Negate 1 input

Sign inverter: outputs -in0 scaled by Scale (default 1) plus an Offset. Flip the phase of an audio signal, invert a control voltage, or build a difference by summing a Negate against another source. Scale at -1 passes the signal through unchanged.

Param	Range	Default	Unit
Scale	-2 – 2	1	—
Offset	-1 – 1	0	—

Reciprocal 1 input

One-over-x: $1 / (\text{in0} + \text{Bias})$, limited to +/-Limit and floored away from zero so it stays finite. Turns a rising CV into a falling one, converts frequency to period (and back), and builds hyperbolic shapes. Bias keeps the denominator off zero when in0 passes through the origin.

Param	Range	Default	Unit
Bias	-2 – 2	0	—
Limit	1 – 64	16	—

Min 2 inputs

Outputs the smaller of in0 and in1 (after adding the In B offset to in1). Pairs with Max to build hard clippers, envelope floors, and CV limiters. An unconnected in1 reads 0, so In B sets the constant floor in that case.

Param	Range	Default	Unit
In B	-2 – 2	0	—

Max 2 inputs

Outputs the larger of in0 and in1 (after adding the In B offset to in1). Max against 0 is a half-wave rectifier / positive-only clamp; against a constant it is an envelope ceiling. An unconnected in1 reads 0, so In B sets the constant ceiling in that case.

Param	Range	Default	Unit
In B	-2 – 2	0	—

Clamp 1 input

Hard range limiter: constrains in0 to the [Lo, Hi] window with no saturation curve (unlike Clip, which rounds the knee). Use it to bound CV, keep a feedback signal in range, or window a modulator. If Lo > Hi the bounds swap so the window is always valid.

Param	Range	Default	Unit
Lo	-2 - 2	-1	—
Hi	-2 - 2	1	—

Constant 0 inputs

DC source: emits a fixed Value every sample with no input. Patch it into a maths block as a constant operand, into a parameter mod input as a manual offset, or sum it onto a signal as bias. The simplest way to add a tunable number to the graph.

Param	Range	Default	Unit
Value	-1 - 1	1	—

Crossfade 3 inputs

Blends from in0 to in1 by a fade position of Mix plus the in2 CV (clamped 0..1). Mode 0 is a linear (constant-sum) fade; Mode 1 is an equal-power (constant-energy) fade for click-free audio crossfades. Patch in2 to automate the morph; leave it unpatched to set the blend by hand with Mix.

Param	Range	Default	Unit
Mix	0 - 1	0.5	—
Mode	Linear · Equal Power		—

Select 3 inputs

Two-way signal switch: routes in0 when the in2 selector (plus the Threshold) is below 0.5 and in1 when it is at or above, so a gate or LFO can hard-flip between two sources. Unlike Crossfade there is no blend - it is a clean A/B route for muxing oscillators or CV paths.

Param	Range	Default	Unit
Threshold	-1 - 1	0	—

Power 1 input

Sign-preserving exponent: outputs $\text{sign}(\text{in0}) * |\text{in0}|^{\text{Exp}}$, so it reshapes a normalised signal's curve without changing its polarity or its 0 and +/-1 endpoints. Exp > 1 bends a CV toward the extremes (snappier envelopes), Exp < 1 expands the middle (gentler control feel).

Param	Range	Default	Unit
Exp	0.1 - 8	2	—

Sign 1 input

Sign extractor with a dead zone: outputs -1, 0, or +1 depending on whether in0 sits below, inside, or above a +/-Dead band around zero. Turns any signal into a bipolar gate, derives direction from a slope, or squares an oscillator into a pulse. Dead at 0 gives a pure two-state comparator.

Param	Range	Default	Unit
Dead	0 – 1	0	—

Quantize 1 input

Rounds in0 to the nearest of Steps evenly-spaced levels across the [-1, 1] range, turning a smooth signal into a stepped one. Drop the resolution of a sweep into discrete plateaus, build sample-rate-style bit-crush on CV, or snap a glide into a staircase. Steps at 2 gives a bipolar square; higher Steps gives finer stairs.

Param	Range	Default	Unit
Steps	2 – 64	8	—

Fold 1 input

Triangle wavefolder: when $|in0 * Drive|$ exceeds Threshold the excess is reflected back inward, repeatedly, so a loud signal folds onto itself and sprouts bright upper harmonics (Serge / Buchla west-coast timbre). Unlike a clipper it keeps adding partials with more Drive instead of flattening into a square.

Param	Range	Default	Unit
Drive	1 – 16	1	—
Threshold	0.1 – 1	1	—

Wrap 1 input

Phase-style wrap: folds in0 back into the [-Range, Range] window by wrapping (modulo) rather than clamping, so a value leaving one edge re-enters from the opposite edge. Ideal for keeping a phasor or free-running ramp bounded, or for hard sync-style discontinuities.

Param	Range	Default	Unit
Range	0.1 – 2	1	—

Modulo 1 input

Euclidean remainder: outputs in0 wrapped into [0, Mod), the non-negative remainder of $in0 / Mod$. Generate ramps from a rising counter, build polyrhythmic resets, or derive a phasor from accumulated CV. Differs from Wrap, which keeps a bipolar +/-Range window centred on zero.

Param	Range	Default	Unit
Mod	0.01 – 2	1	—

Compare 1 input

Schmitt comparator: outputs +1 (high) when in0 rises above Threshold + Hyst and -1 (low) when it falls below Threshold - Hyst, holding its state in between so noise near the edge cannot chatter. Turns an audio or control signal into a clean bipolar gate; Hyst at 0 is an instantaneous comparator.

Param	Range	Default	Unit
Threshold	-1 – 1	0	—
Hyst	0 – 0.5	0.02	—

Logic 2 inputs

Boolean gate combiner: treats in0 and in1 as logic highs when they exceed 0.5 and applies AND, OR, XOR, NAND, NOR, or XNOR (Op), emitting +1 for true and -1 for false. Build trigger conditioners, gate masks, and rhythmic combinations from clocks, gates, and square LFOs.

Param	Range	Default	Unit
Op	AND · OR · XOR · NAND · NOR · XNOR		—

Map 1 input

Affine range mapper: rescales in0 from its [In Lo, In Hi] range onto a [Out Lo, Out Hi] range, then optionally clamps to the output bounds. The general-purpose glue for matching signal ranges - convert a 0..1 envelope to a +/-1 CV, narrow an LFO's swing, or invert a range by setting Out Lo > Out Hi.

Param	Range	Default	Unit
In Lo	-2 - 2	-1	—
In Hi	-2 - 2	1	—
Out Lo	-2 - 2	0	—
Out Hi	-2 - 2	1	—
Clamp	Off · On		—

Mux 4 inputs

Four-way input selector: routes one of in0..in3 to the output based on the Sel index (plus the in-built Sel knob), so a stepped CV or sequencer can hard-switch between four sources. Where Select chooses between two inputs, Mux scales to four - a clean N-way router with no crossfade. The selector is floored to an integer and wrapped, so an out-of-range index stays valid.

Param	Range	Default	Unit
Sel	In 0 · In 1 · In 2 · In 3	—	—

Average 4 inputs

Mean of the four signal inputs, weighted only by how many are non-zero so an unpatched input does not drag the result toward zero. Use it to find the centre of several CVs, average detuned oscillators, or build a simple low-pass on a set of parallel signals. With one input patched it passes that signal through unchanged.

Window 1 input

Window comparator: outputs High when in0 sits inside the [Lo, Hi] band and Low when it is outside, the two-sided counterpart to Compare's single threshold. Detect when a CV enters a range, build a band-gate, or fire a trigger only while a modulator is within bounds. If Lo > Hi the bounds swap so the window is always valid.

Param	Range	Default	Unit
Lo	-1 - 1	-0.5	—
Hi	-1 - 1	0.5	—
High	-1 - 1	1	—
Low	-1 - 1	-1	—

Round 1 input

Integer rounding with a selectable Mode (nearest, floor, ceil, truncate) applied at a chosen Step size, so a smooth signal snaps to a grid of that spacing. Unlike Quantize (which maps onto a fixed number of levels across [-1,1]), Round works at an absolute step and direction - quantise a control voltage to semitone-sized steps, or floor a phasor to whole cycles.

Param	Range	Default	Unit
Step	0.01 – 1	0.1	—
Mode	Nearest · Floor · Ceil · Trunc		—

Smoothstep 1 input

Hermite smoothstep ramp: maps in0 across the [Edge0, Edge1] range through the classic $3t^2 - 2t^3$ S-curve (zero slope at both ends), clamped outside the edges. Where Power bends a curve around zero, Smoothstep eases between two thresholds - de-zipper a control, soften a gate into a fade, or shape a crossfade position so it accelerates and decelerates smoothly.

Param	Range	Default	Unit
Edge0	-1 – 1	0	—
Edge1	-1 – 1	1	—

Hypot 2 inputs

Vector magnitude: outputs $\sqrt{\text{in0}^2 + \text{in1}^2}$ scaled by Gain - the Euclidean length of the two inputs treated as X and Y. Combine two quadrature signals into one amplitude, derive an envelope from a stereo pair, or build a distance/radius control from two CVs. Always non-negative regardless of input sign.

Param	Range	Default	Unit
Gain	0 – 4	1	—

Frac 1 input

Fractional part: outputs in0 minus its floor, the [0, 1) remainder after the whole number is removed. A rising ramp becomes a repeating 0..1 sawtooth, so it is the quickest way to turn an accumulator or phasor into a wrapped phase - feed it a slow rising CV to generate a tempo ramp, or take the Frac of a scaled signal to build stepped repeats. Negative inputs still map to [0, 1).

Param	Range	Default	Unit
Scale	0 – 16	1	—

Atan2 2 inputs

Two-argument arctangent: outputs the angle of the (in0, in1) vector mapped to [-1, 1] ($= \text{atan2}(\text{in1}, \text{in0}) / \pi$). The phase counterpart to Hypot's magnitude - together they convert a quadrature (X/Y) pair to polar form. Derive a rotation angle from two CVs, recover phase from a sine/cosine pair, or build a circular panner control. Output is bipolar and wraps at the +/-1 edges.

Exp 1 input

Linear-to-exponential converter: outputs $2^{(\text{in0} * \text{Range})}$, turning a linear control voltage into the exponential frequency or time multiplier that pitch and filter cutoffs expect. in0 of 1 at Range 1 gives x2 (one octave up); 0 gives x1. The companion to Log - feed a linear envelope or LFO here so it tracks musical per-octave intervals instead of a flat sweep.

Param	Range	Default	Unit
Range	-8 – 8	1	—

Log 1 input

Exponential-to-linear converter: outputs $\log_2(\text{in0}) / \text{Range}$, the inverse of Exp - recover a linear control voltage from a frequency or time multiplier. in0 of 2 at Range 1 gives 1 (one octave). The input is floored at a tiny positive value so a zero or negative multiplier stays finite rather than diverging.

Param	Range	Default	Unit
Range	-8 - 8	1	—

Threshold 1 input

Unipolar threshold gate: outputs High when in0 is at or above Threshold and Low otherwise - a hard one-sided switch. Where Compare returns a bipolar +/-1 with hysteresis and Window tests a two-sided band, Threshold is the plain 'is it past this level' detector: turn an envelope or LFO into a clean 0/1 gate, or trigger an event when a signal crosses a level.

Param	Range	Default	Unit
Threshold	-1 - 1	0	—
High	-1 - 1	1	—
Low	-1 - 1	0	—

Skew 1 input

Asymmetric curve bender: warps in0 across the [0,1] range so its midpoint slides toward 0 or 1 by Skew, bending a linear ramp into an ease-in or ease-out while leaving the 0 and 1 endpoints fixed. Where Power is symmetric around zero, Skew pivots one-sided over the unit range - shape an envelope segment, bias an LFO ramp, or curve a crossfade. Skew at 0 passes the value through unchanged.

Param	Range	Default	Unit
Skew	-1 - 1	0	—

ScaleQuant 1 input

Musical scale / pitch quantizer: snaps in0 - read as a pitch in semitones (the synth's Note source, a sequencer, or any CV scaled so 1.0 = one semitone) - to the nearest note of the selected Scale rooted on Root. Where Quantize maps onto N evenly-spaced levels across [-1,1], ScaleQuant snaps to a musical key, so a random or smoothly-drifting CV driving an oscillator's pitch always lands in tune. Output is the quantised semitone value - patch it into an oscillator's pitch / Note input.

Param	Range	Default	Unit
Root	C · C# · D · D# · E · F · F# · G · G# · A · A# · B		—
Scale	Chromatic · Major · Minor · Dorian · Phrygian · Lydian · Mixolydian · Locrian · Min Penta · Maj Penta · Blues · Whole Tone		—

Abs 1 input

Absolute value / full-wave rectifier: outputs $|\text{in0}| * \text{Gain} + \text{Offset}$, so every sample is folded to the positive side. Where Sign keeps only the polarity and Negate flips it, Abs discards it - turn a bipolar LFO or audio signal into a unipolar envelope, derive a level from a waveform, or build a full-wave rectifier (Max against 0 is only half-wave).

Param	Range	Default	Unit
Gain	0 - 4	1	—
Offset	-2 - 2	0	—

AbsDiff 2 inputs

Unsigned difference: outputs $|in0 - in1| * Scale$, the distance between the two inputs on the number line regardless of which is larger. Where Subtract keeps the sign and Hypot is the 2-D vector length, AbsDiff is the 1-D magnitude of an error - a rectified side-chain, a change detector, or a symmetric closeness measure for two CVs. Both inputs must be patched.

Param	Range	Default	Unit
Scale	0 – 4	1	—

Equal 2 inputs

Tolerance comparator: outputs High when in0 sits within +/-Tol of (in1 plus Offset) and Low otherwise. Where Compare tests a rising/falling threshold crossing and Window tests a fixed band on one signal, Equal asks whether two moving signals currently match - detect a sequencer landing on a value, gate when two CVs coincide, or build a quantizer-hit indicator. With in1 unpatched it tests in0 against the constant Offset.

Param	Range	Default	Unit
Offset	-2 – 2	0	—
Tol	0 – 2	0.01	—
High	-1 – 1	1	—
Low	-1 – 1	0	—

Complement 1 input

Unit-range inverter: outputs $Center - in0 * Scale$, defaulting to $1 - in0$ so a normalised 0..1 control is flipped end-for-end. Where Negate mirrors a bipolar signal about zero, Complement mirrors a unipolar one about its Center - derive a dry amount from a wet 0..1 knob, invert an envelope's shape, or turn a rising 0..1 ramp into a falling one. Set Center to 2 with Scale 1 to reflect about 1.

Param	Range	Default	Unit
Center	-2 – 2	1	—
Scale	-2 – 2	1	—

Decibels 1 input

Decibel <-> linear-gain converter. In dB->Gain mode it outputs $10^{(in0 / 20)}$, turning a decibel value at in0 into the linear amplitude multiplier a VCA wants; in Gain->dB mode it outputs $20 * \log_{10}(|in0|)$, recovering the dB level of a linear gain (floored so a zero input stays finite). The result is limited to +/-Limit. Where Exp/Log work in per-octave powers of two, Decibels is the audio-level ($20 * \log_{10}$) law.

Param	Range	Default	Unit
Mode	dB->Gain · Gain->dB	—	—
Limit	1 – 256	64	—

Semitones 1 input

Semitone-to-ratio pitch converter: outputs $2^{((in0 * Range) / 12)}$, turning a semitone interval at in0 into the frequency multiplier that retunes an oscillator, FM ratio, delay time, or sample playback rate. in0 of 12 at Range 1 gives x2 (one octave); 7 gives a perfect fifth (~1.498). Range scales the incoming interval. The 12-tone-equal-temperament counterpart to Exp's per-octave power.

Param	Range	Default	Unit
Range	-24 – 24	1	—

AbsMax 2 inputs

Largest magnitude: outputs the greater of $|in0|$ and $|in1|$ times Gain - the loudest of the two inputs regardless of sign (the L-infinity norm). Where Max compares signed values, Hypot is the L2 vector length and AbsDiff is the gap between them, AbsMax is the peak of the pair - combine two level followers into a single peak meter, or take the worse-case of two error signals. Always non-negative.

Param	Range	Default	Unit
Gain	0 – 4	1	—

GeoMean 2 inputs

Geometric mean of two inputs: outputs $\text{sign}(in0 \cdot in1) \cdot \sqrt{|in0 \cdot in1|} \cdot \text{Scale}$. Where Average is the arithmetic mean (sum/2), the geometric mean is the multiplicative centre - it pulls toward the smaller magnitude and collapses to zero if either input is zero, which makes it a soft AND for two level/CV signals. The sign follows the product, so two negatives give a positive root.

Param	Range	Default	Unit
Scale	0 – 4	1	—

Deadband 1 input

Centre dead-zone: passes $in0$ straight through once it moves more than Width from zero, and outputs 0 inside the $\pm$ Width band. With Subtract on, the surviving signal is shifted back so it leaves the band continuously (no step) instead of jumping by Width. Unlike Sign's dead-zone (which emits ± 1) this keeps the signal itself - a noise gate for CV, a joystick centre detent, or a way to ignore small wobble on a control.

Param	Range	Default	Unit
Width	0 – 1	0.1	—
Subtract	Off · On		—

Gaussian 1 input

Bell-curve shaper: outputs $\exp(-((in0 - \text{Centre})^2) / (2 * \text{Width}^2))$, a smooth hump that peaks at 1 when $in0 = \text{Centre}$ and falls off symmetrically on both sides, never quite reaching 0. Width sets how broad the bell is. A ready-made windowing/proximity response - turn distance-from-a-target into a fade, build a resonant-looking CV peak, or weight a value by how close it sits to Centre. Unlike Smoothstep's one-sided ramp this is a two-sided peak.

Param	Range	Default	Unit
Centre	-2 – 2	0	—
Width	0.01 – 2	0.4	—

Quadratic 1 input

Second-order polynomial shaper: outputs $A \cdot in0^2 + B \cdot in0 + C$, a full parabola with adjustable curvature (A), slope (B) and offset (C). Where Power gives a single sign-preserving exponent, Quadratic mixes a squared term with a linear one for asymmetric curves, gentle pedal/expo CV responses, or a parabolic envelope segment. $A=0$ leaves a plain affine $B \cdot in0 + C$; $B=0$ is a pure square bowl.

Param	Range	Default	Unit
A	-4 – 4	1	—
B	-4 – 4	0	—
C	-2 – 2	0	—

Logistic 1 input

Logistic sigmoid: outputs $1 / (1 + \exp(-\text{Gain} * (\text{in0} - \text{Centre})))$, the classic S-curve that eases from 0 up to 1 with its steepest slope at Centre and Gain setting the sharpness. Unlike Smoothstep (a clamped Hermite that hits exactly 0 and 1 over a finite window) the logistic asymptotes and never fully reaches the rails, so it is the natural soft gate, compander curve, or probability mapping for an unbounded input. High Gain approaches a hard 0/1 switch.

Param	Range	Default	Unit
Gain	0.1 – 32	6	—
Centre	-2 – 2	0	—

Median3 3 inputs

Median of three inputs: outputs the middle value of in0, in1, in2, discarding the highest and lowest. Where Average is pulled by an outlier and Min/Max take the extreme, the median rejects a single spike entirely - de-glitch a noisy CV, pick the consensus of three sensors or LFOs, or smooth a control without the lag of a filter. Unconnected inputs read 0, so feed all three.

Spread 3 inputs

Dispersion of three inputs: outputs $(\max - \min)$ of in0, in1, in2 times Gain - how far apart the most extreme pair sits. Where AbsDiff is the gap between two signals, Spread is the full range across three, a one-number measure of how much a set of CVs disagree: drive a 'tightness' control, detect when voices diverge, or derive a width amount. Always non-negative; unconnected inputs read 0.

Param	Range	Default	Unit
Gain	0 – 4	1	—

Manhattan 2 inputs

Taxicab (L1) magnitude: outputs $(|\text{in0}| + |\text{in1}|) * \text{Gain}$, the sum of the two inputs' magnitudes. The third member of the norm family beside Hypot (the L2 straight-line length) and AbsMax (the L-infinity peak) - L1 grows fastest and is cheap, useful as a combined level/activity measure of two signals or an energy estimate. Always non-negative regardless of input sign.

Param	Range	Default	Unit
Gain	0 – 4	1	—

Sinh 1 input

Hyperbolic-sine shaper: warps in0 through a normalised $\sinh(\text{in0} * \text{Drive}) / \sinh(\text{Drive})$, an odd (sign-preserving) exponential curve that passes 0 and reaches +/-1 at +/-1 while bowing the middle. Low Drive is nearly linear; high Drive is steeply exponential near the rails. The symmetric, both-sides counterpart to Exp's one-sided per-octave curve - an expo response for bipolar CV or a soft expander curve.

Param	Range	Default	Unit
Drive	0.01 – 6	2	—

Cosh 1 input

Hyperbolic-cosine bowl: warps in0 through a normalised $(\cosh(\text{in0} * \text{Drive}) - 1) / (\cosh(\text{Drive}) - 1)$, an even (symmetric) U-shaped curve that is 0 at the centre and rises to 1 at +/-1 - the catenary/parabola-like counterpart to the odd Sinh. Turn a bipolar CV into a centre-dipped magnitude, build a smile/frown response, or weight by distance from zero with a steeper-than-square edge as Drive rises.

Param	Range	Default	Unit
Drive	0.01 – 6	2	—

Sinc 1 input

Normalised sinc: outputs $\sin(\pi * \text{in0} * \text{Scale}) / (\pi * \text{in0} * \text{Scale})$, the cardinal sine that is 1 at the origin and rings out in decaying side-lobes on either side, crossing zero at every integer of $\text{in0} * \text{Scale}$. The impulse response behind ideal interpolation and brick-wall filtering - use it as a ripply symmetric window, a decaying-oscillation CV shape, or a damped bipolar response around a centre.

Param	Range	Default	Unit
Scale	0 – 16	1	—

Atan 1 input

Arctangent shaper: outputs $(2/\pi) * \text{atan}(\text{Drive} * \text{in0})$, a smooth odd S-curve that asymptotes toward ± 1 without ever clipping, with Drive setting how hard it leans on the signal. The single-argument companion to Atan2 (which takes an X/Y pair) and a gentler always-on alternative to the threshold-based Clip - soft-saturate a CV, tame a hot modulator, or compress an audio signal with a rounded knee that never reaches the rail.

Param	Range	Default	Unit
Drive	0.1 – 32	4	—

Asinh 1 input

Inverse-hyperbolic-sine compander: outputs $\text{asinh}(\text{Drive} * \text{in0}) / \text{asinh}(\text{Drive})$, a normalised odd curve that is near-linear for small inputs and rolls into a logarithmic slope for large ones, reaching ± 1 at ± 1 . The bipolar, through-zero counterpart to Log (which is positive-only and diverges at 0) and the inverse of Sinh - a soft dynamics compander for CV or audio that keeps small detail while taming peaks, with no discontinuity at the origin.

Param	Range	Default	Unit
Drive	0.01 – 16	3	—

Atanh 1 input

Inverse-hyperbolic-tangent expander: clamps in0 to $\pm \text{Edge}$ then outputs $\text{atanh}(\text{in0}) / \text{atanh}(\text{Edge})$, an odd curve that is gentle near zero and steepens sharply toward the rails - the inverse of a tanh saturator, so it undoes soft clipping and exaggerates dynamics instead of taming them. Edge sets how close to 1 the input may get before the slope runs away (the clamp keeps it finite). Use it to add bite, expand a squashed envelope, or pre-distort before a saturator.

Param	Range	Default	Unit
Edge	0.5 – 0.999	0.95	—

ArgMax 4 inputs

Winner-take-all index: outputs which of $\text{in0}..\text{in3}$ is currently the largest, as the integer 0, 1, 2 or 3. Where Mux routes a chosen input by a selector, ArgMax generates that selector - patch its output into a Mux Sel to follow the loudest source, or use it to detect which of several envelopes/LFOs is on top. Ties resolve to the lowest index; unconnected inputs read 0.

RMS3 3 inputs

Root-mean-square of three inputs: outputs $\sqrt{(\text{in0}^2 + \text{in1}^2 + \text{in2}^2) / 3} * \text{Gain}$, the power-average level of the trio. Where Average is the arithmetic mean (which cancels when signals are out of phase) and Hypot is the unnormalised two-input vector length, RMS3 is the normalised energy of three signals - a combined loudness/activity estimate that ignores sign and does not null out. Always non-negative; unconnected inputs read 0.

Param	Range	Default	Unit
Gain	0 – 4	1	—

PhaseDiff 2 inputs

Wrapped phase distance: treats in0 and in1 as phases on a [-1, 1] cycle and outputs their shortest signed difference, folded back into [-1, 1] so a value near the +1/-1 seam still reads as a small gap instead of a near-2 jump. Where Subtract gives the raw (unwrapped) difference and AbsDiff its unsigned size, PhaseDiff respects the circular wrap - compare two LFO/oscillator phases, measure how far a phasor has drifted, or build a phase-lock error signal.

Asin 1 input

Arcsine shaper: clamps in0 to [-1, 1] then outputs $\text{asin}(\text{in0}) * 2/\pi$, an odd curve that is gentle through the centre and steepens sharply as it nears the rails - the circular cousin of Atanh (which steepens harder) and the inverse of a sine-shaped saturator. Use it to expand the mid-range of a normalised CV, add edge near the extremes, or pre-warp a value before a sine lookup. Input outside +/-1 is clamped.

Acos 1 input

Arccosine shaper: clamps in0 to [-1, 1] then outputs $\text{acos}(\text{in0}) / \pi$, a smooth monotone curve that falls from 1 (at in0 = -1) to 0 (at in0 = +1), passing 0.5 at the centre. Where Complement inverts a unit range linearly, Acos inverts it along a cosine arc - eased at both ends and steepest in the middle. A drop-in curved 'one minus' for a normalised control, or the falling partner to Asin's rising shape.

Erf 1 input

Error-function sigmoid: outputs $\text{erf}(\text{Drive} * \text{in0})$, the Gaussian-integral S-curve that eases through zero and saturates smoothly toward +/-1 as the input grows. Drive sets the steepness. It sits between Atan (a gentle, slowly-saturating knee) and a hard tanh clip - a distinct, statistically-flavoured soft saturator whose shoulder rolls off faster than the logistic. Soft-limit a CV or warm an audio signal with a rounded, never-clipping ceiling.

Param	Range	Default	Unit
Drive	0.1 – 8	2	—

Logit 1 input

Logit (inverse-logistic) expander: clamps in0 to [1-Edge, Edge] then outputs $\log(p / (1 - p))$ normalised so the Edge maps to +/-1. The exact inverse of Logistic - it takes a squashed 0..1 probability-like signal and stretches it back out, gentle near the 0.5 centre and steepening toward the 0/1 ends. Edge sets how close to 0/1 the input may get before the slope runs away (the clamp keeps it finite). Expand a compressed envelope or undo a logistic soft-knee.

Param	Range	Default	Unit
Edge	0.5 – 0.999	0.99	—

TriShape 1 input

Phasor-to-triangle shaper: treats in0 as a phase and outputs a triangle that rises 0 -> 1 over the first half of each cycle and falls 1 -> 0 over the second, repeating every 1/Freq of input. Where Frac turns a rising ramp into a sawtooth, TriShape turns it into a symmetric triangle - reshape a phasor or accumulator into a triangle LFO, or fold a linear sweep into an up-down contour without an oscillator. Freq multiplies the cycle count.

Param	Range	Default	Unit
Freq	0 – 16	1	—

Sum 4 inputs

Summing node: adds inputs in 1..4 into a single output with no per-input gain - the quick way to merge oscillators, parallel chains or control signals without wiring up a full Mixer.

Analyzer 2 inputs

Signal analyzer: passes in 1 straight through (out = in 1, transparent in the chain) while its detail panel shows a live 3D FFT waterfall + oscilloscope of in 1, and a Lissajous (X-Y) phase scope of in 1 against in 2. A measurement tool, not an effect - drop it inline to see the spectrum feeding a filter, the harmonics a distortion adds, what an envelope follower tracks, or the stereo/phase relationship of two signals.

ClockDiv 1 input

Clock divider: passes only every Div-th rising edge of the in0 trigger, turning a fast gate or clock into a slower one for polyrhythms and rhythmic gating. The output mirrors the input pulse on the pulses it lets through and is silent otherwise, and the count stays phase-locked to the input.

Param	Range	Default	Unit
Div	2 – 16	2	—

Counter 2 inputs

Pulse counter / staircase: each rising edge of the in0 trigger advances an internal count, and the output is that count scaled to 0..1 over Steps before wrapping to zero - a stepped CV ramp for sequences and rhythmic builds. A rising edge on in1 resets the count so it can be re-synced to a bar.

Param	Range	Default	Unit
Steps	2 – 32	8	—

TrigToGate 1 input

Trigger-to-gate: each rising edge of the in0 trigger opens a gate that stays high for a fixed Length, then closes - turning a momentary pulse (a clock tick, a Compare edge) into a sustained gate of predictable duration. Retriggering restarts the timer, so a stream of pulses keeps the gate open. Drive envelopes, VCAs or effects from short triggers.

Param	Range	Default	Unit
Length	1 – 1000	50	ms

Hysteresis 1 input

Schmitt trigger with memory: the output latches High once in0 rises above the upper threshold (Center + Width) and stays High until in0 falls below the lower threshold (Center - Width), then latches Low. Because it remembers its last state, a noisy or slowly-drifting signal around one level produces a clean debounced gate instead of chattering - unlike the stateless Compare, which has no memory.

Param	Range	Default	Unit
Center	-1 - 1	0	—
Width	0 - 1	0.1	—
High	-1 - 1	1	—
Low	-1 - 1	-1	—

PeakHold 2 inputs

Peak / hold detector: latches the extreme value in0 reaches and holds it, optionally bleeding back toward zero at Decay (Decay 0 = hold forever). Mode picks the maximum (Peak), the minimum (Trough), or the largest magnitude (Abs Peak). A rising edge on the in1 reset clears the held value to the current input. Where EnvFollow smooths the amplitude with attack/release, PeakHold captures the strict extreme - peak-metering CV, grabbing the loudest transient, or a max-tracking sample-and-hold.

Param	Range	Default	Unit
Decay	0 - 1	0	—
Mode	Peak · Trough · Abs Peak		—

Latch 2 inputs

Bistable latch / flip-flop: remembers a High/Low state across time. In Toggle mode each rising edge on in0 flips the state (a clock divide-by-two, or tap-to-toggle). In Set/Reset mode a rising edge on in0 sets the output High and a rising edge on in1 sets it Low (a one-bit memory cell driven by two triggers). Outputs High when set and Low when clear - the missing fundamental logic block alongside Compare, Logic and Hysteresis.

Param	Range	Default	Unit
Mode	Toggle · Set/Reset		—
High	-1 - 1	1	—
Low	-1 - 1	0	—

CrossTrig 1 input

Crossing detector: emits a one-sample High pulse each time in0 crosses the Level (zero by default), on the Rising, Falling or Both edge as selected. Turns an oscillator or LFO into a clock (a zero-crossing tick), derives a trigger from a gate's edge, or fires an event when a signal passes a threshold. Pairs with TrigToGate to stretch the pulse into a sustained gate.

Param	Range	Default	Unit
Level	-1 - 1	0	—
Edge	Rising · Falling · Both		—

Switch 4 inputs

Clocked sequential switch: routes one of the signal inputs in0..in2 to the output and advances to the next on every rising edge of the in3 clock - an addressed router (Doepfer A-151 style) that steps through your sources in time. Steps sets how many inputs are in the rotation (2 or 3). Where Mux selects an input by a manual index and Select chooses between two by level, Switch cycles automatically on a clock.

Param	Range	Default	Unit
Steps	2 – 3	3	—

Bernoulli 1 input

Bernoulli gate (coin-flip router): on each rising edge of the in0 trigger it flips a weighted coin and either passes that gate through or blocks it, with Probability setting the odds. A stream of clocks becomes a randomly thinned-out pattern - the building block for generative rhythms and stochastic sequencing. Unlike the deterministic ClockDiv, every pulse is an independent chance event; the decision latches until the next trigger.

Param	Range	Default	Unit
Probability	0 – 1	0.5	—

Burst 1 input

Burst generator: a rising edge on in0 fires a burst of Count evenly-spaced trigger pulses at Rate, then falls silent until the next input edge - turn one hit into a ratchet, a drum-roll trigger or a stutter clock. Count sets how many pulses and Rate their spacing; re-triggering mid-burst restarts the count. Patch its output into envelopes, sample-and-holds or sequencer clocks.

Param	Range	Default	Unit
Count	1 – 16	4	—
Rate	1 – 100	20	Hz

Integrator 2 inputs

Leaky integrator / accumulator (CV): sums in0 over time so a constant input ramps and a periodic input is smoothed and phase-shifted a quarter cycle. Rate scales the input before accumulation, Leak bleeds the accumulator toward zero (0 = a near-perfect integrator that holds, higher = a shorter-memory smoother), and a rising edge on in1 resets the sum to zero. The running output is safety-limited. Distinct from Slew (which rate-limits a signal) and Counter (which counts edges).

Param	Range	Default	Unit
Rate	-8 – 8	1	—
Leak	0 – 1	0.3	—

Slope 1 input

Differentiator / rate-of-change detector (CV): outputs how fast in0 is moving - positive while it rises, negative while it falls, zero when steady - so it reads the slope of an envelope, LFO or any control signal. Scale sets the sensitivity and Smooth low-passes the result to tame noise on fast signals. Pair it with Compare to fire a trigger on a rising or falling edge. The inverse complement of the Integrator.

Param	Range	Default	Unit
Scale	0 – 1	0.05	—
Smooth	0 – 1	0.2	—

TrackHold 2 inputs

Track-and-hold (Doepfer A-148): while the gate at in1 is high the output follows the in0 signal; when the gate goes low it freezes and holds the last value until the gate returns. Unlike Sample-and-hold (which grabs one instant on a trigger), it tracks continuously through the whole gate - use it to freeze a slow LFO or envelope, glide-and-hold pitch, or window a control signal. A core CV utility.

PhaseInvert 1 input

Flips the polarity of the signal, negating every sample when engaged and passing it untouched otherwise. Invert toggles the inversion on or off. Use it to correct out-of-phase tracks or to null one source against another.

Param	Range	Default	Unit
Invert	0 – 1	1	—

DCBlock 1 input

Removes DC offset and sub-sonic bias with a one-pole high-pass that subtracts the previous input and feeds back a 0.9995 coefficient, passing audio while draining any constant component. It takes no parameters and always runs at the same fixed corner. Place it after asymmetric distortion or rectification stages to keep the signal centred.

Entropy 1 input

Information-theoretic modulation source (a first as a synth block): outputs a running Shannon-entropy estimate of the input - near 0 for silence or a steady tone, rising toward 1 for noisy, unpredictable signals. It folds recent samples into a decaying 16-bin amplitude histogram and reports the normalised entropy of that distribution, answering 'how disordered is this signal right now?' as a control voltage. Window sets the memory (higher = slower, smoother). Patch it so an effect opens up only when the input turns chaotic.

Param	Range	Default	Unit
Window	0.001 – 1	0.05	—

Stochastic 1 input

Stochastic-resonance detector (a first as a synth block): the real phenomenon where adding the RIGHT amount of noise lets a weak, sub-threshold signal cross a detector it otherwise never could - too little and the signal stays buried, too much and it drowns. Noise sets the injected level, Threshold the detector, Smooth the output ballistic, and the output is the recovered crossing envelope. Sweep Noise to find the resonance sweet spot where a rhythm hidden below the threshold suddenly emerges.

Param	Range	Default	Unit
Threshold	0 – 1	0.5	—
Noise	0 – 1	0.3	—
Smooth	0.001 – 1	0.05	—

Surprise 1 input

Adaptive novelty detector (a first as a synth block): keeps a self-tuning one-tap linear predictor of the input and outputs the magnitude of its prediction error - a spike whenever the signal does something the recent past did not predict, near silence while it stays predictable. Unlike Slope (the raw derivative, which stays high through any steady ramp) Surprise LEARNS the trend and fires only on genuine novelty: a new note, a sudden jump, a broken pattern. Rate sets how fast it adapts, Smooth the output ballistic.

Param	Range	Default	Unit
Rate	0 – 1	0.3	—
Smooth	0.001 – 1	0.05	—

Correlate 2 inputs

Two-signal relatedness meter (a first as a synth block): outputs a running normalised cross-correlation of in0 against in1 - how strongly the two signals move together - as a control voltage. About +1 when they rise and fall in lock-step, 0 when independent, -1 when they mirror each other. Window sets the averaging time. Detect when two modulators line up, gate an effect on the agreement of two sources, or measure stereo/phase coherence. Both inputs must vary (a constant has no correlation).

Param	Range	Default	Unit
Window	0 – 1	0.02	—

Periodicity 1 input

Self-similarity detector (a first as a synth block): writes the input to a delay line and outputs the running normalised correlation between the signal now and the signal one Period ago - how strongly it repeats itself at that lag. About 1 for a steady tone or loop whose cycle matches Period, near 0 for noise or a one-shot, so it reports the STRENGTH of repetition rather than the pitch (set Period to the cycle you are listening for). Window sets the averaging time. Drive an effect by how locked-in a groove is.

Param	Range	Default	Unit
Period	0.1 – 100	10	ms
Window	0 – 1	0.02	—

Dither 1 input

Bit-depth reducer with triangular (TPDF) dither: builds a quantization step from Bits, adds triangular noise from two xorshift draws scaled by Amount, then rounds to the step. Bits sets the target word length from 8 to 24 and Amount scales the dither level. The dither decorrelates quantization error so low-level detail survives reduction instead of distorting.

Param	Range	Default	Unit
Bits	8 – 24	16	—
Amount	0 – 1	1	—

CVBridge 1 input

Control-voltage utility that conditions a single input value in one of three Mode settings: attenuvert (scale by Scale plus Offset), note-to-1V/oct pitch conversion, or bipolar-to-unipolar remap. Scale provides positive or negative gain including inversion, and Offset shifts the result up or down. Useful for routing and rescaling modulation or pitch control between modules.

Param	Range	Default	Unit
Mode	0 – 2	0	—
Scale	-2 – 2	1	—
Offset	-1 – 1	0	—

Meter 1 input

Analysis meter that tracks the running mean-square of the input and outputs its RMS level as a scalar control value. Smooth sets the one-pole averaging coefficient, with higher values giving slower, steadier readings and lower values reacting faster to transients. Used to derive a level signal for metering or to drive other modules from program loudness.

Param	Range	Default	Unit
Smooth	0 – 0.999	0.99	—

NullTest 2 inputs

Null tester that subtracts the second input from the first and outputs the residual difference, scaled by Gain. When two signals are identical the output is silence, and any audible residual reveals the difference between them. Gain amplifies that residual so small mismatches become easy to hear, making this a tool for verifying bit-for-bit equivalence between processing chains.

Param	Range	Default	Unit
Gain	0 – 4	1	—

EQ

3 modules

Baxandall 1 input

Baxandall negative-feedback tone network: the classic hi-fi Bass/Treble - a gentle low shelf (~100 Hz) and high shelf (~10 kHz) that boost or cut smoothly and symmetrically, with none of the midrange scoop of a passive stack. Bass and Treble in dB.

Param	Range	Default	Unit
Bass	-15 – 15	0	dB
Treble	-15 – 15	0	dB

MassivePassive 1 input

Manley Massive Passive-style EQ: broad, gentle passive-network shelf and bell curves with wide, interactive Q - boosts feel airy and musical rather than surgical, the reason it's a mastering staple. Low shelf, a wide high bell at HiFreq, and an Air shelf on top, each in dB.

Param	Range	Default	Unit
Low	-11 – 11	0	dB
High	-11 – 11	0	dB
HiFreq	2000 – 16000	8000	Hz
Air	0 – 8	0	dB

Sontec 1 input

Sontec MEP-250-style mastering parametric EQ: clean Class-A bell bands with a musical, selectable Q - precise without sounding clinical, the mastering standard. Two parametric bells (Lo/Hi freq + gain) sharing the Q control.

Param	Range	Default	Unit
LoFreq	40 – 1200	200	Hz
LoGain	-10 – 10	0	dB
HiFreq	1500 – 18000	6000	Hz
HiGain	-10 – 10	0	dB
Q	0.4 – 3	1	—

Effect

7 modules

SynthDelay 3 inputs

Delay line with feedback and a modulation input.

Param	Range	Default	Unit
Time	1 – 2000	250	ms
Feedback	0 – 0.95	0.35	—
Mod	0 – 500	0	ms

SynthRingMod 2 inputs

Ring and amplitude modulation of two inputs.

Param	Range	Default	Unit
Amount	0 – 1	1	—

SynthDrive 1 input

Saturation/overdrive with output trim.

Param	Range	Default	Unit
Drive	0 – 40	12	dB
Output	-24 – 6	-6	dB

SynthWaveshaper 1 input

Waveshaper: tanh/arctan/polynomial/sine transfer curves.

Param	Range	Default	Unit
Drive	1 – 20	2	—
Shape	0 – 3	0	—
Mix	0 – 1	1	—

Crusher 1 input

Bit-depth reduction plus downsampling (bitcrush).

Param	Range	Default	Unit
Bits	1 – 16	8	—
Downsample	1 – 64	1	—
Mix	0 – 1	1	—

Folder 1 input

Wavefolder: west-coast sine folding with bias.

Param	Range	Default	Unit
Fold	1 – 12	2	—
Bias	-1 – 1	0	—
Mix	0 – 1	1	—

Lockhart 1 input

Lockhart-style transistor wavefolder (west-coast): the signal reflects off the rails through several folding stages, multiplying harmonics as Fold rises - the Buchla/Serge timbre. 4x-oversampled.

Param	Range	Default	Unit
Fold	1 – 16	3	—
Symmetry	-1 – 1	0	—
Mix	0 – 1	1	—

Envelope

1 modules

Env 1 input

DAHDSR envelope generator triggered by the gate input. Delay waits before the attack; Hold sustains the peak before the decay (both 0 = classic ADSR).

Param	Range	Default	Unit
Attack	0 – 5	0.005	s
Decay	0 – 5	0.2	s
Sustain	0 – 1	0.7	—
Release	0 – 5	0.3	s
Hold	0 – 5	0	s
Delay	0 – 5	0	s

Sequencing

184 modules

Automaton 2 inputs

Elementary cellular-automaton sequencer (a first as a synth block): a 16-cell row of one-bit cells evolves one generation on every rising edge of the in0 clock, each cell's next state set by the 8-bit Rule applied to its left/centre/right neighbours (Wolfram's elementary CA, wrapped). Rule 90 paints a fractal Sierpinski pattern, 30 is chaotic, 110 is Turing-complete. The output is the live cell density (0..1 stepped CV) - a deterministic yet endlessly evolving pattern, unlike the random shift-register of Turing or Marbles. A rising edge on in1 reseeds a single centre cell.

Param	Range	Default	Unit
Rule	0 – 255	90	—

Markov 2 inputs

Style-learning sequencer (a first as a synth block): quantises the input at in0 into 4 states and continuously learns the transition probabilities between them (a Markov chain; Memory sets how fast old transitions fade). On each rising edge of the in1 clock it samples the learned transitions from its current state to pick the next, improvising a NEW sequence in the style of what it heard - feed it a melody or CV and it riffs endless variations that obey the same move-to-move statistics. Output is the generated state as a 0..1 stepped CV.

Param	Range	Default	Unit
Memory	0.9 – 1	0.999	—

GameOfLife 2 inputs

Conway's Game of Life sequencer (a first as a synth block): a 32-cell torus grid (8 wide x 4 tall) evolves one Conway generation on every rising edge of the in0 clock - a live cell with 2 or 3 neighbours survives, a dead cell with exactly 3 is born (the classic B3/S23) - and the output is the live population density as a 0..1 stepped CV. Where the 1-D Automaton runs a row rule, this is full 2-D Life with its gliders, blinkers and still-lives: the population breathes, oscillates or settles into stable colonies. A rising edge on in1 reseeds the grid; Seed sets the random fill. Genuinely alive modulation.

Param	Range	Default	Unit
Seed	0 – 1	0.35	—

BrianBrain 2 inputs

Brian's Brain cellular automaton (a first as a synth block): a 4x4 torus grid of three-state cells - ready, firing, dying - evolves on every rising edge of the in0 clock. A ready cell ignites only when exactly two neighbours are firing; a firing cell always passes to dying; a dying cell falls back to ready. Where Conway's Game of Life has two states, this refractory third state keeps structures restless - they almost never settle, throwing off gliders and sparks. Output is the firing-cell density as a 0..1 stepped CV. A rising edge on in1 reseeds; Seed sets the random fill.

Param	Range	Default	Unit
Seed	0 – 1	0.3	—

Langton 2 inputs

Langton's Ant turmite sequencer (a first as a synth block): a single 'ant' walks an 8x4 torus of cells - turning right on a white cell, left on a black one, flipping each cell it leaves, then stepping forward. From a blank grid it scribbles chaotically for a while, then abruptly locks into a periodic 'highway' - emergent order from one trivial rule. Each rising edge of the in0 clock advances the ant Steps cells; the output is the ant's column position as a 0..1 stepped CV. A rising edge on in1 resets the ant to the centre of a clear grid. Unlike the cellular automata this is a single moving agent, not a population.

Param	Range	Default	Unit
Steps	1 – 16	1	—

Collatz 2 inputs

Collatz-conjecture sequencer (a first as a synth block): walks the famous $3n+1$ trajectory - on each rising edge of the in0 clock the number halves if even or becomes $3n+1$ if odd, and the output is its altitude ($\log_2$ of the value, 0..1) so you hear the trajectory's jagged climbs and plunges. When it reaches 1 the tour restarts from Start. A rising edge on in1 also restarts. Different Start values give wildly different melodic contours from one deterministic rule.

Param	Range	Default	Unit
Start	1 – 2000	27	—

ThueMorse 2 inputs

Thue-Morse sequencer (a first as a synth block): outputs the Thue-Morse sequence - the parity of the number of 1-bits in a step counter - as a 0/1 gate that advances on every rising edge of the in0 clock (0,1,1,0,1,0,0,1...). It is the canonical fair-share, cube-free, self-similar binary sequence: never periodic yet never random, full of nested symmetry. Use it as an aperiodic rhythm, a generative on/off mask, or a melodic gate that feels structured but never repeats. A rising edge on in1 resets the counter.

RudinShapiro 2 inputs

Rudin-Shapiro sequencer (a first as a synth block): outputs the Rudin-Shapiro sequence as a 0/1 gate - high when the number of overlapping '11' pairs in the step counter's binary is even (the +1 term), low when odd. Its defining property is remarkable: it is fully deterministic yet its power spectrum is FLAT, like white noise - the +/-1 sequence whose partial sums stay bounded by $\sqrt{n}$, the closest a structured pattern gets to acoustically random. A rhythm that sounds scattered and unpredictable but never actually repeats, distinct from the nested symmetry of Thue-Morse. A rising edge on in1 resets the counter.

Cantor 2 inputs

Cantor-set rhythm gate (a first as a synth block): the step counter is read in base 3 and the output goes high only when NO ternary digit is 1 - exactly the membership rule of the Cantor middle-thirds set. The result is a self-similar fractal on/off pattern: dense clusters of hits separated by silences of every scale, the same gappy structure repeating inside itself at 1/3, 1/9, 1/27... A deeply uneven yet deterministic rhythm with built-in fractal phrasing, unlike the even fair-share of Thue-Morse. A rising edge on in1 resets the counter.

FibWord 2 inputs

Fibonacci-word sequencer (a first as a synth block): the infinite Fibonacci word over {0,1} grown by the substitution $0 \rightarrow 01, 1 \rightarrow 0$ (its own fixed point) - 0,1,0,0,1,0,1,0,0,1,0,0,1... - emitted as a 0/1 gate, one symbol per in0 clock. It is the canonical Sturmian sequence: the 1s fall at the golden-ratio spacing, the quasicrystal of one dimension, balanced so any two windows of equal length hold the same count of 1s to within one. A structured aperiodic gate built on phi, distinct from the self-describing Kolakoski. A rising edge on in1 resets; the first 256 symbols then loop.

Champernowne 2 inputs

Champernowne-constant sequencer (a first as a synth block): walks the decimal digits of Champernowne's constant 0.123456789101112131415... - simply the counting numbers written out and concatenated - emitting one digit per in0 clock as a 0..1 stepped melody (digit / 9). Because the constant is provably NORMAL, the digit stream eventually contains every finite pattern, every motif, infinitely often: it climbs 1..9 then folds into the rolling two- and three-digit runs of 10,11,12..., a melody that is rigidly deterministic yet endlessly inventive. Distinct from the integer sequences - this plays the digits themselves. A rising edge on in1 resets.

Pascal 2 inputs

Pascal's-triangle sequencer (a first as a synth block): reads Pascal's triangle row by row - 1; 1,1; 1,2,1; 1,3,3,1... - taking each binomial coefficient modulo Base and emitting it as a 0..1 stepped value, one per in0 clock. Modulo 2 the pattern IS the Sierpinski triangle (Lucas' theorem): a self-similar fractal of 1s and 0s; higher Base reveals the richer nested structure of binomials mod a number. A fractal melody/gate from the most elementary combinatorics, distinct from the cellular-automaton fractals. A rising edge on in1 resets to the top of the triangle.

Param	Range	Default	Unit
Base	2 – 8	2	—

Prime 2 inputs

Prime-step gate (a first as a synth block): a step counter advances on each rising edge of the in0 clock and the output goes high only when the count is a prime number (2,3,5,7,11,13...). Because primes thin out as they grow, the gate fires densely at first then ever more sparsely and irregularly - a deterministic yet non-repeating rhythm with no built-in period. Use it to trigger events, mask a sequence, or gate an effect on the primes. A rising edge on in1 resets the counter.

Happy 2 inputs

Happy-number gate (a first as a synth block): a step counter advances on each in0 clock and the output goes high when the count is a happy number - one that reaches 1 when you repeatedly replace it by the sum of the squares of its digits (7 -> 49 -> 97 -> 130 -> 10 -> 1). The unhappy numbers instead fall forever into the cycle 4,16,37,58,89,145,42,20. The happy numbers are scattered with no pattern (1,7,10,13,19,23,28,31,32,44...), so the gate fires on a sparse, irregular, deterministic rhythm distinct from the primes. A rising edge on in1 resets.

Harshad 2 inputs

Harshad-number gate (a first as a synth block): a step counter advances on each in0 clock and the output goes high when the count is a Harshad (Niven) number - one divisible by the sum of its own digits (18 is, since $1+8=9$ divides 18; 19 is not). Harshad numbers are common among small values then thin out gradually, so the gate fires densely at first and slowly sparsens - a denser, more regular cousin of the prime gate, with little runs and gaps tied to the decimal digits. A rising edge on in1 resets.

Abundant 2 inputs

Abundant-number gate (a first as a synth block): a step counter advances on each in0 clock and the output goes high when the count is an abundant number - one whose proper divisors sum to MORE than itself (12: $1+2+3+4+6 = 16 > 12$). These are the 'over-rich' integers, the opposite of the deficient majority, with the perfect numbers sitting exactly on the boundary. Abundant numbers cluster around the highly-composite values, so the gate fires in factor-rich bursts - a number-theoretic rhythm tied to divisor structure, distinct from the digit-based Harshad. A rising edge on in1 resets.

Lucky 2 inputs

Lucky-number gate (a first as a synth block): the output goes high when the step count is a 'lucky number' (1,3,7,9,13,15,21,25,31...) - survivors of a sieve like the prime sieve but which crosses out by POSITION instead of by value: keep every number, delete every 2nd, then from the survivors delete every 3rd, then every 7th, and so on. Astonishingly the luckies share many statistical traits with the primes (twin luckies, a similar density) despite being built from counting alone, not divisibility. A different deterministic sparse rhythm from the prime gate. A rising edge on in1 resets; the first 256-step phrase then loops.

DigitSum 2 inputs

Digit-sum sequencer (a first as a synth block): each in0 clock advances the step n and outputs the sum of n 's digits in base Base, folded into Steps as a 0..1 melody. As n counts up, the digit sum climbs steadily then drops sharply at every carry (9 -> 10, 99 -> 100), tracing a self-similar sawtooth whose teeth get longer in higher place values - the fingerprint of positional notation itself. Base reshapes the pattern (base 2 gives the popcount/binary-weight contour, base 10 the familiar decimal roll). A digit-structure melody distinct from the value-based recurrences. A rising edge on in1 resets.

Param	Range	Default	Unit
Base	2 – 16	10	—
Steps	4 – 32	12	—

DigitReverse 2 inputs

Digit-reversal sequencer (a first as a synth block): each in0 clock advances the step n and outputs n with its decimal digits written backwards (123 -> 321, 250 -> 52), folded into Steps. The reversal scrambles the steady count into a jagged, self-similar permutation - the ones digit becomes the loudest place, so the melody leaps by big jumps within each block of ten and resets at each new power of ten. A digit-permutation contour distinct from the digit-SUM and the value-based sequences. A rising edge on in1 resets.

Param	Range	Default	Unit
Steps	4 – 32	16	—

DigitProduct 2 inputs

Digit-product sequencer (a first as a synth block): each in0 clock advances the step n and outputs the product of n 's decimal digits (24 -> $2*4 = 8$, 39 -> 27), folded into Steps. Unlike the steadily-climbing digit SUM, the product swings wildly and collapses to ZERO whenever any digit is 0 (every tenth number, all the hundreds...), punching rhythmic holes into the melody. A multiplicative digit contour with its own jagged, gap-ridden shape distinct from the digit sum and reversal. A rising edge on in1 resets.

Param	Range	Default	Unit
Steps	4 – 32	16	—

MultRoot 2 inputs

Multiplicative-digital-root sequencer (a first as a synth block): each in0 clock advances the step n and repeatedly replaces it by the product of its digits until a single digit is left - the multiplicative digital root (39 -> 27 -> 14 -> 4). The output is that final digit over 9. Because any number containing a 0 (or that ever reaches one) collapses to 0, the melody is dotted with zeros among the 1..9 values, a sparse fingerprint of the multiplicative structure of each integer. Distinct from the additive digital root, which simply cycles 1..9. A rising edge on in1 resets.

Kaprekar 2 inputs

Kaprekar-routine sequencer (a first as a synth block): each in0 clock advances the step n and counts how many rounds of Kaprekar's routine its four-digit form takes to reach 6174 - the magic Kaprekar constant. One round sorts the digits up and down and subtracts the smaller from the larger (e.g. 3524 -> 5432 - 2345 = 3087 -> ... -> 6174); EVERY four-digit number with at least two different digits funnels to 6174 in at most seven rounds, while repeated-digit numbers fall to zero. The output is that round count over 7, a strange staircase rising and falling with no obvious pattern. A rising edge on in1 resets.

Keith 2 inputs

Keith-number gate (a first as a synth block): the output goes high when the step n is a Keith number (14, 19, 28, 47, 61, 75, 197...). To test n , seed a Fibonacci-like sequence with its digits - then each new term is the sum of the previous (digit-count) terms; n is a Keith number if that sequence eventually lands exactly on n . They are far rarer than the primes (only a hundred or so below a trillion), so the gate fires on a very sparse, irregular pulse - a deep, hard-to-find numerical rhythm distinct from the prime, happy and lucky gates. A rising edge on in1 resets.

Totient 2 inputs

Euler-totient sequencer (a first as a synth block): each rising edge of the in0 clock advances the step n and outputs $\phi(n)/n$ - the fraction of integers up to n that share no factor with it. The contour is jagged and number-aware: it sits high near primes ($\phi(p)/p = 1 - 1/p$, almost 1) and plunges at the highly-composite, factor-rich numbers (6, 30, 210...), tracing the multiplicative texture of the integers as a melody. A deterministic line whose ups and downs encode coprimality itself, unlike the additive Stern or the word sequences. A rising edge on in1 resets.

Mobius 2 inputs

Mobius-function source (a first as a synth block): each rising edge of the in0 clock advances the step n and outputs the Mobius function $\mu(n)$ as a three-level CV - high for a squarefree number with an even count of prime factors (+1), centre for a non-squarefree number (0, it has a repeated prime), low for squarefree with an odd count (-1). The squarefree/squareful flicker and the parity swings produce a jittery three-state pattern that encodes the prime-factor structure of each integer. A number-theoretic trit stream found nowhere else in modular gear. A rising edge on in1 resets.

Liouville 2 inputs

Liouville-function gate (a first as a synth block): each rising edge of the in0 clock advances the step n and the output goes high when $\lambda(n) = +1$, low when $-1 - \lambda$ being (-1) raised to the TOTAL number of prime factors of n counted with multiplicity. It is the multiplicative cousin of Thue-Morse (which counts bits, this counts prime factors): a deeply structured, conjecturally-balanced two-level pattern tied to the Riemann hypothesis through its running sum. An aperiodic gate built from the deepest texture of the primes. A rising edge on in1 resets.

Divisors 2 inputs

Divisor-count sequencer (a first as a synth block): each rising edge of the in0 clock advances the step n and outputs $d(n)$, the number of divisors of n , folded into Steps as a 0..1 melody. Primes drop to the floor ($d = 2$) while the highly-composite numbers (12, 24, 36, 48, 60...) leap up with many divisors, so the line is a spiky staircase that spotlights the most factor-rich integers - the arithmetic opposite of the Totient dip. A melody driven by raw divisibility, distinct from the additive and word sequences. A rising edge on in1 resets.

Param	Range	Default	Unit
Steps	4 – 32	12	—

Zeckendorf 2 inputs

Zeckendorf-weight sequencer (a first as a synth block): every positive integer has a UNIQUE representation as a sum of non-consecutive Fibonacci numbers (Zeckendorf's theorem); each rising edge of the in0 clock advances the step n and outputs how many Fibonacci terms that representation needs, folded into Steps. The weight rises and falls as the greedy Fibonacci-base digits shuffle - the 'Fibonacci-base digit sum', a self-similar contour related to the golden ratio and the Fibonacci word. A melody from base-phi numeration, distinct from the decimal Champernowne. A rising edge on in1 resets.

Param	Range	Default	Unit
Steps	4 – 32	12	—

Partition 2 inputs

Partition-number sequencer (a first as a synth block): outputs $p(n)$, the number of ways to write n as a sum of positive integers ($4 = 4, 3+1, 2+2, 2+1+1, 1+1+1+1$ so $p(4)=5$), taken modulo Base, one per in0 clock. The partition numbers - 1,1,2,3,5,7,11,15,22,30,42... - grow almost as fast as the primes thin out (Hardy-Ramanujan), and are built here by Euler's pentagonal-number recurrence (additions and subtractions only), so mod Base is exact. A melody from the deep additive combinatorics of the integers, distinct from the multiplicative divisor functions. A rising edge on in1 resets.

Param	Range	Default	Unit
Base	2 – 12	7	—

Sigma 2 inputs

Divisor-sum sequencer (a first as a synth block): each in0 clock advances the step n and outputs $\sigma(n)$, the sum of ALL of n 's divisors ($\sigma(6) = 1+2+3+6 = 12$), folded into Steps as a 0..1 melody. Where the Totient counts what is coprime to n and Divisors counts how many divisors it has, Sigma adds them all up - so it climbs highest at the abundant, factor-rich numbers and sits low ($\sigma(p) = p+1$) at the primes. The perfect numbers are exactly where $\sigma(n) = 2n$. A melody from the total weight of divisibility, distinct from the other arithmetic functions. A rising edge on in1 resets.

Param	Range	Default	Unit
Steps	4 – 32	16	—

Mertens 2 inputs

Mertens-function source (a first as a synth block): each in0 clock advances the step n and outputs the running sum of the Mobius function, $M(n) = \mu(1) + \mu(2) + \dots + \mu(n)$ - a famous slowly-staggering walk that creeps up and down by ± 1 and 0 as squarefree numbers of even and odd prime-count alternate. Whether it stays bounded by $\sqrt{n}$ (the disproved Mertens conjecture) is tied to the Riemann hypothesis. The output is centred on 0.5, drifting either side like a number-theoretic Brownian path that is in fact fully deterministic. Distinct from the per-step Mobius block. A rising edge on in1 resets.

PrimePi 2 inputs

Prime-counting sequencer (a first as a synth block): each in0 clock advances the step n and outputs $\pi(n)$, the number of primes less than or equal to n , folded into Steps. Unlike the prime GATE (which only fires on the primes themselves) this plays the running TALLY - a monotone staircase that steps up by one at every prime and holds flat across the composite runs, the function at the heart of the Prime Number Theorem ($\pi(n) \sim n / \ln n$). The widening flat steps trace how the primes thin out. A counting melody distinct from the prime gate. A rising edge on in1 resets.

Param	Range	Default	Unit
Steps	4 – 32	16	—

Triangular 2 inputs

Triangular-number gate (a first as a synth block): the output goes high when the step n is a triangular number - one that counts the dots in a filled triangle, 0,1,3,6,10,15,21,28,36,45... ($n(n+1)/2$). The test is the classic one: n is triangular exactly when $8n+1$ is a perfect square. Because the gaps between successive triangular numbers grow by one each time, the gate fires on a steadily-rarefying rhythm - dense at first, then ever-longer silences - a smoothly-decelerating pulse distinct from the irregular prime and happy-number gates. A rising edge on in1 resets.

Square 2 inputs

Perfect-square gate (a first as a synth block): the output goes high when the step n is a perfect square - 0, 1, 4, 9, 16, 25, 36... The gaps between squares grow by the odd numbers (1, 3, 5, 7...), so the gate fires on a smoothly and quadratically decelerating rhythm: a quick flurry at the start that stretches into ever-longer silences faster than the triangular gate. A perfect-square pulse distinct from the triangular and prime gates. A rising edge on in1 resets.

Pentagonal 2 inputs

Pentagonal-number gate (a first as a synth block): the output goes high when the step n is a pentagonal number - 1, 5, 12, 22, 35, 51, 70... ($k(3k-1)/2$), the dots that tile a growing pentagon. These are the very numbers in Euler's pentagonal-number theorem that drive the partition function, and n is one exactly when $24n+1$ is a perfect square whose root is one less than a multiple of 6. The gate fires on a decelerating but sparser rhythm than the squares or triangulars. A rising edge on in1 resets.

Pronic 2 inputs

Pronic-number gate (a first as a synth block): the output goes high when the step n is a pronic (oblong) number - the product of two consecutive integers, 0, 2, 6, 12, 20, 30, 42... ($k(k+1)$), the count of dots in a rectangle one taller than it is wide, and exactly twice each triangular number. n is pronic when $4n+1$ is a perfect square. The gate fires on a decelerating rhythm interleaved exactly halfway between the squares. A rising edge on in1 resets.

Fibonacci 2 inputs

Fibonacci-number gate (a first as a synth block): the output goes high when the step n is itself a Fibonacci number - 0, 1, 2, 3, 5, 8, 13, 21, 34, 55... Because each Fibonacci number is roughly the golden ratio times the last, the gaps grow GEOMETRICALLY, so the gate fires fast and densely at first then thins out exponentially - a self-similar, golden-spaced rhythm quite unlike the polynomial slow-down of the square and triangular gates. The test is Gessel's: n is Fibonacci exactly when $5n^2+4$ or $5n^2-4$ is a perfect square. A rising edge on in1 resets.

Power2 2 inputs

Power-of-two gate (a first as a synth block): the output goes high only when the step n is an exact power of two - 1, 2, 4, 8, 16, 32, 64... Each gap is double the last, so the gate fires on the sparsest possible self-similar rhythm: it hits, waits twice as long, hits, waits twice as long again, forever halving in density. The test is the classic bit trick $(n \text{ AND } n-1) == 0$. The most extreme decelerando of the number gates, ideal for octave-spaced accents and bar-doubling structure. A rising edge on in1 resets.

Beatty 2 inputs

Golden-ratio (Beatty) rhythm gate (a first as a synth block): fires high when the clock step lands on $\text{floor}(n * \text{Ratio})$ for $n = 1, 2, 3...$ With Ratio at the golden mean (~ 1.618) the hits fall on 1, 3, 4, 6, 8, 9... - a quasi-periodic pattern that never settles into a loop, the irrational cousin of the Euclidean rhythm. By Beatty's theorem two such sequences with complementary ratios partition every beat exactly once. Use it for organic, non-repeating grooves. A rising edge on in1 resets.

Param	Range	Default	Unit
Ratio	1.05 - 4	1.618	—

GrayCode 2 inputs

Gray-code sequencer (a first as a synth block): advances a counter on each rising edge of the in0 clock and outputs its reflected binary Gray code ($n \text{ XOR } n \gg 1$), folded into 16 steps - so successive values always differ by exactly one bit. The result is a stepped CV that walks 0, 1, 3, 2, 6, 7, 5, 4... visiting levels in an order where every move is a single minimal change, giving smooth one-step contours that still tour the whole range without repeating until the cycle completes. A rising edge on in1 resets.

VanDerCorput 2 inputs

Van der Corput low-discrepancy sequencer (a first as a synth block): outputs the radical-inverse sequence - reverse the bits of the step counter and read them as a fraction - so each new value lands in the largest remaining gap (0.5, 0.25, 0.75, 0.125, 0.625...). The result spreads evenly across the range yet never repeats a spacing, the quasi-random-but-balanced pattern used in quasi-Monte-Carlo sampling. Advance on the in0 clock; a rising edge on in1 resets. A more even, less clumpy alternative to a random source.

DeBruijn 2 inputs

De Bruijn sequence gate (a first as a synth block): plays a cyclic binary sequence in which every possible 5-bit pattern appears exactly once before the 32-step cycle repeats - the shortest sequence that tours all sub-patterns. Advancing on the in0 clock it fires a dense, structured, maximally-varied rhythm that never presents the same five-step window twice within a bar. Where Euclid spreads pulses evenly and ThueMorse is self-similar, the de Bruijn cycle is exhaustive. A rising edge on in1 resets to the start.

Pisano 2 inputs

Fibonacci-modulo sequencer (a first as a synth block): runs the Fibonacci recurrence modulo M, so the output is (...0,1,1,2,3,5,8,13...) wrapped into M levels - a melodic loop whose length is the Pisano period of M (often surprisingly long and lopsided). Advance on the in0 clock; Modulus sets the wrap and therefore the loop length and contour. Deterministic and exactly repeating yet rarely an obvious cycle - the additive cousin of the multiplicative sequencers. A rising edge on in1 resets to 0,1.

Param	Range	Default	Unit
Modulus	2 – 32	8	—

Wireworld 2 inputs

Wireworld cellular automaton (a first as a synth block): a 4x4 grid of cells in four states - empty, electron head, electron tail, conductor - evolves on each rising edge of the in0 clock. A head becomes a tail, a tail becomes a conductor, and a conductor becomes a head only if exactly one or two neighbours are heads, so electrons stream along the conductor wires like current in a circuit. The output is the electron-head density as a stepped CV - sparks chasing each other around closed loops. Seed sets the initial spark density; a rising edge on in1 reseeds.

Param	Range	Default	Unit
Seed	0 – 1	0.2	—

TentMap 2 inputs

Tent-map chaos sequencer (a first as a synth block): iterates $x' = R * \min(x, 1-x)$, the simplest piecewise-linear chaotic map - a symmetric tent that stretches and folds the unit interval. At R near 2 it is fully chaotic, producing a stepped CV that never repeats yet stays evenly spread across the range (unlike the lopsided logistic Chaos). Each rising edge of the in0 clock takes one iteration; R below 2 tames it toward order. A rising edge on in1 reseeds. The angular, uniform cousin of the smooth attractors.

Param	Range	Default	Unit
R	1 – 2	1.99	—

Sandpile 2 inputs

Abelian sandpile (a first as a synth block): drops a grain on the centre of a 4x4 pile each clock; when any cell reaches four grains it topples, sending one to each neighbour and possibly starting a chain reaction off the open edges. The output is the avalanche size - how many topplings that grain triggered - which is silent most of the time then spikes, following the power-law statistics of self-organised criticality. Most hits do nothing; occasionally one grain unleashes a cascade. A model of avalanches, forest fires and earthquakes as a rhythmic CV. A rising edge on in1 clears the pile.

Dragon 2 inputs

Dragon-curve (paperfolding) sequencer (a first as a synth block): outputs the regular paperfolding sequence - the fold directions of the famous dragon-curve fractal - as a 0/1 gate that advances on the in0 clock (1,1,0,1,1,0,0,1...). Strip the trailing zeros from the step counter and the next bit decides the turn, giving a sequence that is perfectly self-similar (it contains scaled copies of itself) yet never periodic. Between Euclid's even spread and ThueMorse's parity, the dragon fold is the fractal one. A rising edge on in1 resets.

Ikeda 2 inputs

Ikeda-map chaos sequencer (a first as a synth block): iterates the Ikeda map - a model of light circulating in a nonlinear optical cavity - whose state rotates by an angle that itself depends on the state, folding the plane into an intricate strange attractor. On each rising edge of the in0 clock it advances one step and outputs the x coordinate as a stepped CV that hops chaotically around the attractor's hooked arms. U sets how deep into chaos it runs. A different flavour again from the angular Henon, smooth Rossler or tumbling pendulum. A rising edge on in1 reseeds.

Param	Range	Default	Unit
U	0.7 – 0.92	0.9	—

CyclicCA 2 inputs

Cyclic cellular automaton (a first as a synth block): every cell holds one of N colours arranged in a cycle, and a cell advances to the next colour if at least Threshold neighbours already wear it - rock beats scissors beats paper beats rock. From noise it self-organises into rotating spiral waves that endlessly consume one another. Each rising edge of the in0 clock steps one generation; the output is the grid's mean colour as a smoothly cycling CV. A demolition-derby of colours quite unlike the Game-of-Life family. A rising edge on in1 reseeds.

Param	Range	Default	Unit
States	3 – 6	4	—
Threshold	1 – 3	1	—

BakSneppen 2 inputs

Bak-Sneppen evolution source (a first as a synth block): sixteen species sit in a ring, each with a random fitness; every clock the least-fit species - and its two neighbours - are wiped out and replaced with new random fitnesses. Driving out the weakest drags the whole ecosystem upward in fits and starts: long calm stretches of slow improvement shattered by sudden cascades of extinction, the punctuated equilibrium of self-organised criticality. The output is the current minimum fitness, climbing toward a critical threshold then collapsing. A rising edge on in1 reseeds.

Stern 2 inputs

Stern diatomic sequencer (a first as a synth block): outputs Stern's diatomic sequence (the fusc function) - 0,1,1,2,1,3,2,3,1,4,3,5... - in which consecutive pairs enumerate every positive rational number exactly once, the arithmetic behind the Stern-Brocot tree. Each rising edge of the in0 clock advances one term, folded into 16 steps; the contour rises and falls in a self-similar fractal staircase that never repeats. A deterministic, number-theoretic melody distinct from the Fibonacci-based Pisano. A rising edge on in1 resets.

Tribonacci 2 inputs

Tribonacci-word sequencer (a first as a synth block): the three-symbol cousin of the Fibonacci word, grown by the substitution a -> a b, b -> a c, c -> a (its own fixed point) and emitted as a three-level CV (0, 0.5, 1), one symbol per in0 clock. Where the Fibonacci word is built on the golden ratio, this is built on the tribonacci constant ~1.839 - a balanced, aperiodic THREE-state pattern that never repeats and tiles the line like a 1-D quasicrystal. A richer alphabet than the binary words, distinct from the Fibonacci word and Kolakoski. A rising edge on in1 resets; the first 256 symbols then loop.

Padovan 2 inputs

Padovan-sequence sequencer (a first as a synth block): the plastic-number analogue of Fibonacci - $P(n) = P(n-2) + P(n-3)$, giving 1,1,1,2,2,3,4,5,7,9,12,16,21... - whose ratio of successive terms tends to the plastic number ~1.3247 instead of the golden ratio. Each rising edge of the in0 clock advances one term, taken modulo Steps so it stays in range. Its slower, gentler growth makes a melody that climbs more lazily than Fibonacci, distinct from the Pisano (Fibonacci mod m) and Perrin recurrences. A rising edge on in1 resets.

Param	Range	Default	Unit
Steps	4 – 32	16	—

Perrin 2 inputs

Perrin-sequence sequencer (a first as a synth block): the recurrence $P(n) = P(n-2) + P(n-3)$ seeded 3,0,2 - giving 3,0,2,3,2,5,5,7,10,12,17,22,29... - famous for the Perrin pseudoprime test (n divides P(n) almost exactly when n is prime). Each rising edge of the in0 clock advances one term, taken modulo Steps. It shares Padovan's plastic-number growth but a different opening, so its melody starts on a falling

figure (3,0,2) before climbing - a distinct number-theoretic line from Padovan or the Fibonacci-based Pisano. A rising edge on in1 resets.

Param	Range	Default	Unit
Steps	4 – 32	16	—

Bell 2 inputs

Bell-number sequencer (a first as a synth block): outputs the Bell numbers - 1,1,2,5,15,52,203,877... - the count of ways to partition a set of n elements, taken modulo Base, one per in0 clock. They are built here by Aitken's array (pure additions, the analogue of Pascal's triangle for set partitions), so taking them mod Base is exact. The values explode combinatorially, so mod Base they scramble into a busy, evenly-spread pattern - a melody from the deep counting numbers of combinatorics, distinct from the additive recurrences. A rising edge on in1 resets to the top.

Param	Range	Default	Unit
Base	2 – 12	7	—

GoldenAngle 2 inputs

Golden-angle (phyllotaxis) sequencer (a first as a synth block): each in0 clock adds the golden ratio's fractional part (~0.618) to a running position wrapped into 0..1 - the additive recurrence that places sunflower seeds and pinecone scales at the 137.5-degree golden angle, the most irrational rotation there is. The result is a maximally even, low-discrepancy melody: successive notes land as far from all previous notes as possible, filling the range without ever repeating or clustering. The optimal spread, distinct from the bit-reversal Van der Corput. A rising edge on in1 resets to the start.

Jacobsthal 2 inputs

Jacobsthal-sequence sequencer (a first as a synth block): the recurrence $J(n) = J(n-1) + 2J(n-2)$ - giving 0,1,1,3,5,11,21,43,85,171... - a Fibonacci cousin whose doubled second term makes successive ratios tend to 2 instead of the golden ratio. Each rising edge of the in0 clock advances one term, taken modulo Steps. Its faster, near-doubling growth gives a melody that leaps up the scale more aggressively than Fibonacci, a distinct integer-recurrence line from the Pisano, Padovan and Pell sequencers. A rising edge on in1 resets.

Param	Range	Default	Unit
Steps	4 – 32	16	—

Pell 2 inputs

Pell-sequence sequencer (a first as a synth block): the recurrence $P(n) = 2P(n-1) + P(n-2)$ - giving 0,1,2,5,12,29,70,169,408... - whose ratios converge on the SILVER ratio ($1 + \sqrt{2} \sim 2.414$), the silver-mean analogue of Fibonacci's golden. The Pell numbers also give the best rational approximations to $\sqrt{2}$ (the side-and-diagonal numbers). Each rising edge of the in0 clock advances one term, taken modulo Steps - a faster-climbing, silver-ratio melody distinct from the golden Fibonacci/Pisano and the plastic Padovan. A rising edge on in1 resets.

Param	Range	Default	Unit
Steps	4 – 32	16	—

Narayana 2 inputs

Narayana's-cows sequencer (a first as a synth block): the 14th-century recurrence $N(n) = N(n-1) + N(n-3)$ - giving 1,1,1,2,3,4,6,9,13,19,28... - from the Indian problem of how a herd grows if each cow bears a calf every year from its fourth year on. Its ratios converge on the SUPERGOLDEN ratio (~1.4656). Each rising edge of the in0 clock advances one term, taken modulo Steps. Its three-step memory gives a gentler, more loping climb than Fibonacci, distinct from the Padovan ($N(n-2)+N(n-3)$) it is often confused with. A rising edge on in1 resets.

Param	Range	Default	Unit
Steps	4 – 32	16	—

Catalan 2 inputs

Catalan-number sequencer (a first as a synth block): outputs the Catalan numbers - 1,1,2,5,14,42,132,429,1430... - taken modulo Base, one per in0 clock. They count an astonishing range of things (the ways to balance n pairs of brackets, to triangulate a polygon, the paths that never cross the diagonal) and are built here by the convolution $C(n) = \sum C(i)C(n-1-i)$, so taking them mod Base is exact. They explode faster than 2^n , so mod Base they scramble into a busy pattern - a melody from the most ubiquitous numbers in combinatorics, distinct from the Bell and Motzkin counts. A rising edge on in1 resets.

Param	Range	Default	Unit
Base	2 – 12	7	—

Motzkin 2 inputs

Motzkin-number sequencer (a first as a synth block): outputs the Motzkin numbers - 1,1,2,4,9,21,51,127,323... - taken modulo Base, one per in0 clock. They count the ways to draw non-crossing chords between points on a circle (the chords-and-non-edges cousin of the bracket-counting Catalan), and are built here by the convolution $M(n) = M(n-1) + \sum M(i)M(n-2-i)$, so mod Base is exact. A combinatorial melody with its own growth rate ($\sim 3^n$), distinct from the Catalan and Bell counts. A rising edge on in1 resets.

Param	Range	Default	Unit
Base	2 – 12	7	—

Ruler 2 inputs

Ruler-sequence sequencer (a first as a synth block): outputs the 2-adic valuation of the step n - the number of times 2 divides it, equivalently the count of trailing zeros in its binary - folded into Steps, one per in0 clock. The result is exactly the tick pattern on a ruler: 0,1,0,2,0,1,0,3,0,1,0,2... where every other mark is short, every fourth is taller, every eighth taller still, forever self-similar. A perfectly nested fractal staircase that drives rhythmic accents and melodic emphasis, distinct from the digit-sum and popcount contours. A rising edge on in1 resets.

Param	Range	Default	Unit
Steps	4 – 32	16	—

Popcount 2 inputs

Popcount sequencer (a first as a synth block): outputs the Hamming weight of the step n - the number of 1-bits in its binary representation - folded into Steps, one per in0 clock. As n counts up the bit-count rises and falls in a self-similar fractal pattern (0,1,1,2,1,2,2,3...), climbing on dense numbers and dropping to 1 at every power of two. Where Thue-Morse keeps only the PARITY of this count, Popcount plays the full value - a binary-weight melody distinct from the digit-sum and ruler contours. A rising edge on in1 resets.

Param	Range	Default	Unit
Steps	4 – 32	16	—

Gould 2 inputs

Gould-sequence sequencer (a first as a synth block): outputs Gould's sequence - 2 raised to the number of 1-bits of the step n - folded into Steps, one per in0 clock. By Kummer's theorem this is exactly the count of ODD entries in row n of Pascal's triangle, so plotting it draws the Sierpinski-triangle skyline: it leaps to a new power of two on the dense numbers and collapses to 1 at the powers of two. A doubling, self-similar contour distinct from the plain popcount it is built on. A rising edge on in1 resets.

Param	Range	Default	Unit
Steps	4 – 32	16	—

BaumSweet 2 inputs

Baum-Sweet gate (a first as a synth block): the output goes high when the step n is a Baum-Sweet number - one whose binary representation contains NO block of consecutive 0s of odd length ($n = 0$ counts as high by convention). It is one of the classic automatic sequences (computable by a tiny finite automaton reading n 's bits) yet its pattern of 1s and 0s looks irregular and unpredictable - 1,1,0,1,1,0,0,1,0,1... A deterministic, self-similar gate built purely from the shape of the zero-runs in binary, distinct from the bit-COUNT sequences. A rising edge on in1 resets.

Fibbinary 2 inputs

Fibbinary gate (a first as a synth block): the output goes high when the step n is a fibbinary number - one whose binary representation has NO two adjacent 1-bits (0,1,2,4,5,8,9,10,16...). These are exactly the numbers expressible as a sum of non-consecutive powers of two, the base-2 mirror of Zeckendorf's non-consecutive Fibonacci sums. The gate fires in a sparse, self-similar pattern whose density follows the golden ratio, distinct from the prime and happy-number gates. A rising edge on in1 resets.

Recaman 2 inputs

Recaman's-sequence sequencer (a first as a synth block): walks the famous OEIS A005132 path $a(n) = a(n-1) - n$ when that step lands on a positive integer not already visited, else $a(n-1) + n$ - the jump-back-or-leap-forward self-avoiding walk that is renowned for sounding musical (the Numberphile melody). Each rising edge of the in0 clock advances one term, folded into Steps to keep it in range; the contour darts down and springs up with none of the symmetry of a periodic pattern. Distinct from the monotone hailstone of Collatz. A rising edge on in1 resets to the 256-term phrase start.

Param	Range	Default	Unit
Steps	4 – 32	16	—

Kolakoski 2 inputs

Kolakoski self-describing sequencer (a first as a synth block): the unique sequence over $\{1,2\}$ that is its OWN run-length encoding - 1,2,2,1,1,2,1,2,2,1,2,2... - so reading off the lengths of its own runs reproduces it exactly. Each rising edge of the in0 clock emits the next symbol as a two-level CV (1 low, 2 high): a quasi-periodic gate pattern that never settles into a loop and has no underlying period, aperiodic yet fully deterministic. A rhythmic counterpart to the number-theoretic pitch sequences. A rising edge on in1 resets; the first 256 symbols are generated, then the phrase repeats.

Hofstadter 2 inputs

Hofstadter Q-sequence source (a first as a synth block): the chaotic meta-Fibonacci recurrence $Q(n) = Q(n - Q(n-1)) + Q(n - Q(n-2))$ with $Q(1)=Q(2)=1$ - but where Fibonacci looks a fixed distance back, this looks back by its OWN recent values, so it never settles and wanders erratically around $n/2$. Each rising edge of the in0 clock advances one term; the output is $Q(n)/n$, a turbulent drift hovering near 0.5 with unpredictable excursions - deterministic chaos from pure integer recursion, distinct from the smooth Fibonacci-period Pisano. A rising edge on in1 resets; the first 512 terms then loop.

HofstadterG 2 inputs

Hofstadter G-sequence sequencer (a first as a synth block): the self-referential recurrence $G(n) = n - G(G(n-1))$ with $G(0)=0$, which folds the integers down by feeding the sequence's own output back into itself TWICE. Despite the tangled recursion it grows smoothly at the golden-ratio rate ($G(n)$ is almost exactly $\text{floor}(n/\phi)$), so it makes a gently-climbing staircase with an occasional flat step where the recursion catches up. Each in0 clock advances one term, folded into Steps. A self-referential melody distinct from the chaotic Q-sequence. A rising edge on in1 resets.

Param	Range	Default	Unit
Steps	4 – 32	16	—

HofstadterH 2 inputs

Hofstadter H-sequence sequencer (a first as a synth block): the recurrence $H(n) = n - H(H(n-1))$ with $H(0)=0$ - the same self-referential idea as the G-sequence but nesting the sequence inside itself THREE deep, so it grows at the 'plastic number' rate ($\sim n$ times 0.6823) rather than the golden ratio. The result is a smoothly-climbing staircase that rises a touch slower than G, with its own pattern of flat treads. Each in0 clock advances one term, folded into Steps. A deeper self-recursion distinct from the G-sequence. A rising edge on in1 resets.

Param	Range	Default	Unit
Steps	4 – 32	16	—

Conway 2 inputs

Conway-Hofstadter sequencer (a first as a synth block): the recurrence $a(n) = a(a(n-1)) + a(n - a(n-1))$ with $a(1)=a(2)=1$ - Conway's '\$10000 sequence', so named for the prize he offered to anyone who could tame its wild early behaviour. The output is $a(n)/n$, which lurches chaotically near the start, swinging widely above and below 0.5, then (as Mallows proved) gradually settles toward exactly 0.5 - a turbulence that calms as it goes. Each in0 clock advances one term. A famous self-referential drift distinct from the Hofstadter Q-sequence. A rising edge on in1 resets; the first 512 terms then loop.

Golomb 2 inputs

Golomb self-describing sequencer (a first as a synth block): outputs Golomb's sequence - 1,2,2,3,3,4,4,4,5,5,5... - the unique non-decreasing sequence that DESCRIBES ITSELF: the number n appears in it exactly $a(n)$ times (1 appears once, 2 appears twice, 3 appears three times, and so on). It climbs in ever-longer plateaus whose lengths are given by the sequence's own earlier values. Each in0 clock advances one term, folded into Steps, giving a gently-rising staircase of stretching steps. A self-describing melody distinct from the self-referential Hofstadter recurrences. A rising edge on in1 resets.

Param	Range	Default	Unit
Steps	4 – 32	16	—

FemaleMale 2 inputs

Hofstadter Female-Male sequencer (a first as a synth block): outputs the 'Female' half of Hofstadter's famous pair of INTERTWINED sequences, $F(n) = n - M(F(n-1))$ and $M(n) = n - F(M(n-1))$ - two sequences each defined in terms of the OTHER, an elegant mutual recursion. F grows smoothly (close to the golden-ratio staircase) while leaning on M at every step. Each in0 clock advances one term, folded into Steps. A mutually-recursive melody distinct from the single self-referential Hofstadter sequences. A rising edge on in1 resets.

Param	Range	Default	Unit
Steps	4 – 32	16	—

Tetranacci 2 inputs

Tetranacci sequencer (a first as a synth block): the four-step Fibonacci, $T(n) = T(n-1) + T(n-2) + T(n-3) + T(n-4)$ - 0,0,0,1,1,2,4,8,15,29,56... - each term the sum of the FOUR before it instead of two, so its ratio of successive terms climbs toward ~ 1.9276 rather than the golden 1.618. Each in0 clock advances one term, taken modulo Steps. Its faster four-term growth gives a melody that accelerates up the scale more steeply than Fibonacci or Tribonacci. A higher-order recurrence distinct from the Pisano and Padovan sequencers. A rising edge on in1 resets.

Param	Range	Default	Unit
Steps	4 – 32	16	—

Lucas 2 inputs

Lucas-number sequencer (a first as a synth block): the companion of the Fibonacci sequence - same add-the-last-two rule but seeded 2, 1 instead of 0, 1, giving 2,1,3,4,7,11,18,29,47,76... Its ratios converge on the very same golden ratio, so it shares Fibonacci's growth while landing on different values (the Lucas numbers turn up in primality testing and the closed form for Fibonacci itself). Each in0 clock advances one term, taken modulo Steps - a golden-growth melody with its own distinct contour. A rising edge on in1 resets.

Param	Range	Default	Unit
Steps	4 – 32	16	—

Schroeder 2 inputs

Schroeder-number sequencer (a first as a synth block): outputs the large Schroeder numbers - 1,2,6,22,90,394,1806... - which count the lattice paths from one corner of a grid to the opposite one using east, north and diagonal steps that never cross the diagonal (and a dozen other things in combinatorics). They are built here by the convolution $S(n) = S(n-1) + \sum S(k)S(n-1-k)$, so taking them modulo Base is exact. Growing faster than 5^n , mod Base they scramble into a busy pattern. A lattice-path melody distinct from the Catalan and Motzkin counts. A rising edge on in1 resets.

Param	Range	Default	Unit
Base	2 – 12	7	—

Delannoy 2 inputs

Delannoy-number sequencer (a first as a synth block): outputs the central Delannoy numbers - 1,3,13,63,321,1683... - which count the paths a KING takes across a chessboard from one corner to the other, moving only right, up or diagonally. They are built here by the lattice recurrence $D(i,j) = D(i-1,j) + D(i,j-1) + D(i-1,j-1)$ read down the diagonal, so taking them modulo Base is exact (pure additions). A king-path melody from two-dimensional lattice combinatorics, distinct from the one-dimensional recurrences. A rising edge on in1 resets.

Param	Range	Default	Unit
Base	2 – 12	7	—

CentralBinomial 2 inputs

Central-binomial sequencer (a first as a synth block): outputs the central binomial coefficients $C(2n, n)$ - 1,2,6,20,70,252,924... - the biggest number in each EVEN row of Pascal's triangle (the count of equal-length up/down paths, the heart of the random walk). They are built here straight from Pascal's additive triangle, so modulo Base is exact, and they grow like 4^n so mod Base they scramble busily. The combinatorial spine of Pascal's triangle as a melody, distinct from the Catalan numbers that are derived from them. A rising edge on in1 resets.

Param	Range	Default	Unit
Base	2 – 12	7	—

Stirling2 2 inputs

Stirling-second-kind sequencer (a first as a synth block): reads Stirling's triangle of the SECOND kind row by row - $S(n,k)$, the number of ways to split n objects into exactly k non-empty groups (1; 1,1; 1,3,1; 1,7,6,1...) - taken modulo Base. Built by the recurrence $S(n,k) = S(n-1,k-1) + k*S(n-1,k)$, so mod Base is exact. Each row sums to a Bell number, so this is the partition-counting

triangle behind the Bell sequence. A set-partition triangle melody distinct from the Pascal and Catalan triangles. A rising edge on in1 resets.

Param	Range	Default	Unit
Base	2 – 12	7	—

Stirling1 2 inputs

Stirling-first-kind sequencer (a first as a synth block): reads the UNSIGNED Stirling triangle of the FIRST kind row by row - the number of permutations of n items having exactly k disjoint cycles (1; 1,1; 2,3,1; 6,11,6,1...) - modulo Base. Built by $S(n,k) = S(n-1,k-1) + (n-1)*S(n-1,k)$, so mod Base is exact. Where the second kind counts set partitions, the first kind counts permutation cycles, and each row sums to a factorial. A cycle-counting triangle melody distinct from the second-kind and Pascal triangles. A rising edge on in1 resets.

Param	Range	Default	Unit
Base	2 – 12	7	—

Eulerian 2 inputs

Eulerian-number sequencer (a first as a synth block): reads the triangle of Eulerian numbers row by row - $A(n,k)$, the number of permutations of n items with exactly k ascents (1; 1,1; 1,4,1; 1,11,11,1; 1,26,66,26,1...) - modulo Base. Built by $A(n,k) = (k+1)*A(n-1,k) + (n-k)*A(n-1,k-1)$, so mod Base is exact. The rows are symmetric and sum to a factorial, and they are the coefficients that connect powers to binomial sums. A permutation-ascent triangle melody distinct from the Stirling and Pascal triangles. A rising edge on in1 resets.

Param	Range	Default	Unit
Base	2 – 12	7	—

Lah 2 inputs

Lah-number sequencer (a first as a synth block): reads the triangle of unsigned Lah numbers row by row - $L(n,k)$, the number of ways to partition n items into k non-empty ORDERED lists (1; 2,1; 6,6,1; 24,36,12,1...) - modulo Base. Built by $L(n,k) = (n+k-1)*L(n-1,k) + L(n-1,k-1)$, so mod Base is exact. The Lah numbers are the 'Stirling numbers of the third kind' that convert rising factorials into falling ones. An ordered-partition triangle melody distinct from the two Stirling triangles. A rising edge on in1 resets.

Param	Range	Default	Unit
Base	2 – 12	7	—

Ballot 2 inputs

Ballot-triangle sequencer (a first as a synth block): reads Catalan's ballot triangle row by row - the number of ways a count can stay ahead in an election where each entry is the sum of the one to its left and the one above (1; 1,1; 1,2,2; 1,3,5,5; 1,4,9,14,14...) - modulo Base. Built by $C(n,k) = C(n,k-1) + C(n-1,k)$, so mod Base is exact, and its right edge IS the Catalan numbers. The combinatorics of never-falling-behind paths as a melody, distinct from the Pascal and Stirling triangles. A rising edge on in1 resets.

Param	Range	Default	Unit
Base	2 – 12	7	—

PrimeOmega 2 inputs

Prime-factor-count sequencer (a first as a synth block): each in0 clock advances the step n and outputs big-Omega(n) - the number of prime factors of n counted WITH multiplicity (so $12 = 2^2 \cdot 3$ gives 3) - folded into Steps. It rises to a local peak at the powers and products of small primes and drops to 1 at every prime, climbing on average like $\log\text{-}\log n$. The total weight of a number's prime factorisation as a melody, distinct from the parity-only Liouville and the divisor-counting Divisors. A rising edge on in1 resets.

Param	Range	Default	Unit
Steps	4 – 32	12	—

SmallOmega 2 inputs

Distinct-prime-factor sequencer (a first as a synth block): each in0 clock advances the step n and outputs little-omega(n) - the number of DISTINCT prime factors of n (so $12 = 2^2 \cdot 3$ gives just 2, since only 2 and 3 are distinct) - folded into Steps. It stays at 1 across all the prime powers and rises only when a number draws in a NEW prime, so it climbs very slowly and spotlights the squarefree, many-prime numbers. The count of distinct primes as a melody, distinct from the with-multiplicity PrimeOmega. A rising edge on in1 resets.

Param	Range	Default	Unit
Steps	4 – 32	12	—

Radical 2 inputs

Radical sequencer (a first as a synth block): each in0 clock advances the step n and outputs the RADICAL of n - the product of its distinct prime factors, with all repeats stripped out (so $12 = 2^2 \cdot 3$ has radical $2 \cdot 3 = 6$, and $8 = 2^3$ has radical just 2) - folded into Steps. The radical equals n itself for the squarefree numbers and drops far below it for the prime-power-heavy ones, so the melody dips sharply at every perfect power. The squarefree kernel of a number as a melody, central to the abc conjecture, distinct from the additive divisor functions. A rising edge on in1 resets.

Param	Range	Default	Unit
Steps	4 – 32	16	—

Jordan 2 inputs

Jordan-totient sequencer (a first as a synth block): each in0 clock advances the step n and outputs $J_2(n)/n^2$, where J_2 is Jordan's totient - a generalisation of Euler's totient that counts pairs coprime to n rather than single numbers, given by n^2 times the product of $(1 - 1/p^2)$ over the primes p dividing n . The output rides high (near 1) at the primes and dips at the factor-rich numbers, a SHARPER version of the totient curve because each prime cuts by $1/p^2$. A second-order coprimality melody distinct from the ordinary Totient. A rising edge on in1 resets.

Dedekind 2 inputs

Dedekind-psi sequencer (a first as a synth block): each in0 clock advances the step n and outputs $\psi(n)/n - 1$, where ψ is the Dedekind psi function, n times the product of $(1 + 1/p)$ over the primes p dividing n (the count of points on a certain modular curve). It is the mirror image of the totient: where the totient REMOVES a fraction for each prime, ψ ADDS one, so the output sits near zero at the primes and rises at the factor-rich numbers - the inverse contour of the Jordan and Totient curves. A multiplicative-growth melody distinct from the totient-family dips. A rising edge on in1 resets.

CollatzSteps 2 inputs

Collatz-stopping-time sequencer (a first as a synth block): each in0 clock advances the step n and outputs how many steps n takes to reach 1 under the Collatz rule (halve it if even, triple-plus-one if odd) - the total stopping time, folded into Steps. Whether EVERY number reaches 1 is the famous unsolved Collatz conjecture; the step counts jump around wildly with no pattern ($n=27$ takes 111 steps). Where the Collatz block plays the trajectory itself, this plays its LENGTH - a chaotic-looking melody from an unproven theorem. A rising edge on in1 resets.

Param	Range	Default	Unit
Steps	4 – 32	16	—

CollatzPeak 2 inputs

Collatz-peak sequencer (a first as a synth block): each in0 clock advances the step n and outputs the HIGHEST value the Collatz trajectory of n ever climbs to before falling to 1, folded into Steps. Some starting numbers shoot up to enormous heights before collapsing (27 soars past 9000 on its way down), and the peak heights leap about unpredictably from one n to the next. Where CollatzSteps counts the journey's LENGTH, this measures its greatest ALTITUDE - a different chaotic contour from the same famous iteration. A rising edge on in1 resets.

Param	Range	Default	Unit
Steps	4 – 32	16	—

LookSay 2 inputs

Look-and-say sequencer (a first as a synth block): each in0 clock advances one term of Conway's look-and-say sequence - 1, 11, 21, 1211, 111221, 312211... - where each term is read aloud to make the next ('one 1' -> 11, 'two 1s' -> 21, 'one 2 one 1' -> 1211). The output is the LENGTH of each term, folded into Steps; those lengths grow by Conway's constant (~1.303685) each step, so the melody climbs in a fixed geometric ratio - an 'audioactive' sequence whose only digits are ever 1, 2 and 3. A self-describing melody distinct from the Kolakoski self-description. A rising edge on in1 resets; the first 16 terms then loop.

Param	Range	Default	Unit
Steps	4 – 32	16	—

Persistence 2 inputs

Multiplicative-persistence sequencer (a first as a synth block): each in0 clock advances the step n and outputs its multiplicative persistence - how many times you must replace the number by the product of its digits before reaching a single digit (39 -> 27 -> 14 -> 4 takes three steps, so its persistence is 3). Astonishingly, no number below 10^{233} is known to need more than 11 steps, an unexplained ceiling. Most numbers collapse in one or two steps, so the output is a low, spiky pattern that occasionally jumps. The DEPTH of digit-product collapse, distinct from the MultRoot block that plays its endpoint. A rising edge on in1 resets.

Aliquot 2 inputs

Aliquot-sequence sequencer (a first as a synth block): each in0 clock advances the step n and outputs how many steps its aliquot sequence runs before it terminates - the sequence where each term is the sum of the PROPER divisors of the last (12 -> 16 -> 15 -> 9 -> 4 -> 3 -> 1). Most chains fall to 1, but the PERFECT numbers (6, 28...) are fixed points that never move, the AMICABLE pairs cycle forever, and some chains climb without any known end - one of number theory's great open questions. The output is the chain length (capped), spiking at the perfect and untouchable numbers. A divisor-iteration melody distinct from the single-step Sigma. A rising edge on in1 resets.

Param	Range	Default	Unit
Steps	4 – 32	16	—

Palindrome 2 inputs

Palindrome gate (a first as a synth block): the output goes high when the step n reads the same backwards as forwards in decimal (1, 2, ..., 9, 11, 22, ..., 99, 101, 111, 121...). The palindromes thin out in a self-similar way - dense among the small numbers, then clustering near each new power of ten - so the gate fires in rhythmic bursts that stretch apart as the count climbs. A mirror-symmetry digit rhythm distinct from the divisibility and prime gates. A rising edge on in1 resets.

Automorphic 2 inputs

Automorphic-number gate (a first as a synth block): the output goes high when the step n is automorphic - its SQUARE ends in n itself (5 squared is 25, 6 squared is 36, 25 squared is 625, 76 squared is 5776...). These 'curious' self-reproducing numbers are extraordinarily rare - only a handful below each power of ten - so the gate fires on a very sparse, irregular pulse. A self-squaring digit rhythm distinct from the figurate and prime gates. A rising edge on in1 resets.

Smith 2 inputs

Smith-number gate (a first as a synth block): the output goes high when the step n is a Smith number - a composite whose digits sum to the SAME total as the digits of all its prime factors added up ($4 = 2*2$ has digit sum 4 and factor-digit sum $2+2 = 4$; $22 = 2*11$ gives 4 and $2+1+1 = 4$). Named after a phone number that turned out to have this property, they are scattered irregularly among the integers, so the gate fires on a sparse, unpredictable rhythm tied to the deep link between a number's digits and its factorisation. A rising edge on in1 resets.

Repdigit 2 inputs

Repdigit gate (a first as a synth block): the output goes high when every digit of the step n is the SAME (1, 2, ..., 9, 11, 22, ..., 99, 111, 222, ..., 1111...). These monodigit numbers (the repunits 1, 11, 111... among them) are extremely sparse - only nine in each block of digit-length - so the gate fires in tight little clusters of nine right after each power of ten, then falls silent for a long stretch. The most regular of the rare digit gates, distinct from the palindrome and prime gates. A rising edge on in1 resets.

Undulating 2 inputs

Undulating-number gate (a first as a synth block): the output goes high when the step n has digits that strictly ALTERNATE between two different values in an a-b-a-b-a pattern (101, 121, 131..., 212, 232..., 12121...). These wavy numbers ripple back and forth between two digits, and they are sparse and clustered, so the gate fires in little flurries near each band of three-, then four-, then five-digit numbers. A zig-zag digit rhythm distinct from the palindrome and repdigit gates. A rising edge on in1 resets.

Euler 2 inputs

Euler zigzag sequencer (a first as a synth block): steps through the up-down (alternating-permutation) numbers 1, 1, 1, 2, 5, 16, 61, 272, 1385... that count the ways to arrange $1..n$ so the values go up, down, up, down. Built from the boustrophedon triangle that snakes left-then-right across each row, it is the combinatorial heart shared by the tangent and secant functions. The term is folded modulo Base into a stepped pitch, giving a fast-accelerating melodic climb distinct from the factorial Pisano and Catalan sequencers. A rising edge on in1 resets.

Param	Range	Default	Unit
Base	2 – 32	12	—

Tangent 2 inputs

Tangent-number sequencer (a first as a synth block): steps through the odd-indexed zigzag numbers 1, 2, 16, 272, 7936, 353792... that appear as the coefficients in the Taylor series of $\tan(x)$. These are the alternating permutations of ODD length, and they explode far faster than the secant family, so the melody rockets upward in big leaps once folded modulo Base. A trigonometric-combinatorial climb distinct from the Euler and Catalan sequencers. A rising edge on in1 resets.

Param	Range	Default	Unit
Base	2 – 32	12	—

Secant 2 inputs

Secant-number sequencer (a first as a synth block): steps through the even-indexed zigzag numbers 1, 1, 5, 61, 1385, 50521... that appear as the coefficients in the Taylor series of $\sec(x)$ (the unsigned Euler numbers). These count the alternating permutations of

EVEN length and grow a touch slower than the tangent family, so the melody climbs steeply but a step behind Tangent once folded modulo Base. A trigonometric-combinatorial line distinct from the Euler and Tangent sequencers. A rising edge on in1 resets.

Param	Range	Default	Unit
Base	2 – 32	12	—

Boustrophedon 2 inputs

Boustrophedon-triangle sequencer (a first as a synth block): reads out the Seidel-Entringer-Arnold triangle one cell at a time, row by row. The triangle is built ox-plough style - each row is filled by snaking left-to-right then right-to-left, accumulating the row above - and its diagonal is the Euler zigzag sequence. Walking the interior cells instead of just the diagonal gives a rippling, terraced contour folded modulo Base, distinct from the single-line Euler, Tangent and Secant sequencers that only sample its edges. A rising edge on in1 resets.

Param	Range	Default	Unit
Base	2 – 32	12	—

DistinctParts 2 inputs

Distinct-partition sequencer (a first as a synth block): steps through the number of ways to write n as a sum of DISTINCT positive integers 1, 1, 1, 2, 2, 3, 4, 5, 6, 8, 10, 12, 15... ($4 = 4 = 3+1$, so two ways). Euler proved this equals the count of partitions into ODD parts, a famous identity. It grows smoothly and sub-exponentially, so folded modulo Base it gives a gentle, slowly-rising organic contour distinct from the explosive zigzag sequencers and the unrestricted-partition family. A rising edge on in1 resets.

Param	Range	Default	Unit
Base	2 – 32	12	—

ReverseAdd 2 inputs

Reverse-and-add orbit sequencer (a first as a synth block): walks the famous 196-algorithm - take a number, add it to its own digit-reversal, repeat - which almost always tumbles into a palindrome but for a few stubborn seeds (196, 879...) may climb forever. Each clock takes one reverse-add step and emits the running value folded into a pitch; when it grows large it reseeds to the next small number, so you hear orbit after orbit of this digit-mixing climb. A self-referential digit dynamic distinct from the power-sum orbits. A rising edge on in1 resets.

DigitFactorial 2 inputs

Digit-factorial orbit sequencer (a first as a synth block): iterates the factorion map - replace a number by the sum of the factorials of its digits - whose orbits famously fall into the fixed points 1, 2, 145, 40585 or the cycle 169 -> 363601 -> 1454 -> 169. Each clock advances one step and emits the value folded into a pitch, reseeding every so often to tour fresh orbits. A factorial-of-digits dynamic distinct from the reverse-add and cube orbits. A rising edge on in1 resets.

DigitCube 2 inputs

Digit-cube orbit sequencer (a first as a synth block): iterates the map that replaces a number by the sum of the CUBES of its digits, whose orbits settle on the fixed points 1, 153, 370, 371, 407 (the three-digit narcissistic numbers) or short cycles such as 55 -> 250 -> 133 -> 55 and 136 -> 244 -> 136. Each clock advances one step and emits the value folded into a pitch, reseeding to tour fresh orbits. A cube-of-digits dynamic distinct from the factorial and reverse-add orbits. A rising edge on in1 resets.

Farey 2 inputs

Farey-sequence sequencer (a first as a synth block): walks the Farey sequence of order N - every fraction in 0..1 whose denominator is at most N, listed in increasing size (for N=5: 0, 1/5, 1/4, 1/3, 2/5, 1/2, 3/5, 2/3, 3/4, 4/5, 1). It steps with the exact integer mediant recurrence, so each clock outputs the next fraction directly as a control voltage and loops back at 1. A rising, unevenly-spaced

rational staircase whose gaps mirror the deep structure of the rationals, distinct from the digit-orbit sequencers. A rising edge on in1 resets.

Param	Range	Default	Unit
Order	2 – 32	8	—

ContinuedFraction 2 inputs

Continued-fraction sequencer (a first as a synth block): emits the partial quotients of the square root of N - the integers a0; a1, a2... in $\text{sqrt}(N) = a_0 + 1/(a_1 + 1/(a_2 + \dots))$. For any non-square N this expansion is eventually PERIODIC ($\text{sqrt}(7) = 2; 1,1,1,4$ repeating; $\text{sqrt}(23) = 4; 1,3,1,8\dots$), and the periodic block is palindromic, so the sequencer produces a clean repeating melodic loop whose shape is fixed by N. A number-theoretic period distinct from the digit-orbit and Farey sequencers. A rising edge on in1 resets.

Param	Range	Default	Unit
N	2 – 99	7	—

EKG 2 inputs

EKG-sequence sequencer (a first as a synth block): walks the EKG (or electrocardiogram) sequence 1, 2, 4, 6, 3, 9, 12, 8, 10, 5, 15... where each term is the smallest number not yet used that shares a common factor with the previous one. Plotted, it traces a jagged heartbeat-like trace, and it is a proved permutation of the positive integers with a deep three-line structure. Each clock advances one term, folded modulo Base into a pitch. A gcd-chained permutation distinct from the digit and prime sequencers. A rising edge on in1 resets.

Param	Range	Default	Unit
Base	2 – 32	12	—

Hamming 2 inputs

Hamming-number sequencer (a first as a synth block): steps through the regular or 5-smooth numbers 1, 2, 3, 4, 5, 6, 8, 9, 10, 12, 15, 16, 18, 20, 24, 25, 27... - the integers whose only prime factors are 2, 3 and 5. Famous from Dijkstra's programming exercise, they are exactly the just-intonation frequency ratios, so the sequence is a ladder of harmonically pure intervals. Each clock advances one term, folded modulo Base. A smooth-number ladder distinct from the prime and figurate sequencers. A rising edge on in1 resets.

Param	Range	Default	Unit
Base	2 – 32	12	—

SelfNumber 2 inputs

Self-number sequencer (a first as a synth block): steps through the self (or Colombian) numbers 1, 3, 5, 7, 9, 20, 31, 42, 53, 64, 75, 86, 97, 108... - integers that CANNOT be written as some smaller number plus its own digit sum, so they have no 'generator' and stand alone. Discovered by the Indian mathematician Kaprekar, they thin out in a curious near-periodic pattern tied to base ten. Each clock advances one term, folded modulo Base. A digit-generator gap sequence distinct from the other digit sequencers. A rising edge on in1 resets.

Param	Range	Default	Unit
Base	2 – 32	12	—

Powerful 2 inputs

Powerful-number sequencer (a first as a synth block): steps through the powerful numbers 1, 4, 8, 9, 16, 25, 27, 32, 36, 49, 64, 72, 81, 100... - integers in which every prime factor appears at least squared (so each is a product of a perfect square and a perfect cube). They are sparse and clump near the squares and cubes, giving a sequence that climbs in widening, uneven leaps. Each clock advances one term, folded modulo Base. A factor-exponent class distinct from the squarefree and smooth sequencers. A rising edge on in1 resets.

Param	Range	Default	Unit
Base	2 – 32	12	—

Squarefree 2 inputs

Squarefree-number sequencer (a first as a synth block): steps through the squarefree numbers 1, 2, 3, 5, 6, 7, 10, 11, 13, 14, 15, 17, 19, 21, 22, 23... - integers divisible by no perfect square (no repeated prime factor). They have density $6/\pi^2$, about 61 percent of all integers, so the sequence skips only the few that carry a squared factor (4, 8, 9, 12, 16...), giving a near-chromatic run with occasional gaps. Each clock advances one term, folded modulo Base. The exact complement of the powerful numbers. A rising edge on in1 resets.

Param	Range	Default	Unit
Base	2 – 32	12	—

Leonardo 2 inputs

Leonardo-number sequencer (a first as a synth block): steps through the Leonardo numbers 1, 1, 3, 5, 9, 15, 25, 41, 67, 109... defined by $L(n) = L(n-1) + L(n-2) + 1$ - the Fibonacci rule with a plus-one twist. They are the sizes of the balanced binary trees used in Dijkstra's smoothsort, and each is one less than twice a Fibonacci number. A nearly-Fibonacci climb, slightly steeper, folded modulo Base. A rising edge on in1 resets.

Param	Range	Default	Unit
Base	2 – 32	12	—

PellLucas 2 inputs

Pell-Lucas sequencer (a first as a synth block): steps through the companion Pell numbers 2, 2, 6, 14, 34, 82, 198, 478... defined by $Q(n) = 2*Q(n-1) + Q(n-2)$ - the Pell recurrence (doubling the previous term) seeded to give the numerators of the rational approximations to the square root of 2. They grow by the silver ratio $1 + \sqrt{2}$, a faster climb than the golden Fibonacci. Folded modulo Base. A silver-ratio companion sequence distinct from the Pell and Fibonacci sequencers. A rising edge on in1 resets.

Param	Range	Default	Unit
Base	2 – 32	12	—

JacobsthalLucas 2 inputs

Jacobsthal-Lucas sequencer (a first as a synth block): steps through the Jacobsthal-Lucas numbers 2, 1, 5, 7, 17, 31, 65, 127, 257... defined by $a(n) = a(n-1) + 2*a(n-2)$, the companion to the Jacobsthal sequence. Their values hug the powers of two (each is 2^n plus or minus one), so the sequence is a wobble around a clean octave doubling. Folded modulo Base. A near-power-of-two companion distinct from the Jacobsthal and Mersenne sequencers. A rising edge on in1 resets.

Param	Range	Default	Unit
Base	2 – 32	12	—

Pentanacci 2 inputs

Pentanacci sequencer (a first as a synth block): the five-step Fibonacci - each term is the sum of the previous FIVE: 0, 0, 0, 0, 1, 1, 2, 4, 8, 16, 31, 61, 120... For the first several terms it doubles exactly (powers of two) before the deeper memory pulls it just below, its growth ratio approaching 2 as the number of summed terms rises. A deep-memory recurrence distinct from the two-term Fibonacci and three-term Tribonacci. Folded modulo Base. A rising edge on in1 resets.

Param	Range	Default	Unit
Base	2 – 32	12	—

Wedderburn 2 inputs

Wedderburn-Etherington sequencer (a first as a synth block): steps through the Wedderburn-Etherington numbers 1, 1, 1, 2, 3, 6, 11, 23, 46, 98, 207, 451... which count the number of distinct ways to bracket a product of n identical items when the operation is commutative but NOT associative - equivalently the unordered binary trees with n leaves. They arise in counting hydrocarbon isomers and phylogenetic trees, and grow by a deep nonlinear self-convolution. Folded modulo Base. A combinatorial tree count distinct from the linear recurrences. A rising edge on in1 resets.

Param	Range	Default	Unit
Base	2 – 32	12	—

CopelandErdos 2 inputs

Copeland-Erdos sequencer (a first as a synth block): reads out the digits of the Copeland-Erdos constant 0.235711131719232931... - formed by writing the prime numbers one after another. Copeland and Erdos proved this constant is normal in base ten (its digits are perfectly evenly distributed), so the melody is a statistically uniform digit stream that nonetheless encodes the primes in order. Each clock advances one decimal digit, output as a 0..1 pitch step. A prime-digit stream distinct from the Champernowne and prime-gap sequencers. A rising edge on in1 resets.

FibonacciWord 2 inputs

Fibonacci-word sequencer (a first as a synth block): the infinite binary word grown by the substitution 0 -> 01, 1 -> 0, giving 01001010010010100100100... - the most-ordered aperiodic sequence there is, the symbolic shadow of the golden ratio. The 1s land exactly where the golden-ratio Beatty sequence says, so it is a self-similar two-note rhythm that never repeats yet is utterly deterministic. Each clock advances one symbol, output as a 0 or 1 gate. A golden-ratio word distinct from the Thue-Morse and Rudin-Shapiro sequencers. A rising edge on in1 resets.

TribonacciWord 2 inputs

Tribonacci-word sequencer (a first as a synth block): the infinite three-symbol word grown by the substitution 0 -> 01, 1 -> 02, 2 -> 0, giving 0102010010201010201... - the ternary cousin of the Fibonacci word, the symbolic shadow of the Tribonacci constant. Its three symbols appear in the Tribonacci proportions and tile the line self-similarly, a three-note aperiodic pattern. Each clock advances one symbol, output as 0, 0.5 or 1. A Tribonacci-ratio word distinct from the two-symbol Fibonacci word. A rising edge on in1 resets.

MephistoWaltz 2 inputs

Mephisto-waltz sequencer (a first as a synth block): the binary word grown by the substitution 0 -> 001, 1 -> 110, giving 001001110001001110110... - named after Liszt's waltz for its restless triple-time feel. It is an automatic sequence with curious correlation properties, closely related to the ternary expansion of position. Each clock advances one symbol, output as a 0 or 1 gate, giving a lurching three-against-two rhythm. A triple-time word distinct from the Fibonacci and Thue-Morse sequencers. A rising edge on in1 resets.

BitReversal 2 inputs

Bit-reversal sequencer (a first as a synth block): emits the bit-reversal permutation - count up 0, 1, 2, 3... but flip the binary digits end-for-end before reading the value (with 4 bits: 0, 8, 4, 12, 2, 10, 6, 14...). This is the scrambled order in which the fast Fourier transform visits its data, a perfect shuffle that maximally spreads consecutive indices apart. Bits sets the word length (and so the loop length, 2^{Bits}). A deterministic maximal-spread permutation distinct from the random and quasi-random sources. A rising edge on in1 resets.

Param	Range	Default	Unit
Bits	2 – 12	6	—

Tetrahedral 2 inputs

Tetrahedral-number sequencer (a first as a synth block): steps through the tetrahedral numbers 1, 4, 10, 20, 35, 56, 84, 120... = $n(n+1)(n+2)/6$ - the count of spheres stacked in a triangular pyramid (cannonball-pile shape), the running sums of the triangular numbers. They are the third diagonal of Pascal's triangle. Each clock advances one term, folded modulo Base. A 3-D figurate climb distinct from the flat polygonal and the 4-D pentatope sequencers. A rising edge on in1 resets.

Param	Range	Default	Unit
Base	2 – 32	12	—

Octahedral 2 inputs

Octahedral-number sequencer (a first as a synth block): steps through the octahedral numbers 1, 6, 19, 44, 85, 146... = $n(2n*n + 1)/3$ - the count of spheres stacked into a regular octahedron (two square pyramids base to base). Each is the sum of two consecutive square-pyramidal numbers. Each clock advances one term, folded modulo Base. A different polyhedral packing distinct from the tetrahedral and pyramidal sequencers. A rising edge on in1 resets.

Param	Range	Default	Unit
Base	2 – 32	12	—

SquarePyramidal 2 inputs

Square-pyramidal-number sequencer (a first as a synth block): steps through the square-pyramidal numbers 1, 5, 14, 30, 55, 91... = $n(n+1)(2n+1)/6$ - the count of spheres stacked in a pyramid with a square base, the running sums of the perfect squares. They appear in the classic cannonball problem (only 4900 is both square and square-pyramidal). Each clock advances one term, folded modulo Base. A square-based packing distinct from the triangular tetrahedral sequencer. A rising edge on in1 resets.

Param	Range	Default	Unit
Base	2 – 32	12	—

Pentatope 2 inputs

Pentatope-number sequencer (a first as a synth block): steps through the pentatope (4-simplex) numbers 1, 5, 15, 35, 70, 126... = $n(n+1)(n+2)(n+3)/24$ - the four-dimensional analogue of the tetrahedral numbers, the running sums of the tetrahedral numbers and the fourth diagonal of Pascal's triangle (so every fifth binomial coefficient $C(n,4)$). Each clock advances one term, folded modulo Base. A 4-D figurate climb distinct from the 3-D tetrahedral sequencer. A rising edge on in1 resets.

Param	Range	Default	Unit
Base	2 – 32	12	—

StarNumber 2 inputs

Star-number sequencer (a first as a synth block): steps through the star (or centred-hexagram) numbers 1, 13, 37, 73, 121, 181... = $6n(n-1) + 1$ - dots arranged in a six-pointed Star of David, a centred hexagonal number with six triangular points added. They climb in even, widely-spaced jumps. Each clock advances one term, folded modulo Base. A centred star figurate distinct from the solid polyhedral sequencers. A rising edge on in1 resets.

Param	Range	Default	Unit
Base	2 – 32	12	—

TwinPrime 2 inputs

Twin-prime sequencer (a first as a synth block): steps through the lesser members of the twin-prime pairs 3, 5, 11, 17, 29, 41, 59, 71, 101, 107... - primes p for which $p+2$ is ALSO prime (3 and 5, 11 and 13, 17 and 19). Whether there are infinitely many is one of the oldest open problems in mathematics. They thin out faster than the primes, giving a sparse, irregular climb. Each clock advances one term, folded modulo Base. A prime-pair subset distinct from the full prime sequencer. A rising edge on in1 resets.

Param	Range	Default	Unit
Base	2 – 32	12	—

SophieGermain 2 inputs

Sophie-Germain-prime sequencer (a first as a synth block): steps through the primes p for which $2p+1$ is ALSO prime - 2, 3, 5, 11, 23, 29, 41, 53, 83, 89... Named after the mathematician who used them to attack Fermat's Last Theorem, they are central to cryptography (the safe primes $2p+1$ resist certain attacks). They form a sparse, irregular subset of the primes. Each clock advances one term, folded modulo Base. A cryptographic prime class distinct from the twin-prime and full prime sequencers. A rising edge on in1 resets.

Param	Range	Default	Unit
Base	2 – 32	12	—

ChenPrime 2 inputs

Chen-prime sequencer (a first as a synth block): steps through the Chen primes - primes p for which $p+2$ is either prime OR a product of two primes (a semiprime): 2, 3, 5, 7, 11, 13, 17, 19, 23, 29... Chen Jingrun proved there are infinitely many, a celebrated partial result toward the twin-prime and Goldbach conjectures. They are denser than the twin primes but still a proper prime subset. Each clock advances one term, folded modulo Base. A near-twin prime class distinct from the strict twin-prime sequencer. A rising edge on in1 resets.

Param	Range	Default	Unit
Base	2 – 32	12	—

Primorial 2 inputs

Primorial sequencer (a first as a synth block): steps through the primorials 2, 6, 30, 210, 2310, 30030, 510510... - the running products of the prime numbers (2, then $2*3$, then $2*3*5$, and so on), the prime analogue of the factorial. They explode even faster than factorials and are central to proofs about prime gaps and to Euclid's proof that the primes never end. The exact product is folded modulo Base, giving a fast-cycling pattern as each new prime multiplies in. A prime-product climb distinct from the additive prime sequencers. A rising edge on in1 resets.

Param	Range	Default	Unit
Base	2 – 32	12	—

Mersenne 2 inputs

Mersenne-number sequencer (a first as a synth block): steps through the Mersenne numbers $2^n - 1 = 1, 3, 7, 15, 31, 63, 127, 255...$ - the numbers that are all-ones in binary. When the exponent is prime these are the candidates for Mersenne PRIMES, the form that has held the record for the largest known prime for decades, hunted by the GIMPS distributed search. The exact value $2^n - 1$ is folded modulo Base, giving a doubling-then-wrapping staircase. A power-of-two-minus-one climb distinct from the additive sequencers. A rising edge on in1 resets.

Param	Range	Default	Unit
Base	2 – 32	12	—

Derangement 2 inputs

Derangement sequencer (a first as a synth block): steps through the subfactorials 1, 0, 1, 2, 9, 44, 265, 1854... - the number of ways to shuffle n items so that NONE ends up in its own place (the hat-check problem). They satisfy $D(n) = (n-1)(D(n-1) + D(n-2))$ and the ratio $D(n)/n!$ homes in on $1/e$, so almost exactly 37 percent of all shufflings are derangements. Folded modulo Base. A fixed-point-free permutation count distinct from the factorial and Bell sequencers. A rising edge on in1 resets.

Param	Range	Default	Unit
Base	2 – 32	12	—

Arrangement 2 inputs

Arrangement sequencer (a first as a synth block): steps through the total number of arrangements 1, 2, 5, 16, 65, 326, 1957... - the count of ALL ordered selections (of any length) from n items, equal to the sum of $n!/k!$ over k . It obeys the tidy rule $a(n) = n*a(n-1) + 1$, and the values are the nearest integers to $n!*e$. Folded modulo Base. A partial-permutation total distinct from the full-factorial and derangement sequencers. A rising edge on in1 resets.

Param	Range	Default	Unit
Base	2 – 32	12	—

Fubini 2 inputs

Fubini-number sequencer (a first as a synth block): steps through the ordered Bell (Fubini) numbers 1, 1, 3, 13, 75, 541, 4683... - the number of ways to rank n items allowing ties (the outcomes of a race with photo finishes), equivalently the number of ordered set partitions. They grow faster than the plain Bell numbers because the blocks are ordered. The exact value is folded modulo Base. A weak-ordering count distinct from the unordered Bell sequencer. A rising edge on in1 resets.

Param	Range	Default	Unit
Base	2 – 32	12	—

Necklace 2 inputs

Necklace sequencer (a first as a synth block): steps through the number of distinct two-colour necklaces of n beads 1, 2, 3, 4, 6, 8, 14, 20, 36, 60... - binary strings of length n counted as the same when rotated into one another. Burnside's counting lemma gives the count as the average over rotations, mixing Euler's totient with powers of two. The exact value is folded modulo Base. A cyclic-symmetry count distinct from the linear word sequencers. A rising edge on in1 resets.

Param	Range	Default	Unit
Base	2 – 32	12	—

LyndonWords 2 inputs

Lyndon-word sequencer (a first as a synth block): steps through the number of binary Lyndon words of length n 2, 1, 2, 3, 6, 9, 18, 30, 56, 99... - the aperiodic necklaces, strings that are strictly smallest among all their rotations. Every string factors uniquely into a descending sequence of Lyndon words (the Chen-Fox-Lyndon theorem), making them a basis for free Lie algebras. Counted by Moebius inversion of the powers of two, the exact value is folded modulo Base. A primitive-string count distinct from the full necklace sequencer. A rising edge on in1 resets.

Param	Range	Default	Unit
Base	2 – 32	12	—

Hipparchus 2 inputs

Hipparchus sequencer (a first as a synth block): steps through the little Schroeder (super-Catalan) numbers 1, 1, 3, 11, 45, 197, 903, 4279... which count the ways to insert non-crossing brackets into a row of n items, or to subdivide a polygon by non-crossing diagonals. Plutarch records that Hipparchus counted the tenth term (103049) two thousand years before they were rediscovered. The exact value is folded modulo Base. A bracketing count distinct from the Catalan and Motzkin sequencers. A rising edge on in1 resets.

Param	Range	Default	Unit
Base	2 – 32	12	—

TernaryTree 2 inputs

Ternary-tree sequencer (a first as a synth block): steps through the Fuss-Catalan numbers 1, 1, 3, 12, 55, 273, 1428, 7752... $= C(3n, n)/(2n+1)$ - the count of rooted trees in which every node has exactly zero or three children (the ternary analogue of the Catalan binary trees), and of the ways to triangulate a polygon with non-crossing diagonals into quadrilaterals. The exact value is folded modulo Base. A three-way branching count distinct from the binary Catalan sequencer. A rising edge on in1 resets.

Param	Range	Default	Unit
Base	2 – 32	12	—

Riordan 2 inputs

Riordan-number sequencer (a first as a synth block): steps through the Riordan numbers 1, 0, 1, 1, 3, 6, 15, 36, 91, 232... - the Motzkin paths with no flat steps at ground level, equivalently certain trees and the number of ways to colour cyclic arrangements. They obey $(n+1)a(n) = (n-1)(2a(n-1) + 3a(n-2))$ and are a close relative of the Motzkin and Catalan families. The exact value is folded modulo Base. A flat-step-free path count distinct from the Motzkin sequencer. A rising edge on in1 resets.

Param	Range	Default	Unit
Base	2 – 32	12	—

PlanePartition 2 inputs

Plane-partition sequencer (a first as a synth block): steps through the number of plane partitions of n 1, 1, 3, 6, 13, 24, 48, 86, 160... - stacks of unit cubes pushed into a corner so the heights never increase going away from the walls, the three-dimensional generalisation of ordinary integer partitions. MacMahon's beautiful product formula counts them. The exact value is folded modulo Base. A 3-D partition count distinct from the flat partition and figurate sequencers. A rising edge on in1 resets.

Param	Range	Default	Unit
Base	2 – 32	12	—

Apery 2 inputs

Apery-number sequencer (a first as a synth block): steps through the Apery numbers 1, 5, 73, 1445, 33001, 819005... - the extraordinary integers Roger Apery used in 1979 to prove that $\zeta(3)$ is irrational, a result that had eluded everyone for centuries. They satisfy a deep three-term recurrence with cubic coefficients and are sums of products of squared binomial coefficients. The exact value is folded modulo Base, giving a fast, wide-leaping climb. A zeta-irrationality witness distinct from the polynomial figurate sequencers. A rising edge on in1 resets.

Param	Range	Default	Unit
Base	2 – 32	12	—

MianChowla 2 inputs

Mian-Chowla sequencer (a first as a synth block): the greedy Sidon sequence 1, 2, 4, 8, 13, 21, 31, 45, 66, 81... - built by always taking the smallest next integer such that ALL the pairwise sums stay distinct (no two pairs add to the same value). Sidon sets like this are prized in signal design and radio astronomy for their flat autocorrelation. The terms spread apart ever wider as collisions become harder to avoid. Folded modulo Base. A distinct-sums greedy set distinct from the prime and figurate sequencers. A rising edge on in1 resets.

Param	Range	Default	Unit
Base	2 – 32	12	—

HofstadterQ 2 inputs

Hofstadter Q-sequencer (a first as a synth block): the chaotic meta-Fibonacci sequence $Q(n) = Q(n - Q(n-1)) + Q(n - Q(n-2))$, seeded $Q(1) = Q(2) = 1$, giving 1, 1, 2, 3, 3, 4, 5, 5, 6, 6, 6, 8... Unlike Fibonacci it looks BACK by amounts that the sequence itself decides, so it wobbles unpredictably and it is unknown whether it stays defined forever. A self-indexing, semi-chaotic climb folded modulo Base. A strange-loop recurrence distinct from the smooth linear sequencers. A rising edge on in1 resets.

Param	Range	Default	Unit
Base	2 – 32	12	—

Queens 2 inputs

N-queens sequencer (a first as a synth block): steps through the number of ways to place n non-attacking queens on an n-by-n chessboard 1, 1, 0, 0, 2, 10, 4, 40, 92, 352, 724... - one of the most famous combinatorial search problems. The count is zero for the impossible 2 and 3 boards, then climbs irregularly; the eight-queens puzzle has its celebrated 92 solutions. Each clock recomputes the count by backtracking and folds it modulo Base. A board-packing count distinct from every closed-form sequencer. A rising edge on in1 resets.

Param	Range	Default	Unit
Base	2 – 32	12	—

Stanley 2 inputs

Stanley sequencer (a first as a synth block): steps through the Stanley sequence 0, 1, 3, 4, 9, 10, 12, 13, 27, 28... - built greedily by always adding the smallest integer that creates NO three-term arithmetic progression with any two earlier terms. Remarkably the result is exactly the numbers whose base-three representation contains no digit 2, giving it a clean self-similar Cantor-like structure. Folded modulo Base. A progression-free greedy set distinct from the Sidon (Mian-Chowla) set. A rising edge on in1 resets.

Param	Range	Default	Unit
Base	2 – 32	12	—

PartitionsIntoPrimes 2 inputs

Prime-partition sequencer (a first as a synth block): steps through the number of ways to write n as a sum of PRIME numbers 1, 0, 1, 1, 1, 2, 2, 3, 3, 4, 5, 6, 7... ($7 = 7 = 5+2 = 3+2+2$, so three ways). It restricts the famous partition function to prime parts only, tying the additive world of partitions to the multiplicative primes. Computed by the exact prime-restricted partition recurrence, folded modulo Base. A prime-restricted partition count distinct from the full partition and distinct-parts sequencers. A rising edge on in1 resets.

Param	Range	Default	Unit
Base	2 – 32	12	—

Heptagonal 2 inputs

Heptagonal-number sequencer (a first as a synth block): steps through the seven-sided polygonal numbers 1, 7, 18, 34, 55, 81, 112, 148... $= n(5n-3)/2$ - dots arranged in nested heptagons. Like all polygonal numbers they are second differences constant (here the gaps grow by 5 each time). Each clock advances one term, folded modulo Base. A seven-gon figurate climb distinct from the triangular, square and pentagonal sequencers. A rising edge on in1 resets.

Param	Range	Default	Unit
Base	2 – 32	12	—

Octagonal 2 inputs

Octagonal-number sequencer (a first as a synth block): steps through the eight-sided polygonal numbers 1, 8, 21, 40, 65, 96, 133, 176... $= n(3n-2)$ - dots arranged in nested octagons, the gaps between terms growing by 6 each step. Each clock advances one term, folded modulo Base. An eight-gon figurate climb distinct from the lower polygonal sequencers. A rising edge on in1 resets.

Param	Range	Default	Unit
Base	2 – 32	12	—

CenteredSquare 2 inputs

Centered-square-number sequencer (a first as a synth block): steps through the centered square numbers 1, 5, 13, 25, 41, 61, 85, 113, 145... $= 2n^2 + 2n + 1$ - a central dot ringed by successive square frames (the pattern of a diamond growing outward). They are the sums of two consecutive squares, and the gaps between them run through the multiples of four. Each clock advances one term, folded modulo Base. A centered figurate distinct from the ordinary square sequencer. A rising edge on in1 resets.

Param	Range	Default	Unit
Base	2 – 32	12	—

CenteredTriangular 2 inputs

Centered-triangular-number sequencer (a first as a synth block): steps through the centered triangular numbers 1, 4, 10, 19, 31, 46, 64, 85, 109... $= (3n^2 + 3n + 2)/2$ - a central dot surrounded by growing triangular rings. The gaps between terms run through the multiples of three. Each clock advances one term, folded modulo Base. A centered figurate distinct from the ordinary triangular sequencer. A rising edge on in1 resets.

Param	Range	Default	Unit
Base	2 – 32	12	—

CenteredHexagonal 2 inputs

Centered-hexagonal-number sequencer (a first as a synth block): steps through the centered hexagonal numbers 1, 7, 19, 37, 61, 91, 127, 169, 217... = $3n(n+1) + 1$ - a central dot ringed by hexagons (the hex or honeycomb numbers). Their partial sums are the perfect cubes, and the gaps between them are the multiples of six. Each clock advances one term, folded modulo Base. A honeycomb figurate distinct from the star and pyramidal sequencers. A rising edge on in1 resets.

Param	Range	Default	Unit
Base	2 – 32	12	—

Hexagonal 2 inputs

Hexagonal-number sequencer (a first as a synth block): steps through the six-sided polygonal numbers 1, 6, 15, 28, 45, 66, 91, 120... = $n(2n-1)$ - dots in nested hexagons. They are exactly the odd-indexed triangular numbers, and every perfect number is hexagonal. Each clock advances one term, folded modulo Base. A six-gon figurate climb distinct from the triangular and pentagonal sequencers. A rising edge on in1 resets.

Param	Range	Default	Unit
Base	2 – 32	12	—

Nonagonal 2 inputs

Nonagonal-number sequencer (a first as a synth block): steps through the nine-sided polygonal numbers 1, 9, 24, 46, 75, 111, 154, 204... = $n(7n-5)/2$ - dots arranged in nested nine-gons, the gaps between terms growing by 7 each step. Each clock advances one term, folded modulo Base. A nine-gon figurate climb distinct from the lower polygonal sequencers. A rising edge on in1 resets.

Param	Range	Default	Unit
Base	2 – 32	12	—

Decagonal 2 inputs

Decagonal-number sequencer (a first as a synth block): steps through the ten-sided polygonal numbers 1, 10, 27, 52, 85, 126, 175, 232... = $n(4n-3)$ - dots arranged in nested ten-gons, the gaps between terms growing by 8 each step. Each clock advances one term, folded modulo Base. A ten-gon figurate climb distinct from the octagonal and nonagonal sequencers. A rising edge on in1 resets.

Param	Range	Default	Unit
Base	2 – 32	12	—

Practical 2 inputs

Practical-number sequencer (a first as a synth block): steps through the practical numbers 1, 2, 4, 6, 8, 12, 16, 18, 20, 24... - integers n such that EVERY smaller number can be written as a sum of distinct divisors of n . They behave like little change-making systems (12 is why a dozen is handy), and they are about as dense as the primes yet defined multiplicatively. Each clock advances one term, folded modulo Base. A subset-sum-complete class distinct from the prime and abundant sequencers. A rising edge on in1 resets.

Param	Range	Default	Unit
Base	2 – 32	12	—

HighlyComposite 2 inputs

Highly-composite-number sequencer (a first as a synth block): steps through the highly composite numbers 1, 2, 4, 6, 12, 24, 36, 48, 60, 120... - each having MORE divisors than any smaller number, the record-breakers of factor-richness studied by Ramanujan. They are the anti-primes (12 and 60 and 360 are why clocks and circles are divided as they are). Each clock advances one record-setter, folded modulo Base. A divisor-maximising class distinct from the prime and figurate sequencers. A rising edge on in1 resets.

Param	Range	Default	Unit
Base	2 – 32	12	—

Refactorable 2 inputs

Refactorable-number sequencer (a first as a synth block): steps through the refactorable (or tau) numbers 1, 2, 8, 9, 12, 18, 24, 36, 40, 56... - integers that are divisible by their OWN number of divisors (12 has six divisors and $12/6 = 2$). A self-referential factor property: the count of divisors itself divides the number. Each clock advances one term, folded modulo Base. A self-divisor class distinct from the perfect and abundant sequencers. A rising edge on in1 resets.

Param	Range	Default	Unit
Base	2 – 32	12	—

Achilles 2 inputs

Achilles-number sequencer (a first as a synth block): steps through the Achilles numbers 72, 108, 200, 288, 392, 432, 500, 648... - integers that are POWERFUL (every prime factor appears at least squared) yet are NOT themselves a perfect power. Like the hero, they are strong but imperfect - $72 = 2^3 * 3^2$ cannot be written as a single base to a single exponent. Each clock advances one term, folded modulo Base. A strong-but-imperfect class distinct from the plain powerful sequencer. A rising edge on in1 resets.

Param	Range	Default	Unit
Base	2 – 32	12	—

Amicable 2 inputs

Amicable-number sequencer (a first as a synth block): steps through the members of amicable pairs 220, 284, 1184, 1210, 2620, 2924... - two numbers each equal to the sum of the OTHER'S proper divisors (the divisors of 220 sum to 284 and vice versa). Known to the Pythagoreans as a symbol of friendship, these pairs are rare and were hunted for centuries. Each clock advances one amicable number, folded modulo Base. A mutual-divisor-sum class distinct from the self-referential perfect numbers. A rising edge on in1 resets.

Param	Range	Default	Unit
Base	2 – 32	12	—

SelfConjugate 2 inputs

Self-conjugate-partition sequencer (a first as a synth block): steps through the number of self-conjugate partitions of n 1, 1, 0, 1, 1, 1, 1, 2, 2... - the partitions whose Young diagram is symmetric across its main diagonal (unchanged when rows and columns are swapped). Euler showed these are exactly the partitions into DISTINCT ODD parts. Computed by the distinct-odd-parts recurrence, folded modulo Base. A symmetric-diagram count distinct from the full and distinct partition sequencers. A rising edge on in1 resets.

Param	Range	Default	Unit
Base	2 – 32	12	—

RogersRamanujan1 2 inputs

Rogers-Ramanujan (first) sequencer (a first as a synth block): steps through the partitions of n into parts that leave remainder 1 or 4 when divided by 5: 1, 1, 1, 1, 2, 2, 3, 3, 4, 5... The celebrated first Rogers-Ramanujan identity proves this exactly equals the partitions whose parts differ by at least 2 - a deep and surprising equality between a congruence condition and a gap condition. Folded modulo Base. A gap-condition partition count distinct from the ordinary partition sequencer. A rising edge on in1 resets.

Param	Range	Default	Unit
Base	2 – 32	12	—

RogersRamanujan2 2 inputs

Rogers-Ramanujan (second) sequencer (a first as a synth block): steps through the partitions of n into parts that leave remainder 2 or 3 when divided by 5: 1, 0, 1, 1, 1, 1, 2, 2, 3, 3... The second Rogers-Ramanujan identity proves this equals the partitions whose parts are at least 2 and differ by at least 2. The companion to the first identity, shifted by the smallest allowed part. Folded modulo Base. A shifted gap-condition count distinct from the first Rogers-Ramanujan sequencer. A rising edge on in1 resets.

Param	Range	Default	Unit
Base	2 – 32	12	—

PartitionsIntoSquares 2 inputs

Square-partition sequencer (a first as a synth block): steps through the number of ways to write n as a sum of perfect squares 1, 1, 1, 1, 2, 2, 2, 2, 3, 4... (allowing repeats, order ignored: $4 = 4 = 1+1+1+1$, so two ways). It connects the additive partition world to the squares, and is bounded above by the four-squares theorem that says four are always enough. Folded modulo Base. A square-part partition count distinct from the prime-part and ordinary partition sequencers. A rising edge on in1 resets.

Param	Range	Default	Unit
Base	2 – 32	12	—

PartitionsIntoTriangular 2 inputs

Triangular-partition sequencer (a first as a synth block): steps through the number of ways to write n as a sum of triangular numbers 1, 1, 1, 2, 2, 2, 4, 4, 4, 6... (the parts drawn from 1, 3, 6, 10, 15...). Gauss proved every number is a sum of at most three triangular numbers (his EUREKA theorem), and this sequence counts all the ways. Folded modulo Base. A triangular-part partition count distinct from the square-part and prime-part sequencers. A rising edge on in1 resets.

Param	Range	Default	Unit
Base	2 – 32	12	—

Pernicious 2 inputs

Pernicious-number sequencer (a first as a synth block): steps through the pernicious numbers 3, 5, 6, 7, 9, 10, 11, 12, 13, 14... - integers whose binary representation contains a PRIME number of 1-bits ($3 = 11$ has two ones, $7 = 111$ has three). A property living entirely in the binary digits, bridging primality and bit patterns. Each clock advances one term, folded modulo Base. A binary-popcount-prime class distinct from the decimal-digit sequencers. A rising edge on in1 resets.

Param	Range	Default	Unit
Base	2 – 32	12	—

BinaryPalindrome 2 inputs

Binary-palindrome sequencer (a first as a synth block): steps through the numbers that read the same forwards and backwards in BINARY 0, 1, 3, 5, 7, 9, 15, 17, 21, 27... (5 = 101, 9 = 1001, 21 = 10101). Mirror symmetry in base two rather than base ten, so the pattern is quite different from the decimal palindromes. Each clock advances one term, folded modulo Base. A base-two mirror class distinct from the decimal Palindrome sequencer. A rising edge on in1 resets.

Param	Range	Default	Unit
Base	2 – 32	12	—

Zuckerman 2 inputs

Zuckerman-number sequencer (a first as a synth block): steps through the Zuckerman numbers 1, 2, 3, 4, 5, 6, 7, 8, 9, 11, 12, 15... - integers divisible by the PRODUCT of their own digits (12 has digit product 2 and $12/2 = 6$; numbers containing a zero are excluded). A self-referential multiplicative digit property. Each clock advances one term, folded modulo Base. A digit-product-divisible class distinct from the digit-sum Harshad sequencers. A rising edge on in1 resets.

Param	Range	Default	Unit
Base	2 – 32	12	—

Polydivisible 2 inputs

Polydivisible-number sequencer (a first as a synth block): steps through the polydivisible numbers 1, 2, ..., 9, 10, 12, 14, 16, 18, 20... - numbers where the first digit is divisible by 1, the first two digits form a number divisible by 2, the first three by 3, and so on all the way down. A cascade of divisibility through the digits; there are only finitely many (the longest has 25 digits). Each clock advances one term, folded modulo Base. A prefix-divisibility class distinct from the whole-number divisibility sequencers. A rising edge on in1 resets.

Param	Range	Default	Unit
Base	2 – 32	12	—

CircularPrime 2 inputs

Circular-prime sequencer (a first as a synth block): steps through the circular primes 2, 3, 5, 7, 11, 13, 17, 31, 37, 71, 73, 79, 97, 113... - primes that stay prime under EVERY cyclic rotation of their digits (197 -> 971 -> 719 are all prime). A rare and elegant prime property combining primality with digit rotation. Each clock advances one term, folded modulo Base. A rotation-stable prime class distinct from the twin and Sophie-Germain sequencers. A rising edge on in1 resets.

Param	Range	Default	Unit
Base	2 – 32	12	—

Halton 2 inputs

Halton-sequence sequencer (a first as a synth block): a quasi-random low-discrepancy sequence built from the radical inverse - take the index, write it in base b, then flip the digits across the decimal point (1,2,3 in base 2 give .1, .01, .11 = 0.5, 0.25, 0.75). Unlike true randomness it fills the interval evenly, never clumping or leaving gaps, the workhorse of quasi-Monte-Carlo sampling. As a melody it sounds scattered yet always lands in the holes it left before. Base picks the digit base. A deterministic gap-filling scatter distinct from the pseudo-random generators. A rising edge on in1 resets.

Param	Range	Default	Unit
Base	2 – 12	3	—

Weyl 2 inputs

Weyl-sequence sequencer (a first as a synth block): the additive recurrence $x \leftarrow \text{frac}(x + \alpha)$ with an IRRATIONAL step, whose iterates are equidistributed by Weyl's theorem - they spread out perfectly evenly and never repeat. With the golden ratio as the step it becomes the optimal one-dimensional low-discrepancy sequence, each new note falling in the largest remaining gap. Ratio selects the irrational (golden, root 2, root 3, plastic), each giving a different even scatter. A number-theoretic equidistribution distinct from the chaotic and pseudo-random sources. A rising edge on in1 resets.

Param	Range	Default	Unit
Ratio	Golden · Sqrt2 · Sqrt3 · Plastic	—	—

MiddleSquare 2 inputs

Middle-square-Weyl sequencer (a first as a synth block): a modern repair of von Neumann's 1949 middle-square method - square the number and keep the middle digits - which on its own quickly collapses to zero. Adding a Weyl sequence (a steadily advancing counter) each step cures the collapse, yielding a fast, high-quality pseudo-random stream that passes stringent statistical tests. It plays a deterministic but thoroughly scrambled melody. Bits sets the read-out pitch range. A squared-digit generator distinct from the shift-register and congruential ones. A rising edge on in1 resets.

Param	Range	Default	Unit
Bits	4 – 12	8	—

LaggedFibonacci 2 inputs

Lagged-Fibonacci sequencer (a first as a synth block): generalises the Fibonacci rule by adding two FAR-apart earlier terms - $x(n) = x(n-7) + x(n-10)$ modulo a power of two - keeping a short ring of past values. This classic pseudo-random generator has an enormously long period and the additive, memory-of-the-past structure gives it a subtly different texture from the bit-twiddling generators. It plays a deterministic scrambled melody that drifts and never quite repeats. Bits sets the pitch range. An additive-recurrence noise distinct from the shift-register ones. A rising edge on in1 resets.

Param	Range	Default	Unit
Bits	4 – 12	8	—

PCG 2 inputs

Permuted-congruential sequencer (a first as a synth block): the modern PCG generator - a plain linear-congruential engine (the oldest pseudo-random method) whose weak low bits are fixed by an output permutation that xor-shifts and then bit-rotates the state. The result is small, fast and statistically excellent, far better than the bare congruential generator it is built on. It plays a deterministic, very thoroughly scrambled melody. Bits sets the pitch range. A permuted-LCG noise distinct from the shift-register and squared-digit generators. A rising edge on in1 resets.

Param	Range	Default	Unit
Bits	4 – 12	8	—

Wolfram 2 inputs

Elementary cellular-automaton sequencer (a first as a synth block): runs Stephen Wolfram's one-dimensional 2-state CA on a 31-cell ring - every cell updates from itself and its two neighbours by the 8-bit Rule code (Rule 30 is chaotic and was used as a random generator, Rule 90 draws the Sierpinski triangle, Rule 110 is Turing-complete, Rule 184 models traffic). Each clock advances one generation, reading a 12-cell window of the row out as a stepped CV. The whole zoo of CA behaviour - order, chaos and the complex edge between them - from one integer. A rising edge on in1 resets to a single live cell.

Param	Range	Default	Unit
Rule	0 – 255	30	—

Totalistic 2 inputs

Totalistic cellular-automaton sequencer (a first as a synth block): runs a one-dimensional CA whose next cell depends only on the SUM of its three-cell neighbourhood (0 to 3), not the exact pattern - so the whole rule is a 4-bit Code. This is the symmetric, count-only cousin of the Wolfram CA, the same rule space Wolfram used to study totalistic complexity. Each clock advances one generation on a 31-cell ring, reading a 12-cell window out as a stepped CV. A sum-driven automaton distinct from the pattern-driven Wolfram and second-order CAs. A rising edge on in1 resets to a single live cell.

Param	Range	Default	Unit
Code	0 – 15	10	—

SecondOrder 2 inputs

Second-order (reversible) cellular-automaton sequencer (a first as a synth block): runs an elementary CA with memory - the next row is the Rule applied to the current row XOR-ed with the PREVIOUS row. That backward coupling makes the automaton time-reversible (you can run it backwards to recover any past state), so it conserves information and never settles into a fixed pattern, churning endlessly. Each clock advances one generation on a 31-cell ring, reading a 12-cell window out as a stepped CV. A reversible CA distinct from the dissipative Wolfram and Totalistic ones. A rising edge on in1 resets.

Param	Range	Default	Unit
Rule	0 – 255	90	—

LFSR 2 inputs

Linear-feedback shift-register sequencer (a first as a synth block): clocks a 16-bit Galois LFSR with maximal-length feedback taps, the classic hardware pseudo-random generator behind scramblers, CRCs and chip-tune noise. It marches through all 65535 non-zero states in a fixed, repeatable order before looping, so it sounds random yet plays the exact same long pattern every time - a deterministic random melody. Bits sets how many low bits are read out as the pitch CV. A shift-register noise distinct from the word-mixing Xorshift. A rising edge on in1 resets the register.

Param	Range	Default	Unit
Bits	4 – 12	8	—

Xorshift 2 inputs

Xorshift-PRNG sequencer (a first as a synth block): clocks George Marsaglia's xorshift generator - the running 32-bit word is repeatedly shifted and XOR-ed into itself (left 13, right 17, left 5) - one of the fastest pseudo-random generators known. It visits all $2^{32} - 1$ non-zero states before repeating, so like the LFSR it plays a deterministic but seemingly random melody, with a denser, more thoroughly scrambled feel from mixing the whole word at once. Bits sets the read-out pitch range. A word-mixing noise distinct from the bit-shifting LFSR. A rising edge on in1 resets the seed.

Param	Range	Default	Unit
Bits	4 – 12	8	—

Ulam 2 inputs

Ulam-sequence sequencer (a first as a synth block): the additive sequence 1, 2, 3, 4, 6, 8, 11, 13, 16, 18, 26, 28... where each new term is the smallest integer larger than the last that is the sum of two distinct earlier terms in EXACTLY one way. Despite that purely additive rule it develops a deep, still-unexplained near-periodicity (a hidden spacing near 21.6) that number theory cannot

yet account for. Each rising edge of the in0 clock advances one term, folded into Steps; a quasi-periodic melody distinct from the Fibonacci-based Pisano and the rational-enumerating Stern. A rising edge on in1 resets.

Param	Range	Default	Unit
Steps	4 – 32	16	—

Galton 2 inputs

Galton-board (bean machine) source (a first as a synth block): each rising edge of the in0 clock drops a bean through Rows rows of pins, bouncing left or right on a fair coin at every pin; the output is the bin it settles in, 0..1. Because that bin is the SUM of Rows coin flips it is binomially distributed - clustered around the centre and rare at the edges, the discrete bell curve - unlike the flat, uniform spread of the Random source. Rows sets the curve's sharpness (more rows = tighter central bias). A Gaussian-weighted random step for natural-feeling movement. A rising edge on in1 reseeds.

Param	Range	Default	Unit
Rows	1 – 24	12	—

Spectral

12 modules

SpecGate 1 input

Spectral gate: runs a streaming STFT and, per frame, zeroes any FFT bin whose magnitude falls below Threshold relative to the loudest bin. Threshold sets that cutoff fraction (0 passes everything, 1 keeps only the peak); Mix blends the gated spectrum against the dry input. Keeps only the strongest partials, thinning noise and weak harmonics into a sparser, cleaner tone.

Param	Range	Default	Unit
Threshold	0 – 1	0.2	—
Mix	0 – 1	1	—

SpecFreeze 1 input

Spectral freeze: while Freeze is off it tracks each FFT bin's magnitude per frame; when Freeze is on it holds those captured magnitudes (keeping the live phase) to sustain the last spectrum indefinitely. Freeze is the on/off latch and Mix blends the held spectrum against the dry input. Turns a transient moment into a steady drone or pad.

Param	Range	Default	Unit
Freeze	0 – 1	0	—
Mix	0 – 1	1	—

SpecFilter 1 input

Per-bin spectral filter: runs a streaming STFT and zeroes whole FFT bins on one side of a cutoff bin for a brick-wall response in the frequency domain. Cutoff is the bin index of the edge and Mode selects low-pass (zero bins above) or high-pass (zero bins below); Mix blends against the dry input. The hard bin cut gives a far steeper, ringing-free roll-off than an analog filter.

Param	Range	Default	Unit
Cutoff	1 – 512	256	bin
Mode	0 – 1	0	—
Mix	0 – 1	1	—

SpecComp 1 input

Spectral compressor: per STFT frame it finds the loudest bin, then attenuates every bin above Threshold (relative to that peak) back toward the threshold. Threshold sets the relative knee and Amount scales how hard over-threshold bins are pulled down; Mix blends against the dry input. Evens out spectral dynamics, taming dominant partials for a flatter, denser spectrum.

Param	Range	Default	Unit
Threshold	0 – 1	0.3	—
Amount	0 – 1	0.6	—
Mix	0 – 1	1	—

SpecDelay 1 input

Spectral echo: per STFT frame each bin's magnitude is summed with its previous-frame value scaled by Feedback, and that running magnitude is fed back (live phase retained). Feedback (0..0.95) sets how much each bin's energy persists and decays across frames; Mix blends against the dry input. Produces a smeared, frequency-domain tail closer to a freeze/reverb than a discrete time echo.

Param	Range	Default	Unit
Feedback	0 – 0.95	0.5	—
Mix	0 – 1	1	—

SpecBlur 1 input

Spectral blur: per STFT frame each bin's magnitude is a Blur-weighted mix of its previous value and the current one, smearing magnitudes over time while keeping live phase. Blur (0..1) sets the smoothing amount -- higher values hold past energy longer; Mix blends against the dry input. Softens transients and stretches the sound into a sustained pad.

Param	Range	Default	Unit
Blur	0 – 1	0.5	—
Mix	0 – 1	1	—

SpecMorph 1 input

Spectral smoothing: per STFT frame it runs a one-pole average up the bins, replacing each magnitude with a Smooth-weighted blend of the running value and the current bin (live phase retained). Smooth (0..1) sets how strongly the envelope is smeared across frequency -- higher values flatten spectral detail; Mix blends against the dry input. Blurs sharp formants into a softer, more uniform tone.

Param	Range	Default	Unit
Smooth	0 – 1	0.5	—
Mix	0 – 1	1	—

SpectralReso 1 input

Spectral resonator: per STFT frame each bin is compared with its two neighbors; local peaks are boosted by up to $1 + \text{Amount} \times 2$ while non-peak bins are cut by up to $\text{Amount} \times 0.5$. Amount (0..1) sets both the peak boost and the valley attenuation; Mix blends against the dry input. Sharpens the spectrum into a more resonant, harmonically pronounced, vowel-like tone.

Param	Range	Default	Unit
Amount	0 – 1	0.5	—
Mix	0 – 1	1	—

FormantShift 1 input

Formant shift: per STFT frame it captures the magnitude envelope, resamples it by a ratio of $2^{(\text{Shift}/12)}$, then re-weights each bin to impose the shifted envelope while leaving the underlying partials in place. Shift (+/-12 semitones) moves the formants up or down, so pitch stays fixed while timbre changes; Mix blends against the dry input. Produces vocal gender/size effects without affecting the played note.

Param	Range	Default	Unit
Shift	-12 – 12	0	st
Mix	0 – 1	1	—

Soothe 1 input

Dynamic resonance suppressor: per STFT frame it tracks a smoothed local average up the bins and pulls any bin above that average down toward it, scaled by Amount. Amount (0..1) sets how aggressively over-average peaks are tamed (0 leaves them, 1 flattens them to the average); Mix blends against the dry input. Reduces harsh resonant spikes for a smoother, more even spectrum.

Param	Range	Default	Unit
Amount	0 – 1	0.6	—
Mix	0 – 1	1	—

MatchEQ 1 input

Matching EQ: with Capture on it learns each bin's magnitude as a reference curve; with Capture off it scales every bin toward that stored reference per frame. Capture latches learn vs apply, Amount (0..1) blends between no change and a full match (bin gain clamped to 8x); Mix blends against the dry input. Reshapes the input's tonal balance to resemble a previously captured reference.

Param	Range	Default	Unit
Capture	0 – 1	0	—
Amount	0 – 1	0.7	—
Mix	0 – 1	1	—

LinearEQ 1 input

Linear-phase 3-band EQ: per STFT frame it applies a single magnitude gain to each FFT bin grouped into low (below half/8), mid, and high (from half/2) bands, so the output has zero phase shift. Low, Mid, and High set each band's gain in dB (converted via $10^{(dB/20)}$); Mix blends against the dry input. The linear-phase response avoids the phase smearing of analog EQ, useful on transient or stereo material.

Param	Range	Default	Unit
Low	-18 – 18	0	dB
Mid	-18 – 18	0	dB
High	-18 – 18	0	dB
Mix	0 – 1	1	—

Stereo

17 modules

StereoWidth 2 inputs

Mid/side stereo widener: decomposes the L/R pair into mid (sum) and side (difference), scales the side by Width, then recombines. Width below 1 narrows toward mono and above 1 (up to 2) exaggerates the stereo spread. Out picks L or R, so one instance emits each recombined channel; useful for tightening or opening a stereo bus.

Param	Range	Default	Unit
Width	0 – 2	1	—
Out	0 – 1	0	—

MidSide 2 inputs

Mid/side processor with independent gain on each component: splits L/R into mid and side, applies MidGain and SideGain in dB, then recombines. Lowering MidGain or raising SideGain widens the image while the reverse focuses the centre. Out picks L or R for the recombined channel; handy for level-matching centre versus stereo content.

Param	Range	Default	Unit
MidGain	-24 – 12	0	dB
SideGain	-24 – 12	0	dB
Out	0 – 1	0	—

Rotate 2 inputs

Stereo field rotation: rotates the L/R signal vector by Angle, tilting the whole stereo image (a hard-left source swings toward centre and on toward the right) - a true rotation rather than a width change. Angle sets the rotation in degrees and Out picks L or R. Subtle angles re-balance a lopsided image; +/-90 swaps the channels.

Param	Range	Default	Unit
Angle	-180 – 180	0	deg
Out	0 – 1	0	—

Shuffle 2 inputs

Blumlein shuffler (frequency-dependent stereo width): splits the side (L-R) signal at a crossover and applies Bass width below and Treble width above, so you can collapse the low end toward mono for a tight, phase-safe centre while widening the highs - the classic mastering width control. Freq sets the crossover, Bass/Treble the width on each side of it, and Out picks L or R.

Param	Range	Default	Unit
Freq	80 – 800	250	Hz
Bass	0 – 2	0.4	—
Treble	0 – 2	1.4	—
Out	0 – 1	0	—

Binaural 2 inputs

Binaural 3D panner: positions the (summed) input at an Azimuth around the head using an interaural time delay, an interaural level difference and a head-shadow low-pass on the far ear - a simplified HRTF that places a sound left/right with real depth on headphones, beyond a flat constant-power pan. Azimuth sweeps -90..90 degrees, Distance dims and dulls far sources, and Out picks L or R.

Param	Range	Default	Unit
Azimuth	-90 – 90	0	deg
Distance	0 – 1	0.3	—
Out	0 – 1	0	—

MonoMaker 2 inputs

Bass mono-maker: a one-pole low-pass at Freq splits the spectrum so content below the crossover is summed to mono while the highs stay in their own channel. Freq sets the crossover from 20 to 500 Hz. Out picks L or R, supplying that channel's highs over the shared mono low end; tightens low-frequency phase for vinyl and club playback.

Param	Range	Default	Unit
Freq	20 – 500	150	Hz
Out	0 – 1	0	—

Haas 1 input

Haas widener for a mono source: feeds the input into a delay line and outputs either the dry signal or a copy delayed by a few milliseconds. Delay sets the offset from 0 to 40 ms, exploiting the precedence effect to throw the image to one side. Out picks L (dry) or R (delayed) so the pair forms a wide stereo spread from one mono input.

Param	Range	Default	Unit
Delay	0 – 40	15	ms
Out	0 – 1	0	—

Panner 1 input

Constant-power panner for a mono source: maps Pan (-1 to +1) onto a sine/cosine gain pair so total power stays even across the field. Pan at 0 centres the source; the +/-1 extremes hard-pan it. Out picks L or R, each instance applying the matching sine or cosine leg of the pan law.

Param	Range	Default	Unit
Pan	-1 – 1	0	—
Out	0 – 1	0	—

AutoPan 1 input

LFO auto-panner for a mono source: a sine LFO driven from the shared transport time sweeps a constant-power pan automatically. Rate sets the sweep speed from 0.05 to 10 Hz and Depth scales how far the source travels. Out picks L or R, and because the LFO is phase-coherent across instances the two outputs track in opposition for stable motion.

Param	Range	Default	Unit
Rate	0.05 – 10	1	Hz
Depth	0 – 1	1	—
Out	0 – 1	0	—

Stereoize 1 input

Mono-to-stereo decorrelator: runs the input through a four-stage all-pass chain to phase-shift it without changing tone, then blends that toward the dry signal by Amount. Amount controls how much decorrelated signal is mixed in. Out picks L (dry) or R (decorrelated) so the stereo pair widens a mono source while keeping a centred reference.

Param	Range	Default	Unit
Amount	0 – 1	0.7	—
Out	0 – 1	0	—

StereoImager 2 inputs

Frequency-dependent imager: splits the side signal at Freq with a one-pole low-pass and scales only the high side band by Width, leaving the low side untouched. Width sets the high-band spread and Freq sets the split point from 100 to 2000 Hz. Out picks L or R for the recombined channel; widens air and detail while keeping the low end mono-safe.

Param	Range	Default	Unit
Width	0 – 2	1.3	—
Freq	100 – 2000	400	Hz
Out	0 – 1	0	—

FreqPanner 2 inputs

Frequency-split panner: sums the input to mono, low-passes it at Freq, and applies a constant-power pan to only the high band while the low band stays centred. Freq sets the split from 100 to 4000 Hz and Pan positions the highs from -1 to +1. Out picks L or R, applying the matching pan-law leg to the highs over the shared centred lows.

Param	Range	Default	Unit
Freq	100 – 4000	800	Hz
Pan	-1 – 1	0.5	—
Out	0 – 1	0	—

Pan 1 input

Static constant-power panner for a mono source: places the input at a fixed position from -1 (hard left) to +1 (hard right) using an equal-power (sin/cos) law so perceived loudness stays constant across the sweep. The plain manual counterpart to the LFO-driven AutoPan and the frequency-split FreqPanner - set a source's place in the image once. Out picks L or R, supplying that channel's pan-law leg; centred (Pan 0) sends equal power to both.

Param	Range	Default	Unit
Pan	-1 – 1	0	—
Out	0 – 1	0	—

Balance 2 inputs

Stereo balance for an existing L/R pair: tilts the level between the two input channels without collapsing them to mono - Balance at -1 mutes the right and passes the left, +1 does the reverse, 0 leaves both untouched. Where Pan places a single mono source and StereoWidth changes the mid/side spread, Balance simply re-weights a stereo signal's sides (the classic balance knob). in0 is L, in1 is R; Out picks which channel this instance emits.

Param	Range	Default	Unit
Balance	-1 – 1	0	—

Param	Range	Default	Unit
Out	0 – 1	0	—

AutoWidth 2 inputs

Auto-widening LFO: a sine LFO swings the mid/side width of the L/R pair so the stereo image rhythmically breathes between narrow and wide. Rate sets the sweep speed, Depth how far the width travels around a centred 1.0, and the LFO is driven from the shared transport time so it is phase-coherent across instances. Where AutoPan moves a mono source side to side and StereoWidth sets a fixed spread, AutoWidth animates the width of an existing stereo signal. in0 is L, in1 is R; Out picks the channel.

Param	Range	Default	Unit
Rate	0.05 – 10	0.5	Hz
Depth	0 – 1	0.7	—
Out	0 – 1	0	—

Crossfeed 2 inputs

Headphone crossfeed: low-passes the opposite channel and bleeds it into this one by Amount, emulating how each ear hears both speakers in a room. Amount sets the bleed level from 0 to 1. Out picks L or R, choosing which channel is the destination and which is the filtered source; eases the hard-panned separation that fatigues on headphones.

Param	Range	Default	Unit
Amount	0 – 1	0.3	—
Out	0 – 1	0	—

Doubler 1 input

Vocal doubler: writes the input to a delay line and reads back a single detuned, delayed copy whose time drifts via a slow c.time-phased LFO for a natural double-tracked feel. Detune scales the drift depth, Delay sets the base delay time, and Out shifts the copy slightly shorter or longer (0.9x or 1.1x of the base) to place it as the left or right voice. Layer two instances on opposite Out settings for a widened stereo double.

Param	Range	Default	Unit
Detune	0 – 30	12	cents
Delay	10 – 50	25	ms
Out	0 – 1	0	—